Mathematics

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Notation List

$x \in E,\ x \notin E$	membership / non-membership
$E \subseteq F,\ \emptyset$	subset relation / empty set
$X \times Y,\ f : X \to Y$	Cartesian product / function from X to Y
$\Longrightarrow,\ \Longleftrightarrow$	implies / if and only if
$:=$	defined to be equal to
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	standard number systems
$\mathbb{F}$	field of scalars, usually $\mathbb{R}$ or $\mathbb{C}$
a, b, c, λ, μ	scalars
$\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0}$	vectors and the zero vector
$\mathcal{U}, \mathcal{V}, \mathcal{W}$	vector spaces or subspaces, depending on context
$I_{\mathcal{V}},\ I_n,\ 0$	identity operators or matrices / zero map or matrix
$\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$	finite ordered list, often a basis
$\text{span}(E),\ \dim(\mathcal{V})$	span of E / dimension of $\mathcal{V}$
$\mathcal{U} + \mathcal{W},\ \mathcal{U} \oplus \mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$\mathbf{v} + \mathcal{U},\ \pi(\mathbf{v})$	coset modulo $\mathcal{U}$ / quotient map, when introduced
$T : \mathcal{V} \to \mathcal{W},\ \mathcal{L}(\mathcal{V}, \mathcal{W})$	linear map / space of linear maps
$ST,\ T\|_{\mathcal{U}},\ T/\mathcal{U}$	composition, restriction, quotient operator
$\ker(T),\ \text{range}(T)$	kernel and range of T
$\text{graph}(T),\ [S, T]$	graph of a map / commutator $ST - TS$
$\mathcal{F}',\ \{A\}'$	commutant of a matrix family / commutant of one matrix
$\mathcal{V}',\ \beta',\ T'$	dual space, dual basis, dual map
$\mathcal{U}^0,\ \mathcal{V}'',\ \Lambda_{\mathcal{V}}$	annihilator, double dual, natural evaluation map
$[\mathbf{v}]_\beta,\ [T]_{\gamma \leftarrow \beta}$	coordinate vector / matrix representation
$C_{\gamma \leftarrow \beta}$	change-of-basis matrix from β-coordinates to γ-coordinates
$A, B, C,\ \text{Mat}(m, n, \mathbb{F})$	matrices / $m \times n$ matrices over $\mathbb{F}$
$A_{ij},\ [a_{ij}],\ A\mathbf{x}$	matrix entries / matrix written by entries / matrix action
$A^{-1},\ A^t,\ A^T,\ A^*$	inverse, transpose, transpose notation, adjoint
$\overline{z},\ \text{tr}(A), \det(A)$	complex conjugate, trace, determinant
$\text{col}(A),\ \text{row}(A)$	column space and row space of A
$\text{rank}(A),\ \text{nullity}(A)$	rank and nullity of A
$A \oplus B,\ A \otimes B,\ \text{vec}(A)$	matrix direct sum, Kronecker product, vectorisation
$M/A,\ M/D$	Schur complement of the block A or D in M
$\|z\|,\ \|\mathbf{v}\|,\ \|T\|_{\text{op}}$	modulus or absolute value, norm, operator norm
$\langle \mathbf{u}, \mathbf{v} \rangle,\ \langle A, B \rangle_F$	inner product / Frobenius inner product
$\mathbf{u} \perp \mathbf{v},\ \mathcal{U}^\perp$	orthogonal vectors / orthogonal complement
$\text{proj}_{\mathbf{v}}(\mathbf{u}),\ P_{\mathcal{U}}$	projection onto a line / orthogonal projection onto $\mathcal{U}$
$\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$	often an orthonormal basis
$\mathbf{e} = (1, \ldots, 1)^t$	all-ones vector
$E_\lambda(T),\ \text{spec}(T)$	eigenspace / spectrum
$G(\lambda, T),\ q_T,\ \mu_T$	generalized eigenspace / characteristic polynomial of T / minimal polynomial of T
$J_k(\lambda),\ \rho(A)$	Jordan block of size k / spectral radius of A
$p_A,\ m_A(\lambda)$	characteristic polynomial of A / algebraic multiplicity of λ
$\mathcal{P}(\mathbb{F}),\ p(T)$	polynomials over $\mathbb{F}$ / polynomial applied to an operator

1 Fundamentals of Higher Mathematics

Generally covers four areas: single-variable calculus, linear algebra, multivariable calculus, and differential equations. As this is fundamentals, the focus is generally on computation instead of proofs.

1.1 Single Variable Calculus

Calculus is the branch of mathematics that studies change and accumulation. It has two main branches:

i. Differential Calculus: Concerned with the rate at which things change (e.g. velocity).
ii. Integral Calculus: Concerned with accumulation (e.g. area under a curve).

Calculus underpins physics, engineering, economics, biology, and more.

Definition 1.1.1. The *derivative* is defined as:

$$\lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$$

Definition 1.1.2. A function $f(x)$ is continuous at x_0 when

$$\lim_{x \to x_0} f(x) = f(x_0)$$

Definition 1.1.3. The *right-hand* and *left-hand* limits are as follows:

i. *Right-Hand Limit*: $\lim_{x \to x_0^+} f(x)$ means $\lim_{x \to x_0} f(x)$ for $x > x_0$
ii. *Left-Hand Limit*: $\lim_{x \to x_0^-} f(x)$ means $\lim_{x \to x_0} f(x)$ for $x < x_0$

Definition 1.1.4. A *removable discontinuity* is where

$$\lim_{x \to x_0^+} f(x) = \lim_{x \to x_0^-} f(x)$$

but not at $f(x_0)$, or if $f(x_0)$ is undefined.

Definition 1.1.5. A *jump discontinuity* is where $\lim_{x \to x_0^+} f(x)$ and $\lim_{x \to x_0^-} f(x)$ exists, but not equal.

Definition 1.1.6. A *infinite discontinuity* is where $\lim_{x \to x_0^+} f(x) \neq \lim_{x \to x_0^-} f(x) \in \{+\infty, -\infty\}$

Theorem 1.1.7. *Differentiability Implies Continuity*
If f is differentiable at x_0, then f is continuous at x_0.

Proof.

$$\lim_{x \to x_0} (f(x) - f(x_0)) = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} \right] (x - x_0) = f'(x_0) \cdot 0 = 0$$

□

Definition 1.1.8. *Product Formula*
The derivative of product of two functions.

$$(u \cdot v)' = u' \cdot v + u \cdot v'$$

Definition 1.1.9. *Quotient Formula*
The derivate of ratio of two differentiable functions. Let $v(x) \neq 0$, then

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Definition 1.1.10. *Chain Rule*
The derivative of composition of two functions.

$$\frac{d}{dt} f(g(t)) = f'(g(t))g'(t)$$

May be used to differentiate implicitly defined functions.

Theorem 1.1.11. *Taylor's Theorem*
Let $k \geq 1$ be an integer and let function $f : \mathbb{R} \to \mathbb{R}$ be k-times differentiable at point $a \in \mathbb{R}$. Then there exists a function $h_k : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = \sum_{i=0}^{k} \frac{f^{(i)}(a)}{i!}(x - a)^i + h_k(x)(x - a)^k$$

where $\lim_{x \to a} h_k(x) = 0$ is the Peano form of the remainder.

Theorem 1.1.12. *Mean Value Theorem*
Let $f : [a, b] \to \mathbb{R}$ be a continuous function on closed interval $[a, b]$ and differentiable on open interval (a, b) where $a < b$. Then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem 1.1.13. *First Fundamental Theorem of Calculus*
Let f be continuous real-valued function on closed interval $[a, b]$ Let F be function defined for all $x \in [a, b]$, by

$$F(x) = \int_a^x f(t)dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on open interval (a, b), and

$$F'(x) = f(x)$$

for all x in (a, b) so F is an antiderivative of f.

Remark 1.1.14. *Change of Variable*
If $f(x) = g(u(x))$, then $du = u'(x)dx$, hence

$$\int g(u)du = \int g(u(x))u'(x)dx = \int f(x)u'(x)dx$$

For definite integrals,

$$\int_{x_1}^{x_2} f(x)u'(x)dx = \int_{u_1}^{u_2} g(u)d(u)$$

where $u_1 = u(x_1), u_2 = u(x_2)$.

Theorem 1.1.15. *Second Fundamental Theorem of Calculus*
Let $f : \mathbb{R} \to \mathbb{R}$ on $[a, b]$ and F a continuous function on $[a, b]$ which is an antiderivative of f in (a, b),

$$F'(x) = f(x)$$

If f is Riemann integrable on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Remark 1.1.16. *Integration by Trigonometric Substitution*
Evaluate trigonometric integral, work backwards to original variable by using $\text{trig}(\text{trig}^{-1})$ with right-triangle.

i. $a^2 - x^2$ use $x = a \sin u$ and $dx = a \cos u \; du$
ii. $a^2 + x^2$ use $x = a \tan u$ and $dx = a \sec^2 u \; du$
iii. $x^2 - a^2$ use $x = a \sec u$ and $dx = a \sec u \tan u \; du$

Remark 1.1.17. *Systematic Integration of Rational Functions*
For a rational function P/Q (ratio of polynomials), by method of partial fractions, factor Q completely into linear and quadratic factors, and write P/Q as a sum of simpler terms.

i. If $Q(x) = (x - a)^m(x - b)^n$, then the expression is

$$\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_m}{(x - a)^m} + \frac{B_1}{x - b} + \frac{B_2}{(x - b)^2} + \cdots + \frac{B_n}{(x - b)^n}$$

ii. If there are quadratic factor such as $(Ax^2 + Bx + C)^p$, then there are terms

$$\frac{a_1x + b_1}{Ax^2 + Bx + C} + \frac{a_2x + b_2}{(Ax^2 + Bx + C)^2} + \cdots + \frac{a_px + b_p}{(Ax^2 + Bx + C)^p}$$

for each factor. To integrate, complete the square and make trigonometric substitution.

Remark 1.1.18. *Integration by Parts*

$$\int_b^a uv' \; dx = uv|_a^b - \int_a^b u'v \; dx$$

This is used when $u'v$ is simpler than uv' (often the case when u' is simpler than u).

Remark 1.1.19. *Arc-Length*
Given $ds = \sqrt{dx^2 + dy^2}$, depending on whether to integrate with respect to x, t, y, this is

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \; dx; \quad ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \; dt; \quad ds = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} \; dy$$

Remark 1.1.20. *Surface Area for Surface of Revolution*

i. Around x-axis: (requires formula for $y = y(x)$)

$$2\pi y \, ds = 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

ii. Around y-axis: (requires formula for $x = x(y)$)

$$2\pi x \, ds = 2\pi x \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} \, dy$$

Remark 1.1.21. *Polar Coordinates*

Given $x = r\cos\theta$, $y = r\sin\theta$ (or more rarely $r = \sqrt{x^2 + y^2}\theta$, $\theta = \tan^{-1}(y/x)$)

i. Find the polar equation for a curve from its equation in (x, y) variables by substitution.
ii. Sketch curves given in polar coordinates and find range of θ for integration.
iii. Integration area formula is then

$$\int_{\theta_1}^{\theta_2} \frac{1}{2} r^2 \, d\theta$$

Theorem 1.1.22. *L'Hopital Rule*

For functions f and g defined on open interval I and differentiable on $I\backslash\{c\}$ for a accumulation point c of I, if

$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0 \quad \text{or} \quad \pm\infty$$

and $g'(x) \neq 0$ for all $x \in I\backslash\{c\}$, and $\lim_{x\to c} \frac{f'(x)}{g'(x)}$ exists, then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}$$

2 Linear Algebra

Linear algebra is the branch of mathematics that studies vectors, matrices, and linear transformations. It provides the foundation for solving systems of equations and analysing multidimensional data. Its applications span across machine learning, computer graphics, physics, finance, and engineering.
References from Garcia and Horn (2017) and Axler (2024).

2.1 Structural Linear Algebra

This is the coordinate-free backbone. The purpose is to understand vector spaces and linear maps abstractly.

2.1.1 Vector Spaces and Linear Structure

The guiding idea is that a vector space is not a list of coordinates first. It is a space of objects that can be added and scaled, and coordinates only appear after a basis is chosen.

Definition 2.1.1. *Vector Space*
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set $\mathcal{V}$ together with operations of vector addition and scalar multiplication such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and all $a, b \in \mathbb{F}$, the following axioms hold:

i. vector addition is commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
ii. vector addition is associative: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
iii. there exists an additive identity $\mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{v} = \mathbf{v}$;
iv. for each $\mathbf{v} \in \mathcal{V}$ there exists an additive inverse $-\mathbf{v} \in \mathcal{V}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$;
v. scalar multiplication has identity: $1\mathbf{v} = \mathbf{v}$;
vi. scalar multiplication is compatible with multiplication in $\mathbb{F}$: $(ab)\mathbf{v} = a(b\mathbf{v})$;
vii. scalar multiplication distributes over vector addition: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
viii. scalar multiplication distributes over addition in $\mathbb{F}$: $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

If $\mathcal{V} = \{\mathbf{0}\}$, then $\mathcal{V}$ is a *zero vector space*; if $\mathcal{V} \neq \{\mathbf{0}\}$, then $\mathcal{V}$ is a *nonzero vector space*.

Definition 2.1.2. *Finitely Nonzero Sequence Space*
Let $\mathbb{F}^{(\mathbb{N})}$ denote the set of all sequences $(x_1, x_2, \ldots)$ of elements of $\mathbb{F}$ such that $x_j = 0$ for all but finitely many $j \in \mathbb{N}$. Addition and scalar multiplication are defined componentwise.

Proposition 2.1.3. *Finitely Nonzero Sequences Form a Vector Space*
The set $\mathbb{F}^{(\mathbb{N})}$ is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication.

Proof. The zero sequence belongs to $\mathbb{F}^{(\mathbb{N})}$. If $\mathbf{x} = (x_1, x_2, \ldots)$ and $\mathbf{y} = (y_1, y_2, \ldots)$ are in $\mathbb{F}^{(\mathbb{N})}$ and $a \in \mathbb{F}$, then the nonzero entries of $a\mathbf{x} + \mathbf{y}$ can occur only among the finitely many indices where $\mathbf{x}$ or $\mathbf{y}$ is nonzero. Thus $a\mathbf{x} + \mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$, and the vector space axioms are inherited componentwise from $\mathbb{F}$.
Structural Commentary: Finite support is preserved by finite linear combinations.
Key Insight: Componentwise operations cannot create infinitely many new nonzero entries. □

Theorem 2.1.4. *Basic Consequences of Vector Space Axioms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then the additive identity and additive inverses are unique. Moreover, for all $a \in \mathbb{F}$ and $\mathbf{v} \in \mathcal{V}$, $0\mathbf{v} = \mathbf{0}$, $a\mathbf{0} = \mathbf{0}$, $(-1)\mathbf{v} = -\mathbf{v}$ and $a\mathbf{v} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. If $\mathbf{0}$ and $\mathbf{0}'$ are additive identities, then $\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'$, so the additive identity is unique.
If $\mathbf{w}$ and $\mathbf{w}'$ are additive inverses of $\mathbf{v}$, then $\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{v} + \mathbf{w}') = (\mathbf{w} + \mathbf{v}) + \mathbf{w}' = \mathbf{0} + \mathbf{w}' = \mathbf{w}'$, so additive inverses are unique. Since $0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$, cancellation gives $0\mathbf{v} = \mathbf{0}$.
Similarly, $a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} + a\mathbf{0}$, so $a\mathbf{0} = \mathbf{0}$. Also $\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 - 1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$, hence $(-1)\mathbf{v}$ is the additive inverse of $\mathbf{v}$, and therefore $(-1)\mathbf{v} = -\mathbf{v}$.
Finally, if $a = 0$ or $\mathbf{v} = \mathbf{0}$, then the preceding identities give $a\mathbf{v} = \mathbf{0}$. Conversely, suppose $a\mathbf{v} = \mathbf{0}$ and $a \neq 0$. Then a^{-1} exists in $\mathbb{F}$, and $\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0}$.
Structural Commentary: These identities are the basic algebraic hygiene of vector spaces. They let later arguments use cancellation, zero vectors, and inverses without reopening the axioms each time.
Key Insight: The vector space axioms force the zero and inverse behaviour to be rigid, not optional structure. □

Corollary 2.1.5. *Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.*

Proof. Let $\mathbf{v} \in \mathcal{V}$. We show that $(-1)\mathbf{v} + \mathbf{v} = \mathbf{0}$. By the vector space axioms and Theorem 2.1.4,

$$(-1)\mathbf{v} + \mathbf{v} = (-1)\mathbf{v} + 1\mathbf{v} = (-1 + 1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$$

Thus $(-1)\mathbf{v}$ is the additive inverse of $\mathbf{v}$, so $(-1)\mathbf{v} = -\mathbf{v}$.
Structural Commentary: Scalar multiplication by -1 is exactly same operation as taking additive inverse.
Key Insight: Negation is not an extra operation; it is already encoded by the field action. □

Definition 2.1.6. A *subspace* of a vector space $\mathcal{V}$ over $\mathbb{F}$ is a subset $\mathcal{U} \subseteq \mathcal{V}$ that is itself a vector space over $\mathbb{F}$ with the same vector addition and scalar multiplication as in $\mathcal{V}$.

Theorem 2.1.7. *Subspace Test*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $\mathcal{U}$ be a nonempty subset of $\mathcal{V}$. Then $\mathcal{U}$ is a subspace of $\mathcal{V}$ if and only if $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and all $c \in \mathbb{F}$.

Proof.
($\Rightarrow$) If $\mathcal{U}$ is a subspace, then $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and all $c \in \mathbb{F}$ because $\mathcal{U}$ is closed under scalar multiplication and vector addition.
($\Leftarrow$) Suppose that $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ whenever $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$. We verify that $\mathcal{U}$ is a vector space with the operations inherited from $\mathcal{V}$. Since $\mathcal{U}$ is nonempty, let $\mathbf{u} \in \mathcal{U}$. By Corollary 2.1.5, $\mathbf{0} = (-1)\mathbf{u} + \mathbf{u} \in \mathcal{U}$. If $\mathbf{u} \in \mathcal{U}$, then $-\mathbf{u} = (-1)\mathbf{u} + \mathbf{0} \in \mathcal{U}$. If $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$, then $c\mathbf{u} = c\mathbf{u} + \mathbf{0} \in \mathcal{U}$ and $\mathbf{u} + \mathbf{v} = 1\mathbf{u} + \mathbf{v} \in \mathcal{U}$. Thus $\mathcal{U}$ contains $\mathbf{0}$ and is closed under additive inverses, scalar multiplication, and vector addition; the remaining vector space axioms are inherited from $\mathcal{V}$. Hence $\mathcal{U}$ is a subspace.

Structural Commentary: The theorem compresses the subspace axioms into one closure condition. This is the practical test used whenever a subset is proposed as a linear space.
Key Insight: To prove a subset is a subspace, prove it survives one affine-looking linear combination. □

Proposition 2.1.8. *Intersection of Subspaces*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$. Then $\mathcal{U} \cap \mathcal{W} = \{\mathbf{v} : \mathbf{v} \in \mathcal{U} \text{ and } \mathbf{v} \in \mathcal{W}\}$ is a subspace of $\mathcal{V}$.

Proof. Since $\mathbf{0} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{W}$, we have $\mathbf{0} \in \mathcal{U} \cap \mathcal{W}$, so the intersection is nonempty. If $\mathbf{u}, \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$ and $c \in \mathbb{F}$, then $c\mathbf{u} + \mathbf{v}$ lies in both $\mathcal{U}$ and $\mathcal{W}$ because both are subspaces. Hence $c\mathbf{u} + \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, and Theorem 2.1.7 shows that $\mathcal{U} \cap \mathcal{W}$ is a subspace of $\mathcal{V}$.
Structural Commentary: Intersections preserve every closure condition shared by the subspaces. The subspace test packages this into one check.
Key Insight: Common vectors of two subspaces still form a subspace because the shared part remains closed under linear combinations. □

Definition 2.1.9. *Lists, Linear Combinations, and Span*
A *list* of vectors in $\mathcal{V}$ is a finite ordered sequence $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$; the empty list is denoted $()$.
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is nonempty and $a_1, \ldots, a_m \in \mathbb{F}$, then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m$ is a *linear combination* of β.
For $E \subseteq \mathcal{V}$, the *span* of E is $\operatorname{span}(E) := \{a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m : m \in \mathbb{N},\ a_j \in \mathbb{F},\ \mathbf{v}_j \in E\}$. $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is a nonempty list, then $\operatorname{span}(\beta) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_m) := \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_m\})$.
A subset $E \subseteq \mathcal{V}$ *spans* $\mathcal{V}$, or is a *spanning set* for $\mathcal{V}$, if $\operatorname{span}(E) = \mathcal{V}$.
A list β in $\mathcal{V}$ *spans* $\mathcal{V}$, or is a *spanning list* for $\mathcal{V}$, if $\operatorname{span}(\beta) = \mathcal{V}$.

Theorem 2.1.10. *Span is the Smallest Containing Subspace*
Let $E \subseteq \mathcal{V}$. Then $\operatorname{span}(E)$ is the smallest subspace of $\mathcal{V}$ containing E.

Proof. If $E = \emptyset$, then $\operatorname{span}(E) = \{\mathbf{0}\}$, which is a subspace. Suppose $E \neq \emptyset$. If $\mathbf{x}, \mathbf{y} \in \operatorname{span}(E)$ and $c \in \mathbb{F}$, then $\mathbf{x}$ and $\mathbf{y}$ are finite linear combinations of vectors in E. Hence $c\mathbf{x} + \mathbf{y}$ is also a finite linear combination of vectors in E, so $c\mathbf{x} + \mathbf{y} \in \operatorname{span}(E)$. By Theorem 2.1.7, $\operatorname{span}(E)$ is a subspace of $\mathcal{V}$.
Each $\mathbf{v} \in E$ satisfies $\mathbf{v} = 1\mathbf{v} \in \operatorname{span}(E)$, so $E \subseteq \operatorname{span}(E)$. If $\mathcal{U}$ is a subspace of $\mathcal{V}$ with $E \subseteq \mathcal{U}$, then closure under vector addition and scalar multiplication forces every finite linear combination of vectors in E to lie in $\mathcal{U}$. Thus $\operatorname{span}(E) \subseteq \mathcal{U}$. Therefore $\operatorname{span}(E)$ is the smallest subspace of $\mathcal{V}$ containing E.
Structural Commentary: Span takes a set of vectors and closes it under all finite linear combinations. The result is precisely the smallest subspace forced by those vectors.
Key Insight: Nothing beyond linear closure added. Every containing subspace must already contain the span. □

Proposition 2.1.11. *Monotonicity of Span*
Let $\mathcal{U}$ and $\mathcal{W}$ be subsets of $\mathcal{V}$. If $\mathcal{U} \subseteq \mathcal{W}$, then $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.

Proof. If $\mathcal{U} = \emptyset$, then $\operatorname{span}(\mathcal{U}) = \{\mathbf{0}\} \subseteq \operatorname{span}(\mathcal{W})$.
Suppose $\mathcal{U} \neq \emptyset$. Every vector in $\operatorname{span}(\mathcal{U})$ is a finite linear combination of vectors in $\mathcal{U}$; since $\mathcal{U} \subseteq \mathcal{W}$, the same expression is a finite linear combination of vectors in $\mathcal{W}$. Hence $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.
Structural Commentary: Enlarging the generating set can only enlarge the collection of linear combinations.
Key Insight: Span is monotone with respect to inclusion of generating sets. □

Definition 2.1.12. *Linear Independence*
A list $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *linearly independent* if $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$ implies $a_1 = \cdots = a_m = 0$. Otherwise the list is linearly dependent.

Proposition 2.1.13. *Unique Coefficients for Independent Lists*
Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a linearly independent list in $\mathcal{V}$. Then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = b_1\mathbf{v}_1 + \cdots + b_m\mathbf{v}_m$ if and only if $a_j = b_j$ for each $j = 1, \dots, m$.

Proof. The displayed equality is equivalent to $(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_m - b_m)\mathbf{v}_m = \mathbf{0}$.
If $a_j = b_j$ for every j, this relation is immediate. Conversely, linear independence forces $a_j - b_j = 0$ for every j, hence $a_j = b_j$ for every j.
Structural Commentary: Linear independence turns a vector equality into coefficient equality. It prevents two different coordinate lists from representing the same linear combination.
Key Insight: Independent vectors do not allow hidden cancellation among their coefficients. □

Proposition 2.1.14. *Appending to a Dependent List*
Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a linearly dependent list in $\mathcal{V}$. Then $\forall \mathbf{v} \in \mathcal{V}$, the list $(\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{v})$ is linearly dependent.

Proof. Since $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is linearly dependent, $\exists$ scalars $a_1, \dots, a_m$, not all zero, s.t. $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
For any $\mathbf{v} \in \mathcal{V}$, $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m + 0\mathbf{v} = \mathbf{0}$ and the scalars $a_1, \dots, a_m, 0$ are not all zero.
Structural Commentary: Dependence is inherited by longer lists because the old nontrivial relation survives after assigning coefficient 0 to every new vector.
Key Insight: Once a list has a redundancy, adding more vectors cannot remove it. □

Definition 2.1.15. *Omitting a Vector from a List*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a list in $\mathcal{V}$ with $m \geq 2$. For $j \in \{1, \dots, m\}$, the list obtained from β by omitting $\mathbf{v}_j$ is denoted $(\mathbf{v}_1, \dots, \widehat{\mathbf{v}_j}, \dots, \mathbf{v}_m)$.

Proposition 2.1.16. *Omitting from an Independent List*
Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a linearly independent list in $\mathcal{V}$ with $m \geq 2$. Then for each $j \in \{1, \dots, m\}$, the list $(\mathbf{v}_1, \dots, \widehat{\mathbf{v}_j}, \dots, \mathbf{v}_m)$ is linearly independent.

Proof. Assign coefficient 0 to $\mathbf{v}_j$. Then any zero relation in the sublist becomes a zero relation in the independent full list, so all coefficients vanish.
Structural Commentary: Linear independence passes to shorter lists because deleting a vector cannot create a new cancellation relation.
Key Insight: Every sublist of an independent list remains independent. □

Proposition 2.1.17. *Adjoining and Removing Vectors from Lists*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a list in $\mathcal{V}$.

(i) If β is linearly independent and $\mathbf{v} \notin \operatorname{span}(\beta)$, then $(\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{v})$ is linearly independent.
(ii) If $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$ is a nontrivial relation and $a_j \neq 0$, then $\operatorname{span}(\mathbf{v}_1, \dots, \widehat{\mathbf{v}_j}, \dots, \mathbf{v}_m) = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_m)$. Consequently, if β spans $\mathcal{V}$, then $(\mathbf{v}_1, \dots, \widehat{\mathbf{v}_j}, \dots, \mathbf{v}_m)$ also spans $\mathcal{V}$.

Proof.

(i) Suppose $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m + a\mathbf{v} = \mathbf{0}$.
If $a \neq 0$, then $\mathbf{v} = -a^{-1}(a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m) \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_m)$, a contradiction. Hence $a = 0$, and linear independence of $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ gives $a_1 = \cdots = a_m = 0$.

(ii) Since $a_j \neq 0$, $\mathbf{v}_j = -\sum_{\substack{i=1 \\ i \neq j}}^{m} \frac{a_i}{a_j}\mathbf{v}_i$, so $\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \widehat{\mathbf{v}_j}, \dots, \mathbf{v}_m)$. Thus every linear combination using $\mathbf{v}_j$ can be rewritten without it, while the reverse inclusion is immediate. Hence the two spans are equal, and the spanning conclusion follows.

Structural Commentary: These are the two basic list moves: add a genuinely new direction, or remove a vector already forced by a dependence relation. Together they control how independence and span change.
Key Insight: A vector outside the span increases independence; a vector solved from a dependence relation can be discarded without changing span. □

Lemma 2.1.18. *Replacement Lemma*
Let $\mathcal{V}$ be nonzero, let $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_r)$ be a spanning list of $\mathcal{V}$, and let $\mathbf{v} \in \mathcal{V}$ be nonzero.
Suppose $\mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_r\mathbf{u}_r$.

(i) Some $c_j \neq 0$.
(ii) If $c_j \neq 0$, then $(\mathbf{v}, \mathbf{u}_1, \dots, \widehat{\mathbf{u}_j}, \dots, \mathbf{u}_r)$ spans $\mathcal{V}$; if β is a basis, this replacement list is a basis.
(iii) If $r \geq 2$, β is a basis, and $\mathbf{v} \notin \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ for some $k \in \{1, \dots, r-1\}$, then $\exists\, j \in \{k+1, \dots, r\}$ such that $(\mathbf{v}, \mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \widehat{\mathbf{u}_j}, \dots, \mathbf{u}_r)$ is a basis of $\mathcal{V}$.

Proof.

(i) If every $c_j = 0$, then $\mathbf{v} = \mathbf{0}$, a contradiction.

(ii) Since $c_j \neq 0$, $\mathbf{u}_j = c_j^{-1}\left(\mathbf{v} - \sum_{i\neq j} c_i\mathbf{u}_i\right)$, so replacing $\mathbf{u}_j$ by $\mathbf{v}$ preserves span. If β is a basis and $a\mathbf{v} + \sum_{i\neq j} b_i\mathbf{u}_i = \mathbf{0}$, substituting $\mathbf{v} = \sum_i c_i\mathbf{u}_i$ gives a relation in β whose $\mathbf{u}_j$-coefficient is ac_j. Since $c_j \neq 0$, $a = 0$; then all $b_i = 0$.

(iii) If all $c_{k+1}, \ldots, c_r = 0$, then $\mathbf{v} \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. Thus some $c_j \neq 0$ with $j > k$, (ii) gives desired basis.

Structural Commentary: Replacement swaps one spanning direction without losing span.
Key Insight: A nonzero coefficient identifies the removable vector. □

Lemma 2.1.19. *Linear Dependence Lemma*
If $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then there exists $j \in \{2, \ldots, m\}$ such that $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$ and removing $\mathbf{v}_j$ does not change the span of the list.

Proof. Because the list is dependent, there exist scalars $a_1, \ldots, a_m$, not all zero, such that $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$. Since $\mathbf{v}_1 \neq \mathbf{0}$, not all of $a_2, \ldots, a_m$ can be zero; otherwise $a_1\mathbf{v}_1 = \mathbf{0}$ would force $a_1 = 0$. Choose j to be the largest index such that $a_j \neq 0$. Then $j \geq 2$ and $\mathbf{v}_j = -\frac{a_1}{a_j}\mathbf{v}_1 - \cdots - \frac{a_{j-1}}{a_j}\mathbf{v}_{j-1}$, so $\mathbf{v}_j$ lies in the span of the preceding vectors. Proposition 2.1.17 shows that removing $\mathbf{v}_j$ does not change the span of the list.
Structural Commentary: This lemma is the ordered version of removing a dependent vector. Choosing the largest nonzero coefficient ensures the removable vector occurs after the first vector.
Key Insight: Linear dependence means at least one later vector can be discarded without losing span. □

Theorem 2.1.20. *In a finite-dimensional vector space, the length of every linearly independent list is at most the length of every spanning list.*

Proof. Let $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ be linearly independent and let $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ span $\mathcal{V}$. Suppose $m > n$. Starting from $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$, inductively build a spanning list of length n whose first k entries are $\mathbf{u}_1, \ldots, \mathbf{u}_k$.
For $k < n$, express $\mathbf{u}_{k+1}$ in the current spanning list. Some coefficient after the first k entries is nonzero; otherwise $\mathbf{u}_{k+1} \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. Lemma 2.1.18(ii) replaces that entry by $\mathbf{u}_{k+1}$ while preserving span.
After n steps, $(\mathbf{u}_1, \ldots, \mathbf{u}_n)$ spans $\mathcal{V}$, so $\mathbf{u}_{n+1} \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$, contradicting independence. Hence $m \leq n$.
Structural Commentary: This comparison theorem is the counting engine behind dimension. It says independence and spanning place opposite pressure on list length.
Key Insight: Independent lists cannot be longer than spanning lists in finite dimensions. □

Definition 2.1.21. *Basis*
A list β in $\mathcal{V}$ is a *basis* of $\mathcal{V}$ if β is linearly independent and $\text{span}(\beta) = \mathcal{V}$.

Proposition 2.1.22. *Unique Coordinate Representation*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a list in $\mathcal{V}$. Then β is a basis of $\mathcal{V}$ if and only if every $\mathbf{v} \in \mathcal{V}$ can be written uniquely as

$$\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n, \qquad a_1, \ldots, a_n \in \mathbb{F}.$$

Proof.

($\Rightarrow$) Suppose β is a basis of $\mathcal{V}$. Since $\text{span}(\beta) = \mathcal{V}$, every $\mathbf{v} \in \mathcal{V}$ has a representation of the displayed form. If $a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$, then $(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_n - b_n)\mathbf{v}_n = \mathbf{0}$. Linear independence of β gives $a_j = b_j$ for each j, so the representation is unique.

($\Leftarrow$) Suppose every $\mathbf{v} \in \mathcal{V}$ has a unique representation of the displayed form. Existence gives $\text{span}(\beta) = \mathcal{V}$. If $a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$, then this representation of $\mathbf{0}$ must equal the representation with all coefficients 0. Hence $a_1 = \cdots = a_n = 0$, so β is linearly independent. Therefore β is a basis of $\mathcal{V}$.

Structural Commentary: A basis is a coordinate system: spanning gives existence of coordinates, and linear independence gives uniqueness.
Key Insight: Coordinates are meaningful when every vector has one representation and no hidden alternative. □

Theorem 2.1.23. *Reducing Spanning Lists*
Every spanning list in a finite-dimensional vector space contains a sublist that is a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a spanning list of $\mathcal{V}$. If β is linearly independent, then β is already a basis. Otherwise, remove a redundant vector without changing the span: if $\mathbf{v}_1 = \mathbf{0}$, remove $\mathbf{v}_1$; if $\mathbf{v}_1 \neq \mathbf{0}$, apply Lemma 2.1.19. Repeating this finite process gives a sublist that still spans $\mathcal{V}$ and is linearly independent.
Structural Commentary: A spanning list may contain redundant vectors. Removing only vectors forced by the others preserves span until a basis remains.
Key Insight: Spanning gives enough vectors; reduction deletes exactly the unnecessary ones. □

Theorem 2.1.24. *Extending Independent Lists*
Every linearly independent list in a finite-dimensional vector space can be extended to a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, and choose a spanning list $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$. Starting with $\mathcal{B} = \beta$, inspect $\mathbf{w}_1, \ldots, \mathbf{w}_n$ in order and append $\mathbf{w}_j$ exactly when $\mathbf{w}_j \notin \operatorname{span}(\mathcal{B})$. Each append preserves linear independence by Proposition 2.1.17. After all $\mathbf{w}_j$ have been inspected, every vector in the spanning list lies in $\operatorname{span}(\mathcal{B})$, so $\mathcal{B}$ spans $\mathcal{V}$.
Structural Commentary: An independent list has no redundancy, but may miss directions. Appending only vectors outside its span fills those directions while preserving independence.
Key Insight: A finite-dimensional independent list becomes a basis once enough new directions are added. □

Corollary 2.1.25. *Every finite-dimensional vector space has a basis, and every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Since $\mathcal{V}$ is finite-dimensional, it has a finite spanning list; Theorem 2.1.23 reduces it to a basis. Let $\mathcal{U} \subseteq \mathcal{V}$ be a subspace. If $\mathcal{U} = \{\mathbf{0}\}$, then $\mathcal{U}$ is finite-dimensional. Otherwise, build a linearly independent list in $\mathcal{U}$ by adjoining vectors outside the current span.
Theorem 2.1.20 bounds the length of this list by the length of any spanning list of $\mathcal{V}$, so the process must terminate. At termination, the list spans $\mathcal{U}$ and hence is a basis of $\mathcal{U}$.
Structural Commentary: Finite-dimensional spaces have finite coordinate systems, and subspaces cannot contain longer independent lists than the ambient space permits.
Key Insight: Finite-dimensionality passes downward to subspaces. □

Theorem 2.1.26. *Basis Length Is Well-Defined*
Any two bases of a finite-dimensional vector space have the same length.

Proof. Let $\beta_1 = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\beta_2 = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be bases of $\mathcal{V}$. Since β_1 is linearly independent and β_2 spans $\mathcal{V}$, Theorem 2.1.20 gives $m \leq n$. Reversing the roles of the two bases gives $n \leq m$. Hence $m = n$.
Structural Commentary: Bases are simultaneously independent and spanning, so the comparison theorem traps their lengths in both directions. This makes dimension well-defined.
Key Insight: Dimension is an intrinsic number, not an artefact of a particular basis. □

Definition 2.1.27. *Dimension*
The zero vector space has *dimension* 0. If $\mathcal{V}$ has a basis of length $n > 0 \in \mathbb{Z}$, then $\mathcal{V}$ is *n-dimensional* and $\dim(\mathcal{V}) = n$. A vector space is *finite-dimensional* if it has finite dimension; otherwise it is *infinite-dimensional*.

Proposition 2.1.28. *Dimension Bounds and Basis Criteria*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \subseteq \mathcal{V}$ is a subspace, then $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$, with equality if and only if $\mathcal{U} = \mathcal{V}$. Moreover, a spanning list of length $\dim(\mathcal{V})$ is automatically a basis, and a linearly independent list of length $\dim(\mathcal{V})$ is automatically a basis.

Proof. Let $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ be a basis of $\mathcal{U}$ and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ a basis of $\mathcal{V}$. Theorem 2.1.20 gives $m \leq n$, so $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$. If $\mathcal{U} = \mathcal{V}$, equality is clear. Conversely, if $m = n$, then the basis of $\mathcal{U}$ is an independent list in $\mathcal{V}$ of length $\dim(\mathcal{V})$; extending it to a basis of $\mathcal{V}$ adds no vectors, so it already spans $\mathcal{V}$. Hence $\mathcal{U} = \mathcal{V}$.
A spanning list of length $\dim(\mathcal{V})$ reduces to a basis without removing vectors, and an independent list of that length extends to a basis without adding vectors.
Structural Commentary: Dimension bounds subspaces and detects full-size lists. Equality of dimensions forces a subspace to be the whole space.
Key Insight: In dimension n, n independent or n spanning vectors already form a basis. □

Definition 2.1.29. *Sums of Subspaces*
If $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$, define $\mathcal{U}_1 + \cdots + \mathcal{U}_m := \{\mathbf{u}_1 + \cdots + \mathbf{u}_m : \mathbf{u}_j \in \mathcal{U}_j\}$. In particular, for subspaces $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$, $\mathcal{U} + \mathcal{W} = \operatorname{span}(\mathcal{U} \cup \mathcal{W}) = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in \mathcal{U},\ \mathbf{w} \in \mathcal{W}\}$. If every vector in this sum has a unique such decomposition, then the sum is direct and we write $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.
For two subspaces, $\mathcal{U} + \mathcal{W}$ is direct if and only if $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$; in this case it is denoted $\mathcal{U} \oplus \mathcal{W}$.

Proposition 2.1.30. *Uniqueness in a Direct Sum*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$ with $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$. Then each vector in $\mathcal{U} \oplus \mathcal{W}$ has a unique representation $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.

Proof. Existence follows from the definition of $\mathcal{U} + \mathcal{W}$. For uniqueness, suppose $\mathbf{u}_1 + \mathbf{w}_1 = \mathbf{u}_2 + \mathbf{w}_2$, $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$. Then $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w}_2 - \mathbf{w}_1 \in \mathcal{U}, \mathcal{W}$, hence equals $\mathbf{0}$. Thus $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$.
Structural Commentary: Trivial intersection prevents cancellation between summands.
Key Insight: Directness means unique decomposition. □

Proposition 2.1.31. *Sum of Subspaces Is the Smallest Containing Subspace*
Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \dots, \mathcal{U}_m$.

Proof. Since $\mathbf{0} \in \mathcal{U}_j$ for each j, we have $\mathbf{0} \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{x} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ and $\mathbf{y} = \mathbf{v}_1 + \cdots + \mathbf{v}_m$ lie in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, then for any $c \in \mathbb{F}$, $c\mathbf{x} + \mathbf{y} = (c\mathbf{u}_1 + \mathbf{v}_1) + \cdots + (c\mathbf{u}_m + \mathbf{v}_m) \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$, because $c\mathbf{u}_j + \mathbf{v}_j \in \mathcal{U}_j$ for each j. By Theorem 2.1.7, $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a subspace of $\mathcal{V}$.
Each $\mathcal{U}_j$ is contained in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ by taking all other summands to be $\mathbf{0}$. If $\mathcal{W}$ is any subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \dots, \mathcal{U}_m$, then closure under addition gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m \in \mathcal{W}$ whenever $\mathbf{u}_j \in \mathcal{U}_j$. Hence $\mathcal{U}_1 + \cdots + \mathcal{U}_m \subseteq \mathcal{W}$, so the sum is the smallest such subspace.
Structural Commentary: Sums of subspaces are obtained by closing several subspaces under finite addition. Minimality follows because every containing subspace is already closed under those sums.
Key Insight: The sum adds no extra structure beyond the linear combinations forced by the given subspaces. □

Proposition 2.1.32. *Equivalent Conditions for a Direct Sum*
For subspaces $\mathcal{U}_1, \dots, \mathcal{U}_m \subseteq \mathcal{V}$, the following are equivalent:

i. $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$;
ii. *every* $\mathbf{v} \in \mathcal{V}$ *has a unique representation* $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ *with* $\mathbf{u}_j \in \mathcal{U}_j$;
iii. $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, *and if* $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ *with* $\mathbf{u}_j \in \mathcal{U}_j$, *then* $\mathbf{u}_1 = \cdots = \mathbf{u}_m = \mathbf{0}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) This follows from the definition of direct sum.
(*ii*) $\Rightarrow$ (*iii*) Every vector lies in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, so $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, then this representation of $\mathbf{0}$ must equal $\mathbf{0} + \cdots + \mathbf{0}$ by uniqueness, so each $\mathbf{u}_j = \mathbf{0}$.
(*iii*) $\Rightarrow$ (*i*) Since $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, every $\mathbf{v} \in \mathcal{V}$ has at least one representation. If $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{w}_1 + \cdots + \mathbf{w}_m$, then $(\mathbf{u}_1 - \mathbf{w}_1) + \cdots + (\mathbf{u}_m - \mathbf{w}_m) = \mathbf{0}$. By iii., $\mathbf{u}_j - \mathbf{w}_j = \mathbf{0}$ for every j, so the representation is unique. Hence $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Structural Commentary: Direct sums require both spanning by the summands and no cancellation among them. The zero-representation condition isolates the no-cancellation part.
Key Insight: Uniqueness is equivalent to saying the only way the summands add to zero is trivially. □

Definition 2.1.33. *Product of Vector Spaces*
Let $\mathcal{V}_1, \dots, \mathcal{V}_m$ be vector spaces over $\mathbb{F}$. Their *product* is $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m := \{(\mathbf{v}_1, \dots, \mathbf{v}_m) : \mathbf{v}_j \in \mathcal{V}_j\}$.
Addition, scalar multiplication defined componentwise: $(\mathbf{u}_1, \dots, \mathbf{u}_m) + (\mathbf{v}_1, \dots, \mathbf{v}_m) := (\mathbf{u}_1 + \mathbf{v}_1, \dots, \mathbf{u}_m + \mathbf{v}_m)$ and $a(\mathbf{v}_1, \dots, \mathbf{v}_m) := (a\mathbf{v}_1, \dots, a\mathbf{v}_m)$.

Proposition 2.1.34. *Products Are Vector Spaces*
If $\mathcal{V}_1, \dots, \mathcal{V}_m$ are vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is a vector space over $\mathbb{F}$ under componentwise ops.

Proof. The additive identity is $(\mathbf{0}_1, \dots, \mathbf{0}_m)$, where $\mathbf{0}_j$ is the zero vector of $\mathcal{V}_j$, and additive inverse of $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is $(-\mathbf{v}_1, \dots, -\mathbf{v}_m)$. All remaining vector space axioms hold componentwise because they hold in each $\mathcal{V}_j$.
Structural Commentary: Product spaces collect several vector spaces into one coordinatewise structure.
Key Insight: Every vector space axiom reduces to the same axiom in each component. □

Theorem 2.1.35. *Dimension of a Product*
If $\mathcal{V}_1, \dots, \mathcal{V}_m$ are finite-dimensional vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is finite-dimensional and

$$\dim(\mathcal{V}_1 \times \cdots \times \mathcal{V}_m) = \dim(\mathcal{V}_1) + \cdots + \dim(\mathcal{V}_m)$$

Proof. For each j, let $(\mathbf{v}_{j,1}, \dots, \mathbf{v}_{j,n_j})$ be a basis of $\mathcal{V}_j$. For every j and k, let $\mathbf{e}_{j,k}$ denote the element of $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ whose jth component is $\mathbf{v}_{j,k}$ and whose other components are zero.
The list of all $\mathbf{e}_{j,k}$ spans the product because each component can be expanded in its own basis. If a linear combination of the $\mathbf{e}_{j,k}$ is zero, then looking at the jth component gives a linear relation among $(\mathbf{v}_{j,1}, \dots, \mathbf{v}_{j,n_j})$, so all coefficients in that component are zero. Hence the list is linearly independent and therefore is a basis of the product. Its length is $n_1 + \cdots + n_m$.
Structural Commentary: A basis for a product is formed by placing each component basis into its own slot.
Key Insight: Product dimension counts the independent directions contributed by each factor. □

Theorem 2.1.36. *Dimension Formula for a Sum*
If $\mathcal{U}, \mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$, then $\dim(\mathcal{U} + \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} \cap \mathcal{W})$.

Proof. Choose a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r)$ of $\mathcal{U} \cap \mathcal{W}$, extend it to a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$, and to a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$. Then $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ spans $\mathcal{U} + \mathcal{W}$ because every vector in the sum is the sum of one vector from $\mathcal{U}$ and one vector from $\mathcal{W}$.
To prove linear independence, suppose $a_1\mathbf{e}_1 + \cdots + a_r\mathbf{e}_r + b_1\mathbf{u}_1 + \cdots + b_p\mathbf{u}_p + c_1\mathbf{w}_1 + \cdots + c_q\mathbf{w}_q = \mathbf{0}$. Then $a_1\mathbf{e}_1 + \cdots + a_r\mathbf{e}_r + b_1\mathbf{u}_1 + \cdots + b_p\mathbf{u}_p$ lies in $\mathcal{U} \cap \mathcal{W}$. Since $(\mathbf{e}_1, \ldots, \mathbf{e}_r)$ is a basis of $\mathcal{U} \cap \mathcal{W}$ and $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p)$ is a basis of $\mathcal{U}$, we get $b_1 = \cdots = b_p = 0$. The remaining relation lies in the basis $(\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$, so $a_1 = \cdots = a_r = c_1 = \cdots = c_q = 0$. Thus the displayed list is a basis of $\mathcal{U} + \mathcal{W}$.
Therefore $\dim(\mathcal{U} + \mathcal{W}) = r + p + q = (r + p) + (r + q) - r$.
Structural Commentary: The formula corrects double-counting: vectors in $\mathcal{U} \cap \mathcal{W}$ are counted once in each subspace. Extending an intersection basis separates shared directions from new directions.
Key Insight: The dimension of a sum is the sum of dimensions minus the overlap. □

Corollary 2.1.37. *Dimension of a Direct Sum*
If $\mathcal{U}$ and $\mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$ with $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, then $\dim(\mathcal{U} \oplus \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W})$.

Proof. Since $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, the sum is direct and $\dim(\mathcal{U} \cap \mathcal{W}) = 0$. The result follows from Theorem 2.1.36.
Structural Commentary: When the overlap is trivial, there is no double-counting in the dimension formula. Direct sums are exactly the case where the overlap term vanishes.
Key Insight: Dimensions add exactly for direct sums. □

Theorem 2.1.38. *Direct Sums and Dimension*
Suppose $\mathcal{V}$ is finite-dimensional and $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum if and only if $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m) = \dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_m)$.

Proof. For each j, choose a basis β_j of $\mathcal{U}_j$, and concatenate these bases into a list β.
This list spans $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ and has length $\dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_m)$.

($\Rightarrow$) If the sum is direct, any linear relation among the vectors in β gives a zero decomposition $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$. Directness gives $\mathbf{u}_j = \mathbf{0}$ for every j, and independence of each β_j gives all coefficients zero. Thus β is a basis of the sum, and the dimensions add.

($\Leftarrow$) If the dimensions add, then β is a spanning list whose length equals the dimension of the sum, so Proposition 2.1.28 makes β a basis. A zero decomposition $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ expands to a zero linear relation in this basis, so every $\mathbf{u}_j = \mathbf{0}$. Hence the sum is direct.

Structural Commentary: Direct sums add dimensions without hidden overlap.
Key Insight: Equality of dimensions means the concatenated bases have no linear redundancy. □

Corollary 2.1.39. *Large Subspaces Must Intersect*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of a finite-dimensional vector space $\mathcal{V}$, and let k be a positive integer.

i. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, then $\mathcal{U} \cap \mathcal{W}$ contains a nonzero vector.
ii. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\mathcal{U} \cap \mathcal{W}$ contains k linearly independent vectors.

Proof. Since $\mathcal{U} + \mathcal{W} \subseteq \mathcal{V}$, Theorem 2.1.36 gives

$$\dim(\mathcal{U} \cap \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} + \mathcal{W}) \geq \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{V})$$

If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, the right-hand side is at least 1, so the intersection contains a nonzero vector. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\dim(\mathcal{U} \cap \mathcal{W}) \geq k$, so intersection contains k linearly independent vectors.
Structural Commentary: Excess dimension appears as overlap.
Key Insight: Two large subspaces cannot fit inside $\mathcal{V}$ without sharing directions. □

Proposition 2.1.40. *Complements Exist in Finite Dimensions*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \subseteq \mathcal{V}$ is a subspace, then there exists a subspace $\mathcal{W}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$.
Let $\mathcal{W} = \operatorname{span}(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. Every vector in $\mathcal{V}$ is a linear combination of this extended basis, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$. If $\mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{v}$ has one representation using only $\mathbf{u}_1, \ldots, \mathbf{u}_m$ and another using only $\mathbf{w}_1, \ldots, \mathbf{w}_n$. Uniqueness of coefficients in the basis of $\mathcal{V}$ forces $\mathbf{v} = \mathbf{0}$. Hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.
Structural Commentary: Complements split a space into a subspace and the remaining basis directions.
Key Insight: Extend a basis of $\mathcal{U}$ to split $\mathcal{V}$. □

Definition 2.1.41. *Affine Subset and Parallel Subspaces*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, the subset $\mathbf{v} + \mathcal{U} := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in \mathcal{U}\}$ is an *affine subset* of $\mathcal{V}$ parallel to $\mathcal{U}$.

Proposition 2.1.42. *Parallel Affine Subsets Are Equal or Disjoint*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$ and let $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The following are equivalent:

(i) $\mathbf{v} - \mathbf{w} \in \mathcal{U}$;

(ii) $\mathbf{v}+\mathcal{U}=\mathbf{w}+\mathcal{U}$;
(iii) $(\mathbf{v}+\mathcal{U})\cap(\mathbf{w}+\mathcal{U})\neq\emptyset$.

Proof.

(i) $\Rightarrow$ *(ii)* If $\mathbf{v}-\mathbf{w}\in\mathcal{U}$ and $\mathbf{u}\in\mathcal{U}$, then $\mathbf{v}+\mathbf{u}=\mathbf{w}+((\mathbf{v}-\mathbf{w})+\mathbf{u})\in\mathbf{w}+\mathcal{U}$. Thus $\mathbf{v}+\mathcal{U}\subseteq\mathbf{w}+\mathcal{U}$, and the reverse inclusion is similar.
(ii) $\Rightarrow$ *(iii)* Equal affine subsets have the same elements.
(iii) $\Rightarrow$ *(i)* If $\mathbf{x}\in(\mathbf{v}+\mathcal{U})\cap(\mathbf{w}+\mathcal{U})$, then $\mathbf{x}=\mathbf{v}+\mathbf{u}_1=\mathbf{w}+\mathbf{u}_2$ for some $\mathbf{u}_1,\mathbf{u}_2\in\mathcal{U}$. Hence $\mathbf{v}-\mathbf{w}=\mathbf{u}_2-\mathbf{u}_1\in\mathcal{U}$.

Structural Commentary: Parallel affine subsets either coincide or do not meet.
Key Insight: Intersection is controlled by the difference of base points. □

Definition 2.1.43. *Quotient Space*
Let $\mathcal{U}\subseteq\mathcal{V}$. The *quotient space* $\mathcal{V}/\mathcal{U}$ is the set of all affine subsets of $\mathcal{V}$ parallel to $\mathcal{U}$ $\mathcal{V}/\mathcal{U}:=\{\mathbf{v}+\mathcal{U}:\mathbf{v}\in\mathcal{V}\}$.

Proposition 2.1.44. *Structure of the Quotient*
Addition and scalar multiplication on $\mathcal{V}/\mathcal{U}$ defined by $(\mathbf{v}+\mathcal{U})+(\mathbf{w}+\mathcal{U}):=(\mathbf{v}+\mathbf{w})+\mathcal{U}$, $a(\mathbf{v}+\mathcal{U}):=a\mathbf{v}+\mathcal{U}$ are well-defined, and with these operations $\mathcal{V}/\mathcal{U}$ is a vector space.

Proof. If $\mathbf{v}+\mathcal{U}=\mathbf{v}'+\mathcal{U}$ and $\mathbf{w}+\mathcal{U}=\mathbf{w}'+\mathcal{U}$, then $\mathbf{v}-\mathbf{v}'\in\mathcal{U}$ and $\mathbf{w}-\mathbf{w}'\in\mathcal{U}$.
Since $\mathcal{U}$ is a subspace, $(\mathbf{v}+\mathbf{w})-(\mathbf{v}'+\mathbf{w}')\in\mathcal{U}$, and $a\mathbf{v}-a\mathbf{v}'\in\mathcal{U}$ so the operations do not depend on the chosen representatives. The vector space axioms are inherited from $\mathcal{V}$ under these operations.
Structural Commentary: Quotients preserve operations modulo $\mathcal{U}$.
Key Insight: Cosets identify vectors differing by elements of $\mathcal{U}$. □

Definition 2.1.45. *Quotient Map*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. The *quotient map* is the map $\pi:\mathcal{V}\to\mathcal{V}/\mathcal{U}$ defined by $\pi(\mathbf{v})=\mathbf{v}+\mathcal{U}$.

Proposition 2.1.46. *Properties of the Quotient Map*
The quotient map π is linear, surjective, and $\ker(\pi)=\mathcal{U}$.

Proof. For $\mathbf{v},\mathbf{w}\in\mathcal{V}$ and $a\in\mathbb{F}$, $\pi(a\mathbf{v}+\mathbf{w})=(a\mathbf{v}+\mathbf{w})+\mathcal{U}=a(\mathbf{v}+\mathcal{U})+(\mathbf{w}+\mathcal{U})=a\pi(\mathbf{v})+\pi(\mathbf{w})$ so π is linear. It is surjective because every coset in $\mathcal{V}/\mathcal{U}$ has the form $\mathbf{v}+\mathcal{U}=\pi(\mathbf{v})$. Finally, $\ker(\pi)=\{\mathbf{v}\in\mathcal{V}:\mathbf{v}+\mathcal{U}=\mathcal{U}\}=\mathcal{U}$.
Structural Commentary: The quotient map collapses $\mathcal{U}$ to zero and records cosets.
Key Insight: Quotient maps realise subspaces as kernels. □

Theorem 2.1.47. *Dimension of a Quotient*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace, then $\dim(\mathcal{V}/\mathcal{U})=\dim(\mathcal{V})-\dim(\mathcal{U})$.

Proof. Choose a basis $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1,\dots,\mathbf{u}_m,\mathbf{w}_1,\dots,\mathbf{w}_k)$ of $\mathcal{V}$.
The cosets $\mathbf{w}_1+\mathcal{U},\dots,\mathbf{w}_k+\mathcal{U}$ span $\mathcal{V}/\mathcal{U}$ because the $\mathbf{u}_j$-terms vanish in the quotient.
If $a_1(\mathbf{w}_1+\mathcal{U})+\cdots+a_k(\mathbf{w}_k+\mathcal{U})=\mathcal{U}$, then $a_1\mathbf{w}_1+\cdots+a_k\mathbf{w}_k\in\mathcal{U}$, so a linear relation in the extended basis forces $a_1=\cdots=a_k=0$. Hence these cosets form a basis of $\mathcal{V}/\mathcal{U}$, and its length is $k=\dim(\mathcal{V})-\dim(\mathcal{U})$.
Structural Commentary: Quotiening removes the directions in $\mathcal{U}$ and keeps the added basis directions.
Key Insight: Passing to $\mathcal{V}/\mathcal{U}$ subtracts the collapsed dimension. □

This section develops two complementary ways to organise a vector space: build from smaller pieces by spans and direct sums, or collapse a chosen subspace by passing to a quotient. Bases and dimension measure exactly what is added, split off, or made invisible.

2.1.2 Linear Maps and Representations

This subsection blends the two standard viewpoints. Axler emphasises linear maps as the primary objects and matrices as secondary coordinate descriptions; Garcia and Horn emphasises how matrices expose structure.

Definition 2.1.48. *Coordinate Vector*
Let $\beta=(\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a basis of a finite-dimensional vector space $\mathcal{V}$.
If $\mathbf{v}=c_1\mathbf{v}_1+\cdots+c_n\mathbf{v}_n$, then the *$\beta$-coordinate vector* of $\mathbf{v}$ is

$$[\mathbf{v}]_\beta:=\begin{bmatrix}c_1\\ \vdots\\ c_n\end{bmatrix}\in\mathbb{F}^n.$$

The function $[\cdot]_\beta:\mathcal{V}\to\mathbb{F}^n$ is the *β-basis representation function*.

Definition 2.1.49. *Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces over the same field $\mathbb{F}$.
A function $T : \mathcal{V} \to \mathcal{W}$ is *linear* if for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and all $a, b \in \mathbb{F}$, $T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$.
The set of all linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted $\mathcal{L}(\mathcal{V}, \mathcal{W})$. If $\mathcal{V} = \mathcal{W}$, then $\mathcal{L}(\mathcal{V})$ is an *operator*.

Definition 2.1.50. *Zero and Identity Maps*
The *zero map* from $\mathcal{V}$ to $\mathcal{W}$ is the function $0_{\mathcal{V},\mathcal{W}} : \mathcal{V} \to \mathcal{W}$ defined by $0_{\mathcal{V},\mathcal{W}}\mathbf{v} = \mathbf{0}_{\mathcal{W}}$.
The *identity map* on $\mathcal{V}$ is $I_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}$ defined by $I_{\mathcal{V}}\mathbf{v} = \mathbf{v}$.

Proposition 2.1.51. *Linear Maps Send Zero to Zero*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$.

Proof. By linearity, $T\mathbf{0}_{\mathcal{V}} = T(\mathbf{0}_{\mathcal{V}} + \mathbf{0}_{\mathcal{V}}) = T\mathbf{0}_{\mathcal{V}} + T\mathbf{0}_{\mathcal{V}}$. Adding the additive inverse of $T\mathbf{0}_{\mathcal{V}}$ gives $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$.
Structural Commentary: Linearity preserves zero.
Key Insight: A linear map cannot move zero away from zero. □

Definition 2.1.52. *Operations on Linear Maps*
For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, define $(S + T)\mathbf{v} := S\mathbf{v} + T\mathbf{v}$, $(aT)\mathbf{v} := a(T\mathbf{v})$ for every $\mathbf{v} \in \mathcal{V}$.

Proposition 2.1.53. *Linear Maps Form a Vector Space*
With the operations above, $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is a vector space over $\mathbb{F}$.

Proof. If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $c \in \mathbb{F}$,

$$(aS + T)(c\mathbf{u} + \mathbf{v}) = a(cS\mathbf{u} + S\mathbf{v}) + (cT\mathbf{u} + T\mathbf{v}) = c(aS + T)\mathbf{u} + (aS + T)\mathbf{v}.$$

Thus $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is closed under linear combinations. The zero map is the additive identity, $-T$ is the additive inverse of T, and the remaining vector space axioms hold pointwise because they hold in $\mathcal{W}$.
Structural Commentary: Linear map operations are pointwise.
Key Insight: Pointwise sums and scalar multiples preserve linearity. □

Definition 2.1.54. *Product of Linear Maps*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *product* $ST : \mathcal{U} \to \mathcal{W}$ is composition: $(ST)\mathbf{u} := S(T\mathbf{u})$ for every $\mathbf{u} \in \mathcal{U}$.

Proposition 2.1.55. *Algebra of Linear Map Products*
Products of linear maps are linear and satisfy associativity, identity laws, and distributive laws whenever the domains and codomains make the expressions meaningful.

Proof. If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $\mathbf{u}, \mathbf{x} \in \mathcal{U}$, and $a \in \mathbb{F}$, then $(ST)(a\mathbf{u}+\mathbf{x}) = S(aT\mathbf{u}+T\mathbf{x}) = a(ST)\mathbf{u}+(ST)\mathbf{x}$, so ST is linear. The identities $(T_1T_2)T_3 = T_1(T_2T_3)$, $TI_{\mathcal{U}} = I_{\mathcal{V}}T = T$, $(S_1 + S_2)T = S_1T + S_2T$, and $S(T_1 + T_2) = ST_1 + ST_2$ follow by evaluating both sides on an arbitrary vector and using associativity and distributivity in the relevant codomain.
Structural Commentary: Composition chains linear actions, while addition distributes through evaluation.
Key Insight: Linear map algebra is function composition plus pointwise vector-space algebra. □

Proposition 2.1.56. *Coordinate Maps Are Isomorphisms*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of a finite-dimensional vector space $\mathcal{V}$.
Then the coordinate map $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is an isomorphism.

Proof. If $\mathbf{u} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ and $\mathbf{v} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$, then for $c \in \mathbb{F}$,

$$[c\mathbf{u} + \mathbf{v}]_\beta = \begin{bmatrix} ca_1 + b_1 \\ \vdots \\ ca_n + b_n \end{bmatrix} = c[\mathbf{u}]_\beta + [\mathbf{v}]_\beta$$

Thus $[\cdot]_\beta$ is linear. It is injective because $[\mathbf{v}]_\beta = \mathbf{0}$ forces all basis coefficients of $\mathbf{v}$ to be zero, and it is surjective because every column $(c_1, \ldots, c_n)^t \in \mathbb{F}^n$ equals $[c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n]_\beta$.
Structural Commentary: A basis converts abstract vectors into coordinate columns without losing structure.
Key Insight: Coordinates are an isomorphic copy of the vector space, not extra information. □

Definition 2.1.57. *Graph of a Function*
For a function $T : \mathcal{V} \to \mathcal{W}$, the *graph* of T is the subset of $\mathcal{V} \times \mathcal{W}$ defined by $\text{graph}(T) := \{(\mathbf{v}, T\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}$.

Proposition 2.1.58. *Linearity and Graphs*
A function $T : \mathcal{V} \to \mathcal{W}$ is linear if and only if graph(T) *is a subspace of $\mathcal{V} \times \mathcal{W}$.*

Proof.

($\Rightarrow$) If T is linear, then $(\mathbf{0},\mathbf{0}) \in \text{graph}(T)$.
If $(\mathbf{u},T\mathbf{u}),(\mathbf{v},T\mathbf{v}) \in \text{graph}(T)$ and $a \in \mathbb{F}$, then $a(\mathbf{u},T\mathbf{u})+(\mathbf{v},T\mathbf{v}) = (a\mathbf{u}+\mathbf{v}, T(a\mathbf{u}+\mathbf{v})) \in \text{graph}(T)$.
Thus $\text{graph}(T)$ is a subspace.

($\Leftarrow$) If $\text{graph}(T)$ is a subspace, then for $\mathbf{u},\mathbf{v} \in \mathcal{V}$ and $a \in \mathbb{F}$, $a(\mathbf{u},T\mathbf{u})+(\mathbf{v},T\mathbf{v}) = (a\mathbf{u}+\mathbf{v}, aT\mathbf{u}+T\mathbf{v})$ lies in $\text{graph}(T)$. Hence $T(a\mathbf{u}+\mathbf{v}) = aT\mathbf{u}+T\mathbf{v}$, so T is linear.

Structural Commentary: A graph records a function as a subset of a product space.
Key Insight: Closure of the graph is exactly linearity of the function. □

Principle 2.1.59. *Coordinate-Free Principle*
A linear map is completely determined by what it does to a basis. Coordinates do not create the map; they only encode it once bases have been chosen.

Proposition 2.1.60. *Linear Maps Are Determined by a Basis*
If $(\mathbf{v}_1,\dots,\mathbf{v}_n)$ is a basis of $\mathcal{V}$ and $\mathbf{w}_1,\dots,\mathbf{w}_n \in \mathcal{W}$ are arbitrary, then there exists a unique linear map $T:\mathcal{V}\to\mathcal{W}$ such that $T(\mathbf{v}_j)=\mathbf{w}_j$ for $j=1,\dots,n$.

Proof. Every $\mathbf{v}\in\mathcal{V}$ has a unique expression $\mathbf{v}=a_1\mathbf{v}_1+\cdots+a_n\mathbf{v}_n$. Define $T(\mathbf{v}) := a_1\mathbf{w}_1+\cdots+a_n\mathbf{w}_n$.
This is well-defined as basis coordinates are unique, and linearity follows by applying the definition to coordinate expansions. Any linear map with the prescribed values on the basis must agree with this formula, so it is unique.
Structural Commentary: A basis reduces a linear map to finitely many choices.
Key Insight: Once basis images are fixed, linearity determines every other value. □

Definition 2.1.61. *Kernel and Range*
For $T\in\mathcal{L}(\mathcal{V},\mathcal{W})$, define $\ker(T) := \{\mathbf{v}\in\mathcal{V} : T(\mathbf{v})=\mathbf{0}\}$, $\text{range}(T) := \{T(\mathbf{v}) : \mathbf{v}\in\mathcal{V}\}$.
The kernel is also called the *nullspace* of T.

Definition 2.1.62. *Injective and Surjective Linear Maps*
A function $T:\mathcal{V}\to\mathcal{W}$ is *injective* if $T\mathbf{u}=T\mathbf{v}$ implies $\mathbf{u}=\mathbf{v}$. It is *surjective* if $\text{range}(T)=\mathcal{W}$.

Proposition 2.1.63. *Kernel and Range Are Subspaces*
If $T\in\mathcal{L}(\mathcal{V},\mathcal{W})$, then $\ker(T)$ is a subspace of $\mathcal{V}$ and $\text{range}(T)$ is a subspace of $\mathcal{W}$.

Proof. Since $T\mathbf{0}=\mathbf{0}$, the kernel is nonempty.
If $\mathbf{u},\mathbf{v}\in\ker(T)$ and $a\in\mathbb{F}$, then $T(a\mathbf{u}+\mathbf{v}) = aT\mathbf{u}+T\mathbf{v} = \mathbf{0}$ so $a\mathbf{u}+\mathbf{v}\in\ker(T)$.
For the range, if $T\mathbf{u},T\mathbf{v}\in\text{range}(T)$, then $aT\mathbf{u}+T\mathbf{v} = T(a\mathbf{u}+\mathbf{v})\in\text{range}(T)$. Thus both sets are subspaces.
Structural Commentary: The kernel and range are the two canonical subspaces attached to a linear map.
Key Insight: Linearity makes both preimages of zero and images of vectors closed under linear combinations. □

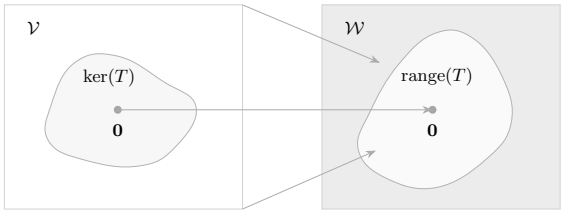


Figure 1: For a linear map $T:\mathcal{V}\to\mathcal{W}$, the range of T may be a proper subspace of $\mathcal{W}$, and the kernel of T may contain nonzero vectors. $\text{range}(T)=\mathcal{W}$ and $\ker(T)=\{\mathbf{0}\}$ characterise surjectivity and injectivity, respectively.

Proposition 2.1.64. *Shift Operators on Finitely Nonzero Sequences*
Define $R,S:\mathbb{F}^{(\mathbb{N})}\to\mathbb{F}^{(\mathbb{N})}$ by $R(x_1,x_2,\dots)=(0,x_1,x_2,\dots)$, $S(x_1,x_2,\dots)=(x_2,x_3,\dots)$. Then $R,S\in\mathcal{L}(\mathbb{F}^{(\mathbb{N})})$. Also, $\ker(R)=\{\mathbf{0}\}$, $\text{range}(R)=\{(x_1,x_2,\dots)\in\mathbb{F}^{(\mathbb{N})} : x_1=0\}$, $\ker(S)=\text{span}((1,0,0,\dots))$, $\text{range}(S)=\mathbb{F}^{(\mathbb{N})}$.

Proof. Both R and S preserve finite support, so they map $\mathbb{F}^{(\mathbb{N})}$ into itself.
Since both maps act componentwise by shifting entries, $R(a\mathbf{x}+\mathbf{y}) = aR\mathbf{x}+R\mathbf{y}$ and $S(a\mathbf{x}+\mathbf{y}) = aS\mathbf{x}+S\mathbf{y}$ for all $\mathbf{x},\mathbf{y}\in\mathbb{F}^{(\mathbb{N})}$ and $a\in\mathbb{F}$. Thus R and S are linear.
The map R has zero kernel as $R\mathbf{x}=\mathbf{0}$ forces every entry of $\mathbf{x}$ to be zero, and its range is exactly the set of finitely nonzero sequences with first entry zero. For S, $S\mathbf{x}=\mathbf{0}$ forces $x_2=x_3=\cdots=0$, so $\ker(S)=\text{span}((1,0,0,\dots))$.
Finally, for any $(y_1,y_2,\dots)\in\mathbb{F}^{(\mathbb{N})}$, we have $S(0,y_1,y_2,\dots)=(y_1,y_2,\dots)$, so S is surjective.
Structural Commentary: Shift operators expose behavior that finite-dimensional maps cannot have.
Key Insight: The right shift is injective but not surjective, while the left shift is surjective but not injective. □

Proposition 2.1.65. *Injectivity and the Kernel*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is injective if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof.

($\Rightarrow$) If T is injective and $\mathbf{v} \in \ker(T)$, then $T\mathbf{v} = \mathbf{0} = T\mathbf{0}$, so $\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $\ker(T) = \{\mathbf{0}\}$ and $T\mathbf{v} = T\mathbf{w}$, then $T(\mathbf{v} - \mathbf{w}) = \mathbf{0}$. Hence $\mathbf{v} - \mathbf{w} \in \ker(T)$, so $\mathbf{v} = \mathbf{w}$.

Structural Commentary: The kernel measures exactly where a linear map collapses vectors to zero.
Key Insight: Injectivity is equivalent to collapsing only the zero vector. □

Proposition 2.1.66. *Direct Sum Map*
Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Define $\Gamma : \mathcal{U}_1 \times \cdots \times \mathcal{U}_m \to \mathcal{U}_1 + \cdots + \mathcal{U}_m$ by $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{u}_1 + \cdots + \mathbf{u}_m$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum if and only if Γ is injective.

Proof. The map Γ is linear, and its kernel consists of all $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ such that $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$.

($\Rightarrow$) If $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum and $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{0}$, uniqueness of the zero decomposition forces each $\mathbf{u}_j = \mathbf{0}$. Hence $\ker(\Gamma) = \{\mathbf{0}\}$, so Γ is injective.
($\Leftarrow$) If Γ is injective and $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, then $(\mathbf{u}_1, \dots, \mathbf{u}_m) \in \ker(\Gamma)$, so every $\mathbf{u}_j = \mathbf{0}$. Thus the only zero decomposition is trivial, which is the direct-sum condition.

Structural Commentary: The sum map packages all decompositions into one linear map.
Key Insight: Directness is injectivity of the map that adds the components. □

Theorem 2.1.67. *Range from Basis Images*
Let $\mathcal{V}$ be finite-dimensional with basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $\operatorname{range}(T) = \operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$.
In particular, $\operatorname{range}(T)$ *is finite-dimensional and* $\dim(\operatorname{range}(T)) \leq n$.

Proof. If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then $T\mathbf{v} = c_1T\mathbf{v}_1 + \cdots + c_nT\mathbf{v}_n$.
Thus every vector in $\operatorname{range}(T)$ lies in $\operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$, and the reverse inclusion is immediate from the definition of range. The dimension bound follows because this spanning list has length n.
Structural Commentary: The range is generated by the images of any basis of the domain.
Key Insight: Linear maps carry a basis-spanning formula directly into a range-spanning formula. □

Theorem 2.1.68. *Rank-Nullity Theorem*
If $\mathcal{V}$ is finite-dim and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $\operatorname{range}(T)$ *is finite-dim and* $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\operatorname{range}(T))$.

Proof. Let $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ be a basis of $\ker(T)$ and extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r)$ of $\mathcal{V}$.
If $\mathbf{v} = \sum_{i=1}^{m} a_i\mathbf{u}_i + \sum_{j=1}^{r} b_j\mathbf{w}_j$, then $T\mathbf{v} = \sum_{j=1}^{r} b_jT\mathbf{w}_j$, so $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ spans $\operatorname{range}(T)$.
If $\sum_{j=1}^{r} c_jT\mathbf{w}_j = \mathbf{0}$, then $\sum_{j=1}^{r} c_j\mathbf{w}_j \in \ker(T)$, so $\sum_{j=1}^{r} c_j\mathbf{w}_j = \sum_{i=1}^{m} d_i\mathbf{u}_i$ for some scalars d_i. Linear independence of the extended basis forces all c_j and d_i to be zero. Hence $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is a basis of $\operatorname{range}(T)$.
Therefore $\dim(\ker(T)) = m$, $\dim(\operatorname{range}(T)) = r$, and $\dim(\mathcal{V}) = m + r$.
Structural Commentary: An adapted basis splits $\mathcal{V}$ into kernel directions and directions detected by T.
Key Insight: Rank-nullity says every domain dimension is either killed by T or appears in its range. □

Corollary 2.1.69. *Maps to Smaller Spaces Are Not Injective*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) > \dim(\mathcal{W})$.
Then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is injective.

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 2.1.68, $\dim(\ker(T)) = \dim(\mathcal{V}) - \dim(\operatorname{range}(T)) \geq \dim(\mathcal{V}) - \dim(\mathcal{W}) > 0$.
Thus $\ker(T)$ contains a nonzero vector, so T is not injective by Proposition 2.1.65.
Structural Commentary: Excess domain dimension forces collapse.
Key Insight: The dimension deficit must appear inside the kernel. □

Corollary 2.1.70. *Maps to Larger Spaces Are Not Surjective*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) < \dim(\mathcal{W})$.
Then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is surjective.

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 2.1.68, $\dim(\operatorname{range}(T)) = \dim(\mathcal{V}) - \dim(\ker(T)) \leq \dim(\mathcal{V}) < \dim(\mathcal{W})$.
Hence $\operatorname{range}(T)$ is a proper subspace of $\mathcal{W}$, so T is not surjective.
Structural Commentary: A smaller domain cannot cover a larger codomain.
Key Insight: The range can never have dimension larger than the domain. □

Definition 2.1.71. *Rank and Nullity*
For a finite-dimensional domain, the rank of T is $\dim(\operatorname{range}(T))$ and the nullity of T is $\dim(\ker(T))$.

Corollary 2.1.72. *Equal-Dimension Kernel-Range Criterion*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) = \dim(\mathcal{W})$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $\ker(T) = \{\mathbf{0}\}$ if and only if $\text{range}(T) = \mathcal{W}$. Equivalently, T is injective if and only if T is surjective.

Proof.
($\Rightarrow$) If $\ker(T) = \{\mathbf{0}\}$, then rank-nullity gives $\dim(\text{range}(T)) = \dim(\mathcal{V}) = \dim(\mathcal{W})$.
Since $\text{range}(T)$ is a subspace of $\mathcal{W}$ with the same dimension, $\text{range}(T) = \mathcal{W}$.
($\Leftarrow$) If $\text{range}(T) = \mathcal{W}$, then rank-nullity gives $\dim(\ker(T)) = \dim(\mathcal{V}) - \dim(\mathcal{W}) = 0$, hence $\ker(T) = \{\mathbf{0}\}$.

Structural Commentary: Equal finite dimensions make injectivity and surjectivity equivalent.
Key Insight: In equal dimensions, having no kernel loss is exactly having full range. □

Definition 2.1.73. *Invertible Linear Map and Isomorphism*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is *invertible* if there exists $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ such that $ST = I_{\mathcal{V}}$ and $TS = I_{\mathcal{W}}$.
An invertible linear map is called an *isomorphism*, and if one exists we write $\mathcal{V} \cong \mathcal{W}$.

Proposition 2.1.74. *Inverse Is Unique*
An invertible linear map has a unique inverse. Thus if T is invertible, its inverse is denoted T^{-1}.

Proof. Suppose $S_1, S_2 \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ are inverses of $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $S_1 = S_1 I_{\mathcal{W}} = S_1(TS_2) = (S_1T)S_2 = I_{\mathcal{V}} S_2 = S_2$. Hence the inverse is unique.
Structural Commentary: A two-sided inverse is forced by the composition identities.
Key Insight: Associativity lets the two candidate inverses meet around T and collapse to the same map. □

Proposition 2.1.75. *Invertibility and Bijectivity*
A linear map is invertible if and only if it is both injective and surjective.

Proof.
($\Rightarrow$) Let S be an inverse of T. If $T\mathbf{v} = T\mathbf{w}$, then $\mathbf{v} = ST\mathbf{v} = ST\mathbf{w} = \mathbf{w}$, so T is injective. Also every $\mathbf{w}$ in the codomain satisfies $\mathbf{w} = T(S\mathbf{w})$, so T is surjective.
($\Leftarrow$) Suppose T is injective and surjective. Define $S\mathbf{w}$ to be the unique vector $\mathbf{v}$ with $T\mathbf{v} = \mathbf{w}$.
Since both sides below map under T to $a\mathbf{w}_1 + \mathbf{w}_2$, $S(a\mathbf{w}_1 + \mathbf{w}_2) = aS\mathbf{w}_1 + S\mathbf{w}_2$. Thus S is linear and satisfies $ST = I$ and $TS = I$, so T is invertible.

Structural Commentary: Bijectivity gives a unique reverse map, and linearity makes it a linear inverse.
Key Insight: A linear map is invertible exactly when it is one-to-one and onto. □

Definition 2.1.76. *Induced Map on a Quotient*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define $\widetilde{T} : \mathcal{V}/\ker(T) \to \mathcal{W}$ by $\widetilde{T}(\mathbf{v} + \ker(T)) = T\mathbf{v}$.

Theorem 2.1.77. *Quotient by the Kernel*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) *$\widetilde{T}$ is a well-defined linear map from $\mathcal{V}/\ker(T)$ to $\mathcal{W}$;*
(ii) *$\widetilde{T}$ is injective;*
(iii) *$\text{range}(\widetilde{T}) = \text{range}(T)$;*
(iv) *$\mathcal{V}/\ker(T) \cong \text{range}(T)$.*

Proof.
(i) If $\mathbf{v} + \ker(T) = \mathbf{v}' + \ker(T)$, then $\mathbf{v} - \mathbf{v}' \in \ker(T)$, so $T\mathbf{v} = T\mathbf{v}'$. Thus $\widetilde{T}$ is well-defined, and linearity follows from the quotient operations.
(ii) If $\widetilde{T}(\mathbf{v} + \ker(T)) = \mathbf{0}$, then $T\mathbf{v} = \mathbf{0}$, so $\mathbf{v} \in \ker(T)$ and $\mathbf{v} + \ker(T) = \ker(T)$. Hence $\ker(\widetilde{T}) = \{\ker(T)\}$.
(iii) By definition, the values of $\widetilde{T}$ are exactly the values of T.
(iv) Parts (ii) and (iii) show that $\widetilde{T}$ is an isomorphism from $\mathcal{V}/\ker(T)$ onto $\text{range}(T)$.

Structural Commentary: Quotienting by the kernel removes exactly the ambiguity of T.
Key Insight: Every linear map factors as a quotient followed by an isomorphism onto its range. □

Theorem 2.1.78. *Finite-Dimensional Invertibility Criteria*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then the following are equivalent:

i. *T is invertible;*
ii. *T is injective;*
iii. *T is surjective.*

Proof.

(i) $\Rightarrow$ (ii) follows from Proposition 2.1.75.
(ii) $\Rightarrow$ (iii): If T is injective, then $\ker(T) = \{\mathbf{0}\}$.
Rank-nullity gives $\dim(\text{range}(T)) = \dim(\mathcal{V})$, hence $\text{range}(T) = \mathcal{V}$, so T is surjective.
(iii) $\Rightarrow$ (i): If T is surjective, rank-nullity gives $\dim(\ker(T)) = 0$, so T is injective. Thus T is bijective, hence invertible by Proposition 2.1.75.

Structural Commentary: For operators on finite-dimensional spaces, kernel and range dimensions trade off exactly. Injectivity and surjectivity therefore force each other.
Key Insight: In equal finite dimensions, losing no information is the same as reaching the whole space. □

Theorem 2.1.79. *Isomorphism and Dimension*
Two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension.

Proof.

($\Rightarrow$) If $T : \mathcal{V} \to \mathcal{W}$ is an isomorphism and $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, then $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is a basis of $\mathcal{W}$.
($\Leftarrow$) If both spaces have dimension n, choose bases $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $(\mathbf{w}_1, \dots, \mathbf{w}_n)$.
By Proposition 2.1.60, there is a linear map T with $T\mathbf{v}_j = \mathbf{w}_j$ for every j. Then $T(\sum a_j \mathbf{v}_j) = \sum a_j \mathbf{w}_j$, so $\ker(T) = \{\mathbf{0}\}$ and $\text{range}(T) = \mathcal{W}$. Hence T is an isomorphism.

Structural Commentary: Dimension classifies finite-dimensional spaces up to isomorphism.
Key Insight: Matching bases give an isomorphism; any isomorphism carries bases to bases. □

Definition 2.1.80. *Matrices, Rows, and Columns*
For positive integers m, n, $\text{Mat}(m, n, \mathbb{F})$ denotes the set of all $m \times n$ matrices with entries in $\mathbb{F}$.
If $A = [A_{j,k}] \in \text{Mat}(m, n, \mathbb{F})$, then $A_{j,k}$ is the entry in row j and column k. The jth row is denoted $A_{j,\cdot}$, and the kth column is denoted $A_{\cdot,k}$. The transpose of A is $A^t \in \text{Mat}(n, m, \mathbb{F})$, defined by $(A^t)_{k,j} = A_{j,k}$.

Definition 2.1.81. *Matrix Operations*
For same size matrices, addition and scalar multiplication are defined $(A+C)_{j,k} = A_{j,k} + C_{j,k}$, $(aA)_{j,k} = aA_{j,k}$.
If $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, p, \mathbb{F})$, then $AB \in \text{Mat}(m, p, \mathbb{F})$ is defined $(AB)_{j,k} := \sum_{r=1}^{n} A_{j,r} B_{r,k}$.

Proposition 2.1.82. *Matrix Spaces*
For positive integers m, n, $\text{Mat}(m, n, \mathbb{F})$ is a vector space over $\mathbb{F}$ with dimension mn.

Proof. Entrywise addition and scalar multiplication satisfy the vector space axioms because $\mathbb{F}$ does. For $1 \le j \le m$ and $1 \le k \le n$, let $E_{j,k}$ be the matrix with 1 in entry (j, k) and 0 elsewhere.

Every $A = [A_{j,k}]$ has the unique expansion $A = \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k} E_{j,k}$.

Thus the mn matrices $E_{j,k}$ form a basis, so $\dim(\text{Mat}(m, n, \mathbb{F})) = mn$.
Structural Commentary: A matrix is a finite array of independent scalar slots.
Key Insight: The standard entry matrices isolate those slots one at a time. □

Proposition 2.1.83. *Matrix Product Entries Are Row Times Column*
If $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, p, \mathbb{F})$, then $(AB)_{j,k} = A_{j,\cdot} B_{\cdot,k}$ for $1 \le j \le m$ and $1 \le k \le p$.

Proof. The row-column product $A_{j,\cdot} B_{\cdot,k}$ is exactly $\sum_{r=1}^{n} A_{j,r} B_{r,k}$, which is the definition of $(AB)_{j,k}$.
Structural Commentary: Matrix multiplication is assembled entry-by-entry from row-column pairings.
Key Insight: Each entry of AB sees one row of A and one column of B. □

Proposition 2.1.84. *Matrix-Vector Products as Column Combinations*
If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \text{Mat}(m, n, \mathbb{F})$ and $\mathbf{x} = (x_1, \dots, x_n)^t \in \mathbb{F}^n$, then $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.

Proof. The jth entry of $A\mathbf{x}$ is $\sum_{k=1}^{n} A_{j,k} x_k$, which is also the jth entry of $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
Structural Commentary: Multiplying by a column selects a linear combination of the columns of A.
Key Insight: The coordinates of $\mathbf{x}$ are precisely the column coefficients. □

Proposition 2.1.85. *Matrix Products by Columns*
If $A \in \text{Mat}(m, r, \mathbb{F})$ and $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \text{Mat}(r, n, \mathbb{F})$, then $AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_n]$.

Proof. For each k, the kth column of AB has entries $\sum_{\ell=1}^{r} A_{j,\ell} B_{\ell,k}$, which are exactly the entries of $A\mathbf{b}_k$.
Structural Commentary: Matrix products apply A to columns of B.
Key Insight: The columns of AB are obtained by applying A to the columns of B. □

Proposition 2.1.86. *Matrix Products as Sums of Outer Products*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_r] \in \mathrm{Mat}(m, r, \mathbb{F})$, *and write* $B \in \mathrm{Mat}(r, n, \mathbb{F})$ *by rows as*

$$B = \begin{bmatrix} \mathbf{b}_1^t \\ \vdots \\ \mathbf{b}_r^t \end{bmatrix}.$$

Then $AB = \mathbf{a}_1\mathbf{b}_1^t + \cdots + \mathbf{a}_r\mathbf{b}_r^t$.

Proof. The (j,k) entry of $\mathbf{a}_\ell\mathbf{b}_\ell^t$ is $A_{j,\ell}B_{\ell,k}$. Summing over $\ell = 1, \ldots, r$ gives $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k} = (AB)_{j,k}$.
Structural Commentary: Column-row products decompose a matrix product into single column-row pieces.
Key Insight: Each shared index ℓ contributes one outer product. □

Proposition 2.1.87. *Transpose Reverses Products*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ *and* $B \in \mathrm{Mat}(n, p, \mathbb{F})$, *then* $(AB)^t = B^tA^t$.

Proof. For $1 \le j \le p$ and $1 \le k \le m$, $((AB)^t)_{j,k} = (AB)_{k,j} = \sum_{\ell=1}^{n} A_{k,\ell}B_{\ell,j} = \sum_{\ell=1}^{n} (B^t)_{j,\ell}(A^t)_{\ell,k} = (B^tA^t)_{j,k}$.
Thus the matrices are equal entrywise.
Structural Commentary: Transposition swaps the order of composition encoded by matrix multiplication.
Key Insight: Rows become columns, so the product order reverses. □

Definition 2.1.88. *Matrix Representation*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ a basis of $\mathcal{W}$. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the matrix of T relative to (β, γ) is $[T]_{\gamma \leftarrow \beta} := \begin{bmatrix} [T(\mathbf{v}_1)]_\gamma & \cdots & [T(\mathbf{v}_n)]_\gamma \end{bmatrix} \in \mathrm{Mat}(m, n, \mathbb{F})$

Definition 2.1.89. *Column Space, Matrix Rank, and Nullity*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{F})$.
The *column space* of A is $\mathrm{col}(A) := \mathrm{span}(\mathbf{a}_1, \ldots, \mathbf{a}_n) \subseteq \mathbb{F}^m$, and the *row space* of A is $\mathrm{row}(A) := \mathrm{col}(A^t) \subseteq \mathbb{F}^n$.
The *column rank* of A is $\dim(\mathrm{col}(A))$, the *row rank* of A is $\dim(\mathrm{row}(A))$, and the *rank* of A is $\mathrm{rank}(A) := \dim(\mathrm{col}(A))$. If $\mathrm{rank}(A) = \min(m, n)$, then A has *full rank*; if $\mathrm{rank}(A) = n$, then A has *full column rank*; if $\mathrm{rank}(A) = m$, then A has *full row rank*.
The *kernel* of A is $\ker(A) := \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{0}\}$, and the *nullity* of A is $\mathrm{nullity}(A) := \dim(\ker(A))$.

Proposition 2.1.90. *Matrices Respect Linear Combinations of Maps*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *be a basis of* $\mathcal{V}$ *and* γ *be a basis of* $\mathcal{W}$.
If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ *and* $a \in \mathbb{F}$, *then* $[aS + T]_{\gamma \leftarrow \beta} = a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.

Proof. The jth column of $[aS + T]_{\gamma \leftarrow \beta}$ is $[(aS + T)\mathbf{v}_j]_\gamma = [aS\mathbf{v}_j + T\mathbf{v}_j]_\gamma = a[S\mathbf{v}_j]_\gamma + [T\mathbf{v}_j]_\gamma$, which is the jth column of $a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.
Structural Commentary: Matrix representation records basis images column-by-column.
Key Insight: Linear combinations of maps become entrywise linear combinations of their matrices. □

Definition 2.1.91. *Linear Map Induced by a Matrix*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$, the *linear map induced by A* is the map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 2.1.92. *Matrix-Induced Maps Are Linear*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$, *then* $T_A : \mathbb{F}^n \to \mathbb{F}^m$ *is linear.*

Proof. For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and $a \in \mathbb{F}$, $T_A(a\mathbf{x} + \mathbf{y}) = A(a\mathbf{x} + \mathbf{y}) = aA\mathbf{x} + A\mathbf{y} = aT_A(\mathbf{x}) + T_A(\mathbf{y})$.
Structural Commentary: Matrix multiplication preserves addition and scalar multiplication.
Key Insight: Every matrix defines a canonical linear map between coordinate spaces. □

Corollary 2.1.93. *Matrix Rank-Nullity*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$. *Then* $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$, $\mathrm{nullity}(A) + \mathrm{rank}(A) = n$.
If $m = n$, *then* $\ker(A) = \{\mathbf{0}\}$ *if and only if* $\mathrm{col}(A) = \mathbb{F}^n$.

Proof. Apply Theorem 2.1.68 to the induced linear map $T_A : \mathbb{F}^n \to \mathbb{F}^m$. By definition, $\ker(T_A) = \ker(A)$ and $\mathrm{range}(T_A) = \mathrm{col}(A)$, giving the displayed identities. When $m = n$, the remaining equivalence follows as:
$(\Rightarrow)$ If $\ker(A) = \{\mathbf{0}\}$, then $\dim(\mathrm{col}(A)) = n$. Since $\mathrm{col}(A) \subseteq \mathbb{F}^n$, Proposition 2.1.28 gives $\mathrm{col}(A) = \mathbb{F}^n$.
$(\Leftarrow)$ If $\mathrm{col}(A) = \mathbb{F}^n$, then $\dim(\mathrm{col}(A)) = n$, so the identity gives $\dim(\ker(A)) = 0$. Hence $\ker(A) = \{\mathbf{0}\}$.
Structural Commentary: Matrix rank-nullity is ordinary rank-nullity written in coordinates.
Key Insight: Kernel loss plus column-space dimension equals the number of columns. □

Theorem 2.1.94. *Row Rank Equals Column Rank*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $\dim(\mathrm{row}(A)) = \dim(\mathrm{col}(A))$. *Equivalently,* $\mathrm{rank}(A^t) = \mathrm{rank}(A)$.

Proof. Let $r = \text{rank}(A)$. If $r = 0$, then $A = 0$, so the result is immediate. Suppose $r > 0$.
Choose columns $\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_r}$ forming a basis of $\text{col}(A)$, and set $B = [\mathbf{a}_{j_1} \ \cdots \ \mathbf{a}_{j_r}] \in \text{Mat}(m, r, \mathbb{F})$.
For each column $\mathbf{a}_k$ of A, there is a unique $\mathbf{x}_k \in \mathbb{F}^r$ such that $\mathbf{a}_k = B\mathbf{x}_k$.
With $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n] \in \text{Mat}(r, n, \mathbb{F})$, we have $A = BX$, and hence $A^t = X^t B^t$. Thus every column of A^t lies in $\text{col}(X^t)$, so $\text{rank}(A^t) \leq \text{rank}(X^t) \leq r = \text{rank}(A)$. Applying the same argument to A^t gives $\text{rank}(A) \leq \text{rank}(A^t)$.
Structural Commentary: Column rank controls row rank by factoring A through a basis of its column space.
Key Insight: Rows and columns measure the same intrinsic rank, viewed from transposed coordinates. □

Theorem 2.1.95. *Full-Rank Products Preserve Rank*
Let $A \in \text{Mat}(m, n, \mathbb{F})$. If $X \in \text{Mat}(p, m, \mathbb{F})$ has full column rank and $Y \in \text{Mat}(n, q, \mathbb{F})$ has full row rank, then $\text{rank}(A) = \text{rank}(XAY)$. This holds when X and Y are square invertible matrices of appropriate sizes.

Proof. Because X has full column rank, the induced map $\mathbf{u} \mapsto X\mathbf{u}$ has kernel $\{\mathbf{0}\}$. Hence $\ker(XA) = \ker(A)$. Since XA and A have the same number of columns, Corollary 2.1.93 gives $\text{rank}(XA) = \text{rank}(A)$.
Because Y has full row rank, Y^t has full column rank. Applying the preceding paragraph to A^t and Y^t, and then using Theorem 2.1.94, gives $\text{rank}(AY) = \text{rank}((AY)^t) = \text{rank}(Y^t A^t) = \text{rank}(A^t) = \text{rank}(A)$.
Combining the two rank-preserving multiplications gives $\text{rank}(XAY) = \text{rank}(A)$.
Structural Commentary: Multiplication by a full-column-rank matrix on the left loses no output information, while full row rank on the right loses no input information.
Key Insight: Rank changes only when multiplication introduces a new kernel or collapses the row side. □

Theorem 2.1.96. *Rank Bounds for Products and Concatenations*
Let $A \in \text{Mat}(m, k, \mathbb{F})$, $B \in \text{Mat}(k, n, \mathbb{F})$, and $C \in \text{Mat}(m, p, \mathbb{F})$. Then $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$ and $\text{rank}(AB) = \text{rank}(A)$ if and only if $\text{col}(AB) = \text{col}(A)$.
Also, $\max(\text{rank}(A), \text{rank}(C)) \leq \text{rank}\begin{bmatrix} A & C \end{bmatrix}$ and $\text{rank}\begin{bmatrix} A & C \end{bmatrix} = \text{rank}(A)$ if and only if $\text{col}(A) = \text{col}(A) + \text{col}(C)$.

Proof. Each column of AB is a linear combination of the columns of A, so $\text{col}(AB) \subseteq \text{col}(A)$. Hence $\text{rank}(AB) \leq \text{rank}(A)$, with equality exactly when the two column spaces are equal. Applying this inequality to $(AB)^t = B^t A^t$ and using Theorem 2.1.94 gives $\text{rank}(AB) \leq \text{rank}(B)$.
For the block matrix, $\text{col}\begin{bmatrix} A & C \end{bmatrix} = \text{col}(A) + \text{col}(C)$, which contains both $\text{col}(A)$ and $\text{col}(C)$. This gives the lower bound, and equality with $\text{rank}(A)$ occurs exactly when adding $\text{col}(C)$ does not enlarge $\text{col}(A)$.
Structural Commentary: Products can only reduce accessible column directions, while horizontal concatenation can only add column directions.
Key Insight: Rank tracks whether a matrix operation preserves, loses, or adds column-space dimension. □

Theorem 2.1.97. *Sylvester Rank Inequality*
Let $A \in \text{Mat}(m, k, \mathbb{F})$ and $B \in \text{Mat}(k, n, \mathbb{F})$. Then $\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$.

Proof. Let $R = \text{col}(B) \subseteq \mathbb{F}^k$. The columns of AB are A applied to the columns of B, so $\text{col}(AB) = A(R)$.
Applying rank-nullity to the restriction of the induced map T_A to R gives

$$\text{rank}(AB) = \dim(A(R)) = \dim(R) - \dim(R \cap \ker(A)) \geq \text{rank}(B) - \dim(\ker(A)).$$

Since A has k columns, Corollary 2.1.93 gives $\dim(\ker(A)) = k - \text{rank}(A)$. Substitution gives desired inequality.
Structural Commentary: Multiplication by A can lose only the part of $\text{col}(B)$ lying in $\ker(A)$. Sylvester's inequality is this loss estimate in dimensions.
Key Insight: The shared middle dimension k bounds how much rank can disappear in the product. □

Theorem 2.1.98. *Rank Triangle Inequality*
If $A, B \in \text{Mat}(m, n, \mathbb{F})$, then $|\text{rank}(A) - \text{rank}(B)| \leq \text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

Proof. Since $\text{col}(A + B) \subseteq \text{col}(A) + \text{col}(B)$, Theorem 2.1.36 gives $\text{rank}(A + B) \leq \dim(\text{col}(A) + \text{col}(B)) \leq \text{rank}(A) + \text{rank}(B)$. Applying this upper bound to $A = (A + B) - B$ gives $\text{rank}(A) \leq \text{rank}(A + B) + \text{rank}(B)$, and applying it to $B = (A + B) - A$ gives $\text{rank}(B) \leq \text{rank}(A + B) + \text{rank}(A)$. These two inequalities are equivalent to the lower bound.
Structural Commentary: The rank of a sum is controlled by the sum of the two column spaces. The reverse bounds come by solving for either summand.
Key Insight: Rank behaves like a dimension-valued size under addition. □

Theorem 2.1.99. *Full-Rank Factorization*
Let $A \in \text{Mat}(m, n, \mathbb{F})$ have rank $r \geq 1$, and let the columns of $X \in \text{Mat}(m, r, \mathbb{F})$ be a basis of $\text{col}(A)$. Then there is a unique $Y \in \text{Mat}(r, n, \mathbb{F})$ such that $A = XY$, and $\text{rank}(X) = \text{rank}(Y) = r$.

Proof. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Since the columns of X form a basis of $\text{col}(A)$, for each j there is a unique $\mathbf{y}_j \in \mathbb{F}^r$ such that $\mathbf{a}_j = X\mathbf{y}_j$. Set $Y = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_n]$. Then $A = [X\mathbf{y}_1 \ \cdots \ X\mathbf{y}_n] = XY$ and uniqueness of the coordinates $\mathbf{y}_j$ gives uniqueness of Y. The columns of X are independent, so $\text{rank}(X) = r$.
By Theorem 2.1.96, $r = \text{rank}(A) = \text{rank}(XY) \leq \text{rank}(Y) \leq r$ because Y has only r rows. Hence $\text{rank}(Y) = r$.
Structural Commentary: A matrix factors through a basis of its column space.
Key Insight: Full-rank factorization separates column directions from their coordinates. □

Definition 2.1.100. *Full-Rank Factorization*
A factorization $A = XY$ with X full column rank and Y full row rank is called a *full-rank factorization* of A.

Theorem 2.1.101. *Completing a Full-Column-Rank Matrix*
If $X \in \text{Mat}(m, n, \mathbb{F})$ has rank $n < m$, then $\exists\ X' \in \text{Mat}(m, m-n, \mathbb{F})$ s.t. $\begin{bmatrix} X & X' \end{bmatrix} \in \text{Mat}(m, m, \mathbb{F})$ is invertible.

Proof. Write $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$. Since $\text{rank}(X) = n$, the columns $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are linearly independent in $\mathbb{F}^m$. By Theorem 2.1.24, extend this list to a basis $(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{x}_{n+1}, \ldots, \mathbf{x}_m)$ of $\mathbb{F}^m$.
Set $X' = [\mathbf{x}_{n+1} \ \cdots \ \mathbf{x}_m]$. Corollary 2.1.113 shows that $\begin{bmatrix} X & X' \end{bmatrix}$ is invertible.
Structural Commentary: Full column rank means the existing columns are independent directions. Extending them to a basis turns the rectangular matrix into an invertible square matrix.
Key Insight: Independent columns can always be completed to a basis in finite dimensions. □

Corollary 2.1.102. *Homogeneous Systems with More Variables Than Equations*
If $A \in \text{Mat}(m, n, \mathbb{F})$ with $n > m$, then the homogeneous system $A\mathbf{x} = \mathbf{0}$ has a nonzero solution.

Proof. The induced map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ has domain dimension larger than codomain dimension, so Corollary 2.1.69 shows that T_A is not injective.
Hence $\ker(A) = \ker(T_A)$ contains a nonzero vector $\mathbf{x}$, and this vector satisfies $A\mathbf{x} = \mathbf{0}$.
Structural Commentary: Extra variables force a nontrivial kernel.
Key Insight: A nontrivial kernel is exactly a nonzero homogeneous solution. □

Corollary 2.1.103. *Inhomogeneous Systems with More Equations Than Variables*
If $A \in \text{Mat}(m, n, \mathbb{F})$ with $m > n$, then there exists $\mathbf{b} \in \mathbb{F}^m$ such that $A\mathbf{x} = \mathbf{b}$ has no solution.

Proof. The induced map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ has domain dimension smaller than codomain dimension, so Corollary 2.1.70 shows that T_A is not surjective. Choose $\mathbf{b} \in \mathbb{F}^m \setminus \text{range}(T_A)$. Then no $\mathbf{x} \in \mathbb{F}^n$ satisfies $A\mathbf{x} = \mathbf{b}$.
Structural Commentary: Too few variables cannot reach every possible right-hand side.
Key Insight: A missing range vector becomes an inconsistent inhomogeneous system. □

Proposition 2.1.104. *Matrix Representation Acts on Coordinates*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, let γ be a basis of $\mathcal{W}$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then for every $\mathbf{v} \in \mathcal{V}$, $[T(\mathbf{v})]_\gamma = [T]_{\gamma \leftarrow \beta}[\mathbf{v}]_\beta$.

Proof. Write $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Then linearity gives $T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \cdots + a_nT(\mathbf{v}_n)$ and the right-hand side is exactly the linear combination of the columns of $[T]_{\gamma \leftarrow \beta}$ with coefficients $a_1, \ldots, a_n$.
Structural Commentary: Matrix representation turns applying T into multiplying coordinate columns.
Key Insight: The columns of $[T]_{\gamma \leftarrow \beta}$ are the coordinate images of the basis vectors. □

Theorem 2.1.105. *Columns of an Invertible Matrix Form a Basis*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \text{Mat}(n, n, \mathbb{F})$ be invertible. Then $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$.

Proof. The columns span $\mathbb{F}^n$ because for any $\mathbf{y} \in \mathbb{F}^n$, writing $\mathbf{x} = A^{-1}\mathbf{y}$ gives $\mathbf{y} = A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
They are linearly independent because if $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{0}$, then $A\mathbf{c} = \mathbf{0}$, so $\mathbf{c} = A^{-1}\mathbf{0} = \mathbf{0}$.
Structural Commentary: An invertible matrix is a coordinate isomorphism of $\mathbb{F}^n$. Its columns are the images of the standard coordinate directions.
Key Insight: Invertibility forces the columns to span and prevents nontrivial column relations. □

Lemma 2.1.106. *Uniqueness of Two-Sided Inverses*
Let $A, B, C \in \text{Mat}(n, n, \mathbb{F})$. If $AB = I$ and $BC = I$, then $A = C$.

Proof. Using $BC = I$ and $AB = I$, $A = AI = A(BC) = (AB)C = IC = C$.
Structural Commentary: A common middle inverse forces the left and right inverses to coincide.
Key Insight: Associativity turns $AB = I = BC$ into $A = C$. □

Theorem 2.1.107. *One-Sided Inverses of Square Matrices*
Let $A, B \in \text{Mat}(n, n, \mathbb{F})$. If $AB = I$, then $BA = I$.

Proof. If $B\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = AB\mathbf{x} = \mathbf{0}$, so the induced map $T_B : \mathbb{F}^n \to \mathbb{F}^n$ is injective. By Theorem 2.1.78, T_B is surjective. For any $\mathbf{y} \in \mathbb{F}^n$, choose $\mathbf{x}$ with $\mathbf{y} = B\mathbf{x}$. Then $BA\mathbf{y} = BAB\mathbf{x} = B\mathbf{x} = \mathbf{y}$.
Thus BA acts as the identity on every vector of $\mathbb{F}^n$, so $BA = I$.
Structural Commentary: In finite dimensions, injectivity and surjectivity coincide for square linear maps. A left inverse makes B injective, hence onto, which forces the missing inverse relation.
Key Insight: A one-sided inverse of a square matrix is automatically two-sided. □

Proposition 2.1.108. *Composition and Matrix Multiplication*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[ST]_{\delta \leftarrow \alpha} = [S]_{\delta \leftarrow \beta}[T]_{\beta \leftarrow \alpha}$ for any bases α of $\mathcal{U}$, β of $\mathcal{V}$, and δ of $\mathcal{W}$.

Proof. For every $\mathbf{u} \in \mathcal{U}$, $[(ST)\mathbf{u}]_\delta = [S]_{\delta \leftarrow \beta}[T\mathbf{u}]_\beta = [S]_{\delta \leftarrow \beta}[T]_{\beta \leftarrow \alpha}[\mathbf{u}]_\alpha$.
Since coordinate columns $[\mathbf{u}]_\alpha$ range over all of $\mathbb{F}^{\dim(\mathcal{U})}$, the representing matrices are equal.
Structural Commentary: Matrix multiplication is the coordinate shadow of composition of linear maps.
Key Insight: Applying T then S becomes multiplying first by the matrix of T, then by the matrix of S. □

Theorem 2.1.109. *Linear Maps and Matrices Are the Same Data*
If $\dim(\mathcal{V}) = n$ and $\dim(\mathcal{W}) = m$, then choosing bases of $\mathcal{V}$ and $\mathcal{W}$ gives isomorphism $\mathcal{L}(\mathcal{V}, \mathcal{W}) \cong \mathrm{Mat}(m, n, \mathbb{F})$. Consequently, $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Proof. Fix bases β of $\mathcal{V}$ and γ of $\mathcal{W}$. The map $T \mapsto [T]_{\gamma \leftarrow \beta}$ is linear. It is injective because a linear map whose matrix is zero sends every basis vector of $\mathcal{V}$ to $\mathbf{0}$, hence is the zero map. It is surjective because, for any $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, there is a unique linear map satisfying $T(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + \cdots + a_{mj}\mathbf{w}_m$, $j = 1, \ldots, n$.
Thus $\mathcal{L}(\mathcal{V}, \mathcal{W}) \cong \mathrm{Mat}(m, n, \mathbb{F})$, whose standard entry matrices form a basis of length mn.
Structural Commentary: Choosing bases converts every linear map into one and only one array of scalars.
Key Insight: A matrix is exactly the list of coordinate images of the domain basis vectors. □

Motivation 2.1.110. *Coordinate Encoding*
This is the key bridge between abstraction and computation. The linear map is basis-free; the matrix is what the map looks like after coordinates have been chosen.

Definition 2.1.111. *Change of Basis Matrix*
Let β and γ be two bases of the same vector space $\mathcal{V}$. The change-of-basis matrix from β-coordinates to γ-coordinates is $C_{\gamma \leftarrow \beta} := [I_\mathcal{V}]_{\gamma \leftarrow \beta}$. Thus $[\mathbf{v}]_\gamma = C_{\gamma \leftarrow \beta}[\mathbf{v}]_\beta$ for every $\mathbf{v} \in \mathcal{V}$.

Theorem 2.1.112. *Change-of-Basis Matrices*
Let $\mathcal{V}$ be n-dimensional and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$, then $C_{\gamma \leftarrow \beta}$ is invertible and $(C_{\gamma \leftarrow \beta})^{-1} = C_{\beta \leftarrow \gamma}$.
(ii) If $S \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, then there is a basis γ of $\mathcal{V}$ such that $S = C_{\gamma \leftarrow \beta}$.

Proof.

(*i*) For any $\mathbf{v} \in \mathcal{V}$, $C_{\beta \leftarrow \gamma}C_{\gamma \leftarrow \beta}[\mathbf{v}]_\beta = [\mathbf{v}]_\beta$, $C_{\gamma \leftarrow \beta}C_{\beta \leftarrow \gamma}[\mathbf{v}]_\gamma = [\mathbf{v}]_\gamma$.
Since $[\cdot]_\beta$ and $[\cdot]_\gamma$ are onto $\mathbb{F}^n$, both products equal I_n. Hence $C_{\gamma \leftarrow \beta}$ is invertible with inverse $C_{\beta \leftarrow \gamma}$.

(*ii*) Write $S^{-1} = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$. For each j, choose the unique $\mathbf{w}_j \in \mathcal{V}$ with $[\mathbf{w}_j]_\beta = \mathbf{s}_j$, and set $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$.
Since columns of S^{-1} form a basis of $\mathbb{F}^n$ and $[\cdot]_\beta$ is an isomorphism, γ is a basis of $\mathcal{V}$.
Moreover $C_{\beta \leftarrow \gamma} = S^{-1}$, so $C_{\gamma \leftarrow \beta} = S$.

Structural Commentary: Change of basis is an invertible coordinate rewrite.
Key Insight: Every basis change is reversible, and every invertible matrix is one. □

Corollary 2.1.113. *Basis Columns Give an Invertible Matrix*
If $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$, then $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible.

Proof. Let $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the standard basis and $\alpha = (\mathbf{a}_1, \ldots, \mathbf{a}_n)$. Then $C_{\epsilon \leftarrow \alpha} = \begin{bmatrix}[\mathbf{a}_1]_\epsilon & \cdots & [\mathbf{a}_n]_\epsilon\end{bmatrix} = A$.
Since α is a basis, Theorem 2.1.112 gives that A is invertible.
Structural Commentary: Basis columns form a change-of-basis matrix.
Key Insight: Changing coordinates is reversible. □

Corollary 2.1.114. *Invertibility and Full Rank*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, A is invertible $\iff \mathrm{rank}(A) = n$.

Proof.

($\Rightarrow$) If A is invertible, its columns form a basis of $\mathbb{F}^n$ by Theorem 2.1.105; hence $\mathrm{rank}(A) = n$.
($\Leftarrow$) If $\mathrm{rank}(A) = n$, then $\mathrm{col}(A) = \mathbb{F}^n$. Thus the columns of A form a basis by Proposition 2.1.28, so Corollary 2.1.113 gives that A is invertible.

Structural Commentary: Full rank is square-matrix invertibility.
Key Insight: Invertibility means the columns fill $\mathbb{F}^n$. □

Corollary 2.1.115. *Transpose Preserves Invertibility*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, the matrix A is invertible if and only if A^t is invertible.

Proof. By Theorem 2.1.94, $\mathrm{rank}(A) = \mathrm{rank}(A^t)$. Thus $\mathrm{rank}(A) = n$ if and only if $\mathrm{rank}(A^t) = n$. Corollary 2.1.114 gives the result.
Structural Commentary: Transposition exchanges rows and columns without changing rank.
Key Insight: A square matrix is invertible exactly when either its rows or columns have full rank. □

Theorem 2.1.116. *Ranks of Matrix Powers Stabilize*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. Then:

(i) $\mathrm{rank}(A^k) \geq \mathrm{rank}(A^{k+1})$ *for* $k = 1, 2, \ldots$;
(ii) if $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+1})$, *then* $\mathrm{col}(A^k) = \mathrm{col}(A^{k+p})$ *and* $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+p})$ *for every* $p = 1, 2, \ldots$;
(iii) if A is not invertible, then there is a least positive integer $q \leq n$ such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Proof.

(i) Since $A^{k+1} = AA^k$, Theorem 2.1.96 gives $\mathrm{rank}(A^{k+1}) \leq \mathrm{rank}(A^k)$.
(ii) The inclusion $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$ and equality of dimensions give $\mathrm{col}(A^{k+1}) = \mathrm{col}(A^k)$. Let $U = \mathrm{col}(A^k)$. Then multiplication by A maps U onto U, so induction gives $\mathrm{col}(A^{k+p}) = U$ for every $p \geq 1$.
(iii) If A is not invertible, then Corollary 2.1.114 gives $\mathrm{rank}(A) \leq n-1$. If all inequalities $\mathrm{rank}(A^k) > \mathrm{rank}(A^{k+1})$ held for $k = 1, \ldots, n$, then $\mathrm{rank}(A^{n+1}) \leq \mathrm{rank}(A) - n < 0$, impossible. Hence equality occurs for some $q \leq n$; choose the least such q.

Structural Commentary: The column spaces of powers form a decreasing chain. In finite dimensions, a decreasing chain cannot keep losing dimension forever.
Key Insight: Once two consecutive ranks agree, the image has stabilized under multiplication by A. □

Definition 2.1.117. *Index of a Matrix*
If $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, define its *index* to be 0. If A is not invertible, its index is the least positive integer q such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Theorem 2.1.118. *Change of Basis Formula*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$ and $S = C_{\gamma \leftarrow \beta}$, then $[T]_{\gamma \leftarrow \gamma} = S[T]_{\beta \leftarrow \beta} S^{-1}$.
(ii) If $S \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, then there is a basis γ of $\mathcal{V}$ such that $[T]_{\gamma \leftarrow \gamma} = S[T]_{\beta \leftarrow \beta} S^{-1}$.

Proof.

(i) Suppose γ is a basis and $S = C_{\gamma \leftarrow \beta}$. By Theorem 2.1.112, $S^{-1} = C_{\beta \leftarrow \gamma}$.
For every $\mathbf{v} \in \mathcal{V}$, $[T\mathbf{v}]_\gamma = S[T\mathbf{v}]_\beta = S[T]_{\beta \leftarrow \beta}[\mathbf{v}]_\beta = S[T]_{\beta \leftarrow \beta} S^{-1}[\mathbf{v}]_\gamma$.
Since $[\cdot]_\gamma$ is onto $\mathbb{F}^n$, the representing matrices are equal.
(ii) If S is invertible, Theorem 2.1.112 supplies a basis γ with $S = C_{\gamma \leftarrow \beta}$, and part (i) applies.

Structural Commentary: Changing basis conjugates operator matrices.
Key Insight: Similarity is a coordinate rewrite. □

Definition 2.1.119. *Similarity*
Matrices $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are similar if there exists an invertible matrix P such that $B = PAP^{-1}$.

Perspective 2.1.120. *Similarity as Same Operator, New Coordinates*
Similarity does not mean two operators are merely related; it means they are the same operator written in two different bases. This is the structural content behind change of basis.

Corollary 2.1.121. *Similarity as One Operator in Two Bases*
For $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, the following are equivalent:

(i) A and B are similar.
(ii) There are an n-dimensional vector space $\mathcal{V}$, bases β, γ of $\mathcal{V}$, and $T \in \mathcal{L}(\mathcal{V})$ s.t. $A = [T]_{\beta \leftarrow \beta}$, $B = [T]_{\gamma \leftarrow \gamma}$.

Proof.

(i) $\Rightarrow$ (ii): Suppose $B = PAP^{-1}$ with P invertible.
Let $\mathcal{V} = \mathbb{F}^n$, let β be the standard basis, and let $T_A(\mathbf{x}) = A\mathbf{x}$. By Theorem 2.1.112, choose γ with $P = C_{\gamma \leftarrow \beta}$. Then $[T_A]_{\beta \leftarrow \beta} = A$, and Theorem 2.1.118 gives $[T_A]_{\gamma \leftarrow \gamma} = B$.
(ii) $\Rightarrow$ (i): If $A = [T]_{\beta \leftarrow \beta}$ and $B = [T]_{\gamma \leftarrow \gamma}$, then $B = C_{\gamma \leftarrow \beta} A C_{\beta \leftarrow \gamma}$ with $C_{\gamma \leftarrow \beta}$ invertible.

Structural Commentary: Similar matrices are one operator in different bases.
Key Insight: Conjugation records a change of basis. □

Theorem 2.1.122. *Similarity and Scalar Shifts*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If A and B are similar, then $A - \lambda I_n$ and $B - \lambda I_n$ are similar for every $\lambda \in \mathbb{F}$.
(ii) If $A - \lambda I_n$ and $B - \lambda I_n$ are similar for some $\lambda \in \mathbb{F}$, then A and B are similar.

Proof.

(i) If $B = PAP^{-1}$ with P invertible, then $B - \lambda I_n = PAP^{-1} - \lambda PP^{-1} = P(A - \lambda I_n)P^{-1}$.
(ii) If $B - \lambda I_n = P(A - \lambda I_n)P^{-1}$ with P invertible, then adding $\lambda I_n = \lambda PP^{-1}$ gives $B = PAP^{-1}$.

Structural Commentary: Scalar shifts commute with change of basis.
Key Insight: The identity matrix is basis-independent. □

Theorem 2.1.123. *Trace and Determinant Are Similarity Invariants*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ are similar, then $\mathrm{tr}(A) = \mathrm{tr}(B)$, $\det(A) = \det(B)$.

Proof. Write $B = PAP^{-1}$ with P invertible. Cyclicity of trace gives $\mathrm{tr}(B) = \mathrm{tr}(PAP^{-1}) = \mathrm{tr}(AP^{-1}P) = \mathrm{tr}(A)$ and multiplicativity of determinant gives $\det(B) = \det(P)\det(A)\det(P^{-1}) = \det(A)$.
Structural Commentary: Trace and determinant ignore the coordinate system used to write an operator.
Key Insight: Conjugating by P inserts P and P^{-1}, which cancel inside trace and determinant. □

Definition 2.1.124. *Equivalence Relation*
A relation is an equivalence relation if it is reflexive, symmetric, and transitive.

Proposition 2.1.125. *Similarity Is an Equivalence Relation*
Similarity is an equivalence relation on $\mathrm{Mat}(n, n, \mathbb{F})$.

Proof. For any A, $A = I_n A I_n^{-1}$, so similarity is reflexive. If $B = PAP^{-1}$, then $A = P^{-1}BP$, so it is symmetric. If $B = PAP^{-1}$ and $C = QBQ^{-1}$, then $C = Q(PAP^{-1})Q^{-1} = (QP)A(QP)^{-1}$ so it is transitive.
Structural Commentary: Similarity groups matrices by change of basis.
Key Insight: Identity, inverse, and composition give the three equivalence properties. □

Definition 2.1.126. *Block Matrices*
Suppose $\mathcal{V} = \mathcal{U}_1 \oplus \mathcal{U}_2$ and $\mathcal{W} = \mathcal{Z}_1 \oplus \mathcal{Z}_2$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, relative to bases adapted to these decompositions, the matrix of T has block form

$$[T] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

where each block records the part of T mapping one summand of the domain into one summand of the codomain.

Definition 2.1.127. *Conformal Block Partitions*
For a block matrix $A = [A_{ij}]$, the submatrices A_{ij} are called *blocks*. Two block partitions are *conformal for addition* when corresponding blocks have the same sizes. They are *conformal for multiplication* when every block product appearing in the usual row-column block multiplication is defined.

Proposition 2.1.128. *Block Matrix Arithmetic*
For conformal 2×2 block matrices,

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{bmatrix}$$

and

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

Proof. Each displayed block consists of the same entries obtained by ordinary entrywise addition or ordinary row-column multiplication, with the common indices grouped according to the partition.
Structural Commentary: This is ordinary matrix arithmetic with indices grouped into rectangular packages.
Key Insight: Conformality is exactly the condition that the grouped row-column products make sense. □

Definition 2.1.129. *Direct Sum of Matrices*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$, the *direct sum* of A and B is

$$A \oplus B := \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \in \mathrm{Mat}(m + p, n + q, \mathbb{F}).$$

Proposition 2.1.130. *Algebra of Matrix Direct Sums*
Whenever the expressions are defined,

(i) $a(A \oplus B) + b(C \oplus D) = (aA + bC) \oplus (aB + bD)$;
(ii) $(A \oplus B)(C \oplus D) = AC \oplus BD$;
(iii) if A *and* B *are square, then* $(A \oplus B)^k = A^k \oplus B^k$ *for every positive integer* k;
(iv) if A *and* B *are square and* p *is a polynomial, then* $p(A \oplus B) = p(A) \oplus p(B)$.

Proof. Parts (i) and (ii) follow immediately from Proposition 2.1.128. Part (iii) follows from (ii) by induction on k, and part (iv) follows by applying (i) to the finite linear combination defining $p(A \oplus B)$.
Structural Commentary: A direct sum matrix keeps the two blocks independent under linear combinations, products, powers, and polynomials.
Key Insight: Block diagonal matrices behave componentwise. □

Lemma 2.1.131. *Identity Blocks Do Not Change Determinant*
Let $A \in \mathrm{Mat}(m, m, \mathbb{F})$ *and let* $n \geq 0$. *Then* $\det(I_n \oplus A) = \det(A) = \det(A \oplus I_n)$.

Proof. For $I_n \oplus A$, expand the determinant along the first n rows. The only nonzero contribution chooses the diagonal entries of I_n, leaving the minor A, so $\det(I_n \oplus A) = \det(A)$. Expanding $A \oplus I_n$ along the last n rows gives the same conclusion. The case $n = 0$ is the empty identity block convention.
Structural Commentary: Identity blocks add directions without changing volume.
Key Insight: In a Laplace expansion through an identity block, exactly one minor survives. □

Theorem 2.1.132. *Determinant of an Upper Block-Triangular Matrix*
Let $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $B \in \mathrm{Mat}(r, s, \mathbb{F})$, *and* $D \in \mathrm{Mat}(s, s, \mathbb{F})$. *Then*

$$\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det(A) \det(D).$$

Proof. Factor

$$\begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} I_r & B \\ 0 & I_s \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & I_s \end{bmatrix}.$$

The first and last factors have determinants $\det(D)$ and $\det(A)$ by Lemma 2.1.131.
The middle factor is upper triangular with every diagonal entry equal to 1, so its determinant is 1. Multiplicativity of determinant gives the formula.
Structural Commentary: Block upper triangular matrices separate determinant contribution into diagonal blocks. The off-diagonal block does not affect the determinant.
Key Insight: Factor the matrix into two block diagonal pieces and one unit upper triangular piece. □

Proposition 2.1.133. *Schur Complement Determinant Formula*
Let

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \qquad A \in \mathrm{Mat}(r, r, \mathbb{F}), \quad D \in \mathrm{Mat}(s, s, \mathbb{F}).$$

If A *is invertible, then the Schur complement of* A *in* M *is* $M/A := D - CA^{-1}B$, $\det(M) = \det(A) \det(M/A)$

Proof. Block Gaussian elimination gives

$$\begin{bmatrix} I_r & 0 \\ -CA^{-1} & I_s \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}.$$

The elimination matrix has determinant 1 by Theorem 2.1.132. Taking determinants and applying Theorem 2.1.132 to the right-hand block triangular matrix gives $\det(M) = \det(A) \det(D - CA^{-1}B)$.
Structural Commentary: Schur complementation is block Gaussian elimination.
Key Insight: Eliminating C splits the determinant into $\det(A)$ times the remaining block determinant. □

Proposition 2.1.134. *Bordered Determinant Formulas*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, *let* $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, *and let* $c \in \mathbb{F}$.

(i) If $c \neq 0$, *then*

$$\det \begin{bmatrix} c & \mathbf{x}^t \\ \mathbf{y} & A \end{bmatrix} = c \det \left(A - \frac{1}{c} \mathbf{y}\mathbf{x}^t \right).$$

(ii) If A *is invertible, then*

$$\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = \left(c - \mathbf{y}^t A^{-1} \mathbf{x} \right) \det(A).$$

Proof.

(i) This is Proposition 2.1.133 with the scalar block c in the upper-left corner.
(ii) This is Proposition 2.1.133 with A as the upper-left block and $D = c$.

Structural Commentary: A bordered determinant is a one-row and one-column Schur complement. The large determinant reduces to a scalar correction or a rank-one correction.
Key Insight: Treat the border as a 1×1 block and apply the Schur complement formula. □

Remark 2.1.135. *Adjugate Form of the Bordered Formula*
If $\operatorname{adj}(A)$ denotes the adjugate matrix, then the second formula in Proposition 2.1.134 can be written as

$$\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = c \det(A) - \mathbf{y}^t \operatorname{adj}(A)\mathbf{x}.$$

This adjugate form remains meaningful even when A is not invertible.

Proposition 2.1.136. *Diagonal Multiplication Scales Rows and Columns*
Let $A = [a_{ij}] \in \operatorname{Mat}(m, n, \mathbb{F})$, let $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, and let $M = \operatorname{diag}(\mu_1, \ldots, \mu_m)$. Then $A\Lambda = [\lambda_j a_{ij}]$ and $MA = [\mu_i a_{ij}]$. Thus right multiplication by Λ scales columns of A, left multiplication by M scales rows of A.

Proof. The (i, j) entry of $A\Lambda$ is $\sum_{r=1}^{n} a_{ir}\Lambda_{rj} = a_{ij}\lambda_j$, and the (i, j) entry of MA is $\sum_{r=1}^{m} M_{ir}a_{rj} = \mu_i a_{ij}$.
Structural Commentary: Diagonal matrices act by independent scalar weights on coordinate directions.
Key Insight: Multiplication on the right scales columns, while multiplication on the left scales rows. □

Proposition 2.1.137. *Inverse of an Upper Block-Triangular Matrix*
If Y and Z are square invertible matrices and X has compatible size, then

$$\begin{bmatrix} Y & X \\ 0 & Z \end{bmatrix}^{-1} = \begin{bmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{bmatrix}.$$

Proof. Multiplying the two displayed block matrices in either order gives

$$\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix},$$

by Proposition 2.1.128. Hence the displayed matrix is the inverse.
Structural Commentary: Upper block-triangular inversion preserves the zero lower-left block. The off-diagonal term is forced by cancelling the original upper-right block.
Key Insight: Inverting a block triangular matrix inverts diagonal blocks and corrects the coupling block. □

Theorem 2.1.138. *Inverse of an Upper Triangular Matrix*
Let $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$ be upper triangular with nonzero diagonal entries. Then A is invertible, A^{-1} is upper triangular, and the diagonal entries of A^{-1} are $a_{11}^{-1}, \ldots, a_{nn}^{-1}$.

Proof. We argue by induction on n. The case $n = 1$ is immediate. For $n > 1$, write

$$A = \begin{bmatrix} B & \mathbf{x} \\ 0 & a_{nn} \end{bmatrix},$$

where B is $(n-1) \times (n-1)$ upper triangular with nonzero diagonal entries. By induction, B^{-1} is upper triangular with diagonal entries $a_{11}^{-1}, \ldots, a_{n-1,n-1}^{-1}$. Proposition 2.1.137 gives

$$A^{-1} = \begin{bmatrix} B^{-1} & -B^{-1}\mathbf{x}a_{nn}^{-1} \\ 0 & a_{nn}^{-1} \end{bmatrix},$$

which is upper triangular and has the asserted diagonal.
Structural Commentary: An upper triangular matrix splits into a smaller upper triangular block and one final scalar block. The block inverse formula preserves this triangular structure.
Key Insight: Nonzero diagonals make blockwise triangular inversion possible. □

Proposition 2.1.139. *Upper Block-Triangular Similarity Shift*
Let $B \in \operatorname{Mat}(m, m, \mathbb{F})$, let $D \in \operatorname{Mat}(n, n, \mathbb{F})$, and let $C, X \in \operatorname{Mat}(m, n, \mathbb{F})$. Then

$$\begin{bmatrix} B & C \\ 0 & D \end{bmatrix} \quad \text{is similar to} \quad \begin{bmatrix} B & C + XD - BX \\ 0 & D \end{bmatrix}.$$

Proof. The matrices $\begin{bmatrix} I & X \\ 0 & I \end{bmatrix}$ and $\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix}$ are inverses, and block multiplication gives

$$\begin{bmatrix} I & X \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} \begin{bmatrix} I & -X \\ 0 & I \end{bmatrix} = \begin{bmatrix} B & C + XD - BX \\ 0 & D \end{bmatrix}.$$

Structural Commentary: Unit upper block-triangular conjugation changes only the coupling block.
Key Insight: Similarity can adjust the connecting block while preserving the diagonal blocks. □

Lemma 2.1.140. *Commuting with a Block Scalar Diagonal Matrix*
Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ *and let* $\Lambda = \lambda_1 I_{n_1} \oplus \cdots \oplus \lambda_k I_{n_k}$.
Partition $A = [A_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ *conformally with* Λ, *so* $A_{ij} \in \mathrm{Mat}(n_i, n_j, \mathbb{F})$.

(i) If A *is block diagonal, then* $A\Lambda = \Lambda A$.
(ii) If $A\Lambda = \Lambda A$ *and* $\lambda_1, \dots, \lambda_k$ *are distinct, then* A *is block diagonal.*

Proof. Block multiplication gives $(A\Lambda)_{ij} = \lambda_j A_{ij}$ and $(\Lambda A)_{ij} = \lambda_i A_{ij}$. If A is block diagonal, then $A_{ij} = 0$ for $i \neq j$, and these two expressions agree for every block. Conversely, if $A\Lambda = \Lambda A$, then $(\lambda_j - \lambda_i)A_{ij} = 0$ for every i, j. Distinctness of the λ_i forces $A_{ij} = 0$ whenever $i \neq j$, so A is block diagonal.
Structural Commentary: Distinct scalar blocks force commuting matrices to respect the block decomposition.
Key Insight: Distinct block scalars make every off-diagonal block visible and force it to vanish. □

Example 2.1.141. *Distinctness Is Necessary*
The distinctness hypothesis in Lemma 2.1.140 cannot be dropped. If $\lambda_1 = \cdots = \lambda_k = 1$, then $\Lambda = I_n$ commutes with every matrix in $\mathrm{Mat}(n, n, \mathbb{F})$, including matrices that are not block diagonal for the chosen partition.

Definition 2.1.142. *Commutator*
For $S, T \in \mathcal{L}(\mathcal{V})$, the *commutator* of S and T is $[S, T] := ST - TS$. For $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, the *matrix commutator* is $[A, B] := AB - BA$.

Proposition 2.1.143. *Trace Obstruction for Commutators*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, *then* $\mathrm{tr}([A, B]) = 0$.
Consequently, if $n \cdot 1_{\mathbb{F}} \neq 0$, *then no nonzero scalar matrix is a commutator.*

Proof. Cyclicity of trace gives $\mathrm{tr}([A, B]) = \mathrm{tr}(AB) - \mathrm{tr}(BA) = 0$.
If $cI_n = [A, B]$, then $nc = \mathrm{tr}(cI_n) = 0$. Under $n \cdot 1_{\mathbb{F}} \neq 0$, this forces $c = 0$.
Structural Commentary: Trace detects the first unavoidable obstruction to being a commutator. Scalar matrices fail this obstruction unless the field characteristic makes n vanish.
Key Insight: The two products AB and BA have the same trace, so their difference has trace zero. □

Lemma 2.1.144. *Scalar Matrices and One-Step Orbits*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ *with* $n \geq 2$.
Then $(\mathbf{x}, A\mathbf{x})$ *is linearly dependent for every* $\mathbf{x} \in \mathbb{F}^n$ *if and only if* A *is a scalar matrix.*

Proof.

(⇒) Suppose $A = cI_n$. Then $A\mathbf{x} = c\mathbf{x}$ for every $\mathbf{x}$, so $(\mathbf{x}, A\mathbf{x})$ is linearly dependent.
(⇐) Suppose $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x}$. For each standard basis vector $\mathbf{e}_i$, there is $a_i \in \mathbb{F}$ such that $A\mathbf{e}_i = a_i\mathbf{e}_i$. For $i \geq 2$, dependence of $\mathbf{e}_1 + \mathbf{e}_i$ and $A(\mathbf{e}_1 + \mathbf{e}_i) = a_1\mathbf{e}_1 + a_i\mathbf{e}_i$ gives $a_1 = a_i$.

Structural Commentary: A scalar matrix preserves every one-dimensional coordinate direction and every diagonal line between two coordinates. These diagonal lines force all coordinate scalars to agree.
Key Insight: If every vector is an eigenvector, then there can be only one eigenvalue. □

Lemma 2.1.145. *Nonscalar Matrices Have a Zero Diagonal Entry After Similarity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ *with* $n \geq 2$.
If A *is not a scalar matrix, then* A *is similar over* $\mathbb{F}$ *to a matrix whose* $(1, 1)$ *entry is* 0.

Proof. By Lemma 2.1.144, there is $\mathbf{x} \in \mathbb{F}^n$ such that $(\mathbf{x}, A\mathbf{x})$ is linearly independent. Extend $(\mathbf{x}, A\mathbf{x})$ to a basis β of $\mathbb{F}^n$. In the basis β, the first column of the matrix of A is the coordinate vector of $A\mathbf{x}$, namely $(0, 1, 0, \dots, 0)^t$. Thus the $(1, 1)$ entry is 0, and this matrix is similar to A.
Structural Commentary: Nonscalarity produces a vector whose first iterate points in a genuinely new direction. Using these two vectors as the start of a basis forces a zero in the first diagonal position.
Key Insight: Put $\mathbf{x}$ first and $A\mathbf{x}$ second in the basis. □

Theorem 2.1.146. *Similarity to a Constant-Diagonal Matrix*
Let $n \geq 1$, *let* $A \in \mathrm{Mat}(n, n, \mathbb{F})$, *and assume* $k \cdot 1_{\mathbb{F}} \neq 0$ *for every* $1 \leq k \leq n$. *Then* A *is similar over* $\mathbb{F}$ *to a matrix whose diagonal entries are all equal to* $\frac{1}{n}\mathrm{tr}(A)$.

Proof. Set $\tau = \frac{1}{n}\operatorname{tr}(A)$ and $B = A - \tau I_n$. Then $\operatorname{tr}(B) = 0$. It is enough to show that every trace-zero matrix is similar to a matrix with zero diagonal, because Theorem 2.1.122 then adds τI_n back to the diagonal.
We prove the trace-zero claim by induction on n. The case $n = 1$ is immediate. Let $n \geq 2$ and let $\operatorname{tr}(B) = 0$. If B is scalar, then $B = cI_n$ and $0 = \operatorname{tr}(B) = nc$, so $c = 0$. If B is not scalar, Lemma 2.1.145 makes B similar to

$$\begin{bmatrix} 0 & * \\ * & B_1 \end{bmatrix}, \qquad \operatorname{tr}(B_1) = 0.$$

By induction, B_1 is similar to a matrix with zero diagonal. Conjugating by a block diagonal matrix that leaves the first coordinate fixed makes the whole matrix have zero diagonal.
Structural Commentary: The proof removes one zero diagonal entry at a time.
Key Insight: Similarity creates the first zero; trace zero handles the smaller block. □

Theorem 2.1.147. *Shoda's Theorem in the Distinct-Diagonal Case*
Let $n \geq 1$ and assume $k \cdot 1_{\mathbb{F}} \neq 0$ for every $1 \leq k \leq n$. For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, the matrix A is a commutator in $\operatorname{Mat}(n, n, \mathbb{F})$ if and only if $\operatorname{tr}(A) = 0$.

Proof.

(⇒) This is Proposition 2.1.143.

(⇐) Suppose $\operatorname{tr}(A) = 0$. By Theorem 2.1.146, there is an invertible S such that $B = S^{-1}AS = [b_{ij}]$ has zero diagonal. Let $X = \operatorname{diag}(1, 2, \ldots, n)$, viewing the integers as scalars in $\mathbb{F}$, and define $Y = [y_{ij}]$ by

$$y_{ij} = \begin{cases} (i-j)^{-1}b_{ij}, & i \neq j, \\ 0, & i = j. \end{cases}$$

The hypothesis ensures that $i - j \neq 0$ in $\mathbb{F}$ whenever $i \neq j$. Thus $XY - YX = [(i-j)y_{ij}] = B$. Hence $A = (SXS^{-1})(SYS^{-1}) - (SYS^{-1})(SXS^{-1})$, so A is a commutator.

Structural Commentary: Similarity reduces trace zero to zero diagonal form.
Key Insight: A distinct diagonal commutator solves off-diagonal entries one at a time. □

Remark 2.1.148. *Field Hypothesis in the Diagonal Proof*
The proof above uses that the integers $1, \ldots, n$ are distinct and that each smaller block size remains nonzero in $\mathbb{F}$. In small characteristic, this diagonal proof must be replaced by a more delicate argument.

Proposition 2.1.149. *Invariant Subspaces and Block Form*
Let $T \in \mathcal{L}(\mathcal{V})$ and suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

(i) If $\mathcal{U}$ is invariant under T, then in a basis adapted to $\mathcal{U} \oplus \mathcal{W}$, matrix of T has upper block-triangular form

$$\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}.$$

(ii) If both $\mathcal{U}$ and $\mathcal{W}$ are invariant under T, then the matrix of T has the block-diagonal form

$$\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}.$$

Proof. Choose basis $(\mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$ and basis $(\mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$, and use concatenation as basis of $\mathcal{V}$.

(i) If $\mathcal{U}$ is invariant, then each $T\mathbf{u}_j$ lies in $\mathcal{U}$, so its coordinates along $\mathbf{w}_1, \ldots, \mathbf{w}_q$ are zero. These entries form the lower-left block.

(ii) If $\mathcal{W}$ is also invariant, then each $T\mathbf{w}_j$ lies in $\mathcal{W}$, so its coordinates along $\mathbf{u}_1, \ldots, \mathbf{u}_p$ are zero. These entries form the upper-right block.

Structural Commentary: Block form records how an operator moves vectors between chosen summands. Invariant summands force the corresponding off-diagonal movement to vanish.
Key Insight: Zero blocks are coordinate evidence of invariant subspaces. □

Perspective 2.1.150. *Matrix Viewpoint on Structure*
Basic block matrices already capture important geometry. Upper block-triangular form records the presence of an invariant subspace; block diagonal form records a genuine decomposition into independent invariant pieces.

2.1.3 Duality and Functional Structure

Duality turns vectors into scalar-valued tests.
The central object is not a new kind of vector, but a linear way to measure vectors.

Definition 2.1.151. *Linear Functional and Dual Space*
A *linear functional* on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$. The *dual space* of $\mathcal{V}$ is $\mathcal{V}' := \mathcal{L}(\mathcal{V}, \mathbb{F})$.

Definition 2.1.152. *Dual Basis*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The *dual basis* of β is the list $\beta' = (\varphi_1, \dots, \varphi_n)$ in $\mathcal{V}'$ defined by

$$\varphi_j(\mathbf{v}_k) = \begin{cases} 1, & k = j, \\ 0, & k \neq j. \end{cases}$$

Existence and uniqueness of each φ_j follow from Proposition 2.1.60.

Theorem 2.1.153. *Dual Basis and Dimension of the Dual Space*
Let $\mathcal{V}$ be finite-dimensional with basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, and let $\beta' = (\varphi_1, \dots, \varphi_n)$ be the dual basis. Then β' is a basis of $\mathcal{V}'$. In particular, $\dim(\mathcal{V}') = \dim(\mathcal{V})$.

Proof. First suppose $a_1\varphi_1 + \cdots + a_n\varphi_n = 0$. Evaluating at $\mathbf{v}_j$ gives $a_j = 0$ for each j, so β' is linearly independent. Now let $\varphi \in \mathcal{V}'$. For $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, $\left(\sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j\right)(\mathbf{v}) = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)c_j = \varphi(\mathbf{v})$.
Thus $\varphi = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j$, so β' spans $\mathcal{V}'$.
Structural Commentary: A basis of $\mathcal{V}$ creates coordinate-reading functionals on $\mathcal{V}$. These functionals form the coordinate system for the dual space.
Key Insight: A linear functional is determined exactly by its values on a basis. □

Definition 2.1.154. *Dual Map*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *dual map* of T is the map $T' : \mathcal{W}' \to \mathcal{V}'$ defined by $T'\varphi := \varphi T$ for every $\varphi \in \mathcal{W}'$.
Equivalently, $(T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v})$.

Proposition 2.1.155. *Dual Maps Are Linear*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$.

Proof. For $\varphi, \psi \in \mathcal{W}'$, $a \in \mathbb{F}$, and $\mathbf{v} \in \mathcal{V}$, $(T'(a\varphi + \psi))(\mathbf{v}) = (a\varphi+\psi)(T\mathbf{v}) = a\varphi(T\mathbf{v})+\psi(T\mathbf{v}) = (aT'\varphi + T'\psi)(\mathbf{v})$.
Structural Commentary: The dual map sends a test on $\mathcal{W}$ to the pulled-back test on $\mathcal{V}$. Linearity is inherited from pointwise linearity of functionals.
Key Insight: Dual maps reverse direction because a functional can test $T\mathbf{v}$ only after T is applied. □

Theorem 2.1.156. *Algebra of Dual Maps*
The dual operation has the following properties:

(i) If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $(aS + T)' = aS' + T'$.
(ii) If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $(ST)' = T'S'$.
(iii) $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Proof.

(i) For $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, $((aS + T)'\varphi)(\mathbf{v}) = \varphi(aS\mathbf{v}+T\mathbf{v}) = a(S'\varphi)(\mathbf{v})+(T'\varphi)(\mathbf{v})$, so $(aS+T)' = aS'+T'$.
(ii) For $\varphi \in \mathcal{W}'$ and $\mathbf{u} \in \mathcal{U}$, $((ST)'\varphi)(\mathbf{u}) = \varphi(S(T\mathbf{u})) = (T'(S'\varphi))(\mathbf{u})$, so $(ST)' = T'S'$.
(iii) For $\varphi \in \mathcal{V}'$ and $\mathbf{v} \in \mathcal{V}$, $(I'_{\mathcal{V}}\varphi)(\mathbf{v}) = \varphi(I_{\mathcal{V}}\mathbf{v}) = \varphi(\mathbf{v})$, so $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Structural Commentary: Duality preserves linear combinations but reverses products. This reversal is the same order reversal seen in transposes.
Key Insight: Pullbacks compose in the opposite order. □

Definition 2.1.157. *Annihilator*
If $\mathcal{U} \subseteq \mathcal{V}$, the *annihilator* of $\mathcal{U}$ is $\mathcal{U}^0 := \{\varphi \in \mathcal{V}' : \varphi(\mathbf{u}) = 0 \text{ for every } \mathbf{u} \in \mathcal{U}\}$.

Proposition 2.1.158. *Annihilators Are Subspaces*
If $\mathcal{U} \subseteq \mathcal{V}$, then $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.

Proof. The zero functional lies in $\mathcal{U}^0$. If $\varphi, \psi \in \mathcal{U}^0$, $a \in \mathbb{F}$, and $\mathbf{u} \in \mathcal{U}$, then $(a\varphi + \psi)(\mathbf{u}) = a\varphi(\mathbf{u}) + \psi(\mathbf{u}) = 0$.
Thus $a\varphi + \psi \in \mathcal{U}^0$, and Theorem 2.1.7 shows that $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.
Structural Commentary: The annihilator collects all linear tests that vanish on a chosen set. Closure follows because vanishing is preserved by linear combinations of tests.
Key Insight: Annihilators are kernels of evaluation constraints. □

Theorem 2.1.159. *Dimension of the Annihilator*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Then $\dim(\mathcal{U}) + \dim(\mathcal{U}^0) = \dim(\mathcal{V})$.

Proof. Choose a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_k)$ of $\mathcal{V}$.
Let $(\varphi_1, \dots, \varphi_m, \psi_1, \dots, \psi_k)$ be the corresponding dual basis.

By Theorem 2.1.153, every $\varphi \in \mathcal{V}'$ has the expansion $\varphi = \sum_{j=1}^{m} \varphi(\mathbf{u}_j)\varphi_j + \sum_{\ell=1}^{k} \varphi(\mathbf{w}_\ell)\psi_\ell$. Thus φ vanishes on $\mathcal{U}$ if and only if the first m coefficients vanish. Hence $\mathcal{U}^0 = \operatorname{span}(\psi_1, \dots, \psi_k)$, so $\dim(\mathcal{U}^0) = k = \dim(\mathcal{V}) - \dim(\mathcal{U})$.
Structural Commentary: Extending a basis of $\mathcal{U}$ separates directions inside $\mathcal{U}$ from complementary directions. The annihilator is dual to the complementary part.
Key Insight: To vanish on $\mathcal{U}$, a functional can only use dual coordinates attached to directions outside $\mathcal{U}$. □

Theorem 2.1.160. *Kernel and Range of the Dual Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then

(i) $\ker(T') = (\operatorname{range}(T))^0$;
(ii) $\dim(\ker(T')) = \dim(\ker(T)) + \dim(\mathcal{W}) - \dim(\mathcal{V})$;
(iii) $\operatorname{range}(T') = (\ker(T))^0$;
(iv) $\dim(\operatorname{range}(T')) = \dim(\operatorname{range}(T))$.

Proof.

(i) For $\varphi \in \mathcal{W}'$, $T'\varphi = 0$ if and only if $\varphi(T\mathbf{v}) = 0$ for every $\mathbf{v} \in \mathcal{V}$, which is equivalent to $\varphi \in (\operatorname{range}(T))^0$.
(ii) By Theorem 2.1.159 and 2.1.68, $\dim(\ker(T')) = \dim(\mathcal{W}) - \dim(\operatorname{range}(T)) = \dim(\mathcal{W}) - \dim(\mathcal{V}) + \dim(\ker(T))$.
(iii) If $\varphi = T'\psi$ and $\mathbf{v} \in \ker(T)$, then $\varphi(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\operatorname{range}(T') \subseteq (\ker(T))^0$. Conversely, suppose $\varphi \in (\ker(T))^0$. Define $\widetilde{\varphi} : \operatorname{range}(T) \to \mathbb{F}$ by $\widetilde{\varphi}(T\mathbf{v}) = \varphi(\mathbf{v})$. This is well-defined because $T\mathbf{v} = T\mathbf{u}$ implies $\mathbf{v} - \mathbf{u} \in \ker(T)$, hence $\varphi(\mathbf{v}) = \varphi(\mathbf{u})$. Extend $\widetilde{\varphi}$ to some $\psi \in \mathcal{W}'$. Then $T'\psi = \varphi$, so $(\ker(T))^0 \subseteq \operatorname{range}(T')$.
(iv) Using Theorem 2.1.153, 2.1.68, and (ii), $\dim(\operatorname{range}(T')) = \dim(\mathcal{W}') - \dim(\ker(T')) = \dim(\operatorname{range}(T))$.

Structural Commentary: Dual kernels are annihilators of ranges, and dual ranges are annihilators of kernels. This is the rank-nullity geometry of T seen through scalar tests.
Key Insight: A functional dies after pulling back by T exactly when it vanishes on everything T can reach. □

Corollary 2.1.161. *Surjectivity and Dual Injectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is surjective if and only if T' is injective.

Proof.

(⇒) Suppose T is surjective. Then $\operatorname{range}(T) = \mathcal{W}$, so Theorem 2.1.160 gives $\ker(T') = \mathcal{W}^0 = \{0\}$.
(⇐) Suppose T' is injective. Then $\ker(T') = \{0\}$, so Theorem 2.1.160 gives $(\operatorname{range}(T))^0 = \{0\}$. By Theorem 2.1.159, $\dim(\operatorname{range}(T)) = \dim(\mathcal{W})$, hence $\operatorname{range}(T) = \mathcal{W}$.

Structural Commentary: A map is onto exactly when no nonzero functional can vanish on its range. Dual injectivity detects this absence of invisible codomain directions.
Key Insight: Surjectivity of T becomes injectivity of the pullback map T'. □

Corollary 2.1.162. *Injectivity and Dual Surjectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is injective if and only if T' is surjective.

Proof.

(⇒) Suppose T is injective. Then $\ker(T) = \{\mathbf{0}\}$, so Theorem 2.1.160 gives $\operatorname{range}(T') = (\ker(T))^0 = \mathcal{V}'$.
(⇐) Suppose T' is surjective. Then $\operatorname{range}(T') = \mathcal{V}'$, so Theorem 2.1.160 gives $(\ker(T))^0 = \mathcal{V}'$. By Theorem 2.1.159, this forces $\dim(\ker(T)) = 0$, hence $\ker(T) = \{\mathbf{0}\}$.

Structural Commentary: Injectivity says there are no nonzero domain directions killed by T. The dual statement says every test on the domain comes by pulling back a test from the codomain.
Key Insight: No kernel directions means no obstruction to extending functionals backward through T. □

Theorem 2.1.163. *Matrix of the Dual Map Is the Transpose*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ be bases of $\mathcal{V}$ and $\mathcal{W}$. If β' and γ' are the corresponding dual bases, then $[T']_{\beta' \leftarrow \gamma'} = ([T]_{\gamma \leftarrow \beta})^t$.

Proof. Let $A = [T]_{\gamma \leftarrow \beta}$ and $C = [T']_{\beta' \leftarrow \gamma'}$. Write $\gamma' = (\psi_1, \dots, \psi_m)$ and $\beta' = (\varphi_1, \dots, \varphi_n)$.
For $1 \le j \le m$, $T'\psi_j = \sum_{r=1}^{n} C_{r,j}\varphi_r$. Evaluating at $\mathbf{v}_k$ gives $C_{k,j} = (T'\psi_j)(\mathbf{v}_k) = \psi_j(T\mathbf{v}_k)$.
Since $T\mathbf{v}_k = \sum_{r=1}^{m} A_{r,k}\mathbf{w}_r$, we get $\psi_j(T\mathbf{v}_k) = A_{j,k}$. Hence $C_{k,j} = A_{j,k}$ for all j, k, so $C = A^t$.
Structural Commentary: Duality turns the matrix of a map into the matrix of its pullback on coordinate functionals. The row-column indexing is exactly transposition.
Key Insight: Coefficients of $T\mathbf{v}_k$ become values of the dual basis functionals on $T\mathbf{v}_k$. □

Definition 2.1.164. *Double Dual Space*
The *double dual space* of $\mathcal{V}$ is $\mathcal{V}'' := (\mathcal{V}')'$. Define $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ by $(\Lambda_{\mathcal{V}}\mathbf{v})(\varphi) = \varphi(\mathbf{v}), \mathbf{v} \in \mathcal{V},\ \varphi \in \mathcal{V}'$.

Theorem 2.1.165. *Natural Map to the Double Dual*
The map $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ satisfies:

(i) $\Lambda_{\mathcal{V}}$ is linear;
(ii) if $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $T'' = (T')'$, then $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}}T$;
(iii) if $\mathcal{V}$ is finite-dimensional, then $\Lambda_{\mathcal{V}}$ is an isomorphism.

Proof.

(i) For fixed $\mathbf{v} \in \mathcal{V}$, the map $\varphi \mapsto \varphi(\mathbf{v})$ is linear on $\mathcal{V}'$, so $\Lambda_{\mathcal{V}}\mathbf{v} \in \mathcal{V}''$.
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $a \in \mathbb{F}$, and $\varphi \in \mathcal{V}'$, $(\Lambda_{\mathcal{V}}(a\mathbf{u} + \mathbf{v}))(\varphi) = \varphi(a\mathbf{u} + \mathbf{v}) = a(\Lambda_{\mathcal{V}}\mathbf{u})(\varphi) + (\Lambda_{\mathcal{V}}\mathbf{v})(\varphi)$, so $\Lambda_{\mathcal{V}}$ is linear.

(ii) If $\psi \in \mathcal{W}'$, then $\left(T''(\Lambda_{\mathcal{V}}\mathbf{v})\right)(\psi) = (\Lambda_{\mathcal{V}}\mathbf{v})(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \left(\Lambda_{\mathcal{W}}(T\mathbf{v})\right)(\psi)$, $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}}T$.

(iii) Suppose $\mathcal{V}$ is finite-dimensional. If $\mathbf{v} \neq \mathbf{0}$, extend $(\mathbf{v})$ to a basis of $\mathcal{V}$ and let φ be the first vector of the corresponding dual basis. Then $\varphi(\mathbf{v}) = 1$, so $\Lambda_{\mathcal{V}}\mathbf{v} \neq 0$. Hence $\Lambda_{\mathcal{V}}$ is injective. Since $\dim(\mathcal{V}'') = \dim(\mathcal{V}') = \dim(\mathcal{V})$, Corollary 2.1.72 shows that $\Lambda_{\mathcal{V}}$ is an isomorphism.

Structural Commentary: The double-dual map sends a vector to its evaluation action on functionals. It is natural because it commutes with every linear map.
Key Insight: Finite-dimensional vectors are completely recovered by how all linear functionals evaluate them. □

Perspective 2.1.166. *Why the Double Dual Is Natural*
A finite-dimensional space $\mathcal{V}$ is isomorphic to $\mathcal{V}'$, but such an isomorphism usually requires choosing a basis. The map $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ requires no choice of basis.

2.2 Geometric Linear Algebra

2.2.1 Inner Product Geometry

Throughout this subsection, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$. Inner products turn linear spaces into geometric spaces: they let us measure length, angle, and orthogonality.

Definition 2.2.1. *Preliminary Terminology*

(i) For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the *dot product* is $\mathbf{x} \cdot \mathbf{y} := x_1y_1 + \cdots + x_ny_n$.
(ii) For $z = a + bi \in \mathbb{C}$, *complex conjugate* is $\overline{z} := a - bi$, *modulus* is $|z| := \sqrt{a^2 + b^2}$. Thus $|z|^2 = z\overline{z}$.
(iii) A map $f : \mathcal{V} \to \mathbb{F}$ is *conjugate linear* if $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$ and $f(\lambda\mathbf{v}) = \overline{\lambda}f(\mathbf{v})$.
(iv) A function $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is *sesquilinear* if it is linear in one slot and conjugate linear in the other.

Motivation 2.2.2. *From Dot Products to Inner Products*
For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the Euclidean length is $\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}$.
The expression under the square root is the dot product of $\mathbf{x}$ with itself. Over $\mathbb{C}$, conjugates are needed to keep squared length real and nonnegative: $\|\mathbf{z}\|^2 = |z_1|^2 + \cdots + |z_n|^2 = z_1\overline{z_1} + \cdots + z_n\overline{z_n}$.
Thus the standard inner product on $\mathbb{F}^n$ is $\langle \mathbf{x}, \mathbf{y} \rangle := x_1\overline{y_1} + \cdots + x_n\overline{y_n}$.

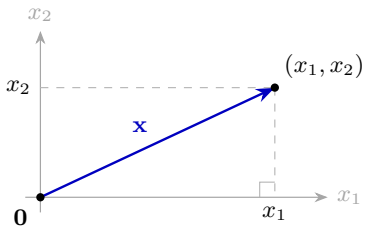


Figure 2: The norm is vector length, with $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$.

Definition 2.2.3. *Inner Product*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. An *inner product* on $\mathcal{V}$ is a function that assigns to each pair $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ a scalar $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{F}$ satisfying, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and all $\lambda \in \mathbb{F}$,

i. *positivity*: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$;
ii. *definiteness*: $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$;

iii. *additivity in the first slot*: $\langle \mathbf{u}+\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{w}\rangle + \langle \mathbf{v}, \mathbf{w}\rangle$;
iv. *homogeneity in the first slot*: $\langle \lambda\mathbf{u}, \mathbf{v}\rangle = \lambda \langle \mathbf{u}, \mathbf{v}\rangle$;
v. *conjugate symmetry*: $\langle \mathbf{u}, \mathbf{v}\rangle = \overline{\langle \mathbf{v}, \mathbf{u}\rangle}$.

In the positivity axiom, $\langle \mathbf{v}, \mathbf{v}\rangle \geq 0$ means that $\langle \mathbf{v}, \mathbf{v}\rangle$ is a real nonnegative scalar.

Definition 2.2.4. *Inner-Product Space*
An *inner-product space* is a vector space $\mathcal{V}$ together with an inner product on $\mathcal{V}$.

Example 2.2.5. *Standard Inner Product on $\mathbb{F}^n$*
For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, the standard inner product is $\langle \mathbf{x}, \mathbf{y}\rangle = \sum_{j=1}^{n} x_j \overline{y_j}$. When $\mathbb{F} = \mathbb{R}$, this is the usual dot product.

Example 2.2.6. *Function and Matrix Inner Products*
If $\mathcal{V} = C_{\mathbb{F}}[a, b]$, then $\langle f, g\rangle = \int_a^b f(t)\overline{g(t)}\, dt$ is an inner product.
If $\mathcal{V} = \mathrm{Mat}(m, n, \mathbb{F})$, then the *Frobenius inner product* is $\langle A, B\rangle_F := \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}\overline{b_{ij}}$.

Proposition 2.2.7. *Basic Inner-Product Identities*
Let $\mathcal{V}$ be an inner-product space. Then for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. for fixed $\mathbf{u}$, the map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u}\rangle$ is linear;
ii. $\langle \mathbf{0}, \mathbf{u}\rangle = 0$ and $\langle \mathbf{u}, \mathbf{0}\rangle = 0$;
iii. $\langle \mathbf{u}, \mathbf{v}+\mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$;
iv. $\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \overline{\lambda}\langle \mathbf{u}, \mathbf{v}\rangle$.

Proof.
(i) This is exactly additivity and homogeneity in the first slot.
(ii) A linear map sends $\mathbf{0}$ to 0, so $\langle \mathbf{0}, \mathbf{u}\rangle = 0$. Conjugate symmetry gives $\langle \mathbf{u}, \mathbf{0}\rangle = \overline{\langle \mathbf{0}, \mathbf{u}\rangle} = 0$.
(iii) By conjugate symmetry and additivity, $\langle \mathbf{u}, \mathbf{v}+\mathbf{w}\rangle = \overline{\langle \mathbf{v}+\mathbf{w}, \mathbf{u}\rangle} = \overline{\langle \mathbf{v}, \mathbf{u}\rangle} + \overline{\langle \mathbf{w}, \mathbf{u}\rangle} = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$.
(iv) By conjugate symmetry and homogeneity, $\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \overline{\langle \lambda\mathbf{v}, \mathbf{u}\rangle} = \overline{\lambda}\,\overline{\langle \mathbf{v}, \mathbf{u}\rangle} = \overline{\lambda}\langle \mathbf{u}, \mathbf{v}\rangle$.
Structural Commentary: First-slot linearity and conjugate symmetry force conjugate linearity in second slot.
Key Insight: Once one slot is linear, the other slot is forced to be conjugate linear. □

Definition 2.2.8. *Norm Induced by an Inner Product*
For $\mathbf{v} \in \mathcal{V}$, the *norm* of $\mathbf{v}$ is $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$.

Example 2.2.9. *Norms from Standard Inner Products*
In $\mathbb{F}^n$, norm from standard inner product is *Euclidean norm* $\|(x_1, \ldots, x_n)\|_2 = \sqrt{|x_1|^2 + \cdots + |x_n|^2}$.
In $C_{\mathbb{F}}[a, b]$, with integral inner product, derived norm is *L^2 norm* $\|f\|_2 = \sqrt{\int_a^b |f(t)|^2\, dt}$.
In $\mathrm{Mat}(m, n, \mathbb{F})$, with Frobenius inner product, *Frobenius norm* is $\|A\|_F = \sqrt{\langle A, A\rangle_F} = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2}$, $A = [a_{ij}]$. The Frobenius norm is also called the *Schur norm* or *Hilbert-Schmidt norm*.

Proposition 2.2.10. *Basic Properties of the Derived Norm*
Let $\mathcal{V}$ be an inner-product space with derived norm $\|\cdot\|$. For all $\mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. $\|\mathbf{v}\|$ is real and nonnegative;
ii. $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$;
iii. $\|\lambda\mathbf{v}\| = |\lambda|\,\|\mathbf{v}\|$.

Proof.
(i) By positivity, $\langle \mathbf{v}, \mathbf{v}\rangle$ is real and nonnegative, so its square root is real and nonnegative.
(ii) By definition, $\|\mathbf{v}\| = 0$ if and only if $\langle \mathbf{v}, \mathbf{v}\rangle = 0$, which holds if and only if $\mathbf{v} = \mathbf{0}$ by definiteness.
(iii) Compute $\|\lambda\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \lambda\mathbf{v}\rangle = \lambda\overline{\lambda}\langle \mathbf{v}, \mathbf{v}\rangle = |\lambda|^2\,\|\mathbf{v}\|^2$. Taking nonnegative square roots, $\|\lambda\mathbf{v}\| = |\lambda|\,\|\mathbf{v}\|$.
Structural Commentary: Positivity gives length; sesquilinearity gives scaling.
Key Insight: Squared length is easier to manipulate than length itself. □

Definition 2.2.11. *Unit Vector and Normalization*
A vector $\mathbf{u} \in \mathcal{V}$ is a *unit vector* if $\|\mathbf{u}\| = 1$. If $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the *normalization* of $\mathbf{v}$.

Proposition 2.2.12. *Normalization Gives a Unit Vector*
If $\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a unit vector in the same direction as $\mathbf{v}$.

Proof. By Proposition 2.2.10, $\|\mathbf{v}\| > 0$ and $\left\| \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\| = \frac{1}{\|\mathbf{v}\|} \|\mathbf{v}\| = 1$.
The vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a positive scalar multiple of $\mathbf{v}$, so it lies in the same one-dimensional direction.
Structural Commentary: Normalization changes scale but preserves direction.
Key Insight: Divide by the length to make the length equal to one. □

Definition 2.2.13. *Orthogonal Vectors*
Vectors $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ are *orthogonal*, written $\mathbf{u} \perp \mathbf{v}$, if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Proposition 2.2.14. *Orthogonality and the Zero Vector*
Let $\mathcal{V}$ be an inner-product space.

i. $\mathbf{u} \perp \mathbf{v}$ *if and only if* $\mathbf{v} \perp \mathbf{u}$.
ii. $\mathbf{0}$ *is orthogonal to every vector in* $\mathcal{V}$.
iii. If $\mathbf{v}$ *is orthogonal to itself, then* $\mathbf{v} = \mathbf{0}$.

Proof.

(i) If $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, then conjugate symmetry gives $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{0} = 0$, and the converse is identical.
(ii) This follows from Proposition 2.2.7, which gives $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ and $\langle \mathbf{u}, \mathbf{0} \rangle = 0$.
(iii) If $\mathbf{v} \perp \mathbf{v}$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, hence $\mathbf{v} = \mathbf{0}$ by definiteness.

Structural Commentary: Conjugate symmetry makes orthogonality symmetric, and definiteness isolates $\mathbf{0}$.
Key Insight: Conjugate symmetry preserves the equation "equals zero." □

Remark 2.2.15. *Cross Terms*
In an expansion such as $\|\mathbf{u} + \mathbf{v}\|^2$, the terms involving both vectors, $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$, are called *cross terms*. The terms $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ are the pure squared-length terms.

Theorem 2.2.16. *Pythagorean Theorem*
If $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \perp \mathbf{v}$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof. Since $\mathbf{u} \perp \mathbf{v}$, Proposition 2.2.14 gives $\langle \mathbf{v}, \mathbf{u} \rangle = 0$. Thus

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 .$$

Structural Commentary: Orthogonality cancels the cross terms in the squared-norm expansion.
Key Insight: Perpendicular components add in squared length, not in length. □

Proposition 2.2.17. *Orthogonal Projection onto a Line*
Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$. Define $c := \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, $\mathbf{w} := \mathbf{u} - c\mathbf{v}$. Then $\mathbf{w} \perp \mathbf{v}$ and $\mathbf{u} = c\mathbf{v} + \mathbf{w}$.
The vector $\text{proj}_{\mathbf{v}}(\mathbf{u}) := c\mathbf{v} = \left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|}$ is the orthogonal projection of $\mathbf{u}$ onto $\text{span}(\mathbf{v})$.

Proof. The decomposition $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ follows from the definition of $\mathbf{w}$.
Moreover, $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u} - c\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - c \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \|\mathbf{v}\|^2 = 0$. Thus $\mathbf{w} \perp \mathbf{v}$.
Structural Commentary: Choose c so the remainder $\mathbf{u} - c\mathbf{v}$ has zero inner product with $\mathbf{v}$.
Key Insight: Projection is the part detected by pairing with the direction vector. □

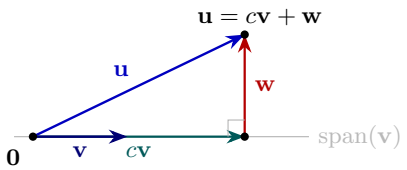


Figure 3: Writing $\mathbf{u}$ as a component parallel to $\mathbf{v}$ plus a component orthogonal to $\mathbf{v}$.

Theorem 2.2.18. *Cauchy-Schwarz Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$. Equality holds if and only if one of $\mathbf{u}, \mathbf{v}$ is a scalar multiple of the other.

Proof. If $\mathbf{v} = \mathbf{0}$, both sides are 0. Assume $\mathbf{v} \neq \mathbf{0}$. By Proposition 2.2.17, $\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} + \mathbf{w}$, $\mathbf{w} \perp \mathbf{v}$.
The Pythagorean Theorem 2.2.16 gives

$$\|\mathbf{u}\|^2 = \left\| \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} \right\|^2 + \|\mathbf{w}\|^2 = \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2} + \|\mathbf{w}\|^2 \geq \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2}.$$

Multiplying by $\|\mathbf{v}\|^2$ and taking square roots proves the inequality. Equality holds exactly when $\|\mathbf{w}\| = 0$, i.e. $\mathbf{w} = \mathbf{0}$, which is equivalent to $\mathbf{u}$ being a scalar multiple of $\mathbf{v}$. The case $\mathbf{v} = \mathbf{0}$ is also covered because $\mathbf{v} = 0\mathbf{u}$.
Structural Commentary: Decompose $\mathbf{u}$ into parallel and orthogonal parts relative to $\mathbf{v}$.
Key Insight: The component of $\mathbf{u}$ along $\mathbf{v}$ cannot be longer than $\mathbf{u}$. □

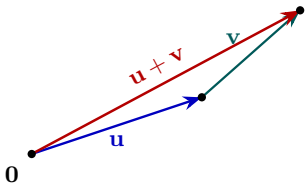


Figure 4: Triangle inequality: $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.

Theorem 2.2.19. *Triangle Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.
Equality holds if and only if one of $\mathbf{u}, \mathbf{v}$ *is a nonnegative scalar multiple of the other.*

Proof. We compute

$$\|\mathbf{u}+\mathbf{v}\|^2 = \langle \mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}\rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + \langle \mathbf{u},\mathbf{v}\rangle + \langle \mathbf{v},\mathbf{u}\rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}\langle \mathbf{u},\mathbf{v}\rangle$$
$$\le \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2|\langle \mathbf{u},\mathbf{v}\rangle| \le \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| = \big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2.$$

Taking square roots gives the inequality. If either vector is $\mathbf{0}$, the equality condition is immediate. Otherwise, equality holds exactly when $\langle \mathbf{u},\mathbf{v}\rangle = |\langle \mathbf{u},\mathbf{v}\rangle|$ and equality holds in Cauchy-Schwarz. Thus $\mathbf{u} = a\mathbf{v}$ for some $a \in \mathbb{F}$ and $a\|\mathbf{v}\|^2 = \langle \mathbf{u},\mathbf{v}\rangle$ is real nonnegative, so $a \ge 0$. Conversely, if $\mathbf{u} = a\mathbf{v}$ with $a \ge 0$, then $\|\mathbf{u}+\mathbf{v}\| = (a+1)\|\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$.
Structural Commentary: Square the norm and bound the cross term by Cauchy-Schwarz.
Key Insight: Equality means the two vectors point in the same direction. □

Definition 2.2.20. *Norm and Normed Vector Space*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. A *norm* on $\mathcal{V}$ is a function $\|\cdot\| : \mathcal{V} \to [0,\infty)$ s.t., for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. $\|\mathbf{u}\| \ge 0$;
ii. $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$;
iii. $\|\lambda\mathbf{u}\| = |\lambda|\,\|\mathbf{u}\|$;
iv. $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.

A *normed vector space* is a vector space equipped with a norm. Its *unit ball* is $\{\mathbf{v} \in \mathcal{V} : \|\mathbf{v}\| \le 1\}$.

Proposition 2.2.21. *Coordinate Norms on* $\mathbb{F}^n$
For $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$, $\|\mathbf{x}\|_1 = \sum_{j=1}^{n} |x_j|$, $\|\mathbf{x}\|_2 = \left(\sum_{j=1}^{n} |x_j|^2\right)^{1/2}$, $\|\mathbf{x}\|_\infty = \max_{1\le j\le n} |x_j|$ *are norms on* $\mathbb{F}^n$.

Proof.

(i) For $\|\cdot\|_1$, nonnegativity, definiteness, and homogeneity follow from the corresponding facts for modulus. The triangle inequality follows from $\|\mathbf{x}+\mathbf{y}\|_1 = \sum_{j=1}^{n} |x_j + y_j| \le \sum_{j=1}^{n} \big(|x_j| + |y_j|\big) = \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$
(ii) For $\|\cdot\|_2$, induced by standard inner product, so axioms follow from Proposition 2.2.10, Theorem 2.2.19.
(iii) For $\|\cdot\|_\infty$, nonnegativity, definiteness, and homogeneity are immediate from the maximum of finitely many moduli. If k is chosen so that $\|\mathbf{x}+\mathbf{y}\|_\infty = |x_k + y_k|$, then $\|\mathbf{x}+\mathbf{y}\|_\infty \le |x_k| + |y_k| \le \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$,

Structural Commentary: The ℓ_1 norm sums coordinate magnitudes, ℓ_∞ keeps only the largest coordinate magnitude, and ℓ_2 comes from the standard inner product.
Key Insight: All three obey the same abstract axioms, even though their unit balls have different shapes. □

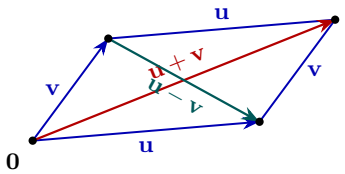


Figure 5: Parallelogram identity: $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$.

Theorem 2.2.22. *Parallelogram Identity*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\big(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\big)$.

Proof. Expanding both squared norms gives

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2, \quad \|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \langle \mathbf{u}, \mathbf{v}\rangle - \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2.$$

Adding the two identities cancels the cross terms and yields the result.
Structural Commentary: Opposite cross terms cancel when the two diagonal norms are added.
Key Insight: Inner-product norms remember parallelogram geometry through cancellation. □

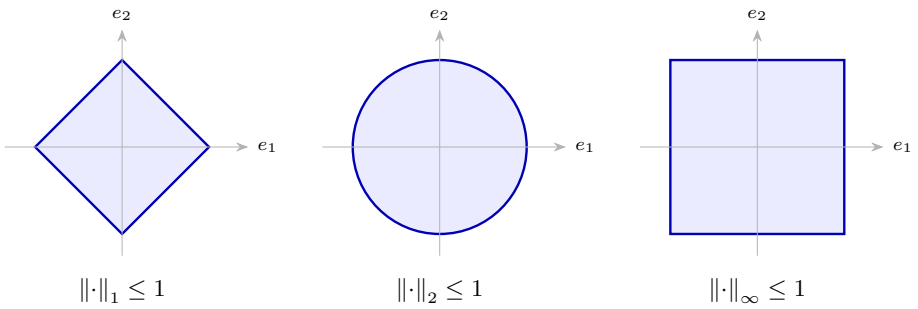


Figure 6: The unit balls of $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ in $\mathbb{R}^2$.

Corollary 2.2.23. *ℓ_1 and ℓ_∞ Are Not Inner-Product Norms*
If $n \ge 2$, then the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $\mathbb{F}^n$ are not induced by any inner product.

Proof. Let $\mathbf{e}_1, \mathbf{e}_2$ be the first two standard coordinate vectors.
For $\|\cdot\|_1$, $\|\mathbf{e}_1+\mathbf{e}_2\|_1^2 + \|\mathbf{e}_1-\mathbf{e}_2\|_1^2 = 4+4 = 8$, but $2\|\mathbf{e}_1\|_1^2 + 2\|\mathbf{e}_2\|_1^2 = 4$.
For $\|\cdot\|_\infty$, $\|\mathbf{e}_1+\mathbf{e}_2\|_\infty^2 + \|\mathbf{e}_1-\mathbf{e}_2\|_\infty^2 = 1+1 = 2$ but $2\|\mathbf{e}_1\|_\infty^2 + 2\|\mathbf{e}_2\|_\infty^2 = 4$.
Both norms violate Theorem 2.2.22, which every inner-product norm must satisfy.
Structural Commentary: The parallelogram identity detects inner-product norms.
Key Insight: Different unit-ball geometry can rule out hidden inner-product structure. □

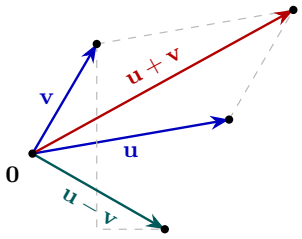


Figure 7: Polarization identity: $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle$.

Theorem 2.2.24. *Polarization Identities*
Let $\mathcal{V}$ be an inner-product space and let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

i. If $\mathbb{F} = \mathbb{R}$, then $\langle \mathbf{u}, \mathbf{v}\rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2\right)$

ii. If $\mathbb{F} = \mathbb{C}$, then $\langle \mathbf{u}, \mathbf{v}\rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 + i\|\mathbf{u}+i\mathbf{v}\|^2 - i\|\mathbf{u}-i\mathbf{v}\|^2\right)$

Proof. Let $z := \langle \mathbf{u}, \mathbf{v}\rangle$.

(i) If $\mathbb{F} = \mathbb{R}$, then expanding squared norms gives $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\operatorname{Re} z = 4z$, proving (i).
(ii) If $\mathbb{F} = \mathbb{C}$, the first expansion gives $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\operatorname{Re} z$. The same expansion with $i\mathbf{v}$ gives $\|\mathbf{u}+i\mathbf{v}\|^2 - \|\mathbf{u}-i\mathbf{v}\|^2 = 4\operatorname{Im} z$. Combining the real and imaginary parts gives (ii).

Structural Commentary: Two norm differences extract the real and imaginary parts.
Key Insight: The norm determines the inner product because it determines both parts of every cross term. □

Principle 2.2.25. *Geometry from Algebra*
Most geometric statements in an inner-product space come from expanding $\|\mathbf{u} \pm \mathbf{v}\|^2$ and then interpreting the cross terms $\langle \mathbf{u}, \mathbf{v}\rangle$ and $\langle \mathbf{v}, \mathbf{u}\rangle$.

2.2.2 Orthogonal Projections and Least Squares

Orthonormal lists are the coordinate systems best adapted to inner-product geometry. They make coefficients visible through inner products and turn least-squares problems into orthogonal decompositions.

Definition 2.2.26. *Orthonormal List*
A list $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ of vectors in an inner-product space $\mathcal{V}$ is *orthonormal* if

$$\langle \mathbf{e}_j, \mathbf{e}_k \rangle = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

Equivalently, every $\mathbf{e}_j$ has norm 1 and distinct vectors in the list are orthogonal.

Proposition 2.2.27. *Norm of an Orthonormal Linear Combination*
If $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ is an orthonormal list in $\mathcal{V}$, then $\|a_1\mathbf{e}_1 + \cdots + a_m\mathbf{e}_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$ for all $a_1, \dots, a_m \in \mathbb{F}$.

Proof. Using orthonormality, $\left\| \sum_{j=1}^{m} a_j \mathbf{e}_j \right\|^2 = \left\langle \sum_{j=1}^{m} a_j \mathbf{e}_j, \sum_{k=1}^{m} a_k \mathbf{e}_k \right\rangle = \sum_{j=1}^{m} \sum_{k=1}^{m} a_j \overline{a_k} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{m} |a_j|^2$.
Structural Commentary: Orthonormality deletes every cross term in the squared-norm expansion.
Key Insight: In orthonormal coordinates, length is computed coefficient-by-coefficient. □

Corollary 2.2.28. *Orthonormal Lists Are Linearly Independent*
Every orthonormal list of vectors in an inner-product space is linearly independent.

Proof. If $a_1\mathbf{e}_1 + \cdots + a_m\mathbf{e}_m = \mathbf{0}$, then Proposition 2.2.27 gives $0 = \|\mathbf{0}\|^2 = \|a_1\mathbf{e}_1 + \cdots + a_m\mathbf{e}_m\|^2 = \sum_{j=1}^{m} |a_j|^2$.
Hence each $a_j = 0$, so the list is linearly independent.
Structural Commentary: The norm identity converts a linear relation into a sum of nonnegative scalars.
Key Insight: A zero orthonormal linear combination can only have zero coefficients. □

Definition 2.2.29. *Orthonormal Basis*
An *orthonormal basis* of $\mathcal{V}$ is a basis of $\mathcal{V}$ that is an orthonormal list.

Proposition 2.2.30. *Right-Length Orthonormal Lists Are Bases*
If $\mathcal{V}$ is finite-dimensional and $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is an orthonormal list in $\mathcal{V}$ with $n = \dim(\mathcal{V})$, then $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is an orthonormal basis of $\mathcal{V}$.

Proof. By Corollary 2.2.28, the list is linearly independent.
Since its length is $\dim(\mathcal{V})$, Proposition 2.1.28 implies that it is a basis.
Structural Commentary: Orthonormality gives independence; the correct length forces spanning.
Key Insight: In finite dimensions, n independent vectors in an n-dimensional space already form a basis. □

Theorem 2.2.31. *Coordinates in an Orthonormal Basis*
Let $\mathcal{V}$ be finite-dimensional and let $\beta = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.
Then for every $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$, $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} |\langle \mathbf{v}, \mathbf{e}_j \rangle|^2$ and

$$[\mathbf{v}]_\beta = \begin{bmatrix} \langle \mathbf{v}, \mathbf{e}_1 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{e}_n \rangle \end{bmatrix}.$$

Proof. Since β is a basis, write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{e}_j$. Taking inner product with $\mathbf{e}_k$ gives $\langle \mathbf{v}, \mathbf{e}_k \rangle = \sum_{j=1}^{n} a_j \langle \mathbf{e}_j, \mathbf{e}_k \rangle = a_k$.
Thus the displayed expansion and coordinate column follow. The norm identity follows from Proposition 2.2.27.
Structural Commentary: An orthonormal basis makes coordinate extraction an inner-product computation.
Key Insight: Pairing with $\mathbf{e}_k$ isolates the kth coordinate. □

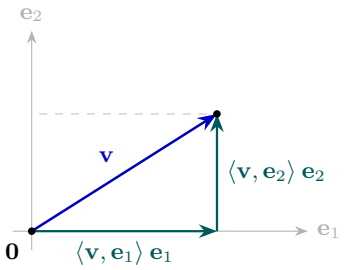


Figure 8: An orthonormal basis resolves a vector into perpendicular coordinate components.

Theorem 2.2.32. *Parseval's Identity*
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of an inner-product space $\mathcal{V}$.
Then, for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle} = \langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$.

Proof. By Theorem 2.2.31, $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{n} \langle \mathbf{w}, \mathbf{e}_k \rangle \mathbf{e}_k$.
Hence $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \sum_{k=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_k \rangle} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle}$.
The last expression is exactly the standard inner product of the coordinate columns in $\mathbb{F}^n$.
Structural Commentary: Orthonormal coordinates make inner products coordinatewise.
Key Insight: Orthonormal coordinates preserve angles and lengths, not just linear combinations. □

Theorem 2.2.33. *Bessel's Inequality*
Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal list in an inner-product space $\mathcal{V}$.
Then, for every $\mathbf{v} \in \mathcal{V}$, $\sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2 \leq \|\mathbf{v}\|^2$.

Proof. Set $\mathbf{u} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$. For each k, orthonormality gives $\langle \mathbf{v} - \mathbf{u}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$.
Thus $\mathbf{v} - \mathbf{u}$ is orthogonal to $\mathbf{u}$. By Pythagorean Theorem 2.2.16 and Proposition 2.2.27,

$$\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v} - \mathbf{u}\|^2 \geq \|\mathbf{u}\|^2 = \sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2 .$$

Structural Commentary: The chosen directions give an orthogonal part of $\mathbf{v}$; the residual is perpendicular.
Key Insight: Partial orthonormal coordinates cannot contain more energy than the whole vector. □

Corollary 2.2.34. *Coefficients Against an Orthonormal Sequence Vanish*
Let $(\mathbf{e}_1, \mathbf{e}_2, \ldots)$ be an orthonormal sequence in an inner-product space $\mathcal{V}$, meaning every finite sublist is orthonormal. Then, for every $\mathbf{v} \in \mathcal{V}$, $\lim_{n \to \infty} \langle \mathbf{v}, \mathbf{e}_n \rangle = 0$.

Proof. Fix $\epsilon > 0$. By Theorem 2.2.33, every finite set of indices satisfying $\left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right| \geq \epsilon$ has cardinality at most $\|\mathbf{v}\|^2 / \epsilon^2$. Hence only finitely many j have $\left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right| \geq \epsilon$. This holds for every $\epsilon > 0$, so $\langle \mathbf{v}, \mathbf{e}_n \rangle \to 0$.
Structural Commentary: Bessel bounds the number of large orthonormal coefficients.
Key Insight: Infinitely many large coefficients would force infinite energy. □

Theorem 2.2.35. *Matrix Entries from an Orthonormal Codomain Basis*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces over $\mathbb{F}$, with $\mathcal{W}$ an inner-product space. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be an orthonormal basis of $\mathcal{W}$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then

$$[T]_{\gamma \leftarrow \beta} = \begin{bmatrix} \langle T\mathbf{v}_1, \mathbf{w}_1 \rangle & \cdots & \langle T\mathbf{v}_n, \mathbf{w}_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle T\mathbf{v}_1, \mathbf{w}_m \rangle & \cdots & \langle T\mathbf{v}_n, \mathbf{w}_m \rangle \end{bmatrix} .$$

Conversely, for each $A \in \mathrm{Mat}(m, n, \mathbb{F})$, there is a unique $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ such that $[T]_{\gamma \leftarrow \beta} = A$.

Proof. The jth column of $[T]_{\gamma \leftarrow \beta}$ is $[T\mathbf{v}_j]_\gamma$. Since γ is orthonormal, Theorem 2.2.31 gives

$$[T\mathbf{v}_j]_\gamma = \begin{bmatrix} \langle T\mathbf{v}_j, \mathbf{w}_1 \rangle \\ \vdots \\ \langle T\mathbf{v}_j, \mathbf{w}_m \rangle \end{bmatrix} .$$

This proves the displayed matrix formula. The converse is exactly the surjectivity and injectivity of the coordinate encoding in Theorem 2.1.109.
Structural Commentary: Columns are basis-image coordinates, read by inner products.
Key Insight: Each entry measures $T\mathbf{v}_j$ against one output direction. □

Corollary 2.2.36. *Inner Products Through an Operator Matrix*
Let $\beta = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthonormal basis of a finite-dimensional inner-product space $\mathcal{V}$, and let $T \in \mathcal{L}(\mathcal{V})$. Then, for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle T\mathbf{v}, \mathbf{w}\rangle = \langle [T]_{\beta\leftarrow\beta}[\mathbf{v}]_\beta, [\mathbf{w}]_\beta\rangle_{\mathbb{F}^n}$.

Proof. By Proposition 2.1.104, $[T\mathbf{v}]_\beta = [T]_{\beta\leftarrow\beta}[\mathbf{v}]_\beta$. Apply Parseval's Identity 2.2.32 to $T\mathbf{v}$ and $\mathbf{w}$ gives result.
Structural Commentary: Orthonormal coordinates turn $\langle T\mathbf{v}, \mathbf{w}\rangle$ into matrix computation.
Key Insight: Orthonormal coordinates make operator geometry computable by matrices. □

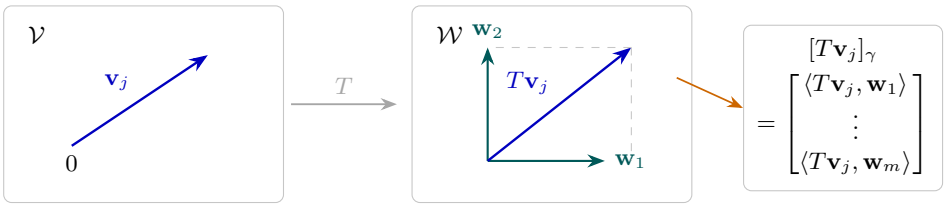


Figure 9: The jth column of $[T]_{\gamma\leftarrow\beta}$ is obtained by resolving $T\mathbf{v}_j$ against the orthonormal basis of $\mathcal{W}$.

Theorem 2.2.37. *Gram–Schmidt Process*
Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a linearly independent list in an inner-product space $\mathcal{V}$.
Define $\mathbf{u}_1 = \mathbf{v}_1$, $\mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$ and, for $2 \le j \le m$, $\mathbf{u}_j = \mathbf{v}_j - \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k\rangle \mathbf{e}_k$, $\mathbf{e}_j = \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}$.
Then $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ is orthonormal and $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) = \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$, $j = 1, \dots, m$.

Proof. Proceed by induction. The case $j = 1$ is normalization of the nonzero vector $\mathbf{v}_1$. Assume the result holds through $j-1$. Since $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}) = \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$ and the $\mathbf{v}$-list is linearly independent, $\mathbf{u}_j \neq \mathbf{0}$.
For $k < j$, $\langle \mathbf{u}_j, \mathbf{e}_k\rangle = \langle \mathbf{v}_j, \mathbf{e}_k\rangle - \sum_{i=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_i\rangle \langle \mathbf{e}_i, \mathbf{e}_k\rangle = \langle \mathbf{v}_j, \mathbf{e}_k\rangle - \langle \mathbf{v}_j, \mathbf{e}_k\rangle = 0$.
Thus $\mathbf{e}_j$ is a unit vector orthogonal to $\mathbf{e}_1, \dots, \mathbf{e}_{j-1}$.
Also $\mathbf{u}_j$ lies in $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$, while $\mathbf{v}_j = \mathbf{u}_j + \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k\rangle \mathbf{e}_k$ lies in $\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$.
Combining this with the induction hypothesis gives equality of the two spans at step j.
Structural Commentary: Subtract the previous-span component, then normalize.
Key Insight: Gram–Schmidt changes the basis vectors but preserves every initial span. □

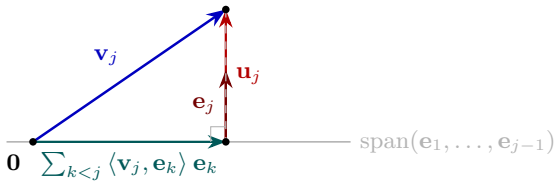


Figure 10: One Gram–Schmidt step: subtract the previous-span component, then normalize.

Lemma 2.2.38. *Gram–Schmidt on Orthogonal Lists*
If $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is a nonzero mutually orthogonal list, then Gram–Schmidt produces $\mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$, $j = 1, \dots, m$.
In particular, if the original list is orthonormal, Gram–Schmidt leaves it unchanged.

Proof. Assume $\mathbf{e}_i = \mathbf{v}_i / \|\mathbf{v}_i\|$ for $i < j$. Since $\mathbf{v}_j \perp \mathbf{v}_i$, we have $\langle \mathbf{v}_j, \mathbf{e}_i\rangle = 0$ for every $i < j$.
Hence the Gram–Schmidt remainder at step j is $\mathbf{u}_j = \mathbf{v}_j$, so $\mathbf{e}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$.
If the original list is orthonormal, then $\|\mathbf{v}_j\| = 1$ for every j.
Structural Commentary: Orthogonal input has no previous-span component to subtract.
Key Insight: Gram–Schmidt only removes nonorthogonal overlap. □

Corollary 2.2.39. *Existence of Orthonormal Bases*
Every finite-dimensional inner-product space has an orthonormal basis.

Proof. Choose any basis of $\mathcal{V}$ and apply Theorem 2.2.37.
The resulting orthonormal list has the same span as the original basis, hence spans $\mathcal{V}$.
Structural Commentary: Gram–Schmidt orthonormalizes a basis without changing its span.
Key Insight: Finite-dimensional inner-product spaces always admit geometry-friendly coordinates. □

Corollary 2.2.40. *Extending an Orthonormal List*
Every orthonormal list in a finite-dimensional inner-product space can be extended to an orthonormal basis.

Proof. Let $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ be orthonormal. It is linearly independent by Corollary 2.2.28, so extend it to a basis $(\mathbf{e}_1, \dots, \mathbf{e}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ by Theorem 2.1.24. Applying Gram–Schmidt to this basis leaves the first m vectors unchanged by Lemma 2.2.38, and orthonormalizes the rest.
Structural Commentary: Extend to a basis, then orthonormalize the new part.
Key Insight: Existing orthonormal directions are already cleaned up, so Gram–Schmidt does not move them. □

Theorem 2.2.41. *Riesz Representation Theorem*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $\varphi : \mathcal{V} \to \mathbb{F}$ be a linear functional. Then there is a unique vector $\mathbf{u} \in \mathcal{V}$ such that $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.
By Theorem 2.2.31, $\varphi(\mathbf{v}) = \varphi\left(\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j\right) = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \varphi(\mathbf{e}_j)$. Set $\mathbf{u} = \sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)}\, \mathbf{e}_j$.
Then orthonormality gives $\langle \mathbf{v}, \mathbf{u} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \varphi(\mathbf{e}_j) = \varphi(\mathbf{v})$. For uniqueness, suppose $\langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle$ for every $\mathbf{v} \in \mathcal{V}$. Taking $\mathbf{v} = \mathbf{u}_1 - \mathbf{u}_2$ gives $\|\mathbf{u}_1 - \mathbf{u}_2\|^2 = 0$, hence $\mathbf{u}_1 = \mathbf{u}_2$.
Structural Commentary: An orthonormal basis packages a functional as one representing vector.
Key Insight: Finite-dimensional functionals are inner products with fixed vectors. □

Perspective 2.2.42. *Riesz as Coordinate Duality*
The vector $\sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)} \mathbf{e}_j$ seems to depend on the orthonormal basis, but Theorem 2.2.41 shows that the resulting representing vector is intrinsic.

Definition 2.2.43. *Riesz Vector*
Let $\mathcal{V}$ be finite-dimensional and let $\varphi : \mathcal{V} \to \mathbb{F}$ be a linear functional. The unique vector $\mathbf{u} \in \mathcal{V}$ satisfying $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ for every $\mathbf{v} \in \mathcal{V}$ is the *Riesz vector* representing φ.

Example 2.2.44. *Trace Functional and the Frobenius Inner Product*
On $\mathrm{Mat}(n, n, \mathbb{F})$ with the Frobenius inner product, the Riesz vector of the trace functional $A \mapsto \mathrm{tr}(A)$ is I_n, since $\langle A, I_n \rangle_F = \sum_{j=1}^{n} a_{jj} = \mathrm{tr}(A)$.

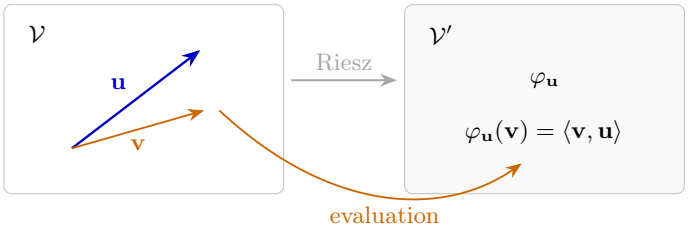


Figure 11: Riesz identifies a functional with its representing vector.

Definition 2.2.45. *Orthogonal Complement*
If $\mathcal{U} \subseteq \mathcal{V}$, then the *orthogonal complement* of $\mathcal{U}$ is $\mathcal{U}^{\perp} := \{\mathbf{v} \in \mathcal{V} : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for every } \mathbf{u} \in \mathcal{U}\}$

Proposition 2.2.46. *Basic Properties of Orthogonal Complements*
For subsets $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$,

i. $\mathcal{U}^{\perp}$ is a subspace of $\mathcal{V}$;
ii. $\{\mathbf{0}\}^{\perp} = \mathcal{V}$;
iii. $\mathcal{V}^{\perp} = \{\mathbf{0}\}$;
iv. $\mathcal{U} \cap \mathcal{U}^{\perp} \subseteq \{\mathbf{0}\}$;
v. if $\mathcal{U} \subseteq \mathcal{W}$, then $\mathcal{W}^{\perp} \subseteq \mathcal{U}^{\perp}$.

Proof.
(i) We have $\mathbf{0} \in \mathcal{U}^{\perp}$. If $\mathbf{v}, \mathbf{w} \in \mathcal{U}^{\perp}$ and $a \in \mathbb{F}$, then $\forall\, \mathbf{u} \in \mathcal{U}$, $\langle a\mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = a \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle = 0$, $a\mathbf{v} + \mathbf{w} \in \mathcal{U}^{\perp}$.

(ii) For every $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{0} \rangle = 0$, so $\mathbf{v} \in \{\mathbf{0}\}^\perp$.
(iii) If $\mathbf{v} \in \mathcal{V}^\perp$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.
(iv) If $\mathbf{v} \in \mathcal{U} \cap \mathcal{U}^\perp$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.
(v) If $\mathbf{v} \in \mathcal{W}^\perp$, then $\mathbf{v}$ is orthogonal to every vector in $\mathcal{W}$, hence to every vector in $\mathcal{U}$.

Structural Commentary: Orthogonal complements are solution spaces of homogeneous inner-product equations. Larger input sets impose more equations, so their complements become smaller.
Key Insight: Orthogonality constraints are linear constraints. □

Theorem 2.2.47. *Direct Sum with the Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal basis of $\mathcal{U}$. For $\mathbf{v} \in \mathcal{V}$, define $\mathbf{u} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$. Clearly $\mathbf{u} \in \mathcal{U}$.

For each k, $\langle \mathbf{w}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$. Thus $\mathbf{w}$ is orthogonal to every vector in $\mathcal{U}$, so $\mathbf{w} \in \mathcal{U}^\perp$ and $\mathbf{v} = \mathbf{u} + \mathbf{w}$. Hence $\mathcal{V} = \mathcal{U} + \mathcal{U}^\perp$. The intersection is $\{\mathbf{0}\}$ by Proposition 2.2.46, so the sum is direct.
Structural Commentary: Each vector splits into a $\mathcal{U}$-component and an orthogonal residual.
Key Insight: Projection plus residual gives the canonical direct-sum decomposition. □

Corollary 2.2.48. *Dimension of the Orthogonal Complement*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$, then $\dim(\mathcal{U}^\perp) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Since $\mathcal{V}$ is finite-dimensional, $\mathcal{U}$ is finite-dimensional by Corollary 2.1.25. Theorem 2.2.47 gives $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$. Thus Corollary 2.1.37 gives $\dim(\mathcal{V}) = \dim(\mathcal{U}) + \dim(\mathcal{U}^\perp)$.
Structural Commentary: Dimensions add across the orthogonal split.
Key Insight: Orthogonal complements contain exactly the dimensions not used by $\mathcal{U}$. □

Corollary 2.2.49. *Double Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{U} = (\mathcal{U}^\perp)^\perp$.

Proof. If $\mathbf{u} \in \mathcal{U}$, then $\mathbf{u}$ is orthogonal to every vector in $\mathcal{U}^\perp$, so $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$. Conversely, let $\mathbf{v} \in (\mathcal{U}^\perp)^\perp$. By Theorem 2.2.47, write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$. Since $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$, we have $\mathbf{w} = \mathbf{v} - \mathbf{u} \in (\mathcal{U}^\perp)^\perp$. Hence $\mathbf{w} \in \mathcal{U}^\perp \cap (\mathcal{U}^\perp)^\perp$, so $\mathbf{w} = \mathbf{0}$ and $\mathbf{v} = \mathbf{u} \in \mathcal{U}$.
Structural Commentary: The orthogonal split leaves no extra direction beyond $\mathcal{U}$.
Key Insight: Taking the orthogonal complement twice recovers the original finite-dimensional subspace. □

Definition 2.2.50. *Orthogonal Projection onto a Subspace*
Let $\mathcal{U}$ be finite-dimensional subspace of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, write unique decomposition $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$ from Theorem 2.2.47. The *orthogonal projection* of $\mathcal{V}$ onto $\mathcal{U}$ is the map $P_\mathcal{U} : \mathcal{V} \to \mathcal{V}$ defined by $P_\mathcal{U}\mathbf{v} = \mathbf{u}$.

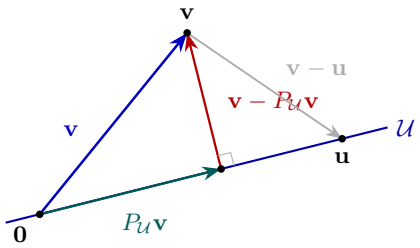


Figure 12: Orthogonal projection decomposes $\mathbf{v}$ into $P_\mathcal{U}\mathbf{v}$ plus an orthogonal residual.

Proposition 2.2.51. *Properties of Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. Then:

i. $P_\mathcal{U} \in \mathcal{L}(\mathcal{V})$;
ii. $P_\mathcal{U}\mathbf{u} = \mathbf{u}$ *for every* $\mathbf{u} \in \mathcal{U}$;
iii. $P_\mathcal{U}\mathbf{w} = \mathbf{0}$ *for every* $\mathbf{w} \in \mathcal{U}^\perp$;
iv. $\text{range}(P_\mathcal{U}) = \mathcal{U}$;
v. $\ker(P_\mathcal{U}) = \mathcal{U}^\perp$, *i.e. the kernel/nullspace of* $P_\mathcal{U}$ *is* $\mathcal{U}^\perp$;
vi. $\mathbf{v} - P_\mathcal{U}\mathbf{v} \in \mathcal{U}^\perp$ *for every* $\mathbf{v} \in \mathcal{V}$;
vii. $P_\mathcal{U}^2 = P_\mathcal{U}$;
viii. $\|P_\mathcal{U}\mathbf{v}\| \leq \|\mathbf{v}\|$ *for every* $\mathbf{v} \in \mathcal{V}$;

ix. if $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *is an orthonormal basis of* $\mathcal{U}$*, then* $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$*,* $\mathbf{v} \in \mathcal{V}$

Proof.

(i) For linearity, write $\mathbf{v}_r = \mathbf{u}_r + \mathbf{w}_r$ with $\mathbf{u}_r \in \mathcal{U}$ and $\mathbf{w}_r \in \mathcal{U}^\perp$ for $r = 1, 2$.
Then $a\mathbf{v}_1 + \mathbf{v}_2 = (a\mathbf{u}_1 + \mathbf{u}_2) + (a\mathbf{w}_1 + \mathbf{w}_2)$, where first summand lies in $\mathcal{U}$ and second in $\mathcal{U}^\perp$.
(ii) This follows from $\mathbf{u} = \mathbf{u} + \mathbf{0}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{U}^\perp$.
(iii) This follows from $\mathbf{w} = \mathbf{0} + \mathbf{w}$ with $\mathbf{0} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$.
(iv) The definition gives $\text{range}(P_{\mathcal{U}}) \subseteq \mathcal{U}$, and (ii) gives $\mathcal{U} \subseteq \text{range}(P_{\mathcal{U}})$.
(v) Part (iii) gives $\mathcal{U}^\perp \subseteq \ker(P_{\mathcal{U}})$.
Conversely, if $P_{\mathcal{U}}\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = \mathbf{0} + (\mathbf{v} - P_{\mathcal{U}}\mathbf{v})$ with $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^\perp$, so $\mathbf{v} \in \mathcal{U}^\perp$.
(vi) This is exactly the residual term in the decomposition $\mathbf{v} = P_{\mathcal{U}}\mathbf{v} + (\mathbf{v} - P_{\mathcal{U}}\mathbf{v})$.
(vii) By (ii), $P_{\mathcal{U}}(P_{\mathcal{U}}\mathbf{v}) = P_{\mathcal{U}}\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.
(viii) Write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$. Then $\mathbf{u} \perp \mathbf{w}$ and $\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{w}\|^2 \geq \|\mathbf{u}\|^2 = \|P_{\mathcal{U}}\mathbf{v}\|^2$
(ix) This is the explicit $\mathcal{U}$-component constructed in the proof of Theorem 2.2.47.

Structural Commentary: Orthogonal projection keeps the $\mathcal{U}$-component and discards the $\mathcal{U}^\perp$-component.
Key Insight: $P_{\mathcal{U}}$ is the identity on $\mathcal{U}$ and zero on $\mathcal{U}^\perp$. □

Theorem 2.2.52. *Orthogonal Projection Minimizes Distance*
Let $\mathcal{U}$ *be a finite-dimensional subspace of* $\mathcal{V}$*. If* $\mathbf{v} \in \mathcal{V}$ *and* $\mathbf{u} \in \mathcal{U}$*, then* $\|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\| \leq \|\mathbf{v} - \mathbf{u}\|$*.*
Equality holds if and only if $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$*.*

Proof. Since $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^\perp$ and $P_{\mathcal{U}}\mathbf{v} - \mathbf{u} \in \mathcal{U}$, the two vectors are orthogonal.
Thus $\|\mathbf{v} - \mathbf{u}\|^2 = \|(\mathbf{v} - P_{\mathcal{U}}\mathbf{v}) + (P_{\mathcal{U}}\mathbf{v} - \mathbf{u})\|^2 = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|^2 + \|P_{\mathcal{U}}\mathbf{v} - \mathbf{u}\|^2$.
Taking square roots gives the inequality. Equality holds exactly when $\|P_{\mathcal{U}}\mathbf{v} - \mathbf{u}\| = 0$, i.e. when $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$.
Structural Commentary: Projection error is orthogonal to every in-subspace correction.
Key Insight: Moving away from the projection adds a perpendicular error term, can only increase distance. □

Principle 2.2.53. *Least Squares as Orthogonal Projection*
In an inner-product space, the best approximation to $\mathbf{v}$ from a finite-dimensional subspace $\mathcal{U}$ is $P_{\mathcal{U}}\mathbf{v}$. The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ is orthogonal to every allowable correction inside $\mathcal{U}$.

Definition 2.2.54. *Distance to a Subspace*
Let $\mathcal{U}$ be a nonempty subset of inner-product space $\mathcal{V}$. The *distance* from $\mathbf{v} \in \mathcal{V}$ to $\mathcal{U}$ is $d(\mathbf{v}, \mathcal{U}) := \inf_{\mathbf{u} \in \mathcal{U}} \|\mathbf{v} - \mathbf{u}\|$.
If $\mathcal{U}$ is finite-dimensional, Theorem 2.2.52 gives $d(\mathbf{v}, \mathcal{U}) = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|$.

Theorem 2.2.55. *Normal Equations for Projection*
Let $\mathcal{U}$ *be a finite-dimensional subspace of an inner-product space* $\mathcal{V}$*, and suppose* $\mathcal{U} = \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$*.*
For $\mathbf{v} \in \mathcal{V}$*, the projection has the form* $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$*, where* $(c_1, \ldots, c_n)$ *solves the normal equations*

$$\begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \langle \mathbf{u}_2, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{u}_1 \rangle \\ \langle \mathbf{v}, \mathbf{u}_2 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{u}_n \rangle \end{bmatrix}.$$

This system is consistent. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ *are linearly independent, then the solution is unique.*

Proof. Since $P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}$, write $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$.
The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ lies in $\mathcal{U}^\perp$, so for each i, $0 = \left\langle \mathbf{v} - \sum_{j=1}^{n} c_j \mathbf{u}_j, \mathbf{u}_i \right\rangle = \langle \mathbf{v}, \mathbf{u}_i \rangle - \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i \rangle$.
These equations are exactly the displayed system, so consistency follows from the existence of $P_{\mathcal{U}}\mathbf{v}$. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, they form a basis of $\mathcal{U}$, so the coordinates of $P_{\mathcal{U}}\mathbf{v}$ in this basis are unique.
Structural Commentary: Normal equations express residual orthogonality.
Key Insight: The projection error is perpendicular to $\mathcal{U}$. □

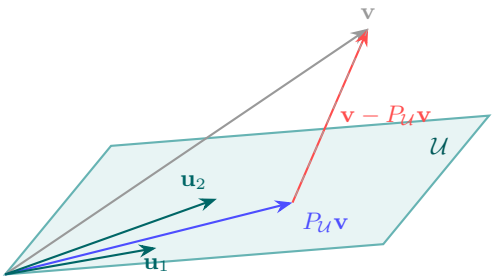


Figure 13: Normal equations impose $\langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i \rangle = 0$ for each spanning vector.

Theorem 2.2.56. *Matrix Least Squares and Normal Equations*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, where A^* denotes the conjugate transpose of A.
A vector $\mathbf{x}_0 \in \mathbb{F}^n$ satisfies $\|\mathbf{y} - A\mathbf{x}_0\|_2 = \min_{\mathbf{x}\in\mathbb{F}^n} \|\mathbf{y} - A\mathbf{x}\|_2$ if and only if $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
The normal equations are always consistent, and their solution is unique if and only if $\mathrm{rank}(A) = n$.

Proof. Let $\mathcal{U} = \mathrm{col}(A)$, and write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Since $\{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathcal{U}$, the vector $\mathbf{x}_0$ is least-squares optimal exactly when $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$, or equivalently when $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.

($\Rightarrow$) If $\mathbf{x}_0$ is least-squares optimal, then $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.
Hence $\langle \mathbf{y} - A\mathbf{x}_0, \mathbf{a}_j \rangle = 0$, $j = 1, \ldots, n$, which is $A^*(\mathbf{y} - A\mathbf{x}_0) = \mathbf{0}$, so $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
($\Leftarrow$) If $A^*A\mathbf{x}_0 = A^*\mathbf{y}$, then $A^*(\mathbf{y} - A\mathbf{x}_0) = \mathbf{0}$, so $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.
Hence $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$, so $\mathbf{x}_0$ is least-squares optimal.

Consistency follows by choosing $\mathbf{x}_0$ such that $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$.
If $\mathbf{x}_0, \mathbf{x}_1$ solve the normal equations, then $\|A(\mathbf{x}_0 - \mathbf{x}_1)\|_2^2 = \langle \mathbf{x}_0 - \mathbf{x}_1, A^*A(\mathbf{x}_0 - \mathbf{x}_1) \rangle = 0$, so $\mathbf{x}_0 - \mathbf{x}_1 \in \ker(A)$.
Thus uniqueness is equivalent to $\ker(A) = \{\mathbf{0}\}$, equivalently $\mathrm{rank}(A) = n$.
Structural Commentary: Least squares is orthogonal projection onto the column space of A.
Key Insight: The normal equations say exactly that the residual is orthogonal to every column of A. □

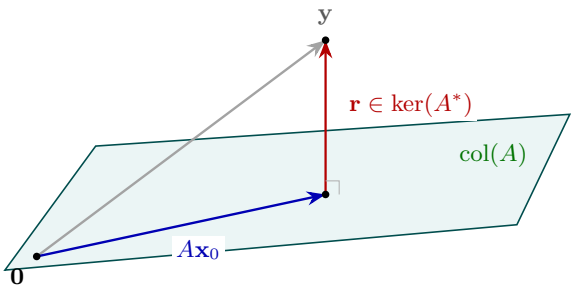


Figure 14: Fitted vector and orthogonal residual: $A\mathbf{x}_0 = P_{\mathrm{col}(A)}\mathbf{y}$ and $\mathbf{r} = \mathbf{y} - A\mathbf{x}_0 \in \mathrm{col}(A)^\perp = \ker(A^*)$.

Corollary 2.2.57. *Projection Formula for Full Column Rank*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have rank n.
Then A^*A is invertible and the orthogonal projection onto $\mathrm{col}(A)$ *is* $P_{\mathrm{col}(A)} = A(A^*A)^{-1}A^*$.

Proof. If $A^*A\mathbf{x} = \mathbf{0}$, then $\|A\mathbf{x}\|_2^2 = \langle \mathbf{x}, A^*A\mathbf{x} \rangle = 0$, so $A\mathbf{x} = \mathbf{0}$.
Since $\mathrm{rank}(A) = n$, we get $\mathbf{x} = \mathbf{0}$. Hence A^*A is invertible. By Theorem 2.2.56, the least-squares solution is $\mathbf{x}_0 = (A^*A)^{-1}A^*\mathbf{y}$. The fitted vector is $P_{\mathrm{col}(A)}\mathbf{y} = A\mathbf{x}_0 = A(A^*A)^{-1}A^*\mathbf{y}$.
Since this holds for every $\mathbf{y}$, the displayed operator formula follows.
Structural Commentary: Full column rank makes the normal-equation coordinates unique.
Key Insight: Solve for coefficients, then map back by A. □

Definition 2.2.58. *Gram Matrix*
For vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in an inner-product space, the *Gram matrix* is

$$G(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \langle \mathbf{u}_2, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix}.$$

The *Gram determinant* is $g(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \det G(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.

Proposition 2.2.59. *Gram Matrices Are Hermitian Positive Semidefinite*
For any vectors $\mathbf{u}_1, \dots, \mathbf{u}_n$, the Gram matrix $G(\mathbf{u}_1, \dots, \mathbf{u}_n)$ is Hermitian and positive semidefinite. If $\mathbf{u}_1, \dots, \mathbf{u}_n$ are linearly independent, then it is positive definite.

Proof. Hermitian symmetry follows from conjugate symmetry of the inner product.
For $\mathbf{c} = [c_1 \ \cdots \ c_n]^t$, $\mathbf{c}^* G(\mathbf{u}_1, \dots, \mathbf{u}_n)\mathbf{c} = \left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2 \geq 0.$
If $\mathbf{u}_1, \dots, \mathbf{u}_n$ are linearly independent, the squared norm above is 0 only when all $c_j = 0$, so G is positive definite.
Structural Commentary: A Gram matrix records every pairwise inner product in a list of vectors.
Key Insight: The quadratic form of a Gram matrix is exactly the squared norm of a linear combination. □

2.2.3 Unitary and Orthogonal Structure

Orthonormal bases allow geometric structure to coexist with coordinate computations. They support triangular forms, and they also turn the dual action of a linear map into its adjoint.

Corollary 2.2.60. *Upper-Triangular Form in an Orthonormal Basis*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. If T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$, then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis for which $[T]_{\beta \leftarrow \beta}$ is upper triangular. By Theorem 2.3.31, each subspace

$$\mathcal{U}_j = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j), \qquad j = 1, \dots, n,$$

is invariant under T. Apply Gram–Schmidt to β and obtain an orthonormal basis $\gamma = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ with

$$\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j) = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) = \mathcal{U}_j.$$

Thus each $\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$ is invariant under T, and Theorem 2.3.31 shows that $[T]_{\gamma \leftarrow \gamma}$ is upper triangular.
Structural Commentary: Upper triangularity is preservation of a nested flag of subspaces. Gram–Schmidt preserves that flag while making the basis orthonormal.
Key Insight: Triangular form depends on the flag, not on the particular basis used inside the flag. □

Theorem 2.2.61. *Schur Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Proceed by induction on $n = \dim(\mathcal{V})$. The cases $n = 0$ and $n = 1$ are immediate.
Suppose $n \geq 2$ and the result holds for smaller dimensions. By Theorem 2.3.23, choose a unit eigenvector $\mathbf{e}_1$ of T, say $T\mathbf{e}_1 = \lambda \mathbf{e}_1$. Let $\mathcal{U} = \operatorname{span}(\mathbf{e}_1)$ and define $S \in \mathcal{L}(\mathcal{U}^\perp)$ by $S\mathbf{w} = P_{\mathcal{U}^\perp}(T\mathbf{w})$.
By induction, choose an orthonormal basis $(\mathbf{e}_2, \dots, \mathbf{e}_n)$ of $\mathcal{U}^\perp$ with respect to which S has an upper-triangular matrix. Then $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is an orthonormal basis of $\mathcal{V}$.
For $j \geq 2$, the coordinates of $T\mathbf{e}_j$ along $\mathbf{e}_2, \dots, \mathbf{e}_n$ are exactly the coordinates of $S\mathbf{e}_j$, hence have no entries below row j. Also $T\mathbf{e}_1 = \lambda \mathbf{e}_1$. Therefore $[T]_{\gamma \leftarrow \gamma}$ is upper triangular for $\gamma = (\mathbf{e}_1, \dots, \mathbf{e}_n)$.
Structural Commentary: Schur triangularisation builds an eigenline first, then triangularises the projected action on its orthogonal complement.
Key Insight: The lower-right block is controlled by the compression $P_{\mathcal{U}^\perp} T|_{\mathcal{U}^\perp}$. □

Theorem 2.2.62. *Matrix Schur Triangularisation with Ordered Diagonal*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ in a prescribed order. If $A\mathbf{x} = \lambda_1 \mathbf{x}$ and $\|\mathbf{x}\|_2 = 1$, then there is a unitary $U = [\mathbf{x} \ U_2]$ such that $U^* A U$ is upper triangular with diagonal $\lambda_1, \dots, \lambda_n$.
If A is real, all $\lambda_i \in \mathbb{R}$, and $\mathbf{x} \in \mathbb{R}^n$, then U may be chosen real orthogonal.

Proof. Induct on n. The case $n = 1$ is immediate. Extend $\mathbf{x}$ to an orthonormal basis and set $V = [\mathbf{x} \ V_2]$.
Since $A\mathbf{x} = \lambda_1 \mathbf{x}$, $V^* A V = \begin{bmatrix} \lambda_1 & * \\ 0 & A' \end{bmatrix}$ and A' has eigenvalues $\lambda_2, \dots, \lambda_n$ by block triangularity.
By induction, choose unitary W with $W^* A' W = T'$ upper triangular and diagonal $\lambda_2, \dots, \lambda_n$.
Then $U := V(I_1 \oplus W)$ has first column $\mathbf{x}$ and $U^* A U = \begin{bmatrix} \lambda_1 & * \\ 0 & T' \end{bmatrix}$.
The real case uses the same argument with real orthonormal extensions.
Structural Commentary: Isolate the chosen eigenline, then repeat on the lower-right block.
Key Insight: The diagonal order is controlled by which eigenline is peeled off first. □

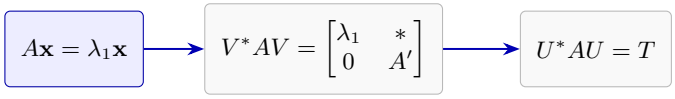


Figure 15: Schur induction: isolate an eigenline, then triangularise the compressed block.

Definition 2.2.63. *Adjoint of a Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
A function $S : \mathcal{W} \to \mathcal{V}$ is an *adjoint* of T if $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, S\mathbf{w}\rangle_{\mathcal{V}}$, $\mathbf{v} \in \mathcal{V},\ \mathbf{w} \in \mathcal{W}$.

Lemma 2.2.64. *Uniqueness of Adjoints*
A linear map has at most one adjoint.

Proof. Suppose $S_1, S_2 : \mathcal{W} \to \mathcal{V}$ are adjoints of T. For each $\mathbf{w} \in \mathcal{W}$ and every $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, S_1\mathbf{w} - S_2\mathbf{w}\rangle_{\mathcal{V}} = 0$.
Taking $\mathbf{v} = S_1\mathbf{w} - S_2\mathbf{w}$ gives $S_1\mathbf{w} = S_2\mathbf{w}$. Hence $S_1 = S_2$.
Structural Commentary: Same tests force candidate adjoints to agree.
Key Insight: Inner products separate vectors. □

Theorem 2.2.65. *Existence of the Adjoint*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $\mathcal{W}$ be an inner-product space, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T has a unique adjoint $T^* : \mathcal{W} \to \mathcal{V}$, and T^* is linear.

Proof. For each $\mathbf{w} \in \mathcal{W}$, the map $\varphi_{\mathbf{w}} : \mathcal{V} \to \mathbb{F}$ defined by $\varphi_{\mathbf{w}}(\mathbf{v}) = \langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}}$ is a linear functional. By Theorem 2.2.41, there is a unique vector $T^*\mathbf{w} \in \mathcal{V}$ such that $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}}$, $\mathbf{v} \in \mathcal{V}$.
Thus T^* is an adjoint, and it is unique by Lemma 2.2.64. For $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$ and $a \in \mathbb{F}$,

$$\langle \mathbf{v}, T^*(a\mathbf{w}_1 + \mathbf{w}_2)\rangle_{\mathcal{V}} = \langle T\mathbf{v}, a\mathbf{w}_1 + \mathbf{w}_2\rangle_{\mathcal{W}} = \overline{a}\,\langle T\mathbf{v}, \mathbf{w}_1\rangle_{\mathcal{W}} + \langle T\mathbf{v}, \mathbf{w}_2\rangle_{\mathcal{W}} = \langle \mathbf{v}, aT^*\mathbf{w}_1 + T^*\mathbf{w}_2\rangle_{\mathcal{V}}$$

for every $\mathbf{v} \in \mathcal{V}$. Hence $T^*(a\mathbf{w}_1 + \mathbf{w}_2) = aT^*\mathbf{w}_1 + T^*\mathbf{w}_2$.
Structural Commentary: Each $\mathbf{w}$ gives a Riesz-represented functional on $\mathcal{V}$.
Key Insight: The adjoint is pointwise Riesz representation for T. □

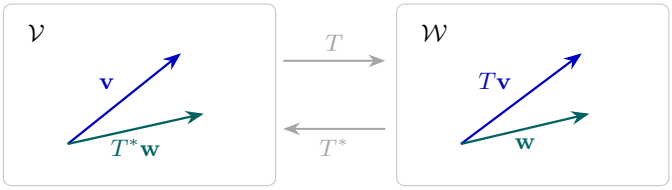


Figure 16: Adjoint relation: $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}}$.

Definition 2.2.66. *Conjugate Transpose*
For $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, the *conjugate transpose* of A is $A^* = [\overline{a_{ji}}] \in \mathrm{Mat}(n, m, \mathbb{F})$.
Over $\mathbb{R}$, this is the ordinary transpose.

Theorem 2.2.67. *Matrix of the Adjoint*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces. Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$ and let $\gamma = (\mathbf{f}_1, \ldots, \mathbf{f}_m)$ be an orthonormal basis of $\mathcal{W}$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[T^*]_{\beta\leftarrow\gamma} = \big([T]_{\gamma\leftarrow\beta}\big)^*$.

Proof. Let $A = [T]_{\gamma\leftarrow\beta}$ and $B = [T^*]_{\beta\leftarrow\gamma}$.
By Theorem 2.2.35, $B_{i,j} = \langle T^*\mathbf{f}_j, \mathbf{e}_i\rangle_{\mathcal{V}} = \overline{\langle \mathbf{e}_i, T^*\mathbf{f}_j\rangle_{\mathcal{V}}} = \overline{\langle T\mathbf{e}_i, \mathbf{f}_j\rangle_{\mathcal{W}}} = \overline{A_{j,i}}$. Thus $B = A^*$.
Structural Commentary: Entries swap by moving T across the inner product.
Key Insight: In orthonormal coordinates, taking adjoints is exactly taking conjugate transposes. □

Proposition 2.2.68. *Basic Adjoint Identities*
For finite-dimensional inner-product spaces and composable linear maps:

i. $(aS + T)^* = \overline{a}S^* + T^*$ *for* $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ *and* $a \in \mathbb{F}$;
ii. $(SR)^* = R^*S^*$ *for* $R \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ *and* $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
iii. $(T^*)^* = T$;
iv. $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Proof. Verify the defining identity; uniqueness gives equality.
(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{w} \in \mathcal{W}$, $\langle (aS + T)\mathbf{v}, \mathbf{w}\rangle = a\,\langle S\mathbf{v}, \mathbf{w}\rangle + \langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, \overline{a}S^*\mathbf{w} + T^*\mathbf{w}\rangle$, so $(aS + T)^* = \overline{a}S^* + T^*$.

(ii) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, $\langle SR\mathbf{u}, \mathbf{w}\rangle = \langle R\mathbf{u}, S^*\mathbf{w}\rangle = \langle \mathbf{u}, R^*S^*\mathbf{w}\rangle$, so $(SR)^* = R^*S^*$.
(iii) Conjugating $\langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, T^*\mathbf{w}\rangle$ gives $\langle T^*\mathbf{w}, \mathbf{v}\rangle = \langle \mathbf{w}, T\mathbf{v}\rangle$, so T is the adjoint of T^*.
(iv) Since $\langle I\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, I\mathbf{w}\rangle$, the identity map is its own adjoint.

Structural Commentary: Put both sides in one inner-product slot, then use uniqueness.
Key Insight: Like matrix transposes, adjoints reverse products and conjugate scalars. □

Proposition 2.2.69. *Adjoint Kernel-Range Orthogonality*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $\ker(T^*) = (\text{range}(T))^\perp$, $\text{range}(T^*) = (\ker(T))^\perp$, *and* $\ker(T) = (\text{range}(T^*))^\perp$, $\text{range}(T) = (\ker(T^*))^\perp$.
Equivalently, the nullspace of an adjoint is the orthogonal complement of the range of the original map.

Proof. For $\mathbf{w} \in \mathcal{W}$,

$$\begin{aligned} \mathbf{w} \in \ker(T^*) &\iff \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V} \\ &\iff \langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V} \\ &\iff \mathbf{w} \in (\text{range}(T))^\perp. \end{aligned}$$

Applying this identity to T^* gives $\ker(T) = (\text{range}(T^*))^\perp$.
Taking orthogonal complements, using Corollary 2.2.49 yields $\text{range}(T) = (\ker(T^*))^\perp$ and $\text{range}(T^*) = (\ker(T))^\perp$.
Structural Commentary: Adjoints trade ranges for kernels, and orthogonal complements reverse that trade.
Key Insight: A vector is killed by T^* exactly when it is orthogonal to everything T can produce. □

Definition 2.2.70. *Self-Adjoint and Hermitian*
An operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space is *self-adjoint* if $T^* = T$. A square matrix A is *Hermitian* if $A^* = A$. In orthonormal coordinates, these are the same condition by Theorem 2.2.67.

Definition 2.2.71. *Orthogonal Projection Matrix*
A square matrix $P \in \text{Mat}(n, n, \mathbb{F})$ is an *orthogonal projection matrix* if $P^2 = P$ and $P^* = P$.

Theorem 2.2.72. *Orthogonal Projections as Self-Adjoint Idempotents*
Let $\mathcal{V}$ be finite-dimensional and let $P \in \mathcal{L}(\mathcal{V})$.
Then P is the orthogonal projection onto $\text{range}(P)$ *if and only if $P^2 = P$ and $P^* = P$.*

Proof.
(⇒) If $P = P_{\mathcal{U}}$ for $\mathcal{U} = \text{range}(P)$, then Proposition 2.2.51 gives $P^2 = P$. For $\mathbf{v} = \mathbf{u} + \mathbf{w}$ and $\mathbf{x} = \mathbf{a} + \mathbf{b}$ with $\mathbf{u}, \mathbf{a} \in \mathcal{U}$ and $\mathbf{w}, \mathbf{b} \in \mathcal{U}^\perp$, $\langle P\mathbf{v}, \mathbf{x}\rangle = \langle \mathbf{u}, \mathbf{a} + \mathbf{b}\rangle = \langle \mathbf{u}, \mathbf{a}\rangle = \langle \mathbf{u} + \mathbf{w}, \mathbf{a}\rangle = \langle \mathbf{v}, P\mathbf{x}\rangle$, so $P^* = P$.
(⇐) Suppose $P^2 = P$ and $P^* = P$. For each $\mathbf{v}$, $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ with $P\mathbf{v} \in \text{range}(P)$ and $P(\mathbf{v} - P\mathbf{v}) = \mathbf{0}$. Thus the residual lies in $\ker(P)$. Since $P = P^*$, Proposition 2.2.69 gives $\ker(P) = (\text{range}(P))^\perp$. Hence P is exactly the orthogonal projection onto $\text{range}(P)$.

Structural Commentary: Idempotence splits by range and kernel; self-adjointness makes the split orthogonal.
Key Insight: An idempotent is orthogonal exactly when $\ker(P) = (\text{range}(P))^\perp$. □

Theorem 2.2.73. *Unitary Coordinate Form of an Orthogonal Projection*
Let $P \in \text{Mat}(n, n, \mathbb{F})$ be an orthogonal projection matrix and let $r = \text{rank}(P)$. Then P is unitarily similar to $I_r \oplus 0_{n-r}$. Equivalently, there is a unitary matrix U such that

$$U^*PU = \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix}$$

Proof. Let $\mathcal{U} = \text{col}(P)$. Identifying P with its induced operator on $\mathbb{F}^n$, Theorem 2.2.72 says P is the orthogonal projection onto $\mathcal{U}$, so P is the identity on $\mathcal{U}$ and zero on $\mathcal{U}^\perp$. Choose an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_r$ of $\mathcal{U}$ and extend it to an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbb{F}^n$. Let $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$.
Then U is unitary, and the matrix of P in this orthonormal basis has the displayed block diagonal form.
Structural Commentary: Orthogonal projection splits the space into range and orthogonal kernel.
Key Insight: In a basis adapted to $\text{col}(P) \oplus \text{col}(P)^\perp$, projection keeps the first coordinates and kills the rest. □

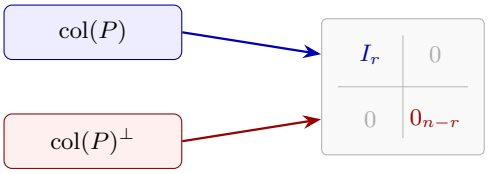


Figure 17: Adapted basis: $U^*PU = I_r \oplus 0_{n-r}$, identity on $\text{col}(P)$ and zero on $\text{col}(P)^\perp$.

Theorem 2.2.74. *Invariant Subspaces, Blocks, and Projections*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, let $\mathcal{U}$ be an r-dimensional subspace of $\mathbb{F}^n$ with $1 \le r \le n-1$, and let $Q = [Q_1\ Q_2]$ be unitary, where the columns of Q_1 and Q_2 are orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$, respectively. Let $P = Q_1 Q_1^*$ be the orthogonal projection onto $\mathcal{U}$. The following are equivalent:

(i) $\mathcal{U}$ *is invariant under* A*;*
(ii) $Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$ *for some compatible matrices* B, X, C*;*
(iii) $PAP = AP$*.*

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$, $E = Q^*PQ = I_r \oplus 0_{n-r}$. Here $Y = Q_2^*AQ_1$ and $\ker(Q_2^*) = \mathrm{col}(Q_1) = \mathcal{U}$.

$(i) \Rightarrow (ii)$ If $A\mathcal{U} \subseteq \mathcal{U}$, then the columns of AQ_1 lie in $\ker(Q_2^*)$, so $Y = 0$.

$(ii) \Rightarrow (iii)$ If $Y = 0$, then $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$.
Thus $Q^*(PAP)Q = EME = ME = Q^*(AP)Q$, so $PAP = AP$.

$(iii) \Rightarrow (i)$ If $PAP = AP$, then $EME = ME$. Since $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $Y = 0$.
Hence AQ_1 has columns in $\ker(Q_2^*) = \mathcal{U}$, so $A\mathcal{U} \subseteq \mathcal{U}$.

Structural Commentary: In adapted coordinates, the lower-left block is the component of $A\mathcal{U}$ in $\mathcal{U}^\perp$.
Key Insight: $PAP = AP$ exactly kills this component. □

Corollary 2.2.75. *Reducing Subspaces and Commuting Projections*
Maintain the notation of Theorem 2.2.74. The following are equivalent:

(i) $\mathcal{U}$ *is invariant under both* A *and* A^**;*
(ii) $Q^*AQ = B \oplus C$ *for some* $B \in \mathrm{Mat}(r,r,\mathbb{F})$ *and* $C \in \mathrm{Mat}(n-r,n-r,\mathbb{F})$*;*
(iii) $PA = AP$*.*

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E = Q^*PQ = I_r \oplus 0_{n-r}$.
Then $Q^*A^*Q = M^*$, $Q^*(PA)Q = EM$, and $Q^*(AP)Q = ME$.

$(i) \Rightarrow (ii)$ By Theorem 2.2.74, A-invariance gives $Y = 0$; apply to A^* gives $X = 0$. Thus $M = B \oplus C$.

$(ii) \Rightarrow (iii)$ If $M = B \oplus C$, then $EM = ME$, hence $PA = AP$.

$(iii) \Rightarrow (i)$ If $PA = AP$, then $EM = ME$. Since $EM = \begin{bmatrix} B & X \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $X = Y = 0$.
Hence $\mathcal{U}$ is invariant under both A and A^*.

Structural Commentary: Reducing subspaces are exactly block diagonal decompositions in adapted coordinates.
Key Insight: Commutation with P kills both off-diagonal blocks. □

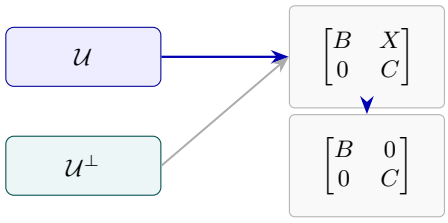


Figure 18: Adapted coordinates: invariance gives upper block form; $PA = AP$ gives block diagonal form.

Definition 2.2.76. *Minimum Norm Solution*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{y} \in \mathrm{col}(A)$.
A solution $\mathbf{s}$ of $A\mathbf{x} = \mathbf{y}$ is a *minimum norm solution* if $\|\mathbf{s}\|_2 \le \|\mathbf{x}\|_2$ for every solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{y}$.

Theorem 2.2.77. *Minimum Norm Solution of a Consistent Linear System*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\mathbf{y} \in \mathbb{F}^m$, and suppose $A\mathbf{x} = \mathbf{y}$ is consistent.

i. There is a unique minimum norm solution $\mathbf{s} \in \mathbb{F}^n$, and $\mathbf{s} \in \mathrm{col}(A^*)$.
ii. $\mathbf{s}$ is the only solution of $A\mathbf{x} = \mathbf{y}$ that lies in $\mathrm{col}(A^*)$.
iii. The system $AA^*\mathbf{u} = \mathbf{y}$ is consistent. If $AA^*\mathbf{u}_0 = \mathbf{y}$, then $\mathbf{s} = A^*\mathbf{u}_0$.

Proof. By Proposition 2.2.69, $\mathrm{col}(A^*) = (\ker A)^\perp$, so $\mathbb{F}^n = \mathrm{col}(A^*) \oplus \ker(A)$ orthogonally.

(i) Choose a solution $\mathbf{x}_0$ of $A\mathbf{x} = \mathbf{y}$ and write $\mathbf{x}_0 = \mathbf{s} + \mathbf{v}$, with $\mathbf{s} \in \mathrm{col}(A^*)$ and $\mathbf{v} \in \ker(A)$. Then $A\mathbf{s} = A\mathbf{x}_0 = \mathbf{y}$. Every solution has the form $\mathbf{s} + \mathbf{w}$ with $\mathbf{w} \in \ker(A)$.
Since $\mathbf{s} \perp \mathbf{w}$, $\|\mathbf{s} + \mathbf{w}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{w}\|_2^2 \geq \|\mathbf{s}\|_2^2$, with equality only when $\mathbf{w} = \mathbf{0}$.

(ii) If $\mathbf{s}' \in \mathrm{col}(A^*)$ is another solution, then $\mathbf{s}' - \mathbf{s} \in \ker(A) \cap \mathrm{col}(A^*) = \{\mathbf{0}\}$, so $\mathbf{s}' = \mathbf{s}$.

(iii) Since $\mathbf{s} \in \mathrm{col}(A^*)$, write $\mathbf{s} = A^*\mathbf{u}_0$. Then $AA^*\mathbf{u}_0 = A\mathbf{s} = \mathbf{y}$.
Conversely, if $AA^*\mathbf{u}_0 = \mathbf{y}$, then $A^*\mathbf{u}_0$ is a solution in $\mathrm{col}(A^*)$, so $A^*\mathbf{u}_0 = \mathbf{s}$ by (ii).

Structural Commentary: Solutions are $\mathbf{s} + \ker(A)$; $\mathrm{col}(A^*)$ meets this set orthogonally.
Key Insight: The minimum norm solution has no nullspace component. □

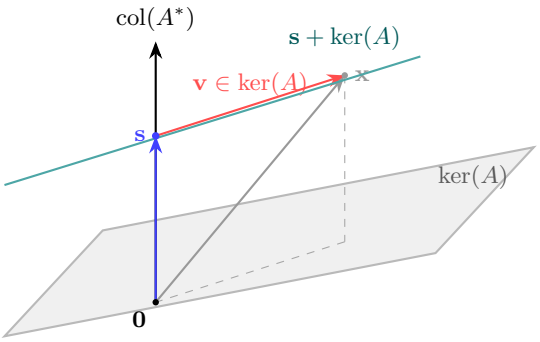


Figure 19: Among all solutions $\mathbf{s} + \ker(A)$, the unique one in $\mathrm{col}(A^*) = (\ker A)^\perp$ has minimum norm.

Theorem 2.2.78. *Eigenvalues of Self-Adjoint Operators Are Real*
Every eigenvalue of a self-adjoint operator on a finite-dimensional inner-product space is real.

Proof. Let $T = T^*$ and $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $\lambda \|\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \langle \mathbf{v}, \lambda\mathbf{v}\rangle = \overline{\lambda} \|\mathbf{v}\|^2$.
Since $\|\mathbf{v}\|^2 > 0$, $\lambda = \overline{\lambda}$, so $\lambda \in \mathbb{R}$.
Structural Commentary: Self-adjointness moves T between inner-product slots without changing the value.
Key Insight: An eigenvalue must equal its conjugate because the eigenvector has positive squared norm. □

Lemma 2.2.79. *Zero Complex Quadratic Form*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
If $\langle T\mathbf{v}, \mathbf{v}\rangle = 0$ for every $\mathbf{v} \in \mathcal{V}$, then $T = 0$.

Proof. Set $Q(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{x}\rangle$.
For $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, expansion gives the polarization identity for the sesquilinear form $(\mathbf{u}, \mathbf{w}) \mapsto \langle T\mathbf{u}, \mathbf{w}\rangle$:

$$\langle T\mathbf{u}, \mathbf{w}\rangle = \frac{Q(\mathbf{u} + \mathbf{w}) - Q(\mathbf{u} - \mathbf{w})}{4} + i\,\frac{Q(\mathbf{u} + i\mathbf{w}) - Q(\mathbf{u} - i\mathbf{w})}{4}.$$

The hypothesis makes every term on the right zero, so $\langle T\mathbf{u}, \mathbf{w}\rangle = 0$ for all $\mathbf{u}, \mathbf{w}$.
Taking $\mathbf{w} = T\mathbf{u}$ gives $\|T\mathbf{u}\|^2 = 0$, hence $T\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$.
Structural Commentary: Over $\mathbb{C}$, diagonal values of a sesquilinear form determine all mixed values.
Key Insight: Polarization turns the one-vector condition into a two-vector condition. □

Proposition 2.2.80. *Self-Adjointness from Real Quadratic Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
Then T is self-adjoint if and only if $\langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. For each $\mathbf{v}$, $\langle (T - T^*)\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle - \langle T^*\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle - \overline{\langle T\mathbf{v}, \mathbf{v}\rangle}$.
If all values $\langle T\mathbf{v}, \mathbf{v}\rangle$ are real, this expression is zero for every $\mathbf{v}$, and Lemma 2.2.79 gives $T - T^* = 0$.
Conversely, if $T = T^*$, then $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \overline{\langle T\mathbf{v}, \mathbf{v}\rangle}$, so $\langle T\mathbf{v}, \mathbf{v}\rangle$ is real.
Structural Commentary: The imaginary part of $\langle T\mathbf{v}, \mathbf{v}\rangle$ is exactly the diagonal trace of $T - T^*$.
Key Insight: Over $\mathbb{C}$, real quadratic values force the operator to equal its adjoint. □

Proposition 2.2.81. *Zero Quadratic Form for Self-Adjoint Operators*
Let T be a self-adjoint operator on a finite-dimensional inner-product space $\mathcal{V}$.
If $\langle T\mathbf{v}, \mathbf{v}\rangle = 0$ for every $\mathbf{v} \in \mathcal{V}$, then $T = 0$.

Proof. If $\mathbb{F} = \mathbb{C}$, this is Lemma 2.2.79. Assume $\mathbb{F} = \mathbb{R}$.
For $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, self-adjointness gives $\langle T\mathbf{w}, \mathbf{u}\rangle = \langle \mathbf{w}, T\mathbf{u}\rangle = \langle T\mathbf{u}, \mathbf{w}\rangle$, so the real polarization identity gives

$$\langle T\mathbf{u}, \mathbf{w}\rangle = \frac{\langle T(\mathbf{u}+\mathbf{w}), \mathbf{u}+\mathbf{w}\rangle - \langle T(\mathbf{u}-\mathbf{w}), \mathbf{u}-\mathbf{w}\rangle}{4} = 0.$$

Taking $\mathbf{w} = T\mathbf{u}$ gives $T\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$.
Structural Commentary: Self-adjointness supplies the missing symmetry needed for real polarization.
Key Insight: A self-adjoint operator is determined by its quadratic form. □

Definition 2.2.82. *Normal Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dim inner-product space is *normal* if it commutes with its adjoint: $TT^* = T^*T$.
Every self-adjoint operator is normal.

Theorem 2.2.83. *Normal Operators and Equal Adjoint Norms*
An operator $T \in \mathcal{L}(\mathcal{V})$ is normal if and only if $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. The operator $A = T^*T - TT^*$ is self-adjoint.
For every $\mathbf{v} \in \mathcal{V}$, $\langle A\mathbf{v}, \mathbf{v}\rangle = \langle T^*T\mathbf{v}, \mathbf{v}\rangle - \langle TT^*\mathbf{v}, \mathbf{v}\rangle = \|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2$.
Thus T is normal exactly when $A = 0$, which by Proposition 2.2.81 is equivalent to the displayed quadratic form being zero for every $\mathbf{v}$. This is equivalent to $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v}$.
Structural Commentary: Normality is measured by the self-adjoint commutator $T^*T - TT^*$.
Key Insight: A normal operator and its adjoint move every vector the same distance. □

Theorem 2.2.84. *Normal Operators Share Eigenvectors with Their Adjoints*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. If $T\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$, then $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Proof. The operator $T - \lambda I_{\mathcal{V}}$ is normal by Proposition 2.2.68 and scalar multiples of $I_{\mathcal{V}}$ commute with T and T^*. Applying Theorem 2.2.83 to $T - \lambda I_{\mathcal{V}}$ gives $0 = \|(T - \lambda I_{\mathcal{V}})\mathbf{v}\| = \|(T - \lambda I_{\mathcal{V}})^*\mathbf{v}\| = \left\|(T^* - \overline{\lambda} I_{\mathcal{V}})\mathbf{v}\right\|$.
Structural Commentary: For normal operators, kernels of $T - \lambda I$ and $T^* - \overline{\lambda} I$ coincide.
Key Insight: Normality makes eigenvector information visible from both T and T^*. □

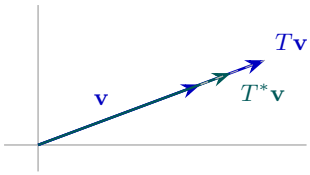


Figure 20: For normal T, $T\mathbf{v} = \lambda\mathbf{v}$ implies $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Theorem 2.2.85. *Orthogonal Eigenvectors for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. Eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose $T\mathbf{u} = \alpha\mathbf{u}$ and $T\mathbf{v} = \beta\mathbf{v}$ with $\alpha \neq \beta$. By Theorem 2.2.84, $T^*\mathbf{v} = \overline{\beta}\mathbf{v}$.
Hence $\alpha\langle \mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T^*\mathbf{v}\rangle = \langle \mathbf{u}, \overline{\beta}\mathbf{v}\rangle = \beta\langle \mathbf{u}, \mathbf{v}\rangle$. Since $\alpha - \beta \neq 0$, $\langle \mathbf{u}, \mathbf{v}\rangle = 0$.
Structural Commentary: Normality lets the operator move from one eigenvector to the other through the adjoint.
Key Insight: Distinct eigenvalue scalings are incompatible unless the inner product is zero. □

Definition 2.2.86. *Isometry*
Let $\mathcal{V}$ be an inner-product space. An operator $T \in \mathcal{L}(\mathcal{V})$ is an *isometry* if $\|T\mathbf{v}\| = \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$.

Proposition 2.2.87. *Basic Properties of Isometries*
Let $\mathcal{V}$ be an inner-product space and let $S, T \in \mathcal{L}(\mathcal{V})$ be isometries.

i. ST is an isometry;
ii. T is injective;
iii. if $\mathcal{V}$ is finite-dimensional, then T is invertible and T^{-1} is an isometry.

Proof.

(i) For every $\mathbf{v}$, $\|ST\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) If $T\mathbf{v} = \mathbf{0}$, then $\|\mathbf{v}\| = \|T\mathbf{v}\| = 0$, so $\mathbf{v} = \mathbf{0}$. Thus $\ker(T) = \{\mathbf{0}\}$.
(iii) In finite dimensions, injectivity implies invertibility by Theorem 2.1.78.
If $\mathbf{u} = T\mathbf{v}$, then $\left\|T^{-1}\mathbf{u}\right\| = \|\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{u}\|$.

Structural Commentary: Isometries preserve length, so they cannot collapse nonzero vectors.
Key Insight: In finite dimensions, length preservation already forces an invertible geometry-preserving map. □

Theorem 2.2.88. *Isometries Preserve Inner Products*
Let $\mathcal{V}$ be an inner-product space, $T \in \mathcal{L}(\mathcal{V})$. Then T is an isometry if and only if $\langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle \ \ \forall \mathbf{u}, \mathbf{v} \in \mathcal{V}$.

Proof.

($\Rightarrow$) Suppose T is an isometry. Then every norm appearing in the polarization identities of Theorem 2.2.24 is unchanged after applying T. Therefore $\langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$.
($\Leftarrow$) Suppose T preserves inner products. Then $\|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle \mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$, so T is an isometry.

Structural Commentary: Polarization converts preservation of norms into preservation of inner products.
Key Insight: Length preservation for all vectors also preserves angle information. □

Theorem 2.2.89. *Isometries and the Adjoint Product*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is an isometry if and only if $T^*T = I_{\mathcal{V}}$.

Proof.

($\Rightarrow$) Suppose T is an isometry. By Theorem 2.2.88, $\langle T^*T\mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$ for all $\mathbf{u}, \mathbf{v}$.
Thus $\langle (T^*T - I_{\mathcal{V}})\mathbf{u}, \mathbf{v}\rangle = 0$ for every $\mathbf{v}$, so $T^*T\mathbf{u} = \mathbf{u}$ for every $\mathbf{u}$.
($\Leftarrow$) Suppose $T^*T = I_{\mathcal{V}}$. Then $\|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle T^*T\mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$ for every $\mathbf{u}$, so T is an isometry.

Structural Commentary: The operator T^*T records inner products after applying T.
Key Insight: T is an isometry exactly when this Gram operator is the identity. □

Definition 2.2.90. *Unitary and Orthogonal*
An operator $U \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space is *unitary* if $U^*U = UU^* = I_{\mathcal{V}}$.
A matrix $U \in \mathrm{Mat}(n, n, \mathbb{F})$ is *unitary* if $U^*U = I_n$. If $\mathbb{F} = \mathbb{R}$, a unitary matrix is called *orthogonal*.

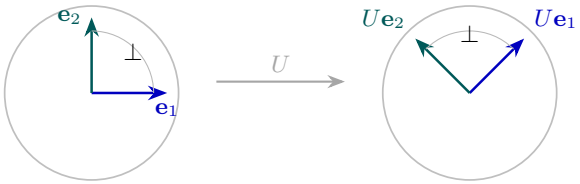


Figure 21: Unitary invariance: $\|U\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle U\mathbf{x}, U\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.

Theorem 2.2.91. *Equivalent Conditions for a Unitary Matrix*
Let $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

i. *U is unitary, i.e. $U^*U = I_n$;*
ii. *the columns $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are orthonormal in $\mathbb{F}^n$;*
iii. *$UU^* = I_n$;*
iv. *U^* is unitary;*
v. *the rows of U are orthonormal;*
vi. *U is invertible and $U^{-1} = U^*$;*
vii. *$\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for every $\mathbf{x} \in \mathbb{F}^n$;*
viii. *$\langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} = \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}$ for every $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$.*

Proof.

$(i) \Rightarrow (ii)$. Since $(U^*U)_{ij} = \mathbf{u}_i^*\mathbf{u}_j = \langle \mathbf{u}_j, \mathbf{u}_i\rangle_{\mathbb{F}^n}$, equation $U^*U = I_n$ states columns of U are orthonormal.
$(ii) \Rightarrow (iii)$. Orthonormal columns give $U^*U = I_n$. Left inverse of a square matrix is right inverse, $UU^* = I_n$.
$(iii) \Rightarrow (iv)$. The equation $UU^* = I_n$ is $(U^*)^*U^* = I_n$, so U^* is unitary.
$(iv) \Rightarrow (v)$. Apply $(i) \Rightarrow (ii)$ to U^*; the columns of U^* are the conjugates of the rows of U.
$(v) \Rightarrow (vi)$. Orthonormal rows give $UU^* = I_n$. Thus U^* is the inverse of U, so $U^{-1} = U^*$.
$(vi) \Rightarrow (vii)$. If $U^{-1} = U^*$, then $U^*U = I_n$.
Therefore $\|U\mathbf{x}\|_2^2 = \langle U\mathbf{x}, U\mathbf{x}\rangle_{\mathbb{F}^n} = \langle U^*U\mathbf{x}, \mathbf{x}\rangle_{\mathbb{F}^n} = \|\mathbf{x}\|_2^2$ for every $\mathbf{x} \in \mathbb{F}^n$.
$(vii) \Rightarrow (viii)$. Apply Theorem 2.2.88 to $\mathbf{x} \mapsto U\mathbf{x}$.
$(viii) \Rightarrow (i)$. For every $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, $\langle (U^*U - I_n)\mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = \langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} - \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = 0$. Thus $U^*U = I_n$.

Structural Commentary: A unitary matrix is an isometry written in orthonormal coordinates.
Key Insight: Columns, rows, inverses, norms, and inner products are different faces of the same condition. □

Proposition 2.2.92. *Algebra of Unitary Matrices*
Let $U, V \in \mathrm{Mat}(n,n,\mathbb{F})$, $W \in \mathrm{Mat}(m,m,\mathbb{F})$, where $V \oplus W$ denotes block diagonal matrix with blocks V and W.

i. If U and V are unitary, then UV is unitary.
ii. $V \oplus W$ is unitary if and only if V and W are unitary.
iii. If U is unitary, then $|\det U| = 1$.

Proof.

(i) Compute $(UV)^*(UV) = V^*U^*UV = V^*V = I_n$.
(ii) Since $(V \oplus W)^*(V \oplus W) = V^*V \oplus W^*W$, this product equals I_{n+m} exactly when $V^*V = I_n$ and $W^*W = I_m$.
(iii) Taking determinants in $U^*U = I_n$ gives $1 = \det(U^*)\det(U) = \overline{\det(U)}\det(U) = |\det U|^2$.

Structural Commentary: Unitary structure is stable under composition and blockwise orthogonal sums.
Key Insight: The equation $U^*U = I$ makes every verification a short adjoint calculation. □

Theorem 2.2.93. *Extending Isometric Columns to a Unitary Matrix*
If $X \in \mathrm{Mat}(m,n,\mathbb{F})$ and $X^*X = I_n$, then there exists $X' \in \mathrm{Mat}(m, m-n, \mathbb{F})$ such that $[X\ X']$ is unitary.

Proof. Write $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_n]$. The condition $X^*X = I_n$ says $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are orthonormal in $\mathbb{F}^m$ by Theorem 2.2.91. By Corollary 2.2.40, extend this list to an orthonormal basis $\mathbf{x}_1, \ldots, \mathbf{x}_m$ of $\mathbb{F}^m$. Let $X' = [\mathbf{x}_{n+1}\ \cdots\ \mathbf{x}_m]$. Then the columns of $[X\ X']$ are orthonormal, so $[X\ X']$ is unitary.
Structural Commentary: Isometric columns are a partial orthonormal basis.
Key Insight: Complete the columns by extending the orthonormal list. □

Definition 2.2.94. *Unitary and Permutation Similarity*
Square matrices $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are *unitarily similar* if $A = UBU^*$ for some unitary U. They are *orthogonally similar* if $\mathbb{F} = \mathbb{R}$ and U can be chosen real orthogonal.
A *permutation matrix* is a square matrix with exactly one entry equal to 1 in each row and each column, and all other entries equal to 0. Matrices are *permutation similar* if $A = PBP^*$ for some permutation matrix P.

Proposition 2.2.95. *Permutation Matrices Are Orthogonal*
If $P = [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}]$ is a permutation matrix, then $P^*P = PP^* = I_n$. Hence P is real orthogonal.

Proof. The columns of P are a rearrangement of the standard orthonormal basis of $\mathbb{R}^n$, hence they are orthonormal. Theorem 2.2.91 gives $P^*P = PP^* = I_n$.
Structural Commentary: A permutation matrix only reorders the standard basis.
Key Insight: Reordering orthonormal vectors preserves orthonormality. □

Proposition 2.2.96. *Permutation Similarity Reorders Diagonal Entries*
Let $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and let $P = [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}]$ be a permutation matrix. Then $P^*\Lambda P = \mathrm{diag}(\lambda_{\sigma(1)}, \ldots, \lambda_{\sigma(n)})$.

Proof. The (i,j) entry of $P^*\Lambda P$ is $\mathbf{e}_{\sigma(i)}^*\Lambda\mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)}\mathbf{e}_{\sigma(i)}^*\mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)}\delta_{ij}$.
Thus all off-diagonal entries are zero, and the jth diagonal entry is $\lambda_{\sigma(j)}$.
Structural Commentary: Permutation similarity applies the same basis reordering to rows and columns.
Key Insight: Diagonal entries move exactly as the standard basis vectors are rearranged. □

Definition 2.2.97. *Rank-One Projection Matrix*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. The *rank-one projection matrix* onto $\mathrm{span}(\mathbf{u})$ is $P_{\mathbf{u}} := \mathbf{u}\mathbf{u}^* \in \mathrm{Mat}(n,n,\mathbb{F})$.
For every $\mathbf{x} \in \mathbb{F}^n$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

Proposition 2.2.98. *Properties of Rank-One Projection Matrices*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. Then:

i. $\mathrm{range}(P_{\mathbf{u}}) = \mathrm{span}(\mathbf{u})$;
ii. $\ker(P_{\mathbf{u}}) = \mathrm{span}(\mathbf{u})^{\perp}$;
iii. $P_{\mathbf{u}}^* = P_{\mathbf{u}}$;
iv. $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.

Proof.

(i) The formula $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle \mathbf{u}$ gives $\mathrm{range}(P_{\mathbf{u}}) \subseteq \mathrm{span}(\mathbf{u})$, and $P_{\mathbf{u}}\mathbf{u} = \mathbf{u}$ gives the reverse inclusion.
(ii) The same formula gives $P_{\mathbf{u}}\mathbf{x} = 0$ exactly when $\langle \mathbf{x}, \mathbf{u} \rangle = 0$.
(iii) Since $P_{\mathbf{u}} = \mathbf{u}\mathbf{u}^*$, $P_{\mathbf{u}}^* = (\mathbf{u}\mathbf{u}^*)^* = \mathbf{u}\mathbf{u}^* = P_{\mathbf{u}}$.
(iv) Because $\mathbf{u}^*\mathbf{u} = \|\mathbf{u}\|_2^2 = 1$, $P_{\mathbf{u}}^2 = \mathbf{u}(\mathbf{u}^*\mathbf{u})\mathbf{u}^* = \mathbf{u}\mathbf{u}^* = P_{\mathbf{u}}$.

Structural Commentary: A rank-one projection keeps the $\mathbf{u}$ direction and kills its orthogonal complement.
Key Insight: The outer product $\mathbf{u}\mathbf{u}^*$ encodes projection onto one unit direction. □

Definition 2.2.99. *Householder Matrix*
Let $\mathbf{w} \in \mathbb{F}^n$ be nonzero and set $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|_2$.
The *Householder matrix* associated to $\mathbf{w}$ is $U_{\mathbf{w}} := I_n - 2P_{\mathbf{u}} = I_n - 2\frac{\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|_2^2}$.
It acts by $\mathbf{x} \mapsto U_{\mathbf{w}}\mathbf{x} = \mathbf{x} - 2\langle \mathbf{x}, \mathbf{u}\rangle_{\mathbb{F}^n} \mathbf{u}$.

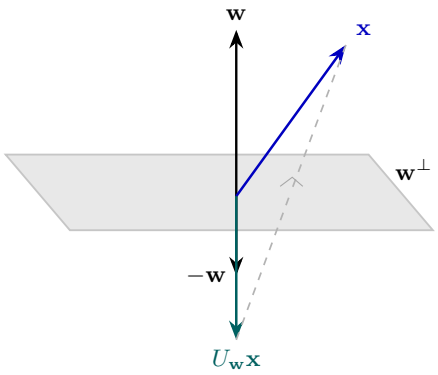


Figure 22: A Householder matrix fixes $\mathbf{w}^\perp$ and flips the span($\mathbf{w}$) component.

Theorem 2.2.100. *Householder Matrices Are Unitary Reflections*
Every Householder matrix $U_{\mathbf{w}}$ is unitary, Hermitian, and involutive: $U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$, $U_{\mathbf{w}}^* = U_{\mathbf{w}}$, $U_{\mathbf{w}}^2 = I_n$.
If $\mathbf{w} \in \mathbb{R}^n$, then $U_{\mathbf{w}}$ is real orthogonal and symmetric.

Proof. Let $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|_2$. By Proposition 2.2.98, $P_{\mathbf{u}}^* = P_{\mathbf{u}}$ and $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.
Therefore $U_{\mathbf{w}}^* = I_n - 2P_{\mathbf{u}} = U_{\mathbf{w}}$, $U_{\mathbf{w}}^2 = (I_n - 2P_{\mathbf{u}})^2 = I_n - 4P_{\mathbf{u}} + 4P_{\mathbf{u}}^2 = I_n$.
Thus $U_{\mathbf{w}}^* U_{\mathbf{w}} = U_{\mathbf{w}}^2 = I_n$, so $U_{\mathbf{w}}$ is unitary. The real case is the same statement with conjugation removed.
Structural Commentary: A Householder matrix reflects across an orthogonal hyperplane.
Key Insight: The identity $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$ makes $(I - 2P_{\mathbf{u}})^2 = I$. □

Theorem 2.2.101. *Householder Map Between Equal-Norm Vectors*
Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ satisfy $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 \neq 0$. Define

$$\sigma = \begin{cases} 1, & \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = 0, \\ -\dfrac{\langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}}{|\langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}|}, & \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} \neq 0, \end{cases} \qquad \mathbf{w} = \mathbf{y} - \sigma\mathbf{x}.$$

Then $\mathbf{w} \neq \mathbf{0}$, $|\sigma| = 1$, and $\sigma U_{\mathbf{w}}\mathbf{x} = \mathbf{y}$.

Proof. Clearly $|\sigma| = 1$. Let $z = \langle \mathbf{x}, \mathbf{y}\rangle$. By construction, $\sigma z = -|z|$, with the convention that this equals 0 when $z = 0$. Hence $\|\mathbf{w}\|_2^2 = \|\mathbf{y} - \sigma\mathbf{x}\|_2^2 = 2\left(\|\mathbf{x}\|_2^2 + |z|\right) > 0$, so $\mathbf{w} \neq \mathbf{0}$.
Since $\mathbf{y} = \mathbf{w} + \sigma\mathbf{x}$, $\sigma U_{\mathbf{w}}\mathbf{x} = \sigma\mathbf{x} - 2\sigma\frac{\langle \mathbf{x}, \mathbf{w}\rangle}{\|\mathbf{w}\|_2^2}\mathbf{w} = \mathbf{y} - \left(1 + 2\sigma\frac{\langle \mathbf{x}, \mathbf{w}\rangle}{\|\mathbf{w}\|_2^2}\right)\mathbf{w}$.
But $\langle \mathbf{x}, \mathbf{w}\rangle = z - \overline{\sigma}\|\mathbf{x}\|_2^2$, so $2\sigma\langle \mathbf{x}, \mathbf{w}\rangle = -2(\|\mathbf{x}\|_2^2 + |z|) = -\|\mathbf{w}\|_2^2$.
Coefficient of $\mathbf{w}$ is hence zero, proving $\sigma U_{\mathbf{w}}\mathbf{x} = \mathbf{y}$.
Structural Commentary: The scalar σ chooses the phase so that $\mathbf{y} - \sigma\mathbf{x}$ gives a stable reflection direction.
Key Insight: Householder reflections connect equal-norm vectors up to a unit scalar. □

Corollary 2.2.102. *Householder Unitary Annihilator*
Let $\mathbf{x} = [x_1 \ \cdots \ x_n]^t \in \mathbb{F}^n$ be nonzero. Define

$$\sigma = \begin{cases} 1, & x_1 = 0, \\ -\dfrac{x_1}{|x_1|}, & x_1 \neq 0, \end{cases} \qquad \mathbf{w} = \|\mathbf{x}\|_2 \mathbf{e}_1 - \sigma\mathbf{x}.$$

Then $\sigma U_{\mathbf{w}}\mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$. If $\mathbf{x}$ is a unit vector, then $\overline{\sigma}U_{\mathbf{w}}$ is unitary and has $\mathbf{x}$ as its first column.

Proof. Apply Theorem 2.2.101 with $\mathbf{y} = \|\mathbf{x}\|_2 \mathbf{e}_1$, noting that $\langle \mathbf{x}, \mathbf{y}\rangle = x_1\|\mathbf{x}\|_2$. If $\|\mathbf{x}\|_2 = 1$, then $(\sigma U_{\mathbf{w}})\mathbf{x} = \mathbf{e}_1$, so $(\sigma U_{\mathbf{w}})^*\mathbf{e}_1 = \mathbf{x}$. Since $(\sigma U_{\mathbf{w}})^* = \overline{\sigma}U_{\mathbf{w}}$, this says the first column of $\overline{\sigma}U_{\mathbf{w}}$ is $\mathbf{x}$.
Structural Commentary: The reflector concentrates all mass of $\mathbf{x}$ into the first coordinate.
Key Insight: Householder matrices produce explicit unitary matrices with a prescribed first column. □

2.3 Spectral Linear Algebra

2.3.1 Eigenstructure and Invariant Subspaces

The guiding idea of spectral linear algebra is to understand an operator by finding subspaces that it maps into themselves. On such subspaces, the operator can be studied without leaving the smaller space.

Definition 2.3.1. *Invariant Subspace*
Let $T \in \mathcal{L}(\mathcal{V})$. A subspace $\mathcal{U}$ of $\mathcal{V}$ is *invariant under T* if $T\mathbf{u} \in \mathcal{U}$ for every $\mathbf{u} \in \mathcal{U}$.

Equivalently, $\mathcal{U}$ is invariant under T exactly when the restriction $T|_{\mathcal{U}}$ is an operator on $\mathcal{U}$.

Proposition 2.3.2. *One-Dimensional Invariant Subspaces and Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, $\mathbf{v} \neq \mathbf{0}$. Then $\operatorname{span}(\mathbf{v})$ *is invariant under T if and only if there exists $\lambda \in \mathbb{F}$ such that $T\mathbf{v} = \lambda\mathbf{v}$.*

Proof.
($\Rightarrow$) If $\operatorname{span}(\mathbf{v})$ is invariant under T, then $T\mathbf{v} \in \operatorname{span}(\mathbf{v})$, so $T\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{F}$.
($\Leftarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{u} = a\mathbf{v} \in \operatorname{span}(\mathbf{v})$, then $T\mathbf{u} = aT\mathbf{v} = a\lambda\mathbf{v} \in \operatorname{span}(\mathbf{v})$.
Structural Commentary: One-dimensional invariance means rescaling one direction.
Key Insight: In dimension one, staying inside the subspace means being a scalar multiple of the same vector. □

Definition 2.3.3. *Eigenvalue and Eigenvector*
Let $T \in \mathcal{L}(\mathcal{V})$. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T if there exists $\mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$ and $T\mathbf{v} = \lambda\mathbf{v}$. Such a vector $\mathbf{v}$ is an *eigenvector* of T corresponding to λ.

Definition 2.3.4. *Eigenspace*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\lambda \in \mathbb{F}$. The *eigenspace* of T corresponding to λ is $E_\lambda(T) := \ker(T - \lambda I_{\mathcal{V}})$. Thus λ is an eigenvalue of T if and only if $E_\lambda(T) \neq \{\mathbf{0}\}$. Equivalently, $E_\lambda(T)$ is the nullspace of the shifted operator $T - \lambda I_{\mathcal{V}}$.

Definition 2.3.5. *Spectrum and Geometric Multiplicity*
Let $T \in \mathcal{L}(\mathcal{V})$. The *spectrum* of T is $\operatorname{spec}(T) := \{\lambda \in \mathbb{F} : \lambda \text{ is an eigenvalue of } T\}$.
The *geometric multiplicity* of $\lambda \in \mathbb{F}$ is $\dim E_\lambda(T)$. For a square matrix $A \in \operatorname{Mat}(n, n, \mathbb{F})$, $E_\lambda(A) := \ker(A - \lambda I_n)$ and $\operatorname{spec}(A)$ denotes the spectrum of the induced operator on $\mathbb{F}^n$.

Proposition 2.3.6. *Eigenvectors as Kernel Vectors*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. Vector $\mathbf{v}$ is eigenvector of T corresponding to λ if and only if $\mathbf{v} \in \ker(T - \lambda I_{\mathcal{V}})$, $\mathbf{v} \neq \mathbf{0}$.

Proof.
($\Rightarrow$) If $\mathbf{v}$ is an eigenvector corresponding to λ, then $\mathbf{v} \neq \mathbf{0}$ and $T\mathbf{v} = \lambda\mathbf{v}$, so $(T - \lambda I_{\mathcal{V}})\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $\mathbf{v} \in \ker(T - \lambda I_{\mathcal{V}})$ and $\mathbf{v} \neq \mathbf{0}$, then $(T - \lambda I_{\mathcal{V}})\mathbf{v} = \mathbf{0}$, so $T\mathbf{v} = \lambda\mathbf{v}$.
Structural Commentary: Eigenvectors are kernels of shifted operators.
Key Insight: Subtract $\lambda I_{\mathcal{V}}$ to convert $T\mathbf{v} = \lambda\mathbf{v}$ into homogeneous form. □

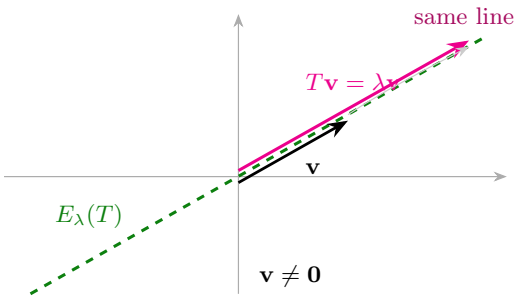


Figure 23: An eigendirection is a one-dimensional invariant subspace.

Theorem 2.3.7. *Equivalent Conditions for an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{F}$. The following are equivalent:

(i) λ is an eigenvalue of T;
(ii) $T - \lambda I_{\mathcal{V}}$ is not injective;
(iii) $T - \lambda I_{\mathcal{V}}$ is not surjective;
(iv) $T - \lambda I_{\mathcal{V}}$ is not invertible.

Proof.

(*i*) $\Rightarrow$ (*ii*) If λ is an eigenvalue, Proposition 2.3.6 gives nonzero vector in $\ker(T-\lambda I_{\mathcal{V}})$, so $T-\lambda I_{\mathcal{V}}$ not injective.
(*ii*) $\Rightarrow$ (*i*) If $T-\lambda I_{\mathcal{V}}$ not injective, then $\ker(T-\lambda I_{\mathcal{V}})$ contains nonzero vector, so λ is eigenvalue (Prop 2.3.6).
(*ii*) $\Rightarrow$ (*iii*) If $T-\lambda I_{\mathcal{V}}$ were surjective, Theorem 2.1.78 would make it injective, a contradiction.
(*iii*) $\Rightarrow$ (*iv*) If $T-\lambda I_{\mathcal{V}}$ were invertible, then it would be surjective, a contradiction.
(*iv*) $\Rightarrow$ (*ii*) If $T-\lambda I_{\mathcal{V}}$ were injective, Theorem 2.1.78 would make it invertible, a contradiction.

Structural Commentary: In finite dimensions, noninvertibility has equivalent forms.
Key Insight: A scalar λ is spectral precisely when $T-\lambda I_{\mathcal{V}}$ has a nontrivial kernel. □

Theorem 2.3.8. *Eigenvectors for Distinct Eigenvalues Are Independent*
Let $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Suppose the list is linearly dependent. Let k be smallest positive integer s.t. $\mathbf{v}_k \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. Then $k \geq 2$, $(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is linearly independent, and $\exists$ scalars $a_1, \ldots, a_{k-1}$ s.t. $\mathbf{v}_k = a_1\mathbf{v}_1 + \cdots + a_{k-1}\mathbf{v}_{k-1}$. Applying T gives $\lambda_k\mathbf{v}_k = a_1\lambda_1\mathbf{v}_1 + \cdots + a_{k-1}\lambda_{k-1}\mathbf{v}_{k-1}$. Multiplying first displayed equation by λ_k and subtracting gives $0 = a_1(\lambda_k - \lambda_1)\mathbf{v}_1 + \cdots + a_{k-1}(\lambda_k - \lambda_{k-1})\mathbf{v}_{k-1}$. Since $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ are linearly independent and $\lambda_k \neq \lambda_i$ for $i < k$, all $a_i = 0$. Hence $\mathbf{v}_k = \mathbf{0}$, contradicting that $\mathbf{v}_k$ is an eigenvector.
Structural Commentary: A smallest dependence isolates the first new eigenvector that depends on earlier ones. Applying T and subtracting separates the distinct eigenvalue factors.
Key Insight: Distinct eigenvalues make the same dependence relation impossible after applying T. □

Corollary 2.3.9. *Number of Distinct Eigenvalues*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then T has at most $\dim(\mathcal{V})$ distinct eigenvalues.

Proof. Choose one eigenvector for each distinct eigenvalue of T. By Theorem 2.3.8, these eigenvectors form a linearly independent list. A linearly independent list in $\mathcal{V}$ has length at most $\dim(\mathcal{V})$ by Theorem 2.1.20.
Structural Commentary: Distinct eigenvalues give independent directions.
Key Insight: Count one eigenvector per eigenspace. □

Corollary 2.3.10. *Invertibility and the Zero Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is invertible if and only if $0 \notin \operatorname{spec}(T)$.

Proof. Apply Theorem 2.3.7 with $\lambda = 0$. Then 0 is an eigenvalue exactly when $T - 0I_{\mathcal{V}} = T$ is not invertible.
Structural Commentary: The zero eigenvalue is the spectral form of singularity.
Key Insight: Invertibility is equivalent to having trivial kernel, and $E_0(T) = \ker T$. □

Definition 2.3.11. *Restriction and Quotient Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T.

(i) The *restriction operator* $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is defined by $T|_{\mathcal{U}}(\mathbf{u}) = T\mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$.
(ii) The *quotient operator* $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ is defined by $(T/\mathcal{U})(\mathbf{v} + \mathcal{U}) = T\mathbf{v} + \mathcal{U}$, $\mathbf{v} \in \mathcal{V}$.

Proposition 2.3.12. *Restriction and Quotient Operators Are Well-Defined*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T. Then $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ and $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$.

Proof. As $\mathcal{U}$ invariant under T, $T\mathbf{u} \in \mathcal{U}$ whenever $\mathbf{u} \in \mathcal{U}$, so $T|_{\mathcal{U}}$ maps $\mathcal{U}$ into itself. Linearity inherited from T. For the quotient operator, suppose $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$. Then $\mathbf{v} - \mathbf{w} \in \mathcal{U}$. Invariance gives $T(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$, so $T\mathbf{v} - T\mathbf{w} \in \mathcal{U}$ and hence $T\mathbf{v} + \mathcal{U} = T\mathbf{w} + \mathcal{U}$. Thus $T/\mathcal{U}$ is well-defined. Linearity follows from the quotient operations and linearity of T.
Structural Commentary: Restriction stays inside $\mathcal{U}$; quotient collapses $\mathcal{U}$.
Key Insight: Invariance makes both operators well-defined. □

Perspective 2.3.13. *Two Smaller Views of an Operator*
When $\mathcal{U}$ is invariant under T, the operators $T|_{\mathcal{U}}$ and $T/\mathcal{U}$ split attention between what happens inside $\mathcal{U}$ and what remains after $\mathcal{U}$ is collapsed. These two views often simplify an operator, even though they need not determine the whole operator by themselves.

2.3.2 Spectral Theory

Spectral theory studies an operator through algebraic expressions in the operator and through bases that reveal invariant flags. Polynomial expressions produce eigenvalues over $\mathbb{C}$, and eigenvalues make upper-triangular representations possible.

Definition 2.3.14. *Powers of an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$. For a positive integer m, $T^m := \underbrace{T \cdots T}_{m \text{ times}}$ and $T^0 := I_{\mathcal{V}}$.
If T is invertible, then $T^{-m} := (T^{-1})^m$ for positive integers m.

Definition 2.3.15. *Polynomial Applied to an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $p \in \mathcal{P}(\mathbb{F})$ be $p(z) = a_0 + a_1 z + \cdots + a_m z^m$. Then $p(T) := a_0 I_{\mathcal{V}} + a_1 T + \cdots + a_m T^m$.

Definition 2.3.16. *Annihilating Polynomial*
Let $T \in \mathcal{L}(\mathcal{V})$. A nonzero polynomial $p \in \mathcal{P}(\mathbb{F})$ *annihilates* T if $p(T) = 0$.

Definition 2.3.17. *Product of Polynomials*
If $p, q \in \mathcal{P}(\mathbb{F})$, their product $pq \in \mathcal{P}(\mathbb{F})$ is defined by $(pq)(z) = p(z)q(z)$ for $z \in \mathbb{F}$.

Proposition 2.3.18. *Multiplicative Properties of Operator Polynomials*
Let $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(\mathcal{V})$. Then

(i) $(pq)(T) = p(T)q(T)$;
(ii) $p(T)q(T) = q(T)p(T)$.

Proof.

(i) Write $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$. Then $(pq)(z) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k z^{j+k}$.
Substituting T gives $(pq)(T) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} = \left(\sum_{j=0}^{m} a_j T^j \right) \left(\sum_{k=0}^{n} b_k T^k \right) = p(T)q(T)$.

(ii) Since polynomial multiplication is commutative, $pq = qp$.
By part (i), $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$.

Structural Commentary: Polynomial evaluation preserves products.
Key Insight: Expanding a polynomial product works the same with T as with z. □

Proposition 2.3.19. *Existence of Annihilating Polynomials*
Let $\mathcal{V}$ be finite-dimensional with $n = \dim(\mathcal{V}) > 0$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then T has a nonconstant annihilating polynomial of degree at most n^2.

Proof. By Theorem 2.1.109, the vector space $\mathcal{L}(\mathcal{V})$ has dimension n^2, so the $n^2 + 1$ operators $I_{\mathcal{V}}, T, T^2, \ldots, T^{n^2}$ are linearly dependent. Hence there are scalars $a_0, \ldots, a_{n^2}$, not all zero, such that $\sum_{j=0}^{n^2} a_j T^j = 0$.
If $a_1 = \cdots = a_{n^2} = 0$, then $a_0 I_{\mathcal{V}} = 0$, forcing $a_0 = 0$ because $\mathcal{V} \neq \{\mathbf{0}\}$, a contradiction. Thus $p(z) = \sum_{j=0}^{n^2} a_j z^j$ is a nonconstant annihilating polynomial.
Structural Commentary: Finite-dimensionality makes the powers of one operator linearly dependent inside $\mathcal{L}(\mathcal{V})$.
Key Insight: A dependence among $I_{\mathcal{V}}, T, \ldots, T^{n^2}$ is exactly a polynomial equation satisfied by T. □

Proposition 2.3.20. *Polynomial Evaluation on Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, let $p \in \mathcal{P}(\mathbb{F})$, and suppose $T\mathbf{v} = \lambda \mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $p(T)\mathbf{v} = p(\lambda)\mathbf{v}$.

Proof. Induct on j to get $T^j \mathbf{v} = \lambda^j \mathbf{v}$ for every $j \geq 0$.
If $p(z) = \sum_{j=0}^{m} a_j z^j$, then $p(T)\mathbf{v} = \sum_{j=0}^{m} a_j T^j \mathbf{v} = \sum_{j=0}^{m} a_j \lambda^j \mathbf{v} = p(\lambda)\mathbf{v}$.
Structural Commentary: On a one-dimensional eigenspace, T acts like the scalar λ.
Key Insight: Polynomial expressions in T reduce to the same polynomial evaluated at λ. □

Theorem 2.3.21. *Annihilating Polynomials Control the Spectrum*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let p be a nonconstant annihilating polynomial for T.

(i) Every eigenvalue of T is a root of p.
(ii) If $\mathbb{F} = \mathbb{C}$, then at least one root of p is an eigenvalue of T.

Proof.

(i) If $T\mathbf{v} = \lambda \mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then Proposition 2.3.20 gives $0 = p(T)\mathbf{v} = p(\lambda)\mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, $p(\lambda) = 0$.

(ii) Factor $p(z) = c(z-\lambda_1)\cdots(z-\lambda_m)$ over $\mathbb{C}$, with $c \neq 0$. Then $0 = p(T) = c(T-\lambda_1 I_\mathcal{V})\cdots(T-\lambda_m I_\mathcal{V})$. At least one factor $T-\lambda_j I_\mathcal{V}$ is not injective; otherwise product would be injective. Hence λ_j is an eigenvalue.

Structural Commentary: Annihilating polynomial restricts spectrum to its roots; over $\mathbb{C}$, one root must appear.
Key Insight: Eigenvectors turn $p(T)$ into $p(\lambda)$. □

Proposition 2.3.22. *Kernel and Range of a Polynomial in an Operator Are Invariant*
Let $T \in \mathcal{L}(\mathcal{V})$ and $p \in \mathcal{P}(\mathbb{F})$. Then $\ker(p(T))$ and $\operatorname{range}(p(T))$ are invariant under T.

Proof. If $\mathbf{v} \in \ker(p(T))$, then $p(T)\mathbf{v} = \mathbf{0}$.
Since $p(T)$ commutes with T by Proposition 2.3.18, $p(T)(T\mathbf{v}) = T(p(T)\mathbf{v}) = T\mathbf{0} = \mathbf{0}$.
If $\mathbf{v} \in \operatorname{range}(p(T))$, write $\mathbf{v} = p(T)\mathbf{u}$. Then $T\mathbf{v} = Tp(T)\mathbf{u} = p(T)(T\mathbf{u})$, so $T\mathbf{v} \in \operatorname{range}(p(T))$.
Structural Commentary: Polynomial kernels and ranges are stable under the original operator.
Key Insight: Since T commutes with $p(T)$, it preserves both $\ker p(T)$ and $\operatorname{range} p(T)$. □

Theorem 2.3.23. *Complex Operators Have Eigenvalues*
Every operator on a finite-dimensional, nonzero complex vector space has an eigenvalue.

Proof. Let $T \in \mathcal{L}(\mathcal{V})$. By Proposition 2.3.19, T has a nonconstant annihilating polynomial. Since $\mathcal{V}$ is complex and nonzero, Theorem 2.3.21 implies that one root of this polynomial is an eigenvalue of T.
Structural Commentary: Finite dimension gives an annihilating polynomial; over $\mathbb{C}$, it factors linearly.
Key Insight: One noninjective factor produces an eigenvector. □

Theorem 2.3.24. *Common Eigenvector for a Commuting Family*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $\mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a nonempty commuting family: $ST = TS$ for all $S, T \in \mathcal{F}$. Then there exists $\mathbf{v} \neq \mathbf{0}$ that is an eigenvector of every operator in $\mathcal{F}$.

Proof. First suppose $\mathcal{F} = \{T_1, \ldots, T_m\}$ is finite, and induct on m. The case $m = 1$ is Theorem 2.3.23.
For $m > 1$, choose $\lambda \in \operatorname{spec}(T_1)$ and set $\mathcal{U} = E_\lambda(T_1)$. If $\mathbf{u} \in \mathcal{U}$, $j \geq 2$, then $(T_1 - \lambda I_\mathcal{V})T_j\mathbf{u} = T_j(T_1 - \lambda I_\mathcal{V})\mathbf{u} = \mathbf{0}$, so $\mathcal{U}$ is invariant under $T_2, \ldots, T_m$. Their commuting restrictions to $\mathcal{U}$ have, by induction, a common eigenvector $\mathbf{v} \in \mathcal{U}$; then $\mathbf{v}$ is also an eigenvector of T_1.
For general $\mathcal{F}$, choose $T_1, \ldots, T_m \in \mathcal{F}$ spanning $\operatorname{span}(\mathcal{F})$. The finite case gives a common eigenvector $\mathbf{v}$ for them. If $S = \sum_j a_j T_j \in \mathcal{F}$ and $T_j\mathbf{v} = \mu_j\mathbf{v}$, then $S\mathbf{v} = (\sum_j a_j\mu_j)\mathbf{v}$, so $\mathbf{v}$ is common to all of $\mathcal{F}$.
Structural Commentary: Commutation makes one eigenspace invariant, reducing the family by one.
Key Insight: Find the common eigenvector inside an eigenspace. □

Definition 2.3.25. *Diagonal and Upper-Triangular Matrices*
The *diagonal* of a square matrix is its upper-left to lower-right entries. A square matrix is *diagonal* if all off-diagonal entries are zero, and *upper triangular* if all entries below the diagonal are zero.

Definition 2.3.26. *Diagonalizable Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ is *diagonalizable* if T has a diagonal matrix with respect to some basis of $\mathcal{V}$.

Proposition 2.3.27. *Eigenspaces for Distinct Eigenvalues Form a Direct Sum*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T. Then $E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)$ is a direct sum. Consequently, $\dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T) \leq \dim(\mathcal{V})$.

Proof. Suppose $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. If some $\mathbf{u}_j$ were nonzero, then the nonzero vectors among $\mathbf{u}_1, \ldots, \mathbf{u}_m$ would be eigenvectors corresponding to distinct eigenvalues, and Theorem 2.3.8 would make them linearly independent. This contradicts their sum being $\mathbf{0}$. Hence every $\mathbf{u}_j = \mathbf{0}$, so the sum is direct.
By Theorem 2.1.38, $\dim(E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.
The left side is the dimension of a subspace of $\mathcal{V}$, so Proposition 2.1.28 gives the stated inequality.
Structural Commentary: Distinct eigenspaces form independent pieces.
Key Insight: A zero sum across them would contradict eigenvector independence. □

Theorem 2.3.28. *Conditions Equivalent to Diagonalizability*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Set $n = \dim(\mathcal{V})$. The following are equivalent:

(i) T is diagonalizable;
(ii) $\mathcal{V}$ has a basis consisting of eigenvectors of T;
(iii) $\exists$ one-dimensional subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_n$ of $\mathcal{V}$, each invariant under T, such that $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$;
(iv) $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$;
(v) $\dim(\mathcal{V}) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If T is diagonalizable, then T has a diagonal matrix with respect to some basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$. Hence $T\mathbf{v}_j = a_j\mathbf{v}_j$ for each j, so this basis consists of eigenvectors of T.

(*ii*) $\Rightarrow$ (*i*) If $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of eigenvectors of T, then $T\mathbf{v}_j = a_j\mathbf{v}_j$ for each j. Thus the matrix of T with respect to this basis is diagonal.

(*ii*) $\Rightarrow$ (*iii*) Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an eigenbasis, and put $\mathcal{U}_j = \operatorname{span}(\mathbf{v}_j)$. Each $\mathcal{U}_j$ is one-dimensional and invariant under T by Proposition 2.3.2; the basis decomposition gives $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$.

(*iii*) $\Rightarrow$ (*ii*) Choose nonzero $\mathbf{v}_j \in \mathcal{U}_j$. Proposition 2.3.2 makes each $\mathbf{v}_j$ an eigenvector of T, and the direct-sum decomposition makes $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ a basis of $\mathcal{V}$.

(*ii*) $\Rightarrow$ (*iv*) Every vector in an eigenbasis lies in one of $E_{\lambda_1}(T), \dots, E_{\lambda_m}(T)$, so these eigenspaces span $\mathcal{V}$. Proposition 2.3.27 shows their sum is direct.

(*iv*) $\Rightarrow$ (*v*) If $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$, then dimensions add by Theorem 2.1.38.

(*v*) $\Rightarrow$ (*ii*) For each j, choose a basis β_j of $E_{\lambda_j}(T)$, and concatenate these bases into a list β. The length of β is $\dim(\mathcal{V})$. If a linear combination of vectors in β is zero, grouping terms by eigenspace gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. Proposition 2.3.27 gives $\mathbf{u}_j = \mathbf{0}$ for every j; since each β_j is independent, all coefficients are zero. Thus β is independent, Proposition 2.1.28 makes β a basis of $\mathcal{V}$ consisting of eigenvectors of T.

Structural Commentary: Diagonalizability means the eigenspaces exhaust $\mathcal{V}$. Directness is automatic for distinct eigenspaces, so the real test is spanning, or equivalently the dimension count.
Key Insight: A diagonal basis is exactly an eigenbasis. □

Perspective 2.3.29. *Diagonalization as Exhaustion by Eigenspaces*
Distinct eigenspaces are already separated by Proposition 2.3.27. Thus diagonalizability is the stronger statement that these separated pieces account for every direction in $\mathcal{V}$.

Corollary 2.3.30. *Enough Eigenvalues Imply Diagonalizability*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$ has $\dim(\mathcal{V})$ distinct eigenvalues, then T is diagonalizable.

Proof. Let $n = \dim(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_n$ be distinct eigenvalues of T. For each j, choose an eigenvector $\mathbf{v}_j$ corresponding to λ_j. By Theorem 2.3.8, $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is linearly independent. Since this list has length n, Proposition 2.1.28 makes it a basis of $\mathcal{V}$. Thus $\mathcal{V}$ has an eigenbasis, so T is diagonalizable by Theorem 2.3.28.
Structural Commentary: One eigenvector from each distinct eigenvalue already gives a full-size independent list.
Key Insight: Once the independent eigenvectors reach $\dim(\mathcal{V})$, they must form a basis. □

Theorem 2.3.31. *Criteria for an Upper-Triangular Operator Matrix*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The following are equivalent:

(i) $[T]_{\beta \leftarrow \beta}$ *is upper triangular;*
(ii) $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ *for each* $j = 1, \dots, n$*;*
(iii) $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ *is invariant under T for each* $j = 1, \dots, n$*.*

Proof.

(*i*) $\Rightarrow$ (*ii*) The jth column of $[T]_{\beta \leftarrow \beta}$ is the coordinate column of $T\mathbf{v}_j$ in the basis β. Since the matrix is upper triangular, this column has zero entries below position j, so $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$.

(*ii*) $\Rightarrow$ (*i*) If $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j, then the jth coordinate column of $T\mathbf{v}_j$ has zero entries below position j. Hence $[T]_{\beta \leftarrow \beta}$ is upper triangular.

(*ii*) $\Rightarrow$ (*iii*) If $\mathbf{u} \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$, then linearity gives $T\mathbf{u}$ as a linear combination of $T\mathbf{v}_1, \dots, T\mathbf{v}_j$, all of which lie in $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$. Hence this span is invariant.

(*iii*) $\Rightarrow$ (*ii*) Apply invariance of $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ to $\mathbf{v}_j$.

Structural Commentary: Upper triangularity records an invariant flag.
Key Insight: Columns below the diagonal vanish exactly when $T\mathbf{v}_j$ uses only earlier basis vectors and itself. □

Theorem 2.3.32. *Complex Upper Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$.

Proof. Induct on $n = \dim(\mathcal{V})$. If $n \leq 1$, the result is immediate.
Suppose $n > 1$ and the result holds in smaller dimensions. Choose an eigenvector $\mathbf{v}_1$ of T by Theorem 2.3.23, and set $\mathcal{U} = \operatorname{span}(\mathbf{v}_1)$. Then $\mathcal{U}$ is T-invariant, so the quotient operator $T/\mathcal{U}$ is defined on $\mathcal{V}/\mathcal{U}$, which has dimension $n - 1$. By induction, choose a basis $(\mathbf{v}_2 + \mathcal{U}, \dots, \mathbf{v}_n + \mathcal{U})$ of $\mathcal{V}/\mathcal{U}$ with respect to which $T/\mathcal{U}$ is upper triangular, and choose representatives $\mathbf{v}_2, \dots, \mathbf{v}_n \in \mathcal{V}$. Then $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$: any relation descends to the quotient basis, forcing all coefficients of $\mathbf{v}_2, \dots, \mathbf{v}_n$ to be zero, and then the coefficient of $\mathbf{v}_1$ is zero.
Now $T\mathbf{v}_1 \in \operatorname{span}(\mathbf{v}_1)$. For $j \geq 2$, upper triangularity of $T/\mathcal{U}$ gives $(T/\mathcal{U})(\mathbf{v}_j + \mathcal{U}) \in \operatorname{span}(\mathbf{v}_2 + \mathcal{U}, \dots, \mathbf{v}_j + \mathcal{U})$, so $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$. Hence Theorem 2.3.31 gives an upper-triangular matrix for T in the basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

Structural Commentary: Lift an upper-triangular quotient basis above an eigenline.
Key Insight: The quotient builds the remaining invariant flag. □

Proposition 2.3.33. *Invertibility from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then T is invertible if and only if $\lambda_j \neq 0$ for every j.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be such a basis.

($\Leftarrow$) If every $\lambda_j \neq 0$, then upper-triangular matrix $[T]_{\beta \leftarrow \beta}$ is invertible by Theorem 2.1.138, so T is invertible.
($\Rightarrow$) Suppose T is invertible. If $\lambda_j = 0$ for some j, set $\mathcal{U}_j = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$. Upper triangularity gives $T(\mathcal{U}_j) \subseteq \mathcal{U}_{j-1}$ because the coefficient of $\mathbf{v}_j$ in $T\mathbf{v}_j$ is $\lambda_j = 0$ and $T\mathbf{v}_i \in \mathcal{U}_{j-1}$ for $i < j$. Since $\dim(\mathcal{U}_j) > \dim(\mathcal{U}_{j-1})$, the restriction $T|_{\mathcal{U}_j} : \mathcal{U}_j \to \mathcal{U}_{j-1}$ is not injective. Hence T has a nonzero kernel vector, contradicting invertibility. Thus every $\lambda_j \neq 0$.

Structural Commentary: Diagonal entries control each flag layer.
Key Insight: A zero diagonal entry forces a kernel vector. □

Proposition 2.3.34. *Eigenvalues from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then the eigenvalues of T are precisely $\lambda_1, \ldots, \lambda_n$.

Proof. For $\lambda \in \mathbb{F}$, the matrix of $T - \lambda I_{\mathcal{V}}$ in the same basis is upper triangular with diagonal entries $\lambda_1 - \lambda, \ldots, \lambda_n - \lambda$. By Proposition 2.3.33, $T - \lambda I_{\mathcal{V}}$ is not invertible if and only if $\lambda = \lambda_j$ for some j. By Theorem 2.3.7, this is equivalent to λ being an eigenvalue of T.
Structural Commentary: Scalar shifts preserve triangular form.
Key Insight: Eigenvalues occur exactly when one shifted diagonal entry becomes zero. □

Perspective 2.3.35. *Why Triangular Form Matters*
Upper-triangular form is the first useful simplification beyond finding one eigenvector. It preserves enough of the operator to read eigenvalues from the diagonal while still allowing nontrivial coupling above the diagonal.

Theorem 2.3.36. *Complex Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

i. T is normal;
ii. $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
iii. T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(i) $\Rightarrow$ (ii) By Schur triangularisation (Theorem 2.2.61), choose an orthonormal basis $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ in which $A = [T]_{\gamma \leftarrow \gamma} = [a_{ij}]$ is upper triangular. Clear rows inductively: once rows $< k$ are cleared, Theorem 2.2.83 gives $|a_{kk}|^2 = \|T\mathbf{e}_k\|^2 = \|T^*\mathbf{e}_k\|^2 = |a_{kk}|^2 + \sum_{j>k} |a_{kj}|^2$.
Hence $a_{kj} = 0$ for $j > k$. Thus A is diagonal, so γ is an orthonormal basis of eigenvectors.
(ii) $\Rightarrow$ (iii) In an orthonormal eigenbasis, the matrix of T is diagonal.
(iii) $\Rightarrow$ (i) If $[T]$ is diagonal in an orthonormal basis, then $[T^*] = [T]^*$ is diagonal as well. Diagonal matrices commute, so $TT^* = T^*T$; hence T is normal.

Structural Commentary: Schur gives triangular form; normality kills the upper entries.
Key Insight: Equal norms of $T\mathbf{e}_j$ and $T^*\mathbf{e}_j$ force each row tail to vanish. □

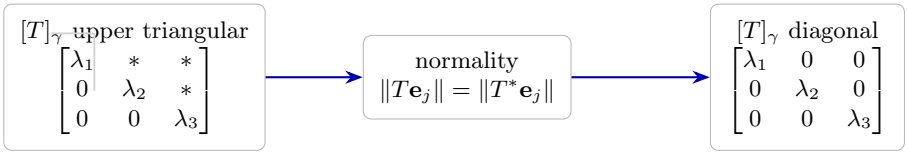


Figure 24: Normality turns Schur triangular form diagonal by forcing row tails to vanish.

Proposition 2.3.37. *Invertible Quadratic Expressions*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint on a finite-dimensional inner-product space.
If $b, c \in \mathbb{R}$ and $b^2 < 4c$, then $T^2 + bT + cI_{\mathcal{V}}$ is invertible.

Proof. Let $S = T^2 + bT + cI_{\mathcal{V}}$. For $\mathbf{v} \neq \mathbf{0}$,

$$\begin{aligned}\langle S\mathbf{v}, \mathbf{v}\rangle &= \|T\mathbf{v}\|^2 + b\langle T\mathbf{v}, \mathbf{v}\rangle + c\|\mathbf{v}\|^2 \geq \|T\mathbf{v}\|^2 - |b|\,\|T\mathbf{v}\|\,\|\mathbf{v}\| + c\|\mathbf{v}\|^2 \\ &= \left(\|T\mathbf{v}\| - \frac{|b|}{2}\|\mathbf{v}\|\right)^2 + \left(c - \frac{b^2}{4}\right)\|\mathbf{v}\|^2 > 0\end{aligned}$$

Thus $S\mathbf{v} \neq \mathbf{0}$ for every nonzero $\mathbf{v}$, so $\ker(S) = \{\mathbf{0}\}$. In finite dimensions, injectivity implies invertibility.
Structural Commentary: Negative discriminant keeps this self-adjoint quadratic positive.
Key Insight: Self-adjointness converts $\langle T^2\mathbf{v}, \mathbf{v}\rangle$ into $\|T\mathbf{v}\|^2$, so completing the square controls the kernel. □

Lemma 2.3.38. *Self-Adjoint Operators Have Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real inner-product space. Every self-adjoint $T \in \mathcal{L}(\mathcal{V})$ has an eigenvalue.

Proof. Choose $\mathbf{v} \neq \mathbf{0}$ and set $n = \dim(\mathcal{V})$. The list $\mathbf{v}, T\mathbf{v}, \ldots, T^n\mathbf{v}$ is linearly dependent, so some nonzero real polynomial p satisfies $p(T)\mathbf{v} = \mathbf{0}$. Factor p over $\mathbb{R}$ as a product of real linear factors and irreducible real quadratic factors: $p(x) = a\prod_{j=1}^{M}(x^2 + b_j x + c_j)\prod_{k=1}^{m}(x - \lambda_k)$, where $a \neq 0$, $b_j^2 < 4c_j$, and $\lambda_k \in \mathbb{R}$.
By Proposition 2.3.37, every quadratic factor $T^2 + b_j T + c_j I_{\mathcal{V}}$ is invertible. If $m = 0$, then $p(T)$ would be invertible, contradicting $p(T)\mathbf{v} = 0$ with $\mathbf{v} \neq 0$. Hence $m \geq 1$.
After removing the invertible quadratic factors, the product $(T - \lambda_1 I_{\mathcal{V}})\cdots(T - \lambda_m I_{\mathcal{V}})$ has nontrivial kernel. Therefore at least one factor $T - \lambda_k I_{\mathcal{V}}$ is not injective, and λ_k is an eigenvalue of T.
Structural Commentary: An annihilating polynomial remains; quadratic factors are invertible.
Key Insight: The nontrivial kernel must come from a real linear factor, hence from an eigenvalue. □

Proposition 2.3.39. *Self-Adjoint Operators and Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint, and let $\mathcal{U}$ be a subspace of $\mathcal{V}$ invariant under T. Then:

i. $\mathcal{U}^\perp$ is invariant under T;
ii. $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is self-adjoint;
iii. $T|_{\mathcal{U}^\perp} \in \mathcal{L}(\mathcal{U}^\perp)$ is self-adjoint.

Proof.

(i) If $\mathbf{v} \in \mathcal{U}^\perp$ and $\mathbf{u} \in \mathcal{U}$, then $\langle T\mathbf{v}, \mathbf{u}\rangle = \langle \mathbf{v}, T\mathbf{u}\rangle = 0$ because $T\mathbf{u} \in \mathcal{U}$. Hence $T\mathbf{v} \in \mathcal{U}^\perp$.
(ii) For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, $\langle (T|_{\mathcal{U}})\mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, (T|_{\mathcal{U}})\mathbf{v}\rangle$. Thus $T|_{\mathcal{U}}$ is self-adjoint.
(iii) By (i), $\mathcal{U}^\perp$ is invariant, so (ii)'s argument applies to $T|_{\mathcal{U}^\perp}$.

Structural Commentary: Self-adjointness preserves orthogonal complements of invariant subspaces.
Key Insight: Move T across the inner product; invariance of $\mathcal{U}$ then forces orthogonality to persist. □

Theorem 2.3.40. *Real Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

i. T is self-adjoint;
ii. $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
iii. T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(i) $\Rightarrow$ *(ii)* Assume $T = T^*$ and induct on $n = \dim(\mathcal{V})$; $n \leq 1$ is clear. For $n \geq 2$, Lemma 2.3.38 gives a unit eigenvector $\mathbf{u}$. Set $\mathcal{U} = \operatorname{span}(\mathbf{u})$. Since $\mathcal{U}$ is invariant, Proposition 2.3.39 makes $\mathcal{U}^\perp$ invariant and $T|_{\mathcal{U}^\perp}$ self-adjoint. Induct on $\mathcal{U}^\perp$ and adjoin $\mathbf{u}$.
(ii) $\Rightarrow$ *(iii)* In an orthonormal eigenbasis, the matrix of T is diagonal.
(iii) $\Rightarrow$ *(i)* In an orthonormal basis where $[T]$ is real diagonal, $[T^*] = [T]^t = [T]$. Hence $T = T^*$.

Structural Commentary: Split off one eigendirection, then restrict to its orthogonal complement.
Key Insight: One eigenvector reduces the dimension. □

Definition 2.3.41. *Positive Operator*
$T \in \mathcal{L}(\mathcal{V})$ is *positive* if $T = T^*$ and $\langle T\mathbf{v}, \mathbf{v}\rangle \geq 0$ for all $\mathbf{v} \in \mathcal{V}$.

Definition 2.3.42. *Square Root of an Operator*
An operator $R \in \mathcal{L}(\mathcal{V})$ is a *square root* of $T \in \mathcal{L}(\mathcal{V})$ if $R^2 = T$.

Theorem 2.3.43. *Characterization of Positive Operators*
For $T \in \mathcal{L}(\mathcal{V})$, the following are equivalent:

i. T is positive;
ii. T is self-adjoint and every eigenvalue of T is nonnegative;
iii. T has a positive square root;

iv. T has a self-adjoint square root;
v. there exists $R \in \mathcal{L}(\mathcal{V})$ such that $T = R^*R$.

Proof.
(*i*) $\Rightarrow$ (*ii*) If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v} \neq 0$, then $\lambda \in \mathbb{R}$ and $0 \leq \langle T\mathbf{v}, \mathbf{v}\rangle = \lambda \|\mathbf{v}\|^2$ so $\lambda \geq 0$.
(*ii*) $\Rightarrow$ (*iii*) By the spectral theorem, choose an orthonormal eigenbasis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $T\mathbf{e}_j = \lambda_j \mathbf{e}_j$ and $\lambda_j \geq 0$. Define $S\mathbf{e}_j = \sqrt{\lambda_j}\mathbf{e}_j$. Then S is positive and $S^2 = T$.
(*iii*) $\Rightarrow$ (*iv*) A positive square root is self-adjoint.
(*iv*) $\Rightarrow$ (*v*) If $T = R^2$ with $R = R^*$, then $T = R^*R$.
(*v*) $\Rightarrow$ (*i*) If $T = R^*R$, then $T^* = T$ and $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle R\mathbf{v}, R\mathbf{v}\rangle = \|R\mathbf{v}\|^2 \geq 0$ for every $\mathbf{v}$, so T is positive.
Structural Commentary: Positivity equals nonnegative spectrum and square-root factorization.
Key Insight: In an eigenbasis, the proof is scalar: $\lambda \geq 0$ iff $\sqrt{\lambda}$ exists. □

Proposition 2.3.44. *Unique Positive Square Root*
Every positive operator on $\mathcal{V}$ has a unique positive square root.

Proof. Existence follows from Theorem 2.3.43. For uniqueness, decompose $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$.
Any positive square root must equal $\sqrt{\lambda_j}I$ on $E_{\lambda_j}(T)$: indeed, if $R^2 = T$, then R commutes with T, so $E_{\lambda_j}(T)$ is R-invariant, and positivity plus the spectral theorem force $R|_{E_{\lambda_j}(T)} = \sqrt{\lambda_j}I$.
Thus two positive square roots agree on every eigenspace.
Structural Commentary: Each eigenspace determines its own scalar root.
Key Insight: Positivity selects the nonnegative root. □

Definition 2.3.45. *Positive Square Root Notation*
If T is positive, then $\sqrt{T}$ denotes the unique positive square root of T.

Theorem 2.3.46. *Description of Complex Isometries*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $S \in \mathcal{L}(\mathcal{V})$. Then S is an isometry if and only if $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of S whose eigenvalues all have absolute value 1.

Proof. If $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$ in an orthonormal basis with $|\lambda_j| = 1$, then for $\mathbf{v} = \sum_j c_j \mathbf{e}_j$,
$\|S\mathbf{v}\|^2 = \sum_j |\lambda_j c_j|^2 = \sum_j |c_j|^2 = \|\mathbf{v}\|^2$, so S is an isometry. Conversely, if S is an isometry, then $S^*S = I_{\mathcal{V}}$ by Theorem 2.2.89. Finite-dimensional invertibility gives $SS^* = I_{\mathcal{V}}$, so S is normal. By Complex Spectral Theorem, choose orthonormal eigenbasis $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$. Since $\mathbf{e}_j$ is unit and S preserves norms, $|\lambda_j| = \|S\mathbf{e}_j\| = 1$.
Structural Commentary: Complex isometries are normal with unit-circle spectrum.
Key Insight: In an eigenbasis, isometry means $|\lambda_j| = 1$. □

2.3.3 Polynomial and Jordan Structure

Generalized eigenvectors repair the main defect of ordinary eigenvectors: an operator may not have enough eigendirections to form a basis. The replacement is to study vectors killed by powers of shifted operators.

Definition 2.3.47. *Generalized Eigenvectors and Generalized Eigenspaces*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. Nonzero vector $\mathbf{v} \in \mathcal{V}$ is a *generalized eigenvector* of T to λ if $(T - \lambda I_{\mathcal{V}})^j \mathbf{v} = \mathbf{0}$ for some positive integer j. The *generalized eigenspace* corresponding to λ is

$$G(\lambda, T) := \{\mathbf{0}\} \cup \{\mathbf{v} \neq \mathbf{0} : \mathbf{v} \text{ is a generalized eigenvector of } T \text{ corresponding to } \lambda\}.$$

Every eigenvector is a generalized eigenvector by taking $j = 1$. The point is that higher powers of $T - \lambda I_{\mathcal{V}}$ may reveal directions that are not eigenvectors but are eventually pushed into the eigenspace.

Proposition 2.3.48. *Kernels of Powers Increase*
Let $T \in \mathcal{L}(\mathcal{V})$. Then $\{\mathbf{0}\} = \ker(T^0) \subseteq \ker(T) \subseteq \ker(T^2) \subseteq \cdots \subseteq \ker(T^k) \subseteq \ker(T^{k+1}) \subseteq \cdots$.

Proof. If $\mathbf{v} \in \ker(T^k)$, then $T^k\mathbf{v} = \mathbf{0}$, hence $T^{k+1}\mathbf{v} = T(T^k\mathbf{v}) = \mathbf{0}$. Thus $\ker(T^k) \subseteq \ker(T^{k+1})$ for every $k \geq 0$.
Structural Commentary: Powers of T form an increasing chain of kernels.
Key Insight: Once a vector is killed by T^k, applying one more T cannot revive it. □

Proposition 2.3.49. *Equality Forces Stabilization*
Let $T \in \mathcal{L}(\mathcal{V})$ and $m \geq 0$. If $\ker(T^m) = \ker(T^{m+1})$, then $\ker(T^m) = \ker(T^{m+1}) = \ker(T^{m+2}) = \cdots$.

Proof. By Proposition 2.3.48, it suffices to prove $\ker(T^{m+k+1}) \subseteq \ker(T^{m+k})$ for every $k \geq 0$.
Take $\mathbf{v} \in \ker(T^{m+k+1})$. Then $T^{m+1}(T^k\mathbf{v}) = T^{m+k+1}\mathbf{v} = \mathbf{0}$, so $T^k\mathbf{v} \in \ker(T^{m+1}) = \ker(T^m)$.
Hence $T^{m+k}\mathbf{v} = T^m(T^k\mathbf{v}) = \mathbf{0}$, proving the reverse inclusion.
Structural Commentary: Stabilization at one step propagates through the whole tail.
Key Insight: A later kernel vector can be shifted down into the already-stable kernel. □

Theorem 2.3.50. *Kernels Stop Growing by $\dim(\mathcal{V})$*
If $\mathcal{V}$ is finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and $n = \dim(\mathcal{V})$, then $\ker(T^n) = \ker(T^{n+1}) = \ker(T^{n+2}) = \cdots$.

Proof. If $\ker(T^n) \neq \ker(T^{n+1})$, Prop 2.3.49 implies all inclusions $\{\mathbf{0}\} = \ker(T^0) \subsetneq \ker(T) \subsetneq \cdots \subsetneq \ker(T^{n+1})$ are strict. Each strict inclusion raises dimension by at least 1, so $\dim \ker(T^{n+1}) \geq n+1$, impossible because $\ker(T^{n+1}) \subseteq \mathcal{V}$. Thus $\ker(T^n) = \ker(T^{n+1})$, and Proposition 2.3.49 gives the tail equality.
Structural Commentary: Strict kernel growth is dimension-limited.
Key Insight: Dimension bounds force the kernel chain to stabilize no later than step $\dim(\mathcal{V})$. □

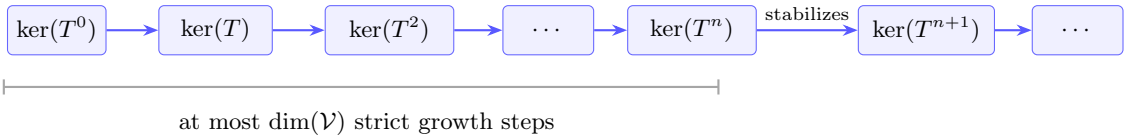


Figure 25: The kernel filtration grows until it stabilizes; after equality appears once, the tail is constant.

Theorem 2.3.51. *Stable Kernel-Range Decomposition*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and set $n = \dim(\mathcal{V})$. Then $\mathcal{V} = \ker(T^n) \oplus \operatorname{range}(T^n)$.

Proof. First show the intersection is trivial. If $\mathbf{v} \in \ker(T^n) \cap \operatorname{range}(T^n)$, then $\mathbf{v} = T^n\mathbf{u}$ and $T^n\mathbf{v} = \mathbf{0}$. Hence $T^{2n}\mathbf{u} = \mathbf{0}$, so $\mathbf{u} \in \ker(T^{2n}) = \ker(T^n)$ by Theorem 2.3.50. Thus $\mathbf{v} = T^n\mathbf{u} = \mathbf{0}$.
By Rank-Nullity Theorem 2.1.68, $\dim \ker(T^n) + \dim \operatorname{range}(T^n) = \dim(\mathcal{V})$.
The zero intersection and the dimension count imply the sum is direct and equals $\mathcal{V}$.
Structural Commentary: At the stable power, killed directions and surviving image directions separate cleanly.
Key Insight: Stabilization turns the usual rank-nullity count into an actual direct-sum decomposition. □

Theorem 2.3.52. *Description of Generalized Eigenspaces*
Let $\mathcal{V}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$, and $n = \dim(\mathcal{V})$. Then $G(\lambda, T) = \ker\big((T - \lambda I_{\mathcal{V}})^n\big)$.
In particular, $G(\lambda, T)$ is a subspace of $\mathcal{V}$.

Proof. Set $S = T - \lambda I_{\mathcal{V}}$.
($\Rightarrow$) If $\mathbf{v} \in G(\lambda, T)$, then $S^j\mathbf{v} = \mathbf{0}$ for some $j \geq 1$. If $j \leq n$, Proposition 2.3.48 gives $\mathbf{v} \in \ker(S^n)$; if $j > n$, Theorem 2.3.50 gives $\ker(S^j) = \ker(S^n)$.
($\Leftarrow$) If $\mathbf{v} \in \ker(S^n)$, then either $\mathbf{v} = \mathbf{0}$ or $\mathbf{v}$ is a generalized eigenvector of T corresponding to λ.
Thus $G(\lambda, T) = \ker(S^n)$, which is a subspace.
Structural Commentary: Generalized eigenspaces are stable shifted kernels.
Key Insight: Finite dimension makes exponent $\dim(\mathcal{V})$ enough. □

Proposition 2.3.53. *Generalized Eigenspaces Are Invariant*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Then $G(\lambda, T)$ is invariant under T for every $\lambda \in \mathbb{F}$.

Proof. By Theorem 2.3.52, $G(\lambda, T) = \ker((T - \lambda I_{\mathcal{V}})^n)$, where $n = \dim(\mathcal{V})$.
This is the kernel of a polynomial in T, so it is invariant under T by Proposition 2.3.22.
Structural Commentary: A generalized eigenspace is cut out by a polynomial equation in T.
Key Insight: Polynomial kernels commute with the operator that defines them. □

Definition 2.3.54. *Nilpotent Operator*
An operator $N \in \mathcal{L}(\mathcal{V})$ is *nilpotent* if $N^m = 0$ for some positive integer m.

Corollary 2.3.55. *Nilpotent Power Bound*
If $\mathcal{V}$ is finite-dimensional, $N \in \mathcal{L}(\mathcal{V})$ is nilpotent, and $n = \dim(\mathcal{V})$, then $N^n = 0$.

Proof. Since N is nilpotent, $\ker(N^m) = \mathcal{V}$ for some m.
The kernel chain has stabilized by the nth step by Theorem 2.3.50, so $\ker(N^n) = \mathcal{V}$. Hence $N^n = 0$.
Structural Commentary: Nilpotence says some kernel in the chain is already the whole space.
Key Insight: The stabilization bound pulls that full-kernel conclusion back to exponent $\dim(\mathcal{V})$. □

Lemma 2.3.56. *Nonzero Scalar Plus Nilpotent Is Invertible*
Let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent and let $a \in \mathbb{F}$ with $a \neq 0$. Then $aI_{\mathcal{V}} + N$ is invertible.

Proof. Choose m such that $N^m = 0$. Then $(aI_{\mathcal{V}} + N)\left(a^{-1}\sum_{r=0}^{m-1}(-a^{-1}N)^r\right) = I_{\mathcal{V}} - (-a^{-1}N)^m = I_{\mathcal{V}}$, and the same computation in the reverse order gives a two-sided inverse.
Structural Commentary: A nilpotent perturbation has a finite geometric-series inverse.
Key Insight: Nilpotence truncates the Neumann series exactly. □

Theorem 2.3.57. *Generalized Eigenvectors for Distinct Eigenvalues Are Independent*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding generalized eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Let $n = \dim(\mathcal{V})$ and suppose $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
Fix $r \in \{1, \ldots, m\}$ and apply $Q_r(T) := \prod_{i \neq r}(T - \lambda_i I_{\mathcal{V}})^n$ to the relation.
For $i \neq r$, Theorem 2.3.52 gives $Q_r(T)\mathbf{v}_i = \mathbf{0}$, so $a_r Q_r(T)\mathbf{v}_r = \mathbf{0}$.
On $G(\lambda_r, T)$, the operator $T - \lambda_i I_{\mathcal{V}}$ equals $(T - \lambda_r I_{\mathcal{V}}) + (\lambda_r - \lambda_i)I_{\mathcal{V}}$. The first term is nilpotent on $G(\lambda_r, T)$ and $\lambda_r - \lambda_i \neq 0$, so Lemma 2.3.56 makes each factor invertible on $G(\lambda_r, T)$. Thus $Q_r(T)\mathbf{v}_r \neq \mathbf{0}$, forcing $a_r = 0$.
Since r was arbitrary, all coefficients vanish.
Structural Commentary: Polynomial filters isolate one generalized eigenspace at a time.
Key Insight: Distinct eigenvalues let polynomial filters kill the other components and preserve the target one. □

Theorem 2.3.58. *Nilpotent Operators and Strictly Upper-Triangular Matrices*
Let $\mathcal{V}$ be finite-dimensional and $N \in \mathcal{L}(\mathcal{V})$. Then N is nilpotent if and only if N has a strictly upper-triangular matrix in some basis.

Proof. Let $n = \dim(\mathcal{V})$. By Corollary 2.3.55, $\{\mathbf{0}\} \subseteq \ker(N) \subseteq \ker(N^2) \subseteq \cdots \subseteq \ker(N^n) = \mathcal{V}$.

(⇒) Choose a basis of $\ker(N)$, extend it to a basis of $\ker(N^2)$, and continue until obtaining a basis of $\mathcal{V}$. If $\mathbf{v}$ first appears in $\ker(N^k)$, then $N\mathbf{v} \in \ker(N^{k-1})$, which is spanned by earlier basis vectors. Hence the matrix of N is strictly upper triangular.

(⇐) If the matrix of N is $n \times n$ strictly upper triangular, then N sends each basis vector into the span of earlier basis vectors. After n applications every basis vector is sent to $\mathbf{0}$, so $N^n = 0$.

Structural Commentary: Kernel filtration gives adapted basis; triangular form records motion to earlier layers.
Key Insight: Nilpotence is strict upper-triangularity in a suitable basis. □

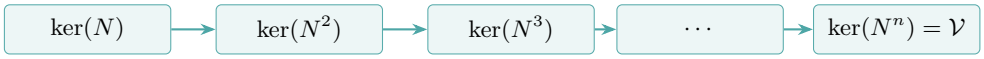


Figure 26: Extending bases through the kernel filtration makes N move vectors to earlier layers.

Perspective 2.3.59. *Eigenvalue Plus Nilpotent Part*
On $G(\lambda, T)$, the operator T behaves as $\lambda I_{\mathcal{V}}$ plus the nilpotent operator $T - \lambda I_{\mathcal{V}}$. Generalized eigenvectors are therefore ordinary eigenvectors enriched by nilpotent chain data.

Theorem 2.3.60. *Primary Decomposition over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. Then

(i) $\mathcal{V} = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$;
(ii) *each $G(\lambda_j, T)$ is T-invariant;*
(iii) *each $(T - \lambda_j I_{\mathcal{V}})|_{G(\lambda_j, T)}$ is nilpotent.*

Proof. Let $n = \dim(\mathcal{V})$.

(i) Prove decomposition by induction on n. The case $n = 1$ is immediate.

(ii) Put $S = T - \lambda_1 I_{\mathcal{V}}$. By Theorem 2.3.51, $\mathcal{V} = \ker(S^n) \oplus \operatorname{range}(S^n) = G(\lambda_1, T) \oplus \mathcal{U}$, $\mathcal{U} := \operatorname{range}(S^n)$.
Here $\mathcal{U}$ is T-invariant by Proposition 2.3.22, and $\dim(\mathcal{U}) < n$. The eigenvalues of $T|_{\mathcal{U}}$ are $\lambda_2, \ldots, \lambda_m$: λ_1 is excluded by $G(\lambda_1, T) \cap \mathcal{U} = \{\mathbf{0}\}$, and if $T\mathbf{v} = \lambda_j\mathbf{v}$ with $j \geq 2$, then $S^n\mathbf{v} = (\lambda_j - \lambda_1)^n\mathbf{v}$, so $\mathbf{v} \in \mathcal{U}$.
If $\mathcal{U} = \{\mathbf{0}\}$, then $m = 1$. Otherwise, induction gives $\mathcal{U} = G(\lambda_2, T|_{\mathcal{U}}) \oplus \cdots \oplus G(\lambda_m, T|_{\mathcal{U}})$.
For $j \geq 2$, $G(\lambda_j, T|_{\mathcal{U}}) = G(\lambda_j, T)$. Indeed, the inclusion $\subseteq$ is clear; conversely, write $\mathbf{v} \in G(\lambda_j, T)$ as $\mathbf{v} = \mathbf{g} + \mathbf{u}$ with $\mathbf{g} \in G(\lambda_1, T)$ and $\mathbf{u} \in \mathcal{U}$. Invariance of both summands gives $\mathbf{g} \in G(\lambda_1, T) \cap G(\lambda_j, T)$, hence $\mathbf{g} = \mathbf{0}$ by Theorem 2.3.57. Thus $\mathbf{v} \in G(\lambda_j, T|_{\mathcal{U}})$, proving (i).

(iii) This is Proposition 2.3.53.

(iv) By Theorem 2.3.52, $(T-\lambda_j I_{\mathcal{V}})^n \mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in G(\lambda_j, T)$; hence the restriction is nilpotent.

Structural Commentary: Peel off one generalized eigenspace, then induct on the invariant range.
Key Insight: Stable kernel-range splitting isolates one eigenvalue at a time. □

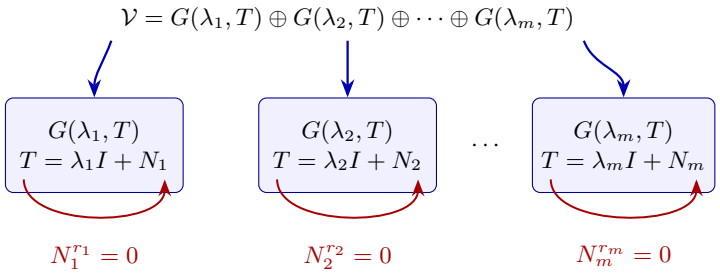


Figure 27: Primary decomposition: on $G(\lambda_j, T)$, $T = \lambda_j I$ plus nilpotent.

Corollary 2.3.61. *Basis of Generalized Eigenvectors*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a generalized-eigenvector basis for T.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 2.3.60, $\mathcal{V} = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. Choose a basis of each generalized eigenspace and concatenate the bases. Directness makes the concatenated list a basis of $\mathcal{V}$, and every nonzero vector in each list is a generalized eigenvector.
Structural Commentary: A direct sum basis is obtained by choosing bases in each summand.
Key Insight: Bases of the direct-sum pieces concatenate.

□

Definition 2.3.62. *Multiplicity of an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$.
The *multiplicity* of an eigenvalue λ of T is $\dim G(\lambda, T) = \dim \ker\big((T-\lambda I_{\mathcal{V}})^{\dim(\mathcal{V})}\big)$.
This is the algebraic multiplicity in generalized-eigenspace language.

Proposition 2.3.63. *Sum of Multiplicities*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then the sum of the multiplicities of all eigenvalues of T equals $\dim(\mathcal{V})$.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 2.3.60, $\mathcal{V} = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. Taking dimensions, by Corollary 2.1.37, $\dim(\mathcal{V}) = \sum_{j=1}^{m} \dim G(\lambda_j, T)$, exactly the sum of the multiplicities.
Structural Commentary: Multiplicity is dimension of the generalized-eigenspace summand.
Key Insight: Multiplicities are dimensions of the direct-sum pieces. □

Motivation 2.3.64. *Two Spectral Polynomials*
The characteristic polynomial records the whole primary decomposition with dimensions.
The minimal polynomial records the shortest polynomial relation satisfied by the operator.

Definition 2.3.65. *Characteristic Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $T \in \mathcal{L}(\mathcal{V})$. If $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T and $d_j = \dim G(\lambda_j, T)$, then the *characteristic polynomial* of T is $q_T(z) := \prod_{j=1}^{m} (z-\lambda_j)^{d_j}$.

Proposition 2.3.66. *Degree and Roots of q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T.

(i) $\deg q_T = \dim(\mathcal{V})$.
(ii) The roots of q_T are precisely the eigenvalues of T.

Proof. Write $q_T(z) = \prod_{j=1}^{m} (z-\lambda_j)^{d_j}$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues and $d_j = \dim G(\lambda_j, T)$.

(i) Proposition 2.3.63 gives $\deg q_T = \sum_j d_j = \dim(\mathcal{V})$.
(ii) The displayed factorization has exactly the roots $\lambda_1, \ldots, \lambda_m$.

Structural Commentary: Each primary summand contributes one root factor.
Key Insight: Multiplicity is dimension. □

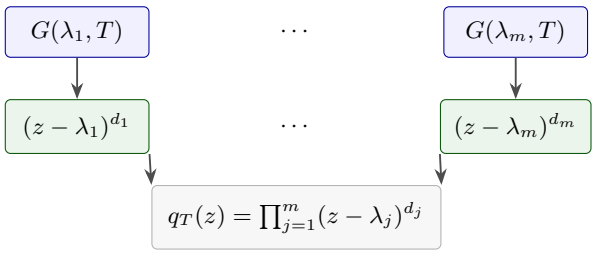


Figure 28: Each primary summand contributes $(z-\lambda_j)^{d_j}$, where $d_j = \dim G(\lambda_j, T)$.

Theorem 2.3.67. *Cayley–Hamilton Theorem for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, q_T be characteristic polynomial of $T \in \mathcal{L}(\mathcal{V})$. Then $q_T(T) = 0$.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues and set $d_j = \dim G(\lambda_j, T)$.
It suffices, by primary decomposition, to show $q_T(T)$ kills each $G(\lambda_j, T)$.
On $G(\lambda_j, T)$, the operator $T - \lambda_j I_{\mathcal{V}}$ is nilpotent, so Corollary 2.3.55 gives $(T - \lambda_j I_{\mathcal{V}})^{d_j} = 0$ on $G(\lambda_j, T)$.
Since the factors of $q_T(T)$ commute, this factor kills the whole product on $G(\lambda_j, T)$. Hence $q_T(T) = 0$.
Structural Commentary: Check the polynomial on each primary summand.
Key Insight: One factor of $q_T(T)$ kills each summand. □

Definition 2.3.68. *Monic Polynomial*
A nonzero polynomial is *monic* if its highest-degree coefficient is 1.

Theorem 2.3.69. *Existence of the Minimal Polynomial*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then T has a unique least-degree monic annihilator μ_T.

Proof. Proposition 2.3.19 gives a nonzero annihilator; scaling makes it monic, so a least-degree monic annihilator exists. sIf p and q are two of them, then $(p-q)(T) = 0$ and $\deg(p-q) < \deg p$. If $p \neq q$, scaling $p - q$ gives a smaller monic annihilator, impossible. Hence $p = q$.
Structural Commentary: Existence comes from finite-dimensional dependence; uniqueness from minimality.
Key Insight: Two least monic annihilators cannot differ. □

Definition 2.3.70. *Minimal Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. The *minimal polynomial* of T, denoted μ_T, is the unique monic polynomial of smallest degree such that $\mu_T(T) = 0$.

Method 2.3.71. *Computing μ_T from a Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be the matrix of T in some basis.

(i) For $m = 1, 2, \ldots$, test the monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1} z^{m-1} + z^m$.
(ii) Solve the linear matrix equation $a_0 I + a_1 A + \cdots + a_{m-1} A^{m-1} = -A^m$ for $a_0, \ldots, a_{m-1}$.

Equivalently, vectorize $\begin{bmatrix} \mathrm{vec}(I) & \mathrm{vec}(A) & \cdots & \mathrm{vec}(A^{m-1}) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{m-1} \end{bmatrix} = -\mathrm{vec}(A^m)$

(iii) If no solution exists, increase m. The first solvable candidate is μ_T.

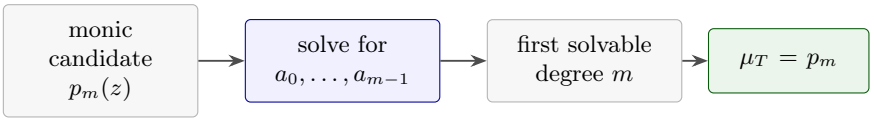


Figure 29: Solve $a_0 I + \cdots + a_{m-1} A^{m-1} = -A^m$; the first solvable monic candidate is μ_T.

Theorem 2.3.72. *Annihilators Are Multiples of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim, $T \in \mathcal{L}(\mathcal{V})$, and $f \in \mathcal{P}(\mathbb{F})$ be nonzero. Then $f(T) = 0$ if and only if μ_T divides f.

Proof.

(⇒) Suppose $f(T) = 0$. Divide $f = s\mu_T + r$, where $r = 0$ or $\deg r < \deg \mu_T$.
Then $r(T) = 0$, so minimality forces $r = 0$. Hence μ_T divides f.
(⇐) If $f = \mu_T s$, then $f(T) = \mu_T(T) s(T) = 0$.

Structural Commentary: Polynomial division reduces every annihilator to its remainder modulo μ_T.
Key Insight: Minimality forces that remainder to be zero. □

Corollary 2.3.73. *q_T Is a Multiple of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then the characteristic polynomial q_T is a polynomial multiple of the minimal polynomial μ_T.

Proof. Theorem 2.3.67 gives $q_T(T) = 0$. Theorem 2.3.72 then implies that μ_T divides q_T.
Structural Commentary: Cayley–Hamilton turns the characteristic polynomial into an annihilator.
Key Insight: The minimal polynomial divides every annihilator, hence divides q_T. □

Theorem 2.3.74. *Roots of μ_T Are Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then the roots of μ_T are precisely the eigenvalues of T.

Proof.
($\Rightarrow$) If $\mu_T(\lambda) = 0$, write $\mu_T(z) = (z - \lambda)g(z)$. Since $\deg g < \deg \mu_T$, minimality gives $g(T) \neq 0$; choose $\mathbf{v}$ with $\mathbf{u} = g(T)\mathbf{v} \neq \mathbf{0}$. Then $(T - \lambda I_{\mathcal{V}})\mathbf{u} = \mu_T(T)\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $0 = \mu_T(T)\mathbf{v} = \mu_T(\lambda)\mathbf{v}$, so $\mu_T(\lambda) = 0$.

Structural Commentary: A root splits off one shifted operator.
Key Insight: The quotient cannot already annihilate T. □

Perspective 2.3.75. *Characteristic vs. Minimal Polynomial*
The characteristic polynomial records every primary block with its dimension. The minimal polynomial records the least polynomial needed to annihilate all blocks at once.

Theorem 2.3.76. *Block Diagonal Form with Upper-Triangular Primary Blocks*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$, with multiplicities $d_1, \ldots, d_m$. Then T has a matrix of the form

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j \in \mathrm{Mat}(d_j, d_j, \mathbb{C})$ is upper triangular with diagonal $\lambda_j, \ldots, \lambda_j$.

Proof. Set $\mathcal{U}_j = G(\lambda_j, T)$ and $N_j = (T - \lambda_j I_{\mathcal{V}})|_{\mathcal{U}_j}$.
By Theorem 2.3.60, the $\mathcal{U}_j$ form a T-invariant direct-sum decomposition and each N_j is nilpotent.
Choose a basis of each $\mathcal{U}_j$ in which N_j is strictly upper triangular. Then $T|_{\mathcal{U}_j} = \lambda_j I_{\mathcal{U}_j} + N_j$ has an upper triangular matrix with diagonal $\lambda_j, \ldots, \lambda_j$. Concatenating these bases gives the desired block diagonal matrix.
Structural Commentary: Primary decomposition gives the blocks; nilpotent triangularisation shapes them.
Key Insight: On each generalized eigenspace, T is eigenvalue plus nilpotent. □

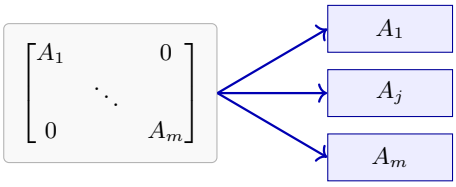


Figure 30: Primary blocks: $A_j = \lambda_j I + N_j$, with N_j strictly upper triangular.

Definition 2.3.77. *Jordan Chains, Blocks, and Matrices*
For $k \geq 1$, the *nilpotent Jordan block* $J_k(0)$ is the $k \times k$ matrix with 1 on the first superdiagonal and 0 elsewhere:

$$J_k(0) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

For $\lambda \in \mathbb{C}$, set $J_k(\lambda) := \lambda I_k + J_k(0)$. A *Jordan matrix* is a direct sum of Jordan blocks. A basis is a *Jordan basis* for an operator T if the matrix of T in that basis is a Jordan matrix.

Theorem 2.3.78. *Nilpotent Jordan Chain Basis*
Let $\mathcal{V}$ be finite-dim, $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $\exists$ vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathcal{V}$ and nonnegative integers $m_1, \dots, m_r$ s.t. $\mathcal{B}_N = \big(N^{m_1}\mathbf{v}_1, \dots, N\mathbf{v}_1, \mathbf{v}_1, \dots, N^{m_r}\mathbf{v}_r, \dots, N\mathbf{v}_r, \mathbf{v}_r\big)$ is a basis of $\mathcal{V}$ and $N^{m_j+1}\mathbf{v}_j = \mathbf{0}$ for every j.

Proof. Induct on $n = \dim(\mathcal{V})$. If $N = 0$, any basis works with all $m_j = 0$.
Now assume $N \neq 0$. Then N is not injective, so $\operatorname{range}(N)$ is a proper subspace, and $N|_{\operatorname{range}(N)}$ is nilpotent. By induction, $\operatorname{range}(N)$ has a chain basis $\big(N^{m_1}\mathbf{u}_1, \dots, N\mathbf{u}_1, \mathbf{u}_1, \dots, N^{m_r}\mathbf{u}_r, \dots, N\mathbf{u}_r, \mathbf{u}_r\big)$ with $N^{m_j+1}\mathbf{u}_j = \mathbf{0}$.
Choose $\mathbf{v}_j \in \mathcal{V}$ such that $N\mathbf{v}_j = \mathbf{u}_j$, and lift the chains to $\mathcal{L} = \big(N^{m_1+1}\mathbf{v}_1, \dots, N\mathbf{v}_1, \mathbf{v}_1, \dots, N^{m_r+1}\mathbf{v}_r, \dots, N\mathbf{v}_r, \mathbf{v}_r\big)$.
This is linearly independent: applying N to any dependence among its vectors gives a dependence among range-chain basis. Only coefficients of $N^{m_j+1}\mathbf{v}_j$ may remain; but $N^{m_j+1}\mathbf{v}_j = N^{m_j}\mathbf{u}_j$, so those coefficients vanish too.
Extend $\mathcal{L}$ to a basis by adding $\mathbf{w}_1, \dots, \mathbf{w}_s$. For each i, choose $\mathbf{x}_i \in \operatorname{span}(\mathcal{L})$ s.t. $N\mathbf{x}_i = N\mathbf{w}_i$, and $\mathbf{z}_i = \mathbf{w}_i - \mathbf{x}_i$.
Then $N\mathbf{z}_i = \mathbf{0}$, and replacing each $\mathbf{w}_i$ by $\mathbf{z}_i$ preserves the span. Thus $\mathcal{L}, \mathbf{z}_1, \dots, \mathbf{z}_s$ is the desired chain basis.
Structural Commentary: Build chains in $\operatorname{range}(N)$, lift them once, then add kernel chains.
Key Insight: Applying N turns lifted-chain dependences into known range-chain dependences. □

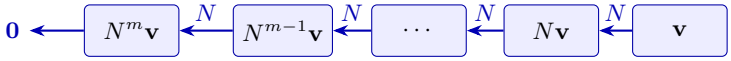


Figure 31: A nilpotent chain shifts left under N; in this basis order it becomes one Jordan block.

Theorem 2.3.79. *Jordan Form over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a Jordan basis for T.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 2.3.60, $\mathcal{V} = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$, and each $N_j := (T - \lambda_j I_{\mathcal{V}})|_{G(\lambda_j, T)}$ is nilpotent. Apply Theorem 2.3.78 to each N_j. In the resulting chain basis of $G(\lambda_j, T)$, the operator N_j has direct-sum matrix blocks $J_k(0)$, so $T|_{G(\lambda_j,T)} = \lambda_j I + N_j$ has the corresponding blocks $J_k(\lambda_j)$. Concatenating the chain bases over all eigenvalues gives a Jordan basis of $\mathcal{V}$.
Structural Commentary: Primary pieces decompose into nilpotent chains.
Key Insight: Jordan form is "eigenvalue plus nilpotent chains" on every generalized eigenspace. □

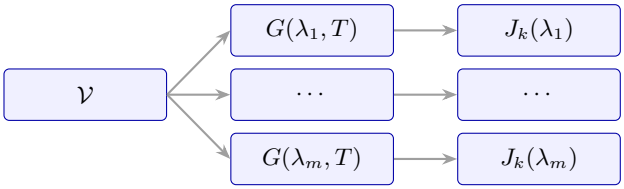


Figure 32: Primary pieces $G(\lambda_j, T)$ give nilpotent chains, hence Jordan blocks.

Lemma 2.3.80. *Powers of a Nilpotent Jordan Block*
For $k \geq 1$ and $p \geq 0$, $\operatorname{rank}\big(J_k(0)^p\big) = \max(k - p, 0)$ and $\dim\ker\big(J_k(0)^p\big) = \min(p, k)$.

Proof. Since $J_k(0)$ is the left-shift, $J_k(0)^p\mathbf{e}_j = \mathbf{0}$ for $j \leq p$ and $J_k(0)^p\mathbf{e}_j = \mathbf{e}_{j-p}$ for $j > p$. Thus the image has dimension $\max(k - p, 0)$, and rank-nullity gives the kernel dimension.
Structural Commentary: A nilpotent Jordan block is a left-shift on the standard basis.
Key Insight: Each power deletes one more leading layer of the chain. □

Definition 2.3.81. *Weyr Characteristic*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\lambda \in \operatorname{spec}(A)$.
For $p \geq 1$, define $w_p(\lambda, A) := \operatorname{rank}\big((A - \lambda I_n)^{p-1}\big) - \operatorname{rank}\big((A - \lambda I_n)^p\big)$.
The positive part of the sequence $w_1, w_2, \dots$ is the *Weyr characteristic* of A associated with λ.

Theorem 2.3.82. *Jordan Block Sizes from Rank Drops*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, $\lambda \in \operatorname{spec}(A)$, and $p \geq 1$. Then $w_p(\lambda, A)$ is the number of λ-Jordan blocks of size at least p; hence $w_p(\lambda, A) - w_{p+1}(\lambda, A)$ is the number of λ-blocks of size exactly p.

Proof. Rank is similarity-invariant, so assume A is in Jordan form.
Blocks $J_k(\mu)$ with $\mu \neq \lambda$ contribute 0 to the rank drop because $J_k(\mu) - \lambda I_k$ is invertible.
A block $J_k(\lambda)$ contributes $\operatorname{rank}\big(J_k(0)^{p-1}\big) - \operatorname{rank}\big(J_k(0)^p\big)$, which is 1 exactly when $p \leq k$ by Lemma 2.3.80.
Summing over blocks proves the first claim; subtracting the $p + 1$ count gives the exact-size count.
Structural Commentary: Rank drops count surviving chain lengths.
Key Insight: A length-k block contributes to drops $1, \dots, k$. □

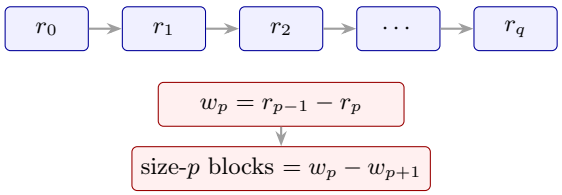


Figure 33: For $r_p = \operatorname{rank}((A - \lambda I)^p)$, drops count surviving chains; $w_p - w_{p+1}$ counts size-p blocks.

Method 2.3.83. *Computing Jordan Blocks by Rank Drops*
For each eigenvalue λ of $A \in \operatorname{Mat}(n, n, \mathbb{C})$:

(i) Compute $r_p = \operatorname{rank}\big((A - \lambda I_n)^p\big)$ for $p = 0, 1, 2, \ldots$ until the differences stabilize at 0.
(ii) Set $w_p = r_{p-1} - r_p$. Then w_p is the number of λ-blocks of size at least p.
(iii) The number of blocks $J_p(\lambda)$ is $w_p - w_{p+1}$.
(iv) Assemble the Jordan form by taking the direct sum over all eigenvalues.

Corollary 2.3.84. *Uniqueness of Jordan Form*
If J and J' are Jordan matrices similar to $A \in \operatorname{Mat}(n, n, \mathbb{C})$, they differ only by a permutation of Jordan blocks.

Proof. Similarity preserves spectra and the ranks of $(X - \lambda I_n)^p$. Thus J and J' have the same Weyr characteristic for each eigenvalue. Theorem 2.3.82 then gives the same block multiplicities, so only block order may differ.
Structural Commentary: Weyr data is similarity-invariant and fixes block multiplicities.
Key Insight: Jordan form is unique modulo block order. □

Definition 2.3.85. *Reversal Matrix*
For $k \geq 1$, the *reversal matrix* $R_k \in \operatorname{Mat}(k, k, \mathbb{C})$ is defined by

$$(R_k)_{ij} = \begin{cases} 1, & i + j = k + 1, \\ 0, & i + j \neq k + 1. \end{cases}$$

Thus $R_k^t = R_k$ and $R_k^2 = I_k$.

Lemma 2.3.86. *Reversal Conjugates a Jordan Block to Its Transpose*
For every $k \geq 1$ and $\lambda \in \mathbb{C}$, $J_k(\lambda)^t = R_k J_k(\lambda) R_k$. Moreover, $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.

Proof. Let $N = J_k(0)$. Since R_k reverses standard basis order, $R_k N R_k = N^t$.
Adding scalar gives $R_k J_k(\lambda) R_k = J_k(\lambda)^t$. As $R_k = R_k^t = R_k^{-1}$, this identity gives $J_k(\lambda)^t R_k = R_k J_k(\lambda)$ and $R_k J_k(\lambda)^t = J_k(\lambda) R_k$, so $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.
Structural Commentary: Reversal changes the superdiagonal shift into the subdiagonal shift.
Key Insight: Transposition reads the same Jordan chain backwards. □

Theorem 2.3.87. *Transpose Similarity and Symmetric Factorisations*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. Then:

(i) A is similar to A^t via a symmetric similarity matrix.
(ii) $A = BC = DE$, where B, C, D, E are symmetric and B, E are invertible.

Proof. Choose a Jordan form $A = SJS^{-1}$, and let R be the direct sum of the reversal matrices matching the Jordan blocks of J. By Lemma 2.3.86, $J^t = RJR$, $R^t = R$, $R^2 = I_n$, and both RJ and JR are symmetric.

(i) Set $P = (S^{-1})^t R S^{-1}$. Then $P^t = P$ and $PAP^{-1} = (S^{-1})^t R J R S^t = (S^{-1})^t J^t S^t = A^t$.
(ii) Using $R^2 = I_n$, $A = (SRS^t)\big((S^{-1})^t R J S^{-1}\big)$ and $A = (SJRS^t)\big((S^{-1})^t R S^{-1}\big)$.
All four factors are symmetric because R, RJ, JR are symmetric; SRS^t and $(S^{-1})^t R S^{-1}$ are invertible.

Structural Commentary: Reduce to Jordan blocks, reverse chains, then conjugate back.
Key Insight: Symmetric involutions produce both transpose similarity and symmetric factors. □

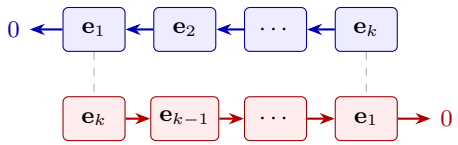


Figure 34: Reversal gives $R_k J_k(0) R_k = J_k(0)^t$: the same chain read backwards.

Theorem 2.3.88. *Conjugate Similarity Criterion*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then A is similar to $\overline{A}$ if and only if each $J_k(\lambda)$ occurs in the Jordan form of A with the same multiplicity as $J_k(\overline{\lambda})$.

Proof. Let J be a Jordan form of A. Then $\overline{J}$ is a Jordan form of $\overline{A}$, obtained by replacing each $J_k(\lambda)$ by $J_k(\overline{\lambda})$.
($\Rightarrow$) If A is similar to $\overline{A}$, then J and $\overline{J}$ are similar.
By Corollary 2.3.84, their Jordan blocks have same multiplicities.
($\Leftarrow$) If the multiplicities agree, then J and $\overline{J}$ differ only by a permutation of blocks. Hence A is similar to $\overline{A}$.
Structural Commentary: Conjugation changes eigenvalue labels but not block sizes.
Key Insight: Non-real Jordan blocks must occur in conjugate pairs. □

Proposition 2.3.89. *Exponential of a Jordan Block*
For $k \geq 1$, $\lambda \in \mathbb{C}$, and $t \in \mathbb{C}$, $e^{tJ_k(\lambda)} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} J_k(0)^r$.

Proof. Since $J_k(\lambda) = \lambda I_k + J_k(0)$ and the two summands commute, $e^{tJ_k(\lambda)} = e^{\lambda t I_k} e^{tJ_k(0)}$.
The first factor is $e^{\lambda t} I_k$, and the second exponential truncates because $J_k(0)^k = 0$.
Structural Commentary: The scalar part supplies $e^{\lambda t}$; the nilpotent part supplies a finite polynomial in t.
Key Insight: Nilpotence turns the exponential series into a finite sum. □

Method 2.3.90. *Solving $\mathbf{x}'(t) = A\mathbf{x}(t)$ by Jordan Form*
To solve $\mathbf{x}'(t) = A\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{x}_0$:
(i) Put $A = SJS^{-1}$, where $J = \bigoplus_{i=1}^{r} J_{k_i}(\lambda_i)$.
(ii) Set $\mathbf{y}(t) = S^{-1}\mathbf{x}(t)$ and $\mathbf{y}_0 = S^{-1}\mathbf{x}_0$. Then $\mathbf{y}'(t) = J\mathbf{y}(t)$ and $\mathbf{x}(t) = S\mathbf{y}(t)$.
(iii) Split $\mathbf{y} = (\mathbf{y}_1, \ldots, \mathbf{y}_r)$ according to the Jordan blocks. Then $\mathbf{y}_i'(t) = J_{k_i}(\lambda_i)\mathbf{y}_i(t)$.
(iv) For each block, use Proposition 2.3.89: $\mathbf{y}_i(t) = e^{tJ_{k_i}(\lambda_i)}\mathbf{y}_i(0) = e^{\lambda_i t} \sum_{q=0}^{k_i-1} \frac{t^q}{q!} J_{k_i}(0)^q \mathbf{y}_i(0)$.
(v) Reassemble $\mathbf{y}(t)$ and transform back: $\mathbf{x}(t) = S\mathbf{y}(t)$.
Thus a size-k block with eigenvalue λ contributes entries of the form $e^{\lambda t}p(t)$, where $\deg p < k$.

Definition 2.3.91. *Spectral Radius and Convergent Matrix*
For $A \in \mathrm{Mat}(n,n,\mathbb{C})$, the *spectral radius* is $\rho(A) := \max\{|\lambda| : \lambda \in \mathrm{spec}(A)\}$.
We say $A^p \to 0$ if every entry of A^p converges to 0 as $p \to \infty$.

Proposition 2.3.92. *Powers of a Jordan Block*
For $p \geq 0$, $J_k(\lambda)^p = \sum_{r=0}^{\min(p,k-1)} \binom{p}{r} \lambda^{p-r} J_k(0)^r$. Thus $J_k(\lambda)^p \to 0$ if and only if $|\lambda| < 1$.

Proof. Formula follows from binomial theorem applied to $J_k(\lambda) = \lambda I_k + J_k(0)$, with truncation from $J_k(0)^k = 0$.
($\Rightarrow$) If $J_k(\lambda)^p \to 0$, then its diagonal entries λ^p tend to 0, so $|\lambda| < 1$.
($\Leftarrow$) If $|\lambda| < 1$, then each coefficient $\binom{p}{r}\lambda^{p-r} \to 0$ for fixed r; hence every term in finite expansion tends to 0.
Structural Commentary: Jordan powers are exponential-polynomial terms.
Key Insight: Polynomial growth cannot defeat exponential decay when $|\lambda| < 1$. □

Theorem 2.3.93. *Matrix Powers Converge to Zero*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then $A^p \to 0$ if and only if $\rho(A) < 1$.

Proof. Choose a Jordan form $A = SJS^{-1}$.
Since S and S^{-1} are fixed, $A^p \to 0$ if and only if every Jordan block power in J^p converges to 0.
($\Rightarrow$) Each block power tends to 0, so Proposition 2.3.92 gives $|\lambda| < 1$ for every eigenvalue. Thus $\rho(A) < 1$.
($\Leftarrow$) If $\rho(A) < 1$, then every eigenvalue satisfies $|\lambda| < 1$. Proposition 2.3.92 makes each block power tend to 0.
Structural Commentary: Similarity reduces matrix-power convergence to Jordan block-power convergence.
Key Insight: The largest eigenvalue modulus controls the asymptotic size of all blocks. □

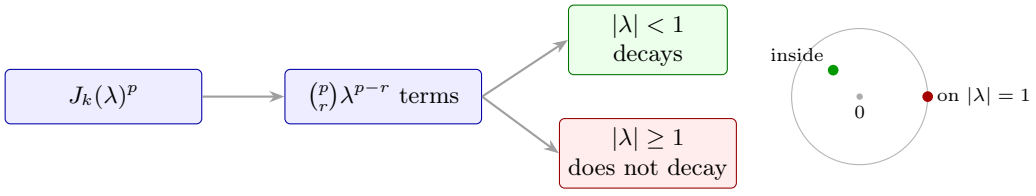


Figure 35: The gray circle is $|\lambda| = 1$: inside it, powers decay; on or outside it, they do not.

Definition 2.3.94. *Power Bounded Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *power bounded* if for each entry position (i, j), the sequence of entries of A^p in that position is bounded for $p = 1, 2, \ldots$.

Theorem 2.3.95. *Power Bounded Jordan Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is power bounded.
(ii) $\rho(A) \leq 1$, and every eigenvalue λ with $|\lambda| = 1$ has only 1×1 Jordan blocks.
(iii) $\rho(A) \leq 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Work in Jordan form.

(*i*) $\Rightarrow$ (*ii*): If $|\lambda| > 1$, the diagonal entries λ^p are unbounded.
If $|\lambda| = 1$ and $J_k(\lambda)$ has $k \geq 2$, then the first superdiagonal contains $p\lambda^{p-1}$, also unbounded.
(*ii*) $\Rightarrow$ (*i*): Blocks with $|\lambda| < 1$ converge to 0 by Proposition 2.3.92; 1×1 blocks with $|\lambda| = 1$ are bounded.
Finite direct sums and fixed similarities preserve boundedness.
(*ii*) $\Leftrightarrow$ (*iii*): For fixed λ, $\dim E_\lambda(A)$ counts Jordan blocks, while $m_A(\lambda)$ counts their total size.
Equality holds exactly when every block has size 1.

Structural Commentary: Unit-circle eigenvalues are harmless only when their nilpotent parts vanish.
Key Insight: Nontrivial Jordan blocks on the unit circle create polynomial growth. □

Definition 2.3.96. *Probability Vectors and Markov Matrices*
A vector $\mathbf{x} \in \mathbb{R}^n$ is a *probability vector* if its entries are nonnegative and sum to 1. Let $\mathbf{e} := (1, \ldots, 1)^t$. A matrix $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is a *Markov matrix* if $a_{ij} \geq 0$ for all i, j and $A\mathbf{e} = \mathbf{e}$; equivalently, every row sum is 1.

Lemma 2.3.97. *Basic Properties of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $1 \in \mathrm{spec}(A)$, and each A^p is Markov with entries in $[0, 1]$.

Proof. Since $A\mathbf{e} = \mathbf{e}$, 1 is an eigenvalue. Inductively, A^p has nonnegative entries and $A^p\mathbf{e} = \mathbf{e}$, so A^p is Markov. Each nonnegative row sums to 1, so all entries lie in $[0, 1]$.
Structural Commentary: Markov structure is closed under powers.
Key Insight: Nonnegative row sums of 1 bound every entry. □

Theorem 2.3.98. *Unit-Circle Spectrum of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $\rho(A) = 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Lemma 2.3.97 makes A power bounded. Theorem 2.3.95 gives $\rho(A) \leq 1$ and $\dim E_\lambda(A) = m_A(\lambda)$ on $|\lambda| = 1$. Since $A\mathbf{e} = \mathbf{e}$, $1 \in \mathrm{spec}(A)$, so $\rho(A) \geq 1$. Thus $\rho(A) = 1$.
Structural Commentary: Bounded Markov powers rule out unit-circle Jordan growth.
Key Insight: $A\mathbf{e} = \mathbf{e}$ forces the eigenvalue 1. □

Theorem 2.3.99. *Positive Markov Matrices Converge to a Stationary Distribution*
Let $n \geq 2$ and let $A \in \mathrm{Mat}(n, n, \mathbb{R})$ be a Markov matrix with positive entries. Then there is a unique probability vector $\mathbf{x}$ with $A^t\mathbf{x} = \mathbf{x}$; its entries are positive and $\lim_{p\to\infty} A^p = \mathbf{e}\mathbf{x}^t$.

Proof. Let $A\mathbf{y} = \lambda\mathbf{y}$ with $|\lambda| = 1$, and choose k with $|y_k| = \|\mathbf{y}\|_\infty$.
Positivity and the kth row sum give $\|\mathbf{y}\|_\infty = |\lambda y_k| = \left|\sum_{j=1}^n a_{kj} y_j\right| \leq \sum_{j=1}^n a_{kj} |y_j| \leq \|\mathbf{y}\|_\infty$.
Equality forces all $|y_j| = \|\mathbf{y}\|_\infty$ and all y_j to have the same argument. Hence $\mathbf{y} = c\mathbf{e}$, and $A\mathbf{e} = \mathbf{e}$ gives $\lambda = 1$. Thus $E_1(A) = \mathrm{span}(\mathbf{e})$ and no other eigenvalue lies on the unit circle.
By Theorem 2.3.98, the Jordan form is $[1] \oplus J$ with $\rho(J) < 1$. Choose $S = [\mathbf{e}\ Y]$ such that $A = S([1] \oplus J)S^{-1}$, and let $\mathbf{x}^t$ be the first row of S^{-1}. Then $\mathbf{x}^t A = \mathbf{x}^t$ and $\mathbf{x}^t\mathbf{e} = 1$. Since $J^p \to 0$, $A^p = S([1] \oplus J^p)S^{-1} \to \mathbf{e}\mathbf{x}^t$.
The nonnegative powers A^p imply $\mathbf{x} \geq 0$; since $A^t\mathbf{x} = \mathbf{x}$ and A^t has positive entries, in fact $\mathbf{x} > 0$. Finally, $\dim E_1(A^t) = \dim E_1(A) = 1$, so $\mathbf{x}^t\mathbf{e} = 1$ gives uniqueness among probability vectors.
Structural Commentary: Positivity leaves one simple unit-circle block; all other blocks decay.
Key Insight: The limit is the rank-one projection $\mathbf{e}\mathbf{x}^t$. □

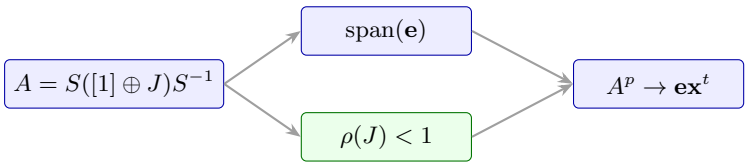


Figure 36: The line span(**e**) remains; all other Jordan directions decay, so $A^p \to \mathbf{e}\mathbf{x}^t$.

Lemma 2.3.100. *Identity Plus Nilpotent Has a Square Root*
Let $\mathcal{V}$ be a complex vector space and let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $I_\mathcal{V} + N$ has a square root.

Proof. Choose $m \geq 1$ such that $N^m = 0$. If $m = 1$, then $N = 0$ and $I_\mathcal{V}$ is a square root.
Otherwise seek $R = I_\mathcal{V} + a_1 N + \cdots + a_{m-1}N^{m-1}$. With $a_0 = 1$, choose $a_1, \ldots, a_{m-1}$ recursively so the coefficients of R^2 match those of $I_\mathcal{V} + N$: $2a_1 = 1$, and $2a_k + \sum_{r=1}^{k-1} a_r a_{k-r} = 0 \quad (2 \leq k \leq m-1)$
Terms of degree $\geq m$ vanish because $N^m = 0$, so $R^2 = I_\mathcal{V} + N$.
Structural Commentary: Nilpotence reduces the square-root series to a finite polynomial.
Key Insight: Match coefficients recursively. □



Figure 37: Nilpotence truncates the square-root series; coefficients are chosen so $R^2 = I + N$.

Theorem 2.3.101. *Invertible Complex Operators Have Square Roots*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$ be invertible. Then T has a square root.

Proof. Let $\mathcal{U}_j = G(\lambda_j, T)$ for the distinct eigenvalues of T. Primary decomposition gives $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$, and invertibility gives $\lambda_j \neq 0$. Put $N_j = (T - \lambda_j I_\mathcal{V})|_{\mathcal{U}_j}$.
Then $T|_{\mathcal{U}_j} = \lambda_j \left(I_{\mathcal{U}_j} + \lambda_j^{-1} N_j\right)$ with $\lambda_j^{-1} N_j$ nilpotent. By Lemma 2.3.100, choose $B_j^2 = I_{\mathcal{U}_j} + \lambda_j^{-1} N_j$; choose $\alpha_j^2 = \lambda_j$ and set $R_j = \alpha_j B_j$. Then $R_j^2 = T|_{\mathcal{U}_j}$. Define $R = R_1 \oplus \cdots \oplus R_m$ on the direct sum. Then $R^2 = T$.
Structural Commentary: Take square roots on primary blocks, then recombine.
Key Insight: Invertibility makes every scalar block nonzero. □

2.4 Matrix Analysis

2.4.1 Matrix Classes and Spectral Invariants

Matrix spectral statements are operator statements written in the standard coordinates of $\mathbb{F}^n$. The coordinate viewpoint is useful because determinants, traces, transposes, and block patterns become visible.

Definition 2.4.1. *Characteristic Polynomial*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, the *characteristic polynomial* of A is $p_A(z) := \det(zI_n - A)$.

Theorem 2.4.2. *Basic Form of the Characteristic Polynomial*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then p_A is monic of degree n and $p_A(z) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots + (-1)^n \det A$. Moreover, for $\lambda \in \mathbb{F}$, $\lambda \in \operatorname{spec}(A)$ if and only if $p_A(\lambda) = 0$.

Proof. The only determinant term of degree greater than $n-2$ comes from the identity permutation:

$$(z - a_{11})(z - a_{22}) \cdots (z - a_{nn}) = z^n - (a_{11} + \cdots + a_{nn})z^{n-1} + \cdots .$$

Thus p_A is monic and the coefficient of z^{n-1} is $-\operatorname{tr} A$; also $p_A(0) = \det(-A) = (-1)^n \det A$. Finally, $p_A(\lambda) = 0$ if and only if $\lambda I_n - A$ is singular, equivalently $\ker(A - \lambda I_n) \neq \{\mathbf{0}\}$.
Structural Commentary: The diagonal product determines the top two coefficients.
Key Insight: Eigenvalues are precisely the roots of $\det(zI_n - A)$. □

Definition 2.4.3. *Algebraic Multiplicity*
For $\lambda \in \mathbb{F}$, let $m_A(\lambda)$ be the multiplicity of λ as a root of p_A, with $m_A(\lambda) = 0$ if $\lambda \notin \operatorname{spec}(A)$.
For $\lambda \in \operatorname{spec}(A)$, this is the *algebraic multiplicity* of λ. Unqualified multiplicity means algebraic multiplicity.

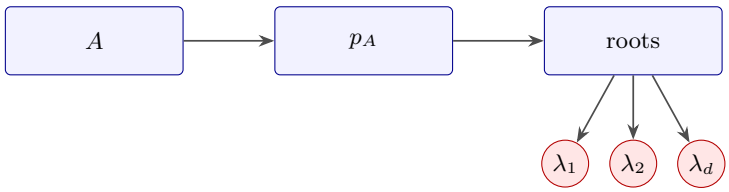


Figure 38: Form $p_A(z) = \det(zI_n - A)$ and factor it; each root λ is counted by $m_A(\lambda)$.

Proposition 2.4.4. *Upper-Triangular Eigenvalues*
If $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ *is upper triangular, then* $p_A(z) = \prod_{j=1}^{n} (z - a_{jj})$.
Thus $\mathrm{spec}(A) = \{a_{jj} : 1 \le j \le n\}$, *and for* $\lambda \in \mathrm{spec}(A)$, $m_A(\lambda) = \#\{j : a_{jj} = \lambda\}$.

Proof. The matrix $zI_n - A$ is upper triangular with diagonal entries $z - a_{11}, \ldots, z - a_{nn}$. Its determinant is the product of those diagonal entries. The roots of this product are precisely the diagonal entries, with repetition.
Structural Commentary: Triangular form removes all off-diagonal determinant contributions.
Key Insight: Once a matrix is triangular, the spectrum is visible on the diagonal. □

Proposition 2.4.5. *Transpose, Conjugate, and Adjoint Spectra*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $p_{A^t}(z) = p_A(z)$, $p_{\overline{A}}(z) = \overline{p_A(\overline{z})}$, *and* $p_{A^*}(z) = \overline{p_A(\overline{z})}$.
Consequently, if $\lambda \in \mathrm{spec}(A)$, *then* $m_{A^t}(\lambda) = m_A(\lambda)$ *and* $m_{\overline{A}}(\overline{\lambda}) = m_{A^*}(\overline{\lambda}) = m_A(\lambda)$.

Proof. Use determinant identities:

$$p_{A^t}(z) = \det(zI_n - A^t) = \det((zI_n - A)^t) = p_A(z),$$
$$p_{\overline{A}}(z) = \det(zI_n - \overline{A}) = \det(\overline{\overline{z}I_n - A}) = \overline{\det(\overline{z}I_n - A)} = \overline{p_A(\overline{z})}.$$

Since $A^* = \overline{A}^{\,t}$, the first identity applied to $\overline{A}$ gives the formula for A^*.
Multiplicity is preserved because conjugating a polynomial sends a factor $(z - \lambda)^k$ to $(z - \overline{\lambda})^k$.
Structural Commentary: Transpose preserves p_A; conjugation conjugates its roots.
Key Insight: The adjoint combines conjugation with transposition, so its spectrum is conjugated. □

Proposition 2.4.6. *Transpose Preserves Spectrum and Geometric Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ *and* $\lambda \in \mathbb{F}$. *Then* $\lambda \in \mathrm{spec}(A) \iff \lambda \in \mathrm{spec}(A^t)$, *and* $\dim E_\lambda(A) = \dim E_\lambda(A^t)$.

Proof. Since $(A - \lambda I_n)^t = A^t - \lambda I_n$, Theorem 2.1.94 gives $\mathrm{rank}(A - \lambda I_n) = \mathrm{rank}(A^t - \lambda I_n)$.
Matrix rank-nullity then gives $\dim E_\lambda(A) = n - \mathrm{rank}(A - \lambda I_n) = n - \mathrm{rank}(A^t - \lambda I_n) = \dim E_\lambda(A^t)$.
($\Rightarrow$) If $\lambda \in \mathrm{spec}(A)$, then $E_\lambda(A) \neq \{\mathbf{0}\}$, so $E_\lambda(A^t) \neq \{\mathbf{0}\}$ and $\lambda \in \mathrm{spec}(A^t)$.
($\Leftarrow$) The same argument with A and A^t interchanged gives the converse.
Structural Commentary: Transposition preserves rank, so shifted kernels have the same dimension.
Key Insight: Spectrum and geometric multiplicity are kernel facts for $A - \lambda I_n$. □

Corollary 2.4.7. *Matrix Invertibility and the Spectrum*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$. *Then* A *is invertible if and only if* $0 \notin \mathrm{spec}(A)$.

Proof.
($\Rightarrow$) If A is invertible, then $\ker(A) = \{\mathbf{0}\}$, so 0 is not an eigenvalue of A.
($\Leftarrow$) If $0 \notin \mathrm{spec}(A)$, then $\ker(A) = \{\mathbf{0}\}$. Corollary 2.1.93 gives full rank, hence A is invertible.
Structural Commentary: Matrix invertibility is the coordinate form of trivial kernel.
Key Insight: A zero eigenvalue is exactly a nonzero solution of $A\mathbf{x} = \mathbf{0}$. □

Proposition 2.4.8. *Trace and Determinant from Eigenvalues*
Let $\lambda_1, \ldots, \lambda_n$ *list the eigenvalues of* $A \in \mathrm{Mat}(n, n, \mathbb{C})$ *according to* m_A. *Then* $\mathrm{tr}\, A = \sum_{j=1}^{n} \lambda_j$ *and* $\det A = \prod_{j=1}^{n} \lambda_j$.

Proof. By Fundamental Theorem of Algebra, $p_A(z) = \prod_{j=1}^{n} (z - \lambda_j) = z^n - \left(\sum_{j=1}^{n} \lambda_j\right) z^{n-1} + \cdots + (-1)^n \prod_{j=1}^{n} \lambda_j$.
Theorem 2.4.2 gives $p_A(z) = z^n - (\mathrm{tr}\, A) z^{n-1} + \cdots + (-1)^n \det A$. Uniqueness of coefficients gives the identities.
Structural Commentary: Coefficients of p_A encode symmetric functions of the eigenvalues.
Key Insight: Trace is the first root sum and determinant is the full root product. □

Proposition 2.4.9. *Eigenvalues of a Polynomial in a Matrix*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $p \in \mathcal{P}(\mathbb{C})$, and let $\lambda_1,\dots,\lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity. Then the eigenvalues of $p(A)$ are $p(\lambda_1),\dots,p(\lambda_n)$, counted with algebraic multiplicity.

Proof. Choose a Schur form $U^*AU = T$ with diagonal $\lambda_1,\dots,\lambda_n$. Then $p(A) = Up(T)U^*$, and $p(T)$ is upper triangular with diagonal $p(\lambda_1),\dots,p(\lambda_n)$. Hence $p(A)$ has precisely these eigenvalues.
Structural Commentary: Schur form reduces the claim to the diagonal.
Key Insight: A polynomial preserves triangular form and applies p to each diagonal entry. □

Corollary 2.4.10. *Polynomial Invertibility Test*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and $p \in \mathcal{P}(\mathbb{C})$. If $p(\lambda) \neq 0$ for every $\lambda \in \mathrm{spec}(A)$, then $p(A)$ is invertible.

Proof. By Proposition 2.4.9, $0 \notin \mathrm{spec}(p(A))$. Corollary 2.4.7 gives invertibility.
Structural Commentary: Invertibility of $p(A)$ is checked on spectral values of A.
Key Insight: Avoiding zero on the spectrum prevents $p(A)$ from having a kernel. □

Corollary 2.4.11. *Eigenvalues of the Inverse*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be invertible with eigenvalues $\lambda_1,\dots,\lambda_n$, counted with algebraic multiplicity. Then A^{-1} has eigenvalues $\lambda_1^{-1},\dots,\lambda_n^{-1}$, counted with algebraic multiplicity.

Proof. Take a Schur form $A = UTU^*$ with diagonal $\lambda_1,\dots,\lambda_n$.
Then $A^{-1} = UT^{-1}U^*$, and T^{-1} is upper triangular with diagonal $\lambda_1^{-1},\dots,\lambda_n^{-1}$.
Structural Commentary: Invert the Schur form and read the diagonal.
Key Insight: The inverse spectrum is the reciprocal spectrum. □

Theorem 2.4.12. *Cayley–Hamilton Theorem*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have characteristic polynomial $p_A(z) = \det(zI_n - A)$. Then $p_A(A) = 0$.

Proof. Choose a Schur form $A = UTU^*$ with diagonal $\lambda_1,\dots,\lambda_n$.
Since $p_A(A) = Up_A(T)U^*$ and $p_A(z) = \prod_{i=1}^n (z-\lambda_i)$, it suffices to prove $(T-\lambda_1 I_n)(T-\lambda_2 I_n)\cdots(T-\lambda_n I_n) = 0$.
Let $P_j := \prod_{i=1}^{j}(T-\lambda_i I_n)$. Prove that first j columns of P_j are zero. This is clear for $j = 1$.
If true for j, then the first $j+1$ columns of $T - \lambda_{j+1}I_n$ lie in $\mathrm{span}(\mathbf{e}_1,\dots,\mathbf{e}_j)$, which P_j kills. Hence P_{j+1} has first $j+1$ zero columns, so $P_n = 0$.
Structural Commentary: Schur form reduces the theorem to a triangular column-killing induction.
Key Insight: Each factor $T - \lambda_i I_n$ kills the next leading column. □

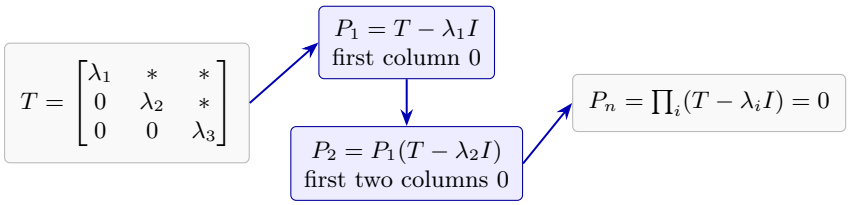


Figure 39: In Schur form, the factors $T - \lambda_i I$ kill the leading columns one layer at a time.

Corollary 2.4.13. *Reduction Modulo the Characteristic Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and let $f \in \mathcal{P}(\mathbb{C})$. If $f = p_A q + r$ with $\deg r < n$, then $f(A) = r(A)$.

Proof. Evaluate division identity at A: $f(A) = p_A(A)q(A) + r(A) = 0q(A) + r(A) = r(A)$.
Theorem 2.4.12 gives $p_A(A) = 0$.
Structural Commentary: Cayley–Hamilton makes polynomial expressions in A reducible modulo p_A.
Key Insight: Every high power of A can be rewritten using powers below n. □

Corollary 2.4.14. *Powers Span at Most n Dimensions*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then $\mathrm{span}(I_n, A, A^2, \dots) = \mathrm{span}(I_n, A, \dots, A^{n-1})$, so this has dimension at most n.

Proof. For each $k \geq 0$, Corollary 2.4.13 applied to $f(z) = z^{n+k}$ gives a polynomial r_k of degree at most $n-1$ such that $A^{n+k} = r_k(A)$. Thus every power of A lies in $\mathrm{span}(I_n, A, \dots, A^{n-1})$.
Structural Commentary: Polynomial reduction collapses the infinite power list to its first n terms.
Key Insight: Cayley–Hamilton gives the recurrence relation for powers of A. □

Corollary 2.4.15. *Inverse from the Characteristic Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have $p_A(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$.
If A is invertible, then $A^{-1} = -c_0^{-1}(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_2 A + c_1 I_n)$.

Proof. Since A is invertible, $0 \notin \mathrm{spec}(A)$, so $c_0 = p_A(0) = (-1)^n \det A \neq 0$.
Cayley–Hamilton gives $A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0$.
Move $c_0 I_n$ to the other side and multiply by $-c_0^{-1}A^{-1}$.
Structural Commentary: The characteristic equation contains an identity term exactly when A is invertible.
Key Insight: Solve Cayley–Hamilton for A^{-1} after factoring out one A. □

Definition 2.4.16. *Minimal Polynomial of a Matrix*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. The *minimal polynomial* of A, denoted μ_A, is the unique monic polynomial of smallest positive degree such that $\mu_A(A) = 0$.

Theorem 2.4.17. *Basic Properties of the Minimal Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let $\lambda_1, \ldots, \lambda_d$ be the distinct eigenvalues of A.

(i) μ_A exists, is unique, and $\deg \mu_A \leq n$.
(ii) If a nonzero polynomial f annihilates A, then μ_A divides f. In particular, μ_A divides p_A.
(iii) There are integers q_i satisfying $1 \leq q_i \leq m_A(\lambda_i)$ such that $\mu_A(z) = \prod_{i=1}^{d}(z-\lambda_i)^{q_i}$.

Proof.

(i) Cayley–Hamilton supplies the monic annihilator p_A, so a least-degree monic annihilator μ_A exists and has degree at most n. If g is another one, write $g = h\mu_A + r$ with $\deg r < \deg \mu_A$. Then $r(A) = 0$, so minimality gives $r = 0$; monicity and equal degree give $g = \mu_A$.
(ii) If $f(A) = 0$, divide $f = q\mu_A + r$ with $\deg r < \deg \mu_A$. Evaluating at A gives $r(A) = 0$, hence $r = 0$ and $\mu_A \mid f$. Taking $f = p_A$ gives $\mu_A \mid p_A$.
(iii) Since $\mu_A \mid p_A = \prod_{i=1}^{d}(z-\lambda_i)^{m_A(\lambda_i)}$, its roots are eigenvalues with exponents at most the algebraic multiplicities. If $A\mathbf{v} = \lambda_i \mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\mu_A(\lambda_i)\mathbf{v} = \mu_A(A)\mathbf{v} = 0$, so every λ_i occurs.

Structural Commentary: Polynomial division makes μ_A the least monic generator of annihilators.
Key Insight: Cayley–Hamilton gives existence; eigenvectors force the roots. □

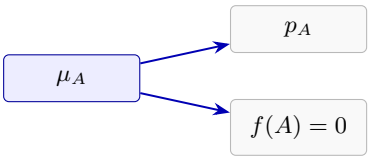


Figure 40: μ_A is the least monic annihilator; arrows indicate that μ_A divides p_A and every annihilator f.

Theorem 2.4.18. *Similarity Preserves the Minimal Polynomial*
If $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$ are similar, then $p_A = p_B$ and $\mu_A = \mu_B$.

Proof. The identity $p_A = p_B$ is Theorem 2.4.25. Let $A = SBS^{-1}$, then $f(A) = Sf(B)S^{-1}$ for every polynomial f, so A and B have the same annihilating polynomials. Hence $\mu_A = \mu_B$.
Structural Commentary: Similarity preserves polynomial equations.
Key Insight: Conjugation commutes with polynomial evaluation. □

Theorem 2.4.19. *Diagonalizability and the Minimal Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_d$. Then A is diagonalizable if and only if

$$\mu_A(z) = (z-\lambda_1)\cdots(z-\lambda_d).$$

Equivalently, A is diagonalizable if and only if μ_A is square-free.

Proof. Set $p(z) := \prod_{i=1}^{d}(z-\lambda_i)$.

(⇒) If $A = S\Lambda S^{-1}$ is diagonalizable, then $p(\Lambda) = 0$, so $p(A) = 0$. Since μ_A contains every factor $z - \lambda_i$ and divides p, we have $\mu_A = p$.
(⇐) Suppose $\mu_A = p$. On each primary summand $G(\lambda_i, A)$, the factors $A - \lambda_j I_n$ with $j \neq i$ are invertible by Lemma 2.3.56. Since $p(A) = 0$, we get $A - \lambda_i I_n = 0$ on $G(\lambda_i, A)$; hence every primary summand is an eigenspace, so A is diagonalizable.

Structural Commentary: Repeated factors of μ_A record nilpotent parts in primary blocks.
Key Insight: Square-free μ_A kills each primary block in one step. □

Corollary 2.4.20. *Square-Free Annihilator Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. If a polynomial f with no repeated roots satisfies $f(A) = 0$, then A is diagonalizable.

Proof. Since $\mu_A \mid f$, the minimal polynomial is square-free; Theorem 2.4.19 applies.
Structural Commentary: A square-free annihilator forces square-free μ_A.
Key Insight: Square-free μ_A means diagonalizable. □

Definition 2.4.21. *Companion Matrix*
Let $f(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$ be monic. For $n \geq 2$, the *companion matrix* of f is

$$C_f := \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}.$$

For $n = 1$, set $C_f = [-c_0]$.

Theorem 2.4.22. *Companion Matrix Realises a Polynomial*
For every monic polynomial f, the companion matrix C_f satisfies $p_{C_f} = f$ and $\mu_{C_f} = f$.

Proof. Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis.
For $j < n$, $C_f^j \mathbf{e}_1 = \mathbf{e}_{j+1}$, and last column is $C_f^n \mathbf{e}_1 = -c_0\mathbf{e}_1 - c_1 C_f \mathbf{e}_1 - \cdots - c_{n-1}C_f^{n-1}\mathbf{e}_1$.
Thus $f(C_f)\mathbf{e}_1 = 0$. Since $\mathbf{e}_j = C_f^{j-1}\mathbf{e}_1$ and $f(C_f)$ commutes with C_f, $f(C_f) = 0$.
If $g(z) = \sum_{i=0}^{m} b_i z^i$ has $m < n$, then $g(C_f)\mathbf{e}_1 = b_0\mathbf{e}_1 + b_1\mathbf{e}_2 + \cdots + b_m\mathbf{e}_{m+1}$, which is zero only when $g = 0$.
Hence no nonzero polynomial of degree $< n$ annihilates C_f. Therefore $\deg \mu_{C_f} = n$ and $\mu_{C_f} \mid f$, so $\mu_{C_f} = f$.
Cayley–Hamilton makes p_{C_f} a monic degree-n annihilator, hence $p_{C_f} = f$.
Structural Commentary: The cyclic vector $\mathbf{e}_1$ generates the basis, and last column imposes relation $f(C_f) = 0$.
Key Insight: The companion matrix makes f the first possible annihilating relation. □

Method 2.4.23. *Testing a Minimal Polynomial from p_A*
Suppose $p_A(z) = \prod_i (z - \lambda_i)^{a_i}$ is known.

i. Write each candidate as $m_{\mathbf{q}}(z) = \prod_i (z - \lambda_i)^{q_i}$, where $1 \leq q_i \leq a_i$.
ii. Test exponent vectors $\mathbf{q} = (q_i)$ in increasing total degree $\sum_i q_i$.
iii. For each $\mathbf{q}$, compute $m_{\mathbf{q}}(A) = \prod_i (A - \lambda_i I)^{q_i}$.
iv. The first candidate with $m_{\mathbf{q}}(A) = 0$ is μ_A, by Theorem 2.4.17.

Proposition 2.4.24. *Block Upper-Triangular Characteristic Polynomial*
Let $A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$, B and D are square. Then $p_A(z) = p_B(z)p_D(z)$, so $m_A(\lambda) = m_B(\lambda) + m_D(\lambda)$ $\forall \lambda \in \mathbb{F}$.

Proof. The shifted matrix is block upper triangular:

$$zI - A = \begin{bmatrix} zI - B & -C \\ 0 & zI - D \end{bmatrix}.$$

Theorem 2.1.132 gives $p_A(z) = \det(zI - B)\det(zI - D) = p_B(z)p_D(z)$.
Roots of a product are the roots of the factors, with multiplicity added.
Structural Commentary: The off-diagonal block couples vectors but not determinant factors.
Key Insight: In block upper-triangular form, spectral data lives on the diagonal blocks. □

Theorem 2.4.25. *Similarity Preserves Characteristic Data*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ and suppose $B = SAS^{-1}$ for some invertible S. Then $p_B = p_A$. In particular, for each $\lambda \in \mathrm{spec}(A)$, $m_B(\lambda) = m_A(\lambda)$ and $\dim E_\lambda(B) = \dim E_\lambda(A)$.

Proof. $p_B(z) = \det(zI_n - SAS^{-1}) = \det(S(zI_n - A)S^{-1}) = \det(S)\det(zI_n - A)\det(S)^{-1} = p_A(z) \Rightarrow$
$m_B(\lambda) = m_A(\lambda)$. Also $B - \lambda I_n = S(A - \lambda I_n)S^{-1}$, multiply by S gives isomorphism $\ker(A - \lambda I_n) \to \ker(B - \lambda I_n)$.
Structural Commentary: Similarity changes coordinates, not the shifted determinant or shifted kernel dimension.
Key Insight: Conjugation inserts S and S^{-1}, which cancel in determinants and identify kernels. □

Theorem 2.4.26. *Geometric Multiplicity Is at Most Algebraic Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ and let $\lambda \in \mathrm{spec}(A)$. Then $\dim E_\lambda(A) \leq m_A(\lambda)$.

Proof. Let $k = \dim E_\lambda(A)$ and choose a basis $\mathbf{x}_1, \dots, \mathbf{x}_k$ of $E_\lambda(A)$. Extend it to a basis of $\mathbb{F}^n$, and let S be the matrix whose columns are this basis. In the new basis,

$$S^{-1}AS = \begin{bmatrix} \lambda I_k & * \\ 0 & C \end{bmatrix}.$$

By Theorem 2.4.25 and Proposition 2.4.24, $p_A(z) = p_{S^{-1}AS}(z) = (z-\lambda)^k p_C(z)$. Thus $m_A(\lambda) \geq k = \dim E_\lambda(A)$.
Structural Commentary: An eigenspace-first basis exposes a λI_k block.
Key Insight: Each independent eigenvector forces one visible factor $z - \lambda$ in the characteristic polynomial. □

Theorem 2.4.27. *Diagonalizability and Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then A is diagonalizable if and only if $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$.

Proof.

(⇒) If A is diagonalizable, then A is similar to a diagonal matrix D. Theorem 2.4.25 gives $m_A(\lambda) = m_D(\lambda)$ and $\dim E_\lambda(A) = \dim E_\lambda(D)$. For diagonal D, both numbers count diagonal entries equal to λ.

(⇐) Assume $\dim E_\lambda(A) = m_A(\lambda)$ for every eigenvalue.

If $\mu_1, \dots, \mu_d$ are distinct eigenvalues, then $\sum_{i=1}^{d} m_A(\mu_i) = n$. Hence $\dim E_{\mu_1}(A) + \cdots + \dim E_{\mu_d}(A) = n$.

Theorem 2.3.28 gives an eigenbasis, so A is diagonalizable.

Structural Commentary: Diagonalizability means root counts match eigenspace dimensions.
Key Insight: No missing eigendirections means an eigenbasis. □

Theorem 2.4.28. *Polynomial Functional Calculus for Diagonalizable Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be diagonalizable and write $A = S\Lambda S^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$.
If $p \in \mathcal{P}(\mathbb{F})$, then $p(A) = S\,\mathrm{diag}(p(\lambda_1), \dots, p(\lambda_n))S^{-1}$.

Proof. For every $j \geq 0$, $A^j = S\Lambda^j S^{-1}$.

If $p(z) = \sum_j c_j z^j$, then $p(A) = \sum_j c_j A^j = S\left(\sum_j c_j \Lambda^j\right) S^{-1} = Sp(\Lambda)S^{-1}$.

Since Λ is diagonal, $p(\Lambda) = \mathrm{diag}(p(\lambda_1), \dots, p(\lambda_n))$.
Structural Commentary: Diagonalization reduces $p(A)$ to scalar evaluation on eigenvalues.
Key Insight: Compute functions of A by applying the function to the eigenvalues. □

Definition 2.4.29. *Commutant*
For $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n, n, \mathbb{F})$, the *commutant* of $\mathcal{F}$ is $\mathcal{F}' := \{X \in \mathrm{Mat}(n, n, \mathbb{F}) : XA = AX \ \forall A \in \mathcal{F}\}$.
For a single matrix A, write $\{A\}'$.

Theorem 2.4.30. *Commutant of a Diagonalizable Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonalizable. Suppose its distinct eigenvalues are $\mu_1, \dots, \mu_d$, and set $n_i = m_A(\mu_i)$.
Write $A = S(\mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d})S^{-1}$. Then $\{A\}' = \{S(X_1 \oplus \cdots \oplus X_d)S^{-1} : X_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})\}$.
In particular, $\dim\{A\}' = n_1^2 + \cdots + n_d^2$.

Proof. Let $\Lambda = \mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d}$. Matrix B commutes with A if and only if $S^{-1}BS$ commutes with Λ.
By Lemma 2.1.140, this happens exactly when $S^{-1}BS = X_1 \oplus \cdots \oplus X_d$ with $X_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})$.
Dimension count follows because conjugation by S is a vector-space isomorphism and $\dim \mathrm{Mat}(n_i, n_i, \mathbb{C}) = n_i^2$.
Structural Commentary: Diagonalizing A reduces commutation to respecting equal-eigenvalue blocks.
Key Insight: Distinct eigenvalue blocks cannot mix under a commuting matrix. □

Corollary 2.4.31. *Simple Spectrum Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ has n distinct eigenvalues, then $\{A\}' = \{p(A) : p \in \mathcal{P}(\mathbb{C})\}$, $\dim\{A\}' = n$.

Proof. Write $A = S\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)S^{-1}$ with distinct λ_i. Theorem 2.4.30 says every $B \in \{A\}'$ has the form $S\,\mathrm{diag}(\xi_1, \dots, \xi_n)S^{-1}$. Choose the Lagrange interpolation polynomial $p(z) = \sum_{i=1}^{n} \xi_i \prod_{j \neq i} \frac{z - \lambda_j}{\lambda_i - \lambda_j}$.

Then $p(\lambda_i) = \xi_i$ for every i, so Theorem 2.4.28 gives $B = p(A)$.
The dimension statement is the case $n_1 = \cdots = n_n = 1$ of Theorem 2.4.30.
Structural Commentary: Simple spectrum leaves one scalar choice per eigendirection.
Key Insight: Interpolation converts arbitrary diagonal data into a polynomial in A. □

2.4.2 Matrix Factorisations and Decompositions

Factorisations expose the geometry hidden inside a matrix or operator. In this section, positive square roots separate stretching from isometric motion, singular values record the stretching scales, and QR factorisation turns arbitrary independent columns into orthonormal coordinates.

Theorem 2.4.32. *Polar Decomposition*
For every $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $\exists$ an isometry $S \in \mathcal{L}(\mathcal{V})$ s.t. $T = S\sqrt{T^*T}$.

Proof. Let $P = \sqrt{T^*T}$. Since T^*T is positive, P is well defined by Proposition 2.3.44.
For every $\mathbf{v} \in \mathcal{V}$, $\|T\mathbf{v}\|^2 = \langle T^*T\mathbf{v}, \mathbf{v}\rangle = \langle P^2\mathbf{v}, \mathbf{v}\rangle = \|P\mathbf{v}\|^2$.
Define $S_1 : \text{range}(P) \to \text{range}(T)$ by $S_1(P\mathbf{v}) = T\mathbf{v}$. The norm identity shows this is well defined: if $P\mathbf{v}_1 = P\mathbf{v}_2$, then $\|T(\mathbf{v}_1 - \mathbf{v}_2)\| = \|P(\mathbf{v}_1 - \mathbf{v}_2)\| = 0$. It also shows S_1 is an isometry onto $\text{range}(T)$.
Choose any isometry from $\text{range}(P)^\perp$ onto $\text{range}(T)^\perp$, possible because the two complements have the same dimension. Combining it orthogonally with S_1 gives an isometry S on all of $\mathcal{V}$.
Then $SP\mathbf{v} = S_1(P\mathbf{v}) = T\mathbf{v}$ for every $\mathbf{v}$, so $T = S\sqrt{T^*T}$.
Structural Commentary: The positive factor $\sqrt{T^*T}$ reproduces exactly the lengths produced by T.
Key Insight: Once the metric stretching has been extracted, the remaining map is forced to be an isometry. □

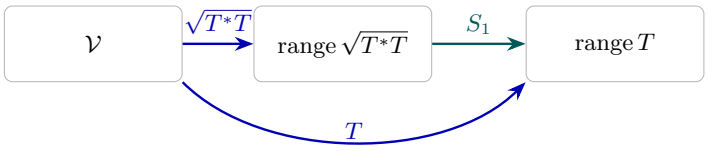


Figure 41: Polar decomposition builds S_1 using $\|T\mathbf{v}\| = \left\|\sqrt{T^*T}\mathbf{v}\right\|$.

Definition 2.4.33. *Singular Values*
For $T \in \mathcal{L}(\mathcal{V})$, *singular values* of T are eigenvalues of $\sqrt{T^*T}$, where each eigenvalue λ is repeated dim $E_\lambda(\sqrt{T^*T})$.

Theorem 2.4.34. *Singular Value Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$ have singular values $s_1, \ldots, s_n$. Then $\exists$ orthonormal bases $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and $(\mathbf{f}_1, \ldots, \mathbf{f}_n)$ of $\mathcal{V}$ s.t. $T\mathbf{v} = s_1 \langle \mathbf{v}, \mathbf{e}_1\rangle \mathbf{f}_1 + \cdots + s_n \langle \mathbf{v}, \mathbf{e}_n\rangle \mathbf{f}_n$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $|T| = \sqrt{T^*T}$. Since $|T|$ is positive, the real or complex spectral theorem gives an orthonormal basis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $|T|\,\mathbf{e}_j = s_j\mathbf{e}_j$. Thus $|T|\,\mathbf{v} = \sum_{j=1}^{n} s_j \langle \mathbf{v}, \mathbf{e}_j\rangle \mathbf{e}_j$.
By Theorem 2.4.32, $T = S\,|T|$ for some isometry S. Put $\mathbf{f}_j = S\mathbf{e}_j$. The vectors $\mathbf{f}_1, \ldots, \mathbf{f}_n$ form an orthonormal basis, and applying S to the displayed expansion gives the desired formula.
Structural Commentary: SVD diagonalises $|T|$, then transports its eigenvectors by S.
Key Insight: The s_j are stretches; the $\mathbf{f}_j$ are their landing directions. □

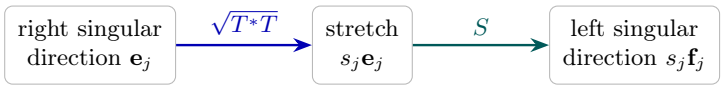


Figure 42: Since $T = S\sqrt{T^*T}$, each $\mathbf{e}_j$ maps to $s_j\mathbf{f}_j$.

Proposition 2.4.35. *Singular Values from T^*T*
The singular values of $T \in \mathcal{L}(\mathcal{V})$ are the nonnegative square roots of the eigenvalues of T^*T, with the same eigenspace multiplicities.

Proof. By the spectral theorem, choose an orthonormal basis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ and nonnegative scalars $\lambda_1, \ldots, \lambda_n$ such that $T^*T\mathbf{e}_j = \lambda_j\mathbf{e}_j$. The unique positive square root of T^*T sends $\mathbf{e}_j$ to $\sqrt{\lambda_j}\mathbf{e}_j$.
Thus the singular values are $\sqrt{\lambda_j}$, with the same eigenspace multiplicities.
Structural Commentary: The positive square root keeps eigenspaces and sends λ to $\sqrt{\lambda}$.
Key Insight: Singular values can be read from T^*T directly. □

Definition 2.4.36. *Wide and Narrow QR Factorisation*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $m \geq n$, a factorisation

$$A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$$

with $V \in \mathrm{Mat}(m,m,\mathbb{F})$ unitary and $R \in \mathrm{Mat}(n,n,\mathbb{F})$ upper triangular is called a *wide QR factorisation*. Writing $V = [Q\ Q']$ gives the *narrow QR factorisation* $A = QR$.

Theorem 2.4.37. *QR Factorisation*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and suppose $m \geq n$.

(i) $\exists$ a unitary V and an upper triangular R with real nonnegative diagonal entries such that

$$A = V \begin{bmatrix} R \\ 0 \end{bmatrix}.$$

Writing $V = [Q\ Q']$ with Q the first n columns gives $A = QR$, with Q orthonormal columns.
(ii) If $\mathrm{rank}(A) = n$, then narrow $A = QR$ is unique when R has positive diagonal entries.

Proof.

(i) At step k, choose a Householder unitary $I_{k-1} \oplus U_k$ that sends the tail of the kth column to $r_{kk}\mathbf{e}_1$ with $r_{kk} \geq 0$, while preserving the earlier triangular columns. After n steps,

$$U_n \cdots U_1 A = \begin{bmatrix} R \\ 0 \end{bmatrix},$$

where R is upper triangular with nonnegative diagonal entries. Let $V = (U_n \cdots U_1)^*$. Then $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, and writing $V = [Q\ Q']$ gives $A = QR$ with Q orthonormal columns. If $\mathrm{rank}(A) = n$, then R is invertible, so its diagonal entries are positive.

(ii) Suppose $A = Q_1R_1 = Q_2R_2$, where $Q_i^*Q_i = I_n$ and the R_i are upper triangular with positive diagonal entries. Then $A^*A = R_1^*R_1 = R_2^*R_2$. Let $D = R_1R_2^{-1}$. Substituting $R_1 = DR_2$ gives $D^*D = I_n$. Thus D is upper triangular and unitary, hence diagonal with entries of modulus 1. Its diagonal entries are positive, so $D = I_n$. Therefore $R_1 = R_2$, and then $Q_1 = Q_2$.

Structural Commentary: Householder steps triangularise columns; full rank makes the diagonal positive.
Key Insight: Positive diagonals remove the sign/phase ambiguity. □

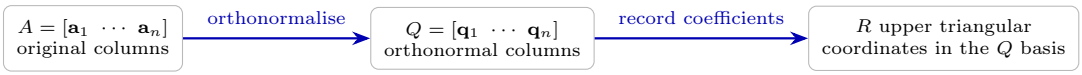


Figure 43: QR writes $A = QR$ while preserving prefix column spaces: $\mathrm{col}(A_k) = \mathrm{col}(Q_k)$.

Corollary 2.4.38. *Prefix Column Spaces in QR Factorisation*
Let $A = QR$ be the narrow QR factorisation of $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $\mathrm{rank}(A) = n$ and positive diagonal entries in R. For $k = 1, \ldots, n$, let A_k and Q_k be the first k columns of A and Q, respectively. Then $\mathrm{col}(A_k) = \mathrm{col}(Q_k)$. Thus the columns of Q_k form an orthonormal basis of $\mathrm{col}(A_k)$.

Proof. Let R_k be the leading $k \times k$ principal submatrix of R. Since R is upper triangular with positive diagonal entries, R_k is invertible. The first k columns of $A = QR$ satisfy $A_k = Q_kR_k$, because upper triangularity prevents later columns of Q from contributing. Hence $\mathrm{col}(A_k) \subseteq \mathrm{col}(Q_k)$ and $Q_k = A_kR_k^{-1}$ gives the reverse inclusion. The columns of Q_k are orthonormal because Q has orthonormal columns.
Structural Commentary: QR preserves prefix column spaces while orthonormalising them.
Key Insight: Invertible R_k only changes coordinates within that space. □

Proposition 2.4.39. *QR Route for Least Squares*
Let $A = QR \in \mathrm{Mat}(m,n,\mathbb{F})$ be a full-column-rank narrow QR factorisation. For $\mathbf{y} \in \mathbb{F}^m$, the least-squares solution solves $R\mathbf{x} = Q^*\mathbf{y}$. Moreover $P_{\mathrm{col}(A)} = QQ^*$.

Proof. The normal equations reduce by $A = QR$ and $Q^*Q = I_n$ to $R^*R\mathbf{x} = R^*Q^*\mathbf{y}$. Since R is invertible, this is equivalent to $R\mathbf{x} = Q^*\mathbf{y}$. Also $\mathrm{col}(A) = \mathrm{col}(Q)$, and the columns of Q are orthonormal, so $P_{\mathrm{col}(A)} = QQ^*$.
Structural Commentary: QR replaces $\mathrm{col}(A)$ by an orthonormal basis Q.
Key Insight: Project onto Q, then solve triangularly. □

2.4.3 Matrix Norms and Numerical Structure

Numerical linear algebra often replaces a matrix by a unitarily similar one with many forced zero entries. Upper Hessenberg form is the standard preliminary reduction before QR-type eigenvalue algorithms.

Definition 2.4.40. *Gersgorin Disks*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. The *deleted absolute row sum* in row k is $R'_k(A) := \sum_{j \neq k} |a_{kj}|$.

The *kth Gersgorin disk* of A is $G_k(A) := \{z \in \mathbb{C} : |z - a_{kk}| \leq R'_k(A)\}$, where $G(A) := \bigcup_{k=1}^{n} G_k(A)$

Theorem 2.4.41. *Gersgorin Disk Theorem*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq G(A) = \bigcup_{k=1}^{n} G_k(A)$.

Proof. Let $\lambda \in \mathrm{spec}(A)$ and choose an eigenvector $\mathbf{x} = [x_1 \ \cdots \ x_n]^t$.
Pick k such that $|x_k| = \max_j |x_j|$; then $x_k \neq 0$. The kth row of $A\mathbf{x} = \lambda\mathbf{x}$ gives $(\lambda - a_{kk})x_k = \sum_{j \neq k} a_{kj} x_j$.

Divide by x_k and use $|x_j/x_k| \leq 1$: $|\lambda - a_{kk}| = \left| \sum_{j \neq k} a_{kj} \frac{x_j}{x_k} \right| \leq \sum_{j \neq k} |a_{kj}| \left| \frac{x_j}{x_k} \right| \leq R'_k(A)$.

Thus $\lambda \in G_k(A) \subseteq G(A)$.
Structural Commentary: Normalize by a largest component; all row ratios have modulus at most one.
Key Insight: Row k traps λ in the disk centered at a_{kk} with radius $R'_k(A)$. □

Method 2.4.42. *Gersgorin Localisation Method*
For $A = [a_{ij}]$, Gersgorin localisation gives a disk enclosure for the spectrum.

(i) For each row k, place a disk at the diagonal entry a_{kk} with radius $R'_k(A) = \sum_{j \neq k} |a_{kj}|$.
The row enclosure is $G(A) = \bigcup_{k=1}^{n} G_k(A)$.
(ii) Theorem 2.4.41 guarantees that every eigenvalue of A lies in $G(A)$.
(iii) For a sharper enclosure, repeat the same construction for A^t. The centers are still a_{kk}, while the radii becomes deleted column sums of A. Then use $\mathrm{spec}(A) \subseteq G(A) \cap G(A^t)$. This is the intersection of two unions of disks; an eigenvalue need not lie in the overlap of each individual row-column disk pair.

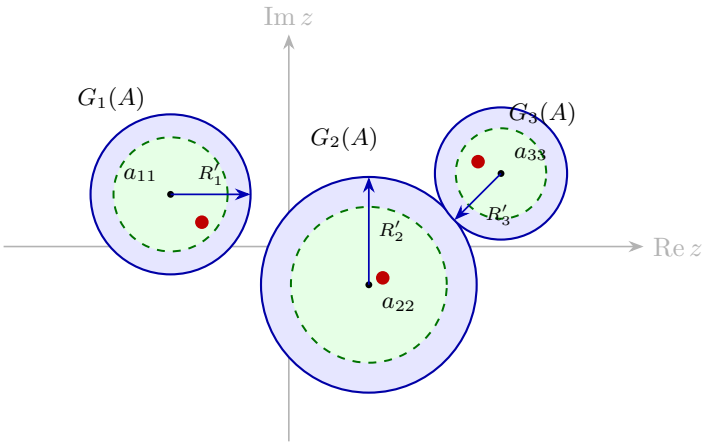


Figure 44: Blue row disks contain the spectrum; green dashed column disks give the sharper row-column enclosure $G(A) \cap G(A^t)$. Red points illustrate possible eigenvalue locations.

Corollary 2.4.43. *Row-Column Gersgorin Refinement*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq G(A) \cap G(A^t)$.
If $R_k(A) = \sum_{j=1}^{n} |a_{kj}|$ *and* $R_{\max}(A) = \max_{1 \leq k \leq n} R_k(A)$, *then* $\mathrm{spec}(A) \subseteq \{z \in \mathbb{C} : |z| \leq \min(R_{\max}(A), R_{\max}(A^t))\}$.

Proof. Theorem 2.4.41 gives $\mathrm{spec}(A) \subseteq G(A)$.
Proposition 2.4.6 gives $\mathrm{spec}(A) = \mathrm{spec}(A^t)$, so applying Gersgorin to A^t gives $\mathrm{spec}(A) \subseteq G(A^t)$.
Also $G_k(A) \subseteq \{z : |z| \leq |a_{kk}| + R'_k(A) = R_k(A)\}$, and similarly for A^t.
Structural Commentary: Row disks and column disks are both valid because A and A^t have the same spectrum.
Key Insight: Intersecting independent enclosures usually sharpens the eigenvalue search region. □

Definition 2.4.44. *Diagonal Dominance*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. Matrix A is *diagonally dominant* if $|a_{kk}| \geq R'_k(A)$ for every k, and *strictly diagonally dominant* if $|a_{kk}| > R'_k(A)$ for every k.

Corollary 2.4.45. *Strict Diagonal Dominance Implies Invertibility*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$.

(i) If A is strictly diagonally dominant, then A is invertible.
(ii) If A has real spectrum and nonnegative diagonal entries, then diagonal dominance gives $\mathrm{spec}(A) \subseteq [0, \infty)$; strict diagonal dominance gives $\mathrm{spec}(A) \subseteq (0, \infty)$.

Proof.

(i) Strict dominance gives $0 \notin G_k(A)$ for every k, since $|a_{kk}| > R'_k(A)$.
Thus $0 \notin \mathrm{spec}(A)$ by Gersgorin, so A is invertible by Corollary 2.4.7.

(ii) If $a_{kk} \geq 0$ and $a_{kk} \geq R'_k(A)$, then each $G_k(A)$ lies in the closed right half-plane. In the strict case, each disk lies in the open right half-plane. Gersgorin and the real-spectrum hypothesis give the claim.

Structural Commentary: Gersgorin converts diagonal dominance into disk placement.
Key Insight: The disks control whether eigenvalues can reach 0 or the left half-plane. □

Theorem 2.4.46. *Weak Diagonal Dominance Criterion*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonally dominant with no zero entries.
If $|a_{kk}| > R'_k(A)$ for some k, then A is invertible.

Proof. Assume $A\mathbf{x} = 0$ with $\mathbf{x} \neq \mathbf{0}$, and choose k with $|x_k| = \|\mathbf{x}\|_\infty$.
Row k gives $|a_{kk}| \, \|\mathbf{x}\|_\infty = \left| -\sum_{j \neq k} a_{kj} x_j \right| \leq \sum_{j \neq k} |a_{kj}| \, |x_j| \leq R'_k(A) \, \|\mathbf{x}\|_\infty \leq |a_{kk}| \, \|\mathbf{x}\|_\infty$.
Hence all inequalities are equalities. Since no a_{kj} vanishes, $|x_j| = \|\mathbf{x}\|_\infty$ for every j. Applying the same chain to any row i gives $|a_{ii}| = R'_i(A)$ for every i, contradicting the strict row. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.
Structural Commentary: A zero vector in the kernel would force equality in every Gersgorin row.
Key Insight: One strict row breaks that equality pattern. □

Definition 2.4.47. *Upper Hessenberg Matrix*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is *upper Hessenberg* if

$$a_{ij} = 0 \qquad \text{whenever } i > j + 1.$$

Equivalently, all entries below the first subdiagonal are zero.

$$\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \end{bmatrix}$$

Figure 45: Upper Hessenberg pattern: red is the first subdiagonal; blue entries below it are forced zero.

Theorem 2.4.48. *Unitary Upper Hessenberg Reduction*
For every $A \in \mathrm{Mat}(n, n, \mathbb{F})$, there exists a unitary matrix U such that U^*AU is upper Hessenberg. Moreover, the subdiagonal entries of U^*AU can be chosen real and nonnegative.

Proof. Construct the unitary similarity column by column. At step $k = 1, \ldots, n-2$, assume entries below the first subdiagonal are already zero in columns $1, \ldots, k-1$. Look at the tail of column k in rows $k+1, \ldots, n$. By Corollary 2.2.102, choose a unitary $V_k \in \mathrm{Mat}(n-k, n-k, \mathbb{F})$ such that V_k^* sends this tail to a vector whose only nonzero entry is its first entry, chosen real and nonnegative.
Set $U_k = I_k \oplus V_k$ and replace the current matrix A_{k-1} by $A_k = U_k^* A_{k-1} U_k$. Because U_k fixes the first k coordinate vectors, right multiplication by U_k does not change the first k columns before the left action is applied. The left action affects only rows $k+1, \ldots, n$, so previously created zeros in columns $1, \ldots, k-1$ stay zero, while column k gains zeros below row $k+1$.
After $n-2$ steps, A_{n-2} is upper Hessenberg with real nonnegative subdiagonal entries. Since each step is a unitary similarity, $A_{n-2} = U^*AU$ for the product $U = U_1 U_2 \cdots U_{n-2}$.
Structural Commentary: Householder similarities annihilate one column tail at a time while fixing the earlier coordinate flag.

Key Insight: Acting only on the trailing coordinates preserves previous zeros and creates the next Hessenberg column. □

Perspective 2.4.49. *Why Hessenberg Form Matters*
Upper Hessenberg matrices are almost triangular but still unitarily similar to the original matrix. This preserves spectral data while making iterative eigenvalue algorithms cheaper because only one subdiagonal must be tracked.

2.4.4 Advanced Matrix Analysis

This subsection collects matrix identities that organise larger linear systems by blocks, tensor products, vectorisation, and rank bookkeeping.

Definition 2.4.50. *Kronecker Product*
Let $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$. The *Kronecker product* of A and B is the block matrix

$$A \otimes B := \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \in \mathrm{Mat}(mp, nq, \mathbb{F}).$$

The Kronecker product is also called the tensor product of matrices.

Proposition 2.4.51. *Basic Kronecker Identities*
Whenever the expressions are defined,

(i) $c(A \otimes B) = (cA) \otimes B = A \otimes (cB)$;
(ii) $(A + B) \otimes C = A \otimes C + B \otimes C$ *and* $A \otimes (B + C) = A \otimes B + A \otimes C$;
(iii) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$;
(iv) $(A \otimes B)^t = A^t \otimes B^t$;
(v) $I_m \otimes I_n = I_{mn}$.

If $\mathbb{F} = \mathbb{C}$, *then also* $\overline{A \otimes B} = \overline{A} \otimes \overline{B}$ *and* $(A \otimes B)^* = A^* \otimes B^*$.

Proof.

(i) The (i, j) block of $c(A \otimes B)$ is $c(a_{ij}B) = (ca_{ij})B = a_{ij}(cB)$, which is (i, j) block of $(cA) \otimes B$ and $A \otimes (cB)$.
(ii) The (i, j) block of $(A + B) \otimes C$ is $(a_{ij} + b_{ij})C = a_{ij}C + b_{ij}C$, so $(A + B) \otimes C = A \otimes C + B \otimes C$. The proof of $A \otimes (B + C) = A \otimes B + A \otimes C$ is identical inside each block.
(iii) If $B = [b_{rs}]$, then both $(A \otimes B) \otimes C$ and $A \otimes (B \otimes C)$ have blocks $a_{ij}b_{rs}C$ in the same relative positions.
(iv) The (j, i) block of $(A \otimes B)^t$ is $(a_{ij}B)^t = a_{ij}B^t$, which is the (j, i) block of $A^t \otimes B^t$.
(v) The block matrix $I_m \otimes I_n$ has I_n on its diagonal blocks and zero elsewhere, hence equals I_{mn}.

If $\mathbb{F} = \mathbb{C}$, conjugation is checked blockwise, adjoint identity follows from conjugation together with part (iv).
Structural Commentary: Kronecker products replace scalar entries by whole scaled blocks. Standard algebraic rules survive because each block is still controlled by scalar distributivity and transposition.
Key Insight: Check identities block-by-block; each block reduces to an ordinary scalar-matrix identity. □

Theorem 2.4.52. *Mixed Product Property*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $B \in \mathrm{Mat}(p, q, \mathbb{F})$, $C \in \mathrm{Mat}(n, r, \mathbb{F})$, $D \in \mathrm{Mat}(q, s, \mathbb{F})$. $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

Proof. Write $A = [a_{ik}]$ and $C = [c_{kj}]$.
The (i, j) block of $(A \otimes B)(C \otimes D)$ is $\sum_{k=1}^{n} (a_{ik}B)(c_{kj}D) = \left(\sum_{k=1}^{n} a_{ik}c_{kj} \right) BD = (AC)_{ij}BD$, which is exactly the (i, j) block of $(AC) \otimes (BD)$.
Structural Commentary: The outer indices multiply as AC, while the inner block action multiplies as BD.
Key Insight: Kronecker multiplication separates into ordinary multiplication in each factor. □

Corollary 2.4.53. *Inverse of a Kronecker Product*
If $A \in \mathrm{Mat}(m, m, \mathbb{F})$ *and* $B \in \mathrm{Mat}(n, n, \mathbb{F})$ *are invertible, then* $A \otimes B$ *is invertible and* $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Proof. By Theorem 2.4.52, $(A \otimes B)(A^{-1} \otimes B^{-1}) = I_m \otimes I_n = I_{mn}$ and the reverse product is identical.
Structural Commentary: Inverting a Kronecker product inverts each independent factor.
Key Insight: The mixed product property turns the inverse check into two ordinary inverse checks. □

Definition 2.4.54. *Vectorisation*
Let $X = [\mathbf{x}_1\ \mathbf{x}_2\ \cdots\ \mathbf{x}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$. The *vectorisation* of X is

$$\mathrm{vec}(X) := \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \in \mathbb{F}^{mn}.$$

Thus vec stacks the columns of X vertically.

Theorem 2.4.55. *Vectorisation of a Triple Product*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $X \in \mathrm{Mat}(n,p,\mathbb{F})$, and $B \in \mathrm{Mat}(p,q,\mathbb{F})$. Then $\mathrm{vec}(AXB) = (B^t \otimes A)\,\mathrm{vec}(X)$.

Proof. Write $B = [\mathbf{b}_1\ \cdots\ \mathbf{b}_q]$ and $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_p]$.
The kth column of AXB is $AX\mathbf{b}_k = A\sum_{i=1}^{p} b_{ik}\mathbf{x}_i = \sum_{i=1}^{p} b_{ik}A\mathbf{x}_i = (\mathbf{b}_k^t \otimes A)\,\mathrm{vec}(X)$.
Stacking these q columns gives $(B^t \otimes A)\,\mathrm{vec}(X)$.
Structural Commentary: Vectorisation rewrites two-sided multiplication as one matrix action.
Key Insight: Left multiplication contributes A; right multiplication contributes B^t. □

Theorem 2.4.56. *Sylvester Linear Matrix Equation*
Let $A \in \mathrm{Mat}(m,m,\mathbb{C})$ and $B \in \mathrm{Mat}(n,n,\mathbb{C})$. If $\mathrm{spec}(A) \cap \mathrm{spec}(B) = \emptyset$, then for every $C \in \mathrm{Mat}(m,n,\mathbb{C})$ the equation $AX - XB = C$ has a unique solution $X \in \mathrm{Mat}(m,n,\mathbb{C})$.

Proof. Let $L(X) = AX - XB$. Since L has equal finite-dimensional domain and codomain, it suffices to show $\ker L = \{0\}$. If $AX = XB$, then $p(A)X = Xp(B)$ for every polynomial p.
With $p = p_B$, Cayley–Hamilton gives $p_B(A)X = Xp_B(B) = 0$. For every $\lambda \in \mathrm{spec}(A)$, disjointness gives $p_B(\lambda) \neq 0$; hence $p_B(A)$ is invertible by Corollary 2.4.10. Thus $X = 0$, so L is bijective.
Structural Commentary: Polynomial calculus pass through homogeneous solutions; disjoint spectra invert $p_B(A)$.
Key Insight: Spectral separation turns an intertwining relation into the zero relation. □

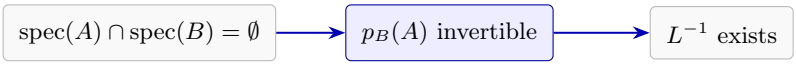


Figure 46: Disjoint spectra make $p_B(A)$ invertible, so the Sylvester map $L(X) = AX - XB$ is invertible.

Corollary 2.4.57. *Removing a Spectrally Separated Off-Diagonal Block*
Let $M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ *and* $\mathrm{spec}(A) \cap \mathrm{spec}(B) = \emptyset$ *Then M is similar to $A \oplus B$.*

Proof. By Theorem 2.4.56, choose X such that $AX - XB = -C$. Then

$$\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix}\begin{bmatrix} A & C \\ 0 & B \end{bmatrix}\begin{bmatrix} I & X \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & AX + C - XB \\ 0 & B \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}.$$

Structural Commentary: A block shear cancels the off-diagonal block via Sylvester's equation.
Key Insight: Spectrally separated diagonal blocks cannot carry essential upper-right coupling. □

Theorem 2.4.58. *Schur Form Grouped by Eigenvalues*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have distinct eigenvalues $\lambda_1,\dots,\lambda_d$ with algebraic multiplicities $n_1,\dots,n_d$. Then A is unitarily similar to a block upper-triangular matrix

$$T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1d} \\ 0 & T_{22} & \cdots & T_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T_{dd} \end{bmatrix}, \qquad T_{ii} \in \mathrm{Mat}(n_i,n_i,\mathbb{C}),$$

where each T_{ii} is upper triangular with all diagonal entries equal to λ_i. Moreover, A is similar to $T_{11} \oplus \cdots \oplus T_{dd}$.

Proof. Order the Schur diagonal as $\lambda_1,\dots,\lambda_1,\lambda_2,\dots,\lambda_2,\dots,\lambda_d,\dots,\lambda_d$.
Theorem 2.2.62 gives the displayed block Schur form. Since $\mathrm{spec}(T_{11})$ is disjoint from the trailing block spectrum, Corollary 2.4.57 splits off T_{11}. Iterating splits off every diagonal block.
Structural Commentary: Grouped Schur form isolates spectral blocks; Sylvester shears split disjoint groups.
Key Insight: Cross-group upper blocks are coordinate artefacts. □

Theorem 2.4.59. *Minimal Polynomial of a Spectrally Disjoint Direct Sum*
Let $A = A_1 \oplus \cdots \oplus A_d$, where the spectra of $A_1, \ldots, A_d$ are pairwise disjoint. Then $\mu_A = \mu_{A_1}\mu_{A_2}\cdots\mu_{A_d}$.

Proof. Let $f = \prod_i \mu_{A_i}$. Since $f(A_i) = 0$ for all i, $f(A) = 0$ and $\mu_A \mid f$. Conversely, $\mu_A(A_i) = 0$, so $\mu_{A_i} \mid \mu_A$ for each i. Disjoint spectra make the μ_{A_i} pairwise relatively prime, hence $f \mid \mu_A$. Thus $\mu_A = f$.
Structural Commentary: Direct sums are annihilated blockwise; disjoint spectra make the block factors coprime.
Key Insight: Each spectral component contributes its own minimal-polynomial factor. □

Corollary 2.4.60. *Commuting with Disjoint Blocks*
Let $A = A_{11} \oplus \cdots \oplus A_{dd}$, where the spectra of $A_{11}, \ldots, A_{dd}$ are pairwise disjoint. Partition $B = [B_{ij}]$ conformally with A. If $AB = BA$, then $B_{ij} = 0$ for all $i \neq j$.

Proof. Equating (i, j) blocks in $AB = BA$ gives $A_{ii}B_{ij} - B_{ij}A_{jj} = 0$.
For $i \neq j$, the spectra of A_{ii} and A_{jj} are disjoint, so Theorem 2.4.56 with $C = 0$ gives $B_{ij} = 0$.
Structural Commentary: A commuting matrix cannot mix direct-sum blocks with disjoint spectra.
Key Insight: The off-diagonal block solves a homogeneous Sylvester equation, whose only solution is zero. □

Definition 2.4.61. *Nonderogatory Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *nonderogatory* if its Jordan form has exactly one Jordan block for each distinct eigenvalue. Equivalently, $\dim E_\lambda(A) = 1$ for every $\lambda \in \mathrm{spec}(A)$.

Lemma 2.4.62. *Commutant of One Jordan Block*
If $J = J_m(\lambda)$ and $XJ = JX$, then $X = p(J)$ for some polynomial p with $\deg p < m$.

Proof. It suffices to treat $N = J_m(0)$, since $J_m(\lambda) = \lambda I_m + N$. The vector $\mathbf{e}_m$ is cyclic for N:

$$(\mathbf{e}_m, N\mathbf{e}_m, \ldots, N^{m-1}\mathbf{e}_m)$$

is the standard basis in reverse order. Choose the unique polynomial p with $\deg p < m$ such that $p(N)\mathbf{e}_m = X\mathbf{e}_m$. For every $r = 0, \ldots, m-1$,

$$p(N)N^r\mathbf{e}_m = N^r p(N)\mathbf{e}_m = N^r X\mathbf{e}_m = XN^r\mathbf{e}_m.$$

Thus $p(N)$ and X agree on a basis, so $X = p(N) = q(J)$ for $q(z) = p(z - \lambda)$.
Structural Commentary: A single Jordan block is generated by one cyclic vector.
Key Insight: A commuting matrix is determined by where it sends the cyclic vector, hence by a polynomial in the block. □

Theorem 2.4.63. *Nonderogatory Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is nonderogatory, then

$$\{A\}' = \{p(A) : p \in \mathcal{P}(\mathbb{C})\}.$$

Proof. The inclusion $\{p(A) : p \in \mathcal{P}(\mathbb{C})\} \subseteq \{A\}'$ is immediate.
For the reverse inclusion, replace A by its Jordan form

$$J = J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_d}(\lambda_d),$$

where the λ_i are distinct. If $XJ = JX$ and $X = [X_{ij}]$ is partitioned conformally with J, Corollary 2.4.60 gives $X_{ij} = 0$ for $i \neq j$. Each diagonal block X_{ii} commutes with $J_{m_i}(\lambda_i)$, so Lemma 2.4.62 gives $X_{ii} = p_i(J_{m_i}(\lambda_i))$ for some $\deg p_i < m_i$.
Because the polynomials $(z - \lambda_i)^{m_i}$ are pairwise coprime, Hermite interpolation gives a polynomial p such that $p \equiv p_i \pmod{(z - \lambda_i)^{m_i}}$ for every i. Then $X = p(J)$. Conjugating back to A proves the claim.
Structural Commentary: Disjoint eigenvalue blocks force block diagonality; one Jordan block per eigenvalue makes each diagonal commutant polynomial.
Key Insight: Hermite interpolation glues the separate block polynomials into one polynomial in A. □

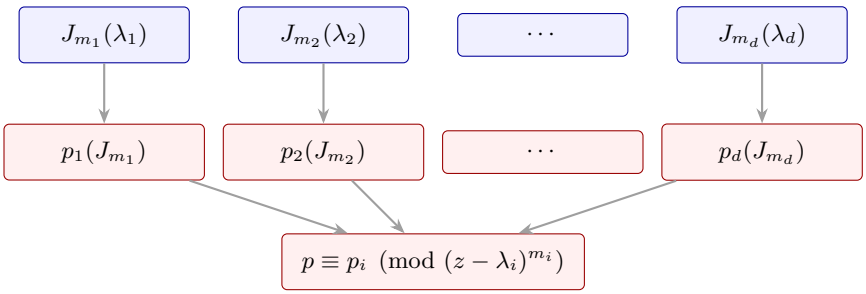


Figure 47: For a nonderogatory matrix, commuting matrices are blockwise polynomials, then one interpolating polynomial glues the blocks.

Theorem 2.4.64. *Simultaneous Schur Triangularisation*
Let $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n,n,\mathbb{C})$ be a commuting family.
Then there is a unitary $U \in \mathrm{Mat}(n,n,\mathbb{C})$ such that U^*AU is upper triangular for every $A \in \mathcal{F}$.
If every matrix in $\mathcal{F}$ is real and has only real eigenvalues, then U can be chosen real orthogonal.

Proof. Induct on n. Choose a common unit eigenvector $\mathbf{x}$, extend it to an orthonormal basis, and write $V = [\mathbf{x}\ V_2]$. Then for every $A \in \mathcal{F}$,

$$V^*AV = \begin{bmatrix} \lambda_A & * \\ 0 & A' \end{bmatrix}.$$

Commutation gives $A'B' = B'A'$ for all $A, B \in \mathcal{F}$, so induction simultaneously triangularises the compressed family.
In the real case, choose a real common eigenvector: if all matrices are scalar, any real unit vector works; otherwise a nonzero proper real eigenspace of some nonscalar $A_0 \in \mathcal{F}$ is invariant under all of $\mathcal{F}$, and induction inside it gives one. Then use real orthonormal extensions.
Structural Commentary: A common eigenline creates the first block; commutation descends to the compression.
Key Insight: Simultaneous triangularisation is Schur induction with a common eigenvector at every step. □

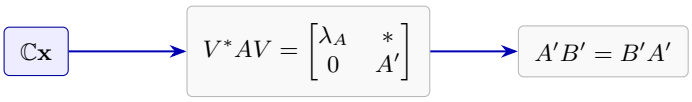


Figure 48: A shared eigenline gives a simultaneous first block; lower-right compressions still commute.

Corollary 2.4.65. *Eigenvalues of Sums and Products of Commuting Matrices*
Let $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$ and $AB = BA$. Their eigenvalues can be ordered as $\lambda_1, \ldots, \lambda_n$ and $\mu_1, \ldots, \mu_n$ so that $A+B$ has eigenvalues $\lambda_i + \mu_i$ and AB has eigenvalues $\lambda_i\mu_i$, counted with multiplicity.

Proof. Choose simultaneous Schur forms $T = U^*AU$ and $R = U^*BU$. Set $\lambda_i = t_{ii}$ and $\mu_i = r_{ii}$. Then $T+R$ and TR are upper triangular with diagonals $\lambda_i + \mu_i$ and $\lambda_i\mu_i$, and are similar to $A+B$ and AB.
Structural Commentary: Simultaneous Schur form aligns both spectra on one diagonal.
Key Insight: In common triangular coordinates, diagonals add and multiply. □

Theorem 2.4.66. *Brauer Rank-One Eigenvalue Shift*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, and let $\mathbf{y} \in \mathbb{C}^n$. If $\lambda, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A with multiplicity, then the eigenvalues of $A + \mathbf{x}\mathbf{y}^*$ are $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \ldots, \lambda_n$.
Moreover, $\mathbf{x}$ is an eigenvector of $A + \mathbf{x}\mathbf{y}^*$ corresponding to $\lambda + \mathbf{y}^*\mathbf{x}$.

Proof. Put $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|_2$. By Schur triangularisation, choose a unitary $U = [\mathbf{u}\ U_2]$ such that $U^*AU = \begin{bmatrix} \lambda & * \\ 0 & T' \end{bmatrix}$, where the diagonal entries of T' are $\lambda_2, \ldots, \lambda_n$. Also $U^*(\mathbf{x}\mathbf{y}^*)U = \begin{bmatrix} \mathbf{y}^*\mathbf{x} & * \\ 0 & 0 \end{bmatrix}$. Therefore $U^*(A + \mathbf{x}\mathbf{y}^*)U$ is upper triangular with diagonal entries $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \ldots, \lambda_n$. Finally, $(A + \mathbf{x}\mathbf{y}^*)\mathbf{x} = A\mathbf{x} + \mathbf{x}(\mathbf{y}^*\mathbf{x}) = (\lambda + \mathbf{y}^*\mathbf{x})\mathbf{x}$.
Structural Commentary: A rank-one eigenvector update changes only the first Schur diagonal entry.
Key Insight: Align the update with the first coordinate; only one diagonal term shifts. □

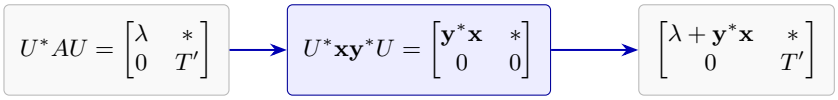


Figure 49: Brauer's update shifts one Schur diagonal entry.

Corollary 2.4.67. *Damped Rank-One Eigenvalue Reset*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenpair $(\lambda, \mathbf{x})$, let $\mathbf{y}^*\mathbf{x} = 1$, and let $\tau \in \mathbb{C}$. If $\lambda, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A, then $A_\tau := \tau A + (1-\tau)\lambda\mathbf{x}\mathbf{y}^*$ has eigenvalues $\lambda, \tau\lambda_2, \ldots, \tau\lambda_n$.

Proof. The eigenvalues of τA are $\tau\lambda, \tau\lambda_2, \ldots, \tau\lambda_n$. Since $A_\tau = \tau A + \mathbf{x}((1-\overline{\tau})\overline{\lambda}\mathbf{y})^*$, Theorem 2.4.66 shifts only $\tau\lambda$ by $(1-\tau)\lambda\mathbf{y}^*\mathbf{x} = (1-\tau)\lambda$. Thus $\tau\lambda$ becomes λ, while the other eigenvalues stay $\tau\lambda_i$.
Structural Commentary: Damping scales the spectrum; Brauer's update restores one eigenvalue.
Key Insight: Brauer's theorem changes one spectral coordinate without moving the rest. □

Proposition 2.4.68. *Schur Complement of the Lower-Right Block*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$.
If D is invertible, then the Schur complement of D in M is $M/D := A - BD^{-1}C$, and $\det(M) = \det(D)\det(M/D)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & -BD^{-1} \\ 0 & I_s \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A - BD^{-1}C & 0 \\ C & D \end{bmatrix}$.
The elimination matrix has determinant 1, and the matrix on the right is lower block-triangular, so Theorem 2.1.132 applied after transposition gives $\det(M) = \det(A - BD^{-1}C)\det(D)$.
Structural Commentary: This is the lower-right version of Schur complementation. Eliminating the upper-right block isolates the determinant contribution of D.
Key Insight: Eliminate by blocks, then read the determinant from the diagonal blocks. □

Theorem 2.4.69. *Sylvester Determinant Identity*
Let $A \in \mathrm{Mat}(n, m, \mathbb{F})$ and $B \in \mathrm{Mat}(m, n, \mathbb{F})$. Then $\det(I_n + AB) = \det(I_m + BA)$.

Proof. Let $M = \begin{bmatrix} I_n & -A \\ B & I_m \end{bmatrix}$.
Using Proposition 2.4.68 with lower-right block I_m gives $\det(M) = \det(I_n + AB)$. Using Proposition 2.1.133 with upper-left block I_n gives $\det(M) = \det(I_m + BA)$. Hence the two determinants are equal.
Structural Commentary: Two Schur complements of one block matrix give Sylvester's identity.
Key Insight: Compare the same determinant after eliminating different identity blocks. □

Theorem 2.4.70. *Nonzero Eigenvalues of AB and BA*
Let $A \in \mathrm{Mat}(m, n, \mathbb{C})$, $B \in \mathrm{Mat}(n, m, \mathbb{C})$, and suppose $n \geq m$. Then $p_{BA}(z) = z^{n-m}p_{AB}(z)$.
Consequently, for $\lambda \neq 0$, $m_{AB}(\lambda) = m_{BA}(\lambda)$. If $m_{AB}(0) = k$, then $m_{BA}(0) = k + n - m$.

Proof. For $z \neq 0$, Sylvester's determinant identity gives

$$\det(zI_n - BA) = z^n \det(I_n - z^{-1}BA) = z^n \det(I_m - z^{-1}AB) = z^{n-m}\det(zI_m - AB).$$

Both sides are polynomials in z, so the identity holds for all z. The claimed multiplicities follow by factoring p_{AB} and comparing with $p_{BA} = z^{n-m}p_{AB}$.
Structural Commentary: Sylvester's determinant identity moves the rectangular product across the determinant.
Key Insight: Passing from AB to BA only adds the necessary extra zero roots when the sizes differ. □

Theorem 2.4.71. *Nonzero Jordan Blocks of AB and BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. For every nonzero $\lambda \in \mathbb{C}$ and every $k \geq 1$, the Jordan forms of AB and BA contain the same number of blocks $J_k(\lambda)$.

Proof. Set

$$X = \begin{bmatrix} AB & A \\ 0 & 0 \end{bmatrix}, \qquad Y = \begin{bmatrix} 0 & A \\ 0 & BA \end{bmatrix}.$$

Then

$$\begin{bmatrix} I_n & 0 \\ B & I_n \end{bmatrix} X \begin{bmatrix} I_n & 0 \\ -B & I_n \end{bmatrix} = Y,$$

so X and Y are similar. For $\lambda \neq 0$ and $p \geq 1$,

$$(X - \lambda I_{2n})^p = \begin{bmatrix} (AB - \lambda I_n)^p & * \\ 0 & (-\lambda)^p I_n \end{bmatrix},$$
$$(Y - \lambda I_{2n})^p = \begin{bmatrix} (-\lambda)^p I_n & * \\ 0 & (BA - \lambda I_n)^p \end{bmatrix}.$$

The invertible diagonal block $(-\lambda)^p I_n$ can be removed by block row and column operations, so

$$\operatorname{rank}\big((X - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((AB - \lambda I_n)^p\big),$$
$$\operatorname{rank}\big((Y - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((BA - \lambda I_n)^p\big).$$

Similarity of X and Y makes the left sides equal. Hence the rank drops defining the Weyr characteristic of AB and BA at λ agree, and Theorem 2.3.82 gives the same nonzero Jordan blocks.
Structural Commentary: Embed AB and BA into similar block matrices, then use $\lambda \neq 0$ to split off an invertible block.
Key Insight: Nonzero Jordan structure is detected by shifted rank drops, and those rank drops are forced to match. □

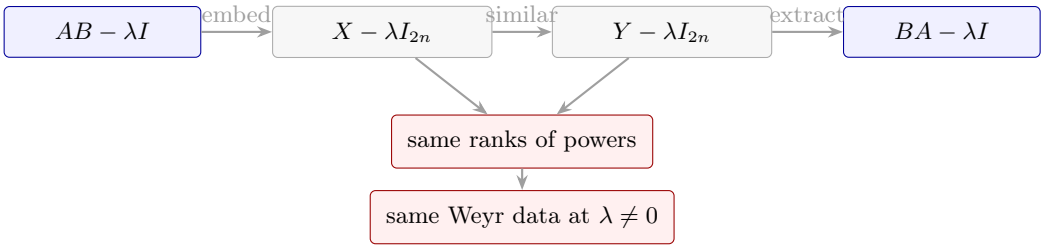


Figure 50: For $\lambda \neq 0$, the auxiliary block similarity makes the shifted ranks of AB and BA agree.

Corollary 2.4.72. *Similarity Criterion for AB and BA*
Let $A, B \in \operatorname{Mat}(n, n, \mathbb{C})$. Then AB is similar to BA if and only if

$$\operatorname{rank}\big((AB)^p\big) = \operatorname{rank}\big((BA)^p\big), \qquad p = 1, \ldots, n.$$

Proof. If AB and BA are similar, then their powers are similar and have equal rank.
Conversely, Theorem 2.4.71 gives equality of all nonzero Jordan blocks. The displayed rank equalities, together with the trivial equality at $p = 0$, give the same Weyr characteristic at $\lambda = 0$ by Theorem 2.3.82. Hence the nilpotent Jordan blocks also agree, so the two Jordan forms are identical up to block order.
Structural Commentary: Nonzero blocks agree automatically; only the nilpotent blocks must be tested.
Key Insight: Powers of AB and BA measure exactly the zero-Jordan-block data. □

Method 2.4.73. *Checking Whether AB and BA Are Similar*
For $A, B \in \operatorname{Mat}(n, n, \mathbb{C})$:

(i) The nonzero Jordan blocks of AB and BA already match.
(ii) Compute $\operatorname{rank}\big((AB)^p\big)$ and $\operatorname{rank}\big((BA)^p\big)$ for $p = 1, \ldots, n$.
(iii) If all ranks agree, then $AB \sim BA$; if some rank differs, then only the nilpotent Jordan structure differs.

Corollary 2.4.74. *Hermitian Products and Conjugate Products*
Let $A, H, K \in \operatorname{Mat}(n, n, \mathbb{C})$, and suppose H and K are Hermitian. Then HK is similar to KH, and $A\overline{A}$ is similar to $\overline{A}A$.

Proof. For each $p = 1, \ldots, n$,

$$\operatorname{rank}\big((HK)^p\big) = \operatorname{rank}\big(((HK)^p)^*\big) = \operatorname{rank}\big((KH)^p\big).$$

Also,

$$\operatorname{rank}\big((A\overline{A})^p\big) = \operatorname{rank}\big(\overline{(A\overline{A})^p}\big) = \operatorname{rank}\big((\overline{A}A)^p\big).$$

Apply Corollary 2.4.72 in both cases.
Structural Commentary: Adjoint and conjugation preserve rank while reversing the product order in these two situations.
Key Insight: The rank-power test turns order reversal into similarity. □

Theorem 2.4.75. *Conjugate Similarity and Hermitian Products*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) *A is similar to $\overline{A}$.*
(ii) *A is similar to A^*.*
(iii) *A is similar to A^* via a Hermitian similarity matrix.*
(iv) *$A = HK = LM$, where H, K, L, M are Hermitian, and H, M are invertible.*
(v) *$A = HK$ for some Hermitian matrices H, K.*

Proof.

(i) $\Rightarrow$ *(ii)* Theorem 2.3.87 applied to $\overline{A}$ gives $\overline{A} \sim (\overline{A})^t = A^*$. Thus $A \sim A^*$.

(ii) $\Rightarrow$ *(iii)* Choose invertible S such that $A = SA^*S^{-1}$, so $AS = SA^*$. For $\theta \in \mathbb{R}$, put $S_\theta = e^{i\theta}S$. Then $AS_\theta = S_\theta A^*$ and, by taking adjoints, $AS_\theta^* = S_\theta^* A^*$. Hence

$$A(S_\theta + S_\theta^*) = (S_\theta + S_\theta^*)A^*.$$

The matrix $H_\theta = S_\theta + S_\theta^*$ is Hermitian. Also

$$S_\theta^{-1} H_\theta = I_n + e^{-2i\theta} S^{-1} S^*.$$

Only finitely many values of $-e^{2i\theta}$ can be eigenvalues of $S^{-1}S^*$, so choose θ for which H_θ is invertible. Then $A = H_\theta A^* H_\theta^{-1}$.

(iii) $\Rightarrow$ *(iv)* Let R be invertible Hermitian and satisfy $A = RA^*R^{-1}$. Then $AR = RA^*$ and $R^{-1}A = A^*R^{-1}$. Thus A^*R^{-1} and RA^* are Hermitian, and

$$A = R(A^*R^{-1}) = (RA^*)R^{-1}.$$

This gives the two asserted Hermitian factorizations, with the first factor in the first product and the second factor in the second product invertible.

(iv) $\Rightarrow$ *(v)* Immediate.

(v) $\Rightarrow$ *(i)* If $A = HK$ with H, K Hermitian, then Corollary 2.4.74 gives $HK \sim KH$. By Theorem 2.3.87,

$$KH \sim (KH)^t = H^t K^t = \overline{H}\,\overline{K} = \overline{A}.$$

Hence $A \sim \overline{A}$.

Structural Commentary: The equivalences move between conjugation, adjoints, and Hermitian products by combining transpose similarity with a Hermitianized similarity matrix.
Key Insight: A phase-rotated average $e^{i\theta}S + e^{-i\theta}S^*$ turns an arbitrary similarity to A^* into a Hermitian one. □

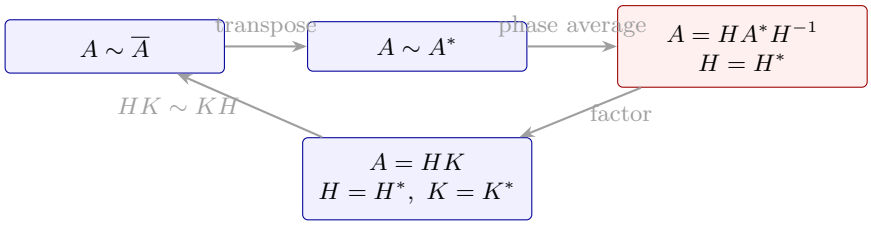


Figure 51: Conjugate similarity, adjoint similarity, and Hermitian product factorisations are equivalent.

Proposition 2.4.76. *Block Centrosymmetry and Commutation*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \in \mathrm{Mat}(2n, 2n, \mathbb{F})$, $L_{2n} = \begin{bmatrix} 0 & I_n \\ I_n & 0 \end{bmatrix}$.
Then M has block centrosymmetric form $A = D$ and $B = C$ if and only if $L_{2n}M = ML_{2n}$.

Proof. Block multiplication gives $L_{2n}M = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$ and $ML_{2n} = \begin{bmatrix} B & A \\ D & C \end{bmatrix}$.
The two matrices are equal exactly when $A = D$ and $B = C$.
Structural Commentary: The matrix L_{2n} swaps the two block coordinates. Commuting with that swap means M treats the two block positions symmetrically.
Key Insight: Compute both block products; equality forces matching opposite blocks. □

Theorem 2.4.77. *Frobenius Rank Inequality*
Let $A \in \mathrm{Mat}(m, k, \mathbb{F})$, $B \in \mathrm{Mat}(k, p, \mathbb{F})$, and $C \in \mathrm{Mat}(p, n, \mathbb{F})$. $\mathrm{rank}(AB) + \mathrm{rank}(BC) \leq \mathrm{rank}(B) + \mathrm{rank}(ABC)$.

Proof. Let $R = \text{col}(B)$ and $S = \text{col}(BC)$. Then $S \subseteq R$. The induced map T_A maps R onto $\text{col}(AB)$ and maps S onto $\text{col}(ABC)$. Rank-nullity on the restrictions $T_A|_R$ and $T_A|_S$ gives

$$\begin{aligned}\text{rank}(AB) &= \dim R - \dim(R \cap \ker A),\\ \text{rank}(ABC) &= \dim S - \dim(S \cap \ker A).\end{aligned}$$

Since $S \cap \ker A \subseteq R \cap \ker A$, we have $\dim(S \cap \ker A) \leq \dim(R \cap \ker A)$.
Substituting $\dim R = \text{rank}(B)$ and $\dim S = \text{rank}(BC)$ gives the stated inequality.
Structural Commentary: Frobenius tracks A on nested column spaces.
Key Insight: The smaller column space cannot lose more dimensions to $\ker A$ than the larger one. □

Corollary 2.4.78. *Rank Inequalities from Frobenius*
The following are special cases of Theorem 2.4.77:

(i) $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$;
(ii) $\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$.

Proof.

(i) Apply Frobenius to generic factors X, Y, Z. Taking $Z = 0$ gives $\text{rank}(XY) \leq \text{rank}(Y)$, and taking $X = 0$ gives $\text{rank}(YZ) \leq \text{rank}(Y)$. Renaming the factors gives $\text{rank}(AB) \leq \text{rank}(B)$ and $\text{rank}(AB) \leq \text{rank}(A)$.
(ii) Taking the middle matrix to be I_k gives $\text{rank}(A)+\text{rank}(B) \leq k+\text{rank}(AB)$, which is Sylvester's inequality.

Structural Commentary: Frobenius is the three-factor parent inequality. The usual upper and lower product-rank bounds appear by making one factor zero or identity.
Key Insight: Zero factors isolate upper bounds; an identity middle factor yields the Sylvester lower bound. □

Proposition 2.4.79. *Sylvester's Law of Nullity*
For a matrix X, let $\text{nullity}(X) = \dim(\ker X)$*; this is the dimension of its nullspace.*

(i) If $A \in \text{Mat}(m, k, \mathbb{F})$ and $B \in \text{Mat}(k, n, \mathbb{F})$, then $\text{nullity}(AB) \leq \text{nullity}(A) + \text{nullity}(B)$
(ii) If $A, B \in \text{Mat}(n, n, \mathbb{F})$, then $\max(\text{nullity}(A), \text{nullity}(B)) \leq \text{nullity}(AB)$

Proof.

(i) By rank-nullity and Theorem 2.1.97,

$$\begin{aligned}\text{nullity}(AB) &= n - \text{rank}(AB) \leq n - \text{rank}(A) - \text{rank}(B) + k\\ &= (k - \text{rank}(A)) + (n - \text{rank}(B)) = \text{nullity}(A) + \text{nullity}(B).\end{aligned}$$

(ii) For square A, B, $\text{rank}(AB) \leq \text{rank}(A)$, $\text{rank}(AB) \leq \text{rank}(B)$. Subtract from n gives desired lower bounds.

Structural Commentary: Nullity is rank loss from the domain.
Key Insight: Use rank-nullity to translate rank bounds into kernel bounds. □

3 Time Series Analysis

3.1 Time Series Analysis

Based on the books by James Douglas Hamilton (1994), and ...

3.1.1 Stationary Time Series

Definition 3.1.1. A *linear first-order difference equation* is defined as

$$y_t = \phi y_{t-1} + w_t$$

where y_t is the target variable (with y_{t-1} the lag 1 variable), w_t is an input variable at time t

The difference equation may be solved by recursive substitution to arrive at

$$y_t = \phi^{t+1} y_{-1} + \phi^t w_0 + \phi^{t-1} w_1 + \phi^{t-2} w_2 + \cdots + \phi w_{t-1} + w_t$$

The *dynamic multiplier* calculates effect of w_t on y_{t+j}, and is given by

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^j$$

If $|\phi| < 1$, the system is stable. If $|\phi| > 1$, then the system is explosive.

We may generalise the process to *p-th order difference equation*, i.e.,

$$y_t = \phi_t y_{t-1} + \phi_2 t_{t-2} + \cdots + \phi_p y_{t-p} + w_t$$

We may rewrite this as a first-order difference equation in a $(p \times 1)$ vector $\boldsymbol{\xi}_t$:

$$\boldsymbol{\xi}_t = \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}$$

Define the $(p \times p)$ matrix $\boldsymbol{F}$ by

$$\boldsymbol{F} = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

Finally, define the $(p \times 1)$ vector $\boldsymbol{v}_t$ by

$$\boldsymbol{v}_t = \begin{bmatrix} w_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Then the system of p equations $\boldsymbol{\xi}_t = \boldsymbol{F}\boldsymbol{\xi}_{t-1} + \boldsymbol{v}_t$ is identical to the first order difference equation, and by recursive substitution, we have the following for the case of p-th order difference equation:

$$\boldsymbol{\xi}_{t+j} = \boldsymbol{F}^{j+1}\boldsymbol{\xi}_{t-1} + \boldsymbol{F}^j \boldsymbol{v}_t + \boldsymbol{F}^{j-1}\boldsymbol{v}_{t+1} + \cdots + \boldsymbol{F}\boldsymbol{v}_{t+j-1} + \boldsymbol{v}_{t+j}$$

Hence the dynamic multiplier is then $\frac{\partial y_{t+j}}{\partial w_t} f_{11}^{(j)}$, where $f_{11}^{(j)}$ is the $(1,1)$ element of $\boldsymbol{F}^j$.

3.1.2 Univariate Time Series Models

Trading activities through an exchange can be described by a sequence of time stamps ('ticks') $t_0 < t_1 < \cdots < t_n$, and 'marks' y_i at time t_i, where t_i denote market open, t_n denote market close. The marks y_i is a characteristic of the order book at time of ith activity. Events with marks associated with the ticks can be described mathematically as a marked point process.

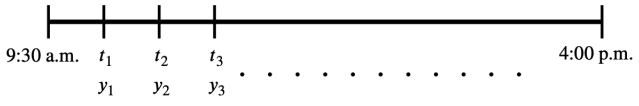


Figure 52: Ticks and Marks

A data aggregation method is where aggregation is conducted when there is a change in marker. An alternative aggregation method would be to divide the time span T for exchange hours into K intervals, so regularly spaced intervals are of size $\Delta t = T/K$. The time series method to be discussed will apply to all aggregated data.

Let $p_{it} = \ln P_{it}$ denote price of ith asset at time t; $p_t = (p_{1t}, p_{2t}, \ldots, p_{nt})$ denote price vector for n assets; y_{it} denote vector of characteristics of ith asset at time t. These quantities are aggregated from high frequency data. Consider r factors $f_t = (f_{1t}, f_{2t}, \ldots, f_{rt})$ that may include market and industry factors, and asset characteristics. Trading rules can be broadly grouped as follows:

i. *Statistical Arbitrage*: $E(p_{i,t+1} \mid p_{i,t}, p_{i,t-1}, \ldots, y_{i,t}, y_{i,t-1}, \ldots)$, that predicts the price of ith stock at $t+1$ based on past trading information (also known as time series momentum)
ii. *Momentum*: $E(p_{t+1} \mid p_t, p_{t-1}, \ldots, y_t, y_{t-1}, \ldots)$, that predicts the cross-sectional momentum of a subset of stocks based on past trading characteristics. For portfolio formation and rebalancing, pairs trading
iii. *Fair Value*: $E(p_{t+1} \mid p_t, p_{t-1}, \ldots, y_t, y_{t-1}, \ldots, f_t, f_{t-1}, \ldots)$, predicts price using all relevant quantities. Factors normally include market, Fama-French; at a more macro level than timescale considered for price prediction, but may still be useful.

Hence price and volatility prediction can be formulated as a time series prediction problem. Autocorrelations and partial autocorrelations can be used to build autoregressive and ARCH models with some predictive power.

Let $Y_1, Y_2, \ldots, Y_T$ be a sequence of random variables with a joint probability distribution. A sequence of observations of stochastic process $\{Y_t, t = 1, \ldots, T\}$ is a realisation of the process.
A time series $\{Y_t\}$ is *stationary* if for every integer m, the set of variables $Y_{t_1}, Y_{t_2}, \ldots, Y_{t_m}$ depends only on the distance between times $t_1, t_2, \ldots, t_m$. Thus $E(Y_t) = \mu$, $Var(Y_t) = \sigma^2$ are constant for all t.
The *auto-covariance* function is defined as

$$\gamma(s) = Cov(Y_t, Y_{t-s}) = E[(Y_t - \mu)(Y_{t-s} - \mu)] \quad \forall s = 0, \pm 1, \ldots$$

The *auto-correlation* of process at lag s is defined as

$$\rho(s) = Corr(Y_t, Y_{t-s}) = \frac{Cov(Y_t, Y_{t-s})}{(Var(Y_t)\ Var(Y_{t-s}))^{1/2}} = \frac{\gamma(s)}{\gamma(0)}, \quad \forall s = 0, \pm 1, \ldots$$

Some examples of stationary stochastic processes are as follows:

Example 3.1.2. *(White Noise)* Sequence of discrete independent random variables (r.v.) $\{\epsilon_t\}$ with $E[\epsilon_t] = 0$ and $E[\epsilon_t^2] = \sigma^2$. Set $Y_t = \mu + \epsilon_t$, then $E[Y_t] = \mu$, $Cov(Y_t, Y_{t-s}) = Var(Y_t) = \sigma^2$ if $s = 0$, and $Cov(Y_t, Y_{t-s}) = 0$ if $s \neq 0$. Hence the process is stationary.

Example 3.1.3. *(Moving Average)* Let $\{\epsilon_t\}$ be independent r.v., with process $\{Y_t\}$ where $Y_t = \mu + \epsilon_t + \epsilon_{t_1}$ for $t = 0, 1, 2, \ldots$, and μ is constant. Then $E[Y_t] = \mu\ \forall t$, and

$$Cov(Y_t, Y_{t-s}) = \gamma(s) = \begin{cases} 2\sigma^2 & \text{if } s = 0 \\ \sigma^2 & \text{if } s = 1 \\ 0 & \text{if } s > 1 \end{cases}$$

Hence $\{Y_t\}$ is stationary, with $\rho(s) = \gamma(s)/\gamma(0)$ such that $\rho(0) = 1$, $\rho(1) = 1/2$, $\rho(s) = 0$ for $|s| > 1$.

Some examples of non-stationary stochastic processes are as follows:

Example 3.1.4. *(Random Walk with Drift)* Let $\{\epsilon_t\}$ be sequence of independent r.v. with $E[\epsilon_t] = 0$, $E[\epsilon_t^2] = \sigma^2$, and define process $\{Y_t\}$ by $Y_t = Y_{t-1} + \delta + \epsilon_t$ with $Y_0 = 0$. Then the process can be summarised as

$$Y_t = \delta t + \sum_{j=1}^{t} \epsilon_j$$

Note that $E[Y_t] = \delta t$ and $Var[Y_t] = t\sigma^2$, hence process $\{Y_t\}$ is not stationary.

Given T observations from stationary process $\{Y_t\}$, the *sample mean* is $\overline{Y} = \frac{1}{T}\sum_{t=1}^{T} Y_t$. The *sample auto-covariance* function is defined by $\hat{\gamma}(s) = \frac{1}{T}\sum_{t=1}^{T-s}(Y_t - \overline{Y})(Y_{T-s} - \overline{Y})$ for $s = 0, 1, \ldots$. The *sample auto-correlation* function (ACF) is defined as $\hat{\rho}(s) = \frac{\hat{\gamma}(s)}{\hat{\gamma}(0)} = r(s)$. The *sample variance* is $Var(\overline{Y}) = \frac{\gamma(0)}{T}\left[1 + 2\sum_{u=1}^{Y-1}\left(\frac{T-u}{T}\right)\rho(u)\right]$; note that it has to account for auto-correlations.

Definition 3.1.5. A stochastic process $\{Y_t\}$ is a *linear process* if it can be represented as

$$Y_t = \mu + \sum_{j=0}^{\infty} \Psi_j \epsilon_{t-j}$$

where ϵ_t are independent with mean 0, variance σ_ϵ^2, and $\sum_{j=0}^{\infty} < \infty$.

4 Mathematical and Statistical Foundation

4.1 Classical Machine Learning

4.1.1 Ensemble Methods

4.2 Deep Learning

4.2.1 Deep Feedforward Networks

Remark 4.2.1. *Deep Feedforward Networks (DFNs)*
Defines a mapping $\boldsymbol{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$ and learns value of parameters $\boldsymbol{\theta}$ that results in best function approximation.

Remark 4.2.2. *Linear Models*
Linear models may fit data efficiently and reliably, either in closed form or with convex optimisation.
Model capacity limited to linear functions, does not model interaction between any two input variables

Remark 4.2.3. *Nonlinear Models*
To represent nonlinear functions of $\boldsymbol{x}$, apply linear model to transformed input $\phi(\boldsymbol{x})$ where ϕ is a nonlinear transformation. Kernel trick may be applied to obtain nonlinear learning algorithm. To choose mapping ϕ:

i. Choose generic ϕ, such as that used by kernel machines based on RBF kernel.
If $\phi(\boldsymbol{x})$ is of high enough dimension, can find enough capacity to fit training set, but generalisations to test set remains poor. Mappings are based on principle of local smoothness and do not encode enough prior information to solve advanced problems.
ii. Manually engineer ϕ. Requires decades of human effort for each task, and practitioners specialising in different domains (i.e, speech recognition, computer vision) with little transfer between domains.
iii. Learn ϕ through deep learning. Model is $y = f(\boldsymbol{x}; \boldsymbol{\theta}, \boldsymbol{w}) = \phi(\boldsymbol{x}; \boldsymbol{\theta})^T \boldsymbol{w}$, where parameters $\boldsymbol{\theta}$ can be used to learn ϕ from broad class of functions, and parameters $\boldsymbol{w}$ that map from $\phi(\boldsymbol{x})$ to desired output.
Do not require training problem to be convex, and only require finding the right general function.

Definition 4.2.4. *Cost Functions*

i. Learning Conditional Distributions with Maximum Likelihood: cost function is negative log-likelihood, which is the cross-entropy between training data and model distribution:
$$J(\boldsymbol{\theta}) = -\mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim\hat{p}_{\text{data}}} \log p_{\text{model}}(\boldsymbol{y} \mid \boldsymbol{x})$$
Specific form of cost function changes from model to model, depending on form of $\log p_{\text{model}}$.
Method removes the burden of designing cost functions for each model, as specifying a model $p(\boldsymbol{y} \mid \boldsymbol{x})$ automatically determines a cost function $\log p(\boldsymbol{y} \mid \boldsymbol{x})$.
Gradient of cost function for neural networks must be large and predictable enough to serve as good guide for the learning algorithm. The negative log-likelihood helps to avoid saturation of the function.
ii. Learning Conditional Statistics: to learn just one conditional statistic of $\boldsymbol{y}$ given $\boldsymbol{x}$.
With sufficiently powerful neural network, this can represent any function f from a wide class of function, limited by features of continuity and boundedness. Hence the cost function is a functional. Learning is choosing a functional rather than a set of parameters.
By calculus of variation , solving the optimisation problem yields the below function,
$$f^* = \arg\min_f \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim p_{\text{data}}} \|\boldsymbol{y} - f(\boldsymbol{x})\|^2$$
$$f^* = \mathbb{E}_{\boldsymbol{y}\sim p_{\text{data}}(\boldsymbol{y} \mid \boldsymbol{x})}[\boldsymbol{y}]$$
If infinitely many samples from the true data-generating distribution could be trained, then minimising the mean squared error cost function gives a function that predicts mean of $\boldsymbol{y}$ for each value of $\boldsymbol{x}$.
A second result from calculus of variations is:
$$f^* = \arg\min_f \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim p_{\text{data}}} \|\boldsymbol{y} - f(\boldsymbol{x})\|_1$$
which is a function that predicts the median value of $\boldsymbol{y}$ for each $\boldsymbol{x}$. This is the mean absolute error.
Note that mean squared error and mean absolute error often lead to poor results when used with gradient-based optimisation. Output units that saturate may produce very small gradients when combined with these cost functions. Hence the reason that cross-entropy cost function is more popular.

Definition 4.2.5. *Output Units*

i. Linear Units: base on affine transformation with no nonlinearity.
Given features $\boldsymbol{h}$, a layer of output units produces vector $\hat{\boldsymbol{y}} = \boldsymbol{W}^T \boldsymbol{h} + \boldsymbol{b}$. Linear output layers are used to produce mean of conditional Gaussian distribution:
$$p(\boldsymbol{y} \mid \boldsymbol{x}) = \mathcal{N}(\boldsymbol{y}; \hat{\boldsymbol{y}}, \boldsymbol{I})$$
Maximising log-likelihood is equivalent to minimising mean squared error.
Linear units do not saturate, hence may be used for wide variety of optimisation algorithms.
ii. Sigmoid Units: define Bernoulli distribution y conditioned on $\boldsymbol{x}$. Neural net to predict $P(y = 1 \mid \boldsymbol{x})$, which lies in interval $[0, 1]$. The sigmoid unit is defined by
$$\hat{y} = \sigma(\boldsymbol{w}^T \boldsymbol{h} + b)$$

where σ is the logistic sigmoid function.
Note that the cost function used with maximum likelihood is $-\log P(y \mid \boldsymbol{x})$, preventing saturation.
The loss function for MLE of Bernoulli parametrised by sigmoid is

$$J(\boldsymbol{\theta}) = -\log P(y \mid \boldsymbol{x}) = -\log \sigma((2y-1)z) = \zeta((1-2y)z)$$

Function saturates only when $(1-2y)z$ is very negative, i.e., when model has the right answer.

iii. Softmax Units: used to represent probability distribution over n different classes.
A linear layer predicts unnormalised log probabilities:

$$\boldsymbol{z} = \boldsymbol{W}^T\boldsymbol{h} + \boldsymbol{b}$$

where $z_i = \log \tilde{P}(y = i \mid \boldsymbol{x})$. Softmax function is then

$$\text{softmax}(\boldsymbol{z})_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

The function is to maximise $\log P(y = i; \boldsymbol{z}) = \log \text{softmax}(\boldsymbol{z})_i = z_i - \log \sum_j \exp(z_j)$.
Note that the input z_i always has direct contribution to cost function, and the term cannot saturate.
Un-regularised maximum likelihood will drive the model to learn parameters that drive the softmax to predict fraction of counts for each outcome observed in the training set:

$$\text{softmax}(z(\boldsymbol{x};\boldsymbol{\theta}))_i \approx \frac{\sum_{j=1}^{m} \mathbf{1}_{y^{(j)}=i, \boldsymbol{x}^{(j)}=x}}{\sum_{j=1}^{m} \mathbf{1}_{\boldsymbol{x}^{(j)}=x}}$$

Note that objective functions other than log-likelihood does not work well with softmax function. Squared error is poor loss function for softmax units, and can fail to train the model to change its output.
Softmax function can saturate, and many functions based on softmax also saturate, unless they are able to invert the saturating activating function.

iv. Other Output Types: generally, given a conditional distribution $p(\boldsymbol{y} \mid \boldsymbol{x};\ \boldsymbol{\theta})$, principle of maximum likelihood suggests using $-\log p(\boldsymbol{y} \mid \boldsymbol{x};\ \boldsymbol{\theta})$ as the cost function.
Neural networks represent a function $f(\boldsymbol{x};\boldsymbol{\theta}) = \boldsymbol{\omega}$. The outputs of the function are not direct predictions of value $\boldsymbol{y}$, but the parameters for distribution over $\boldsymbol{y}$. The loss function is then $-\log p(\boldsymbol{y};\boldsymbol{\omega}(\boldsymbol{x}))$.

Remark 4.2.6. *Learning Distribution Parameters*

i. Heteroscedastic Model: to predict different variance in $\boldsymbol{y}$ for different values of $\boldsymbol{x}$, formulate the Gaussian distribution using precision rather than variance. In multi-variate case, the diagonal precision matrix is used, $\text{diag}(\boldsymbol{\beta})$. The log-likelihood of Gaussian distribution parametrised by $\boldsymbol{\beta}$ involves only multiplication by β_i and addition of $\log \boldsymbol{\beta}_i$. The gradient of these operations are well-behaved.
Let $\boldsymbol{\alpha}$ be raw activation of the model used to determine diagonal precision. The softplus function may be used to obtain a positive precision vector $\boldsymbol{\beta} = \zeta(\boldsymbol{\alpha})$. Same strategy applies equally if using variance variance or standard deviation rather than precision.
If covariance is full and conditional, then parametrisation must be chosen that guarantees positive-definiteness of predicted covariance matrix.

$$\boldsymbol{\Sigma}(\boldsymbol{x}) = \boldsymbol{B}(\boldsymbol{x})\boldsymbol{B}^T(\boldsymbol{x})$$

where $\boldsymbol{B}$ is unconstrained square matrix. Note that if matrix is full rank, then computing likelihood requires $O(d^3)$ a $d \times d$ matrix for the determinant and inverse of $\boldsymbol{\Sigma}(\boldsymbol{x})$.

ii. Mixture Density Networks: to perform multimodal regression (predict real values that come from conditional distribution $p(\boldsymbol{y} \mid \boldsymbol{x})$) that can have several different peaks in $\boldsymbol{y}$ for the same $\boldsymbol{x}$.

$$p(\boldsymbol{y} \mid \boldsymbol{x}) = \sum_{i=1}^{n} p(c = i \mid \boldsymbol{x}) \mathcal{N}(\boldsymbol{y}; \boldsymbol{\mu}^{(i)}(\boldsymbol{x}), \boldsymbol{\Sigma}^{(i)}(\boldsymbol{x}))$$

The neural network will have three outputs:

1. Mixture components $p(c = i \mid \boldsymbol{x})$, forming a multinoulli distribution over n different components with latent variable c, obtained by softmax over n-dimensional vector.
2. Means $\boldsymbol{\mu}^{(i)}(\boldsymbol{x})$
3. Covariances $\boldsymbol{\Sigma}^{(i)}(\boldsymbol{x})$

Gradient-based optimisation of conditional Gaussian mixtures can be unreliable as the divisions can be numerically unstable. May be solved by clipping gradients, or scaling gradients heuristically.

Definition 4.2.7. *Hidden Units*
Even if hidden units are not differentiable at all input points, gradient descent still performs well enough as the training algorithms do not usually arrive at local minimum of cost function, but reduce its value significantly.

Most hidden units has input vector $\boldsymbol{x}$, computes an affine transformation $\boldsymbol{z} = \boldsymbol{W}^T\boldsymbol{x} + \boldsymbol{b}$, then applying element-wise non-linear function $g(\boldsymbol{z})$.

i. Rectified Linear Units (ReLUs): uses activation function $g(z) = \max\{0, z\}$.
Typically used on top of an affine transformation $\boldsymbol{h} = g(\boldsymbol{W}^T\boldsymbol{x} + \boldsymbol{b})$. In initialisation, set all elements of $\boldsymbol{b}$ to small positive values, so that ReLUs will be initially active for most inputs in training set.
Generalisations of ReLUs have non-zero slope α_i for $z_i < 0$: $h_i = \max(0, z_i) + \alpha_i \min(0, z_i)$.
   1. Absolute Value Rectification: sets $\alpha_i = -1$ to obtain $g(z) = |z|$. Used for object recognition from images, to seek features that are invariant under polarity reversal of input illumination.
   2. Leaky ReLU: fixes α_i to small positive value
   3. Parametric ReU: treats α_i as a learnable parameter

ii. Maxout Units: divide $\boldsymbol{z}$ into groups of k values. Each maxout unit then outputs maximum element of one of these groups: $g(\boldsymbol{z})_i = \max_{j \in \mathbb{G}^{(i)}} z_j$, where $\mathbb{G}^{(i)}$ is indices of inputs for group i, which is $\{(i-1)k+1, \ldots, ik\}$.
This allows learning of piecewise linear function that responds to multiple directions in input $\boldsymbol{x}$ space .
Maxout units learn the activation function itself. With large k, maxout unit can learn to approximate any convex function. Each maxout unit is parametrised by k weight vectors, hence need more regularisation than ReLUs. Benefits include:
   1. Can work well without regularisation if training set is large and number of pieces per unit is low.
   2. Can gain statistical and computation advantages by requiring fewer parameters.
   3. Have redundancy that resists catastrophic forgetting, where neural networks forgot how to perform tasks that were trained on in the past.

iii. Logistic Sigmoid and Hyperbolic Tangent: the logistic sigmoid activation function is $g(z) = \sigma(z)$, and the hyperbolic tangent activation function is $g(z) = \tanh(z)$. Note that $\tanh(z) = 2\sigma(2z) - 1$.
Sigmoidal units saturate across most of the domain, which makes gradient-based learning very difficult. Hence the use in hidden units in feedforward networks is now discouraged.
If sigmoidal activation function must be used, hyperbolic tangent activation function performs better, as it resembles identity function more closely. Training $\hat{y} = \boldsymbol{w}^T \tanh(\boldsymbol{U}^T \tanh(\boldsymbol{V}^T\boldsymbol{x}))$ resembles training a linear model $\hat{y} = \boldsymbol{w}^T\boldsymbol{U}^T\boldsymbol{V}^T\boldsymbol{x}$ as long as the activations of the network can be kept small.
Sigmoidal functions are more common in recurrent networks, probabilities models, auto-encoders

iv. Linear Unit: consider neural network layer with n inputs, p outputs, $\boldsymbol{h} = g(\boldsymbol{W}^T\boldsymbol{x} + \boldsymbol{b})$. Replace with two layers, one using weight matrix $\boldsymbol{U}$ and the other using weight matrix $\boldsymbol{V}$. If first layer has no activation function, then the factored approach is to compute $\boldsymbol{h} = g(\boldsymbol{V}^T\boldsymbol{U}^T\boldsymbol{x} + \boldsymbol{b})$. If $\boldsymbol{U}$ produces q outputs, then $\boldsymbol{U}$ and $\boldsymbol{V}$ together only contains $(n+p)q$ parameters, while $\boldsymbol{W}$ contains np parameters. For small q, this is considerable saving in parameters, while the cost is constraining linear transformation to be low rank. This is an efficient way of reducing number of parameters in the model.

v. Softmax: naturally represent probability distribution over discrete variable with k possible values. Used only in more advanced architectures that explicitly learn to manipulate memory.

vi. Radial Basis Function (RBF): function becomes more active as $\boldsymbol{x}$ approaches a template $\boldsymbol{W}_{:,i}$. As it saturates to 0 for most $\boldsymbol{x}$, it can be difficult to optimise.

$$h_t = \exp\left(-\frac{1}{\sigma_i^2}\|\boldsymbol{W}_{:,i} - \boldsymbol{x}\|^2\right)$$

vii. Softplus: smooth version of rectifier for functional approximation and for conditional distributions of undirected probabilistic models. Usage is generally discouraged.

$$g(a) = \zeta(a) = \log(1 + e^a)$$

viii. Hard tanh: shaped similarly to tanh but bounded.

$$g(a) = \max(-1, \min(1, a))$$

Theorem 4.2.8. *Universal Approximation Theorem*
A feedforward network with linear output layer and at least one hidden layer with any 'squashing' activation function can approximate any Borel measurable function from one finite-dimensional space to another with any desired non-zero amount of error, provided the network is given enough hidden units.

Method 4.2.9. *Architecture Design*
Neural network layers are arranged in a chain structure, with each layer being a function of preceding layer.

$$\boldsymbol{h}^{(1)} = g^{(1)}(\boldsymbol{W}^{(1)T}\boldsymbol{x} + \boldsymbol{b}^{(1)})$$
$$\boldsymbol{h}^{(k)} = g^{(k)}(\boldsymbol{W}^{(k)T}\boldsymbol{h}^{(k-1)} + \boldsymbol{b}^{(k)}), \quad k \geq 2$$

The main considerations are the depth of network and width of each layer. Deeper networks use far fewer units per layers and far fewer parameters, and often generalise to the test set, but are harder to optimise.

Remark 4.2.10. *Depth of Network and Universal Approximation Theorem*
A feedforward network with single layer is sufficient to represent any function, but the layer may be infeasibly large and fail to learn and generalise correctly. Using deeper models can reduce number of units required to represent the desired function and can reduce generalisation error.
Shallow networks with broad family of non-polynomial activation functions have universal approximation properties. Piecewise linear networks can represent functions with number of regions that is exponential to depth. The number of linear regions carved out by deep rectifier network with d inputs, depth l, and n units per hidden layer is $O(\binom{n}{d}^{d(d-1)} n^d)$. For maxout networks with k filters per unit, this is $O(k^{(k-1)+d})$.

Remark 4.2.11. *Backpropagation*
To calculate the gradient of loss function with respect to each of individual parameters of the neural network. Model training begins with random initialisation of weights and biases.

i. Forward pass: input is sampled from training data. Nodes receive input vector and passes their value (multiplied by random initial weight) to nodes of first hidden layer. The hidden units take weighted sum of these output values as an input to the activation function, whose output is used for next hidden layer.
ii. Error computation: the final output of the network is compared to the ground truth, difference is calculated for the error value.
iii. Backwards pass: the error value computed earlier is used to compute the gradient of loss function. The gradient is then propagated back through the network, and the weights are updated according to their contribution to the error. The learning rate determines the size of weight updates.
iv. Weights update: the weights are updated in opposite direction of the gradient

Algorithm 1: Backpropagation Learning Algorithm

Require:
A set of training examples $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$
A multilayer network with L layers, weights w_{ij}^l, and activation function f
Loss function $J(y, o)$
Learning rate $0 < \alpha < 1$
Number of epochs *epochs*.

Initialize all weights:
for w_{ij}^l in the network **do**
  $w_{ij}^l \leftarrow$ small random number
end for

for $i = 1$ to *epochs* **do**
  for all training examples $(x, y) \in \mathcal{D}$ **do**

    Propagate the inputs forward to compute the outputs
    for all neurons i in the input layer **do**
      $a_i^0 \leftarrow x_i$
    end for

    for $l = 2$ to L **do**
      for all neurons i in layer l **do**
        $z_i^l \leftarrow \sum_j w_{ij}^l \, a_j^{l-1}$
        $a_i^l \leftarrow f(z_i^l)$
      end for
    end for

    Propagate deltas backward from output layer to input layer
    for all neurone i in the output layer **do**
      Let $o_i \equiv a_i^L$ *be output of neurone* i
      $\delta_i^L \leftarrow \dfrac{\partial J(y_i, o_i)}{\partial o_i} f'(z_i^L)$
    end for

    for $l = L - 1$ to 1 **do**
      for all neurons i in layer l **do**
        $\delta_i^l \leftarrow f'(z_i^l) \sum_j (w_{ji}^{l+1} \, \delta_j^{l+1})$

```
        end for
    end for

    Update weights using the deltas
    for all w_ij^l in the network do
        w_ij^l ← w_ij^l − α δ_i^l a_j^{l−1}
    end for
  end for
end for
```

Example 4.2.12. *Backpropagation Computation Example (Matt Mazur (2015))*

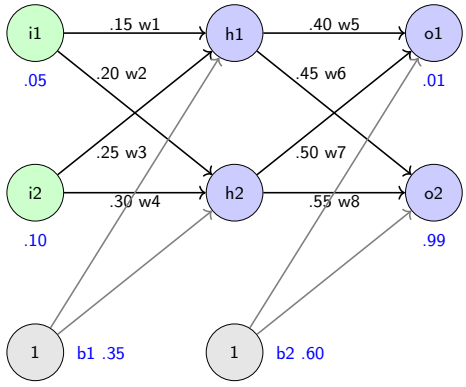


Figure 53: Backpropagation Example

For the forward pass, compute the total net input to each hidden layer neurone, squash total net input with activation function (logistic function), then repeat process with output layer neurones.
The total net input for h_1 is:

$$\begin{aligned}\text{net}_{h1} &= w_1 \cdot i_1 + w_2 \cdot i_2 + b_1 \cdot 1 \\ &= 0.15 \cdot 0.05 + 0.2 \cdot 0.1 + 0.35 \cdot 1 = 0.3775\end{aligned}$$

Squash with logistic function to get output of h_2:

$$\text{out}_{h1} = 1/(1 + e^{-\text{net}_{h1}}) = 1/(1 + e^{-0.3775}) = 0.593269992$$

For h_2, repeating the same process, the output is then $\text{out}_{h2} = 0.596884378$.
Repeating the process for outer layer neurones,

$$\begin{aligned}\text{net}_{o1} &= w_5 \cdot \text{out}_{h1} + w_6 \cdot \text{out}_{h2} + b_2 \cdot 1 \\ &= 0.4 \cdot 0.593269992 + 0.45 \cdot 0.596884378 + 0.6 \cdot 1 = 1.105905967 \\ \text{out}_{o1} &= 1/(1 + e^{-\text{net}_{o1}}) = 1/(1 + e^{-1.105905967}) = 0.75136507\end{aligned}$$

For o_2, repeating the same process, the output is then $\text{out}_{o2} = 0.772928465$.
The total error for each output neurone may be computed using squared error function $E_{\text{total}} = \sum \frac{1}{2}(Y - \hat{Y})^2$.
For o_1, the target output is 0.01, neurone output is 0.75136507, hence error is

$$E_{o1} = \frac{1}{2}(0.01 - 0.75136507)^2 = 0.274811083$$

Repeating the process for o_2, the total error for neural network is then

$$E_{\text{total}} = E_{o1} + E_{o2} = 0.274811083 + 0.023560026 = 0.298371109$$

For the backpropagation step, update each of the weights in network to reduce the error for each output neurone.

$$\begin{aligned}E_{\text{total}} &= \frac{1}{2}(Y_{o1} - \hat{Y}_{o1})^2 + \frac{1}{2}(Y_{o2} - \hat{Y}_{o2})^2 \\ \frac{\partial E_{\text{total}}}{\partial \text{out}_{o1}} &= 2 \cdot \frac{1}{2}(Y_{o1} - \hat{Y}_{o1}) \cdot -1 + 0 = -(0.01 - 0.75136507) = 0.74136507\end{aligned}$$

Also, note that

$$\begin{aligned}
\text{out}_{o1} &= 1/(1+e^{-\text{net}_{o1}}) \\
\frac{\partial \text{out}_{o1}}{\partial \text{net}_{o1}} &= \text{out}_{o1}(1-\text{out}_{o1}) = 0.75136507(1-0.75136507) = 0.186815602 \\
\text{net}_{o1} &= w_5 \cdot \text{out}_{h1} + w_6 \cdot \text{out}_{h2} + b_2 \cdot 1 \\
\frac{\text{net}_{o1}}{w_5} &= 1 \cdot \text{out}_{h1} \cdot w_5^0 + 0 + 0 = \text{out}_{h1} = 0.593269992
\end{aligned}$$

Putting it all together,

$$\begin{aligned}
\frac{\partial E_{\text{total}}}{\partial w_5} &= \frac{\partial E_{\text{total}}}{\partial \text{out}_{o1}} \cdot \frac{\partial \text{out}_{o1}}{\partial \text{net}_{o1}} \cdot \frac{\partial \text{out}_{o1}}{\partial w_5} \\
&= -(Y_{o1} - \hat{Y}_{o1}) \cdot \text{out}_{o1}(1-\text{out}_{o1}) \cdot textout_{o1} \\
&= 0.74136507 \cdot 0.186815602 \cdot 0.593269992 = 0.0821267041
\end{aligned}$$

To decrease the error, subtract the value from current weight multiplied by learning rate (i.e., $\alpha = 0.5$).

$$w_5^+ = w_5 - \alpha \frac{\partial E_{\text{total}}}{\partial w_5} = 0.4 - 0.5 \cdot 0.0821267041$$

Repeating for the rest of the weights, we get

$$w_6^+ = 0.408666186, \quad w_7^+ = 0.511301270, \quad w_8^+ = 0.561370121$$

For the hidden layers, note that

$$\begin{aligned}
\frac{\partial E_{\text{total}}}{\partial w_1} &= \frac{\partial E_{\text{total}}}{\partial \text{out}_{h1}} \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1} \\
&= \left(\frac{\partial E_{o1}}{\partial \text{out}_{h1}} + \frac{\partial E_{o2}}{\partial \text{out}_{h1}} \right) \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1}
\end{aligned}$$

Note that out_{h1} affects both out_{o1} and out_{o2}.

$$\begin{aligned}
\frac{\partial E_{o1}}{\partial \text{out}_{h1}} &= \frac{\partial E_{o1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial \text{out}_{h1}} \\
\frac{\partial E_{o1}}{\partial \text{net}_{h1}} &= \frac{\partial E_{o1}}{\partial \text{out}_{h1}} \cdot \frac{\partial \text{out}_{o1}}{\partial \text{net}_{h1}} = 0.74136507 \cdot 0.186815602 = 0.138498562 \\
\frac{\partial \text{net}_{h1}}{\partial \text{out}_{h1}} &= w_5 = 0.4 \\
\frac{\partial E_{o1}}{\partial \text{out}_{h1}} &= 0.138498562 \cdot 0.4 = 0.055399425
\end{aligned}$$

Following same process, $\frac{\partial E_{o2}}{\partial \text{out}_{h1}} = -0.019049119$. Hence

$$\frac{\partial E_{\text{total}}}{\partial \text{out}_{h1}} = \frac{\partial E_{o1}}{\partial \text{out}_{h1}} + \frac{\partial E_{o2}}{\partial \text{out}_{h1}} = 0.05539425 - 0.019049119 = 0.036350306$$

Next, figure out $\frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}}$ and $\frac{\partial \text{net}_{h1}}{\partial w_1}$.

$$\begin{aligned}
\frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} &= \text{out}_{h1} \cdot (1-\text{out}_{h1}) = 0.59326999(1-0.59326999) = 0.2410.241300709 \\
\frac{\partial \text{net}_{h1}}{\partial w_1} &= i_1 = 0.05
\end{aligned}$$

Putting it together,

$$\frac{\partial E_{\text{total}}}{\partial w_1} = \frac{\partial E_{\text{total}}}{\partial \text{out}_{h1}} \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1} = 0.036350306 \cdot 0.241300709 \cdot 0.05 = 0.000438658$$

This is also written as

$$\frac{\partial E_{\text{total}}}{\partial w_1} = \left(\sum_o \frac{\partial E_{\text{total}}}{\partial \text{out}_o} \cdot \frac{\partial \text{out}_o}{\partial \text{net}_o} \cdot \frac{\partial \text{net}_o}{\partial \text{out}_{h_1}} \right) \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1}$$

Next, update w_1 and rest of the weights

$$\begin{aligned}
w_1^+ &= w_1 - \alpha \frac{\partial E_{\text{total}}}{\partial w_1} = 0.15 - 0.5 \cdot 0.000438658 = 0.149780716 \\
w_2^+ &= 0.19956143, \quad w_3^+ = 0.24975114, \quad w_4^+ = 0.29950229
\end{aligned}$$

Hence all the weights has been updated. After the first round of backpropagation, total error has decreased to 0.291027924. Repeating the process $10,000$ times, the error reduces to 0.0000351085.

4.3 Regularisation for Deep Learning

Definition 4.3.1. *Regularisation* refers to adding a parameter norm penalty $\Omega(\boldsymbol{\theta})$ to the objective function J. The regularised objective function is then

$$\tilde{J}(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) = J(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) + \alpha\Omega(\boldsymbol{\theta})$$

where $\alpha in [0, \infty)$ is a hyper-parameter that weights the contribution of norm penalty term.
For neural networks, the parameter norm penalty Ω is chosen such that it penalises only the weights of the affine transformation at each layer and leaves the biases un-regularised.

Definition 4.3.2. *L^2 (Ridge) Regularisation*
The regularisation term added to objective function is

$$\Omega(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{w}\|_2^2$$

Remark 4.3.3. *Behaviour of Weight Decay (L^2) Regularisation*
Assuming no bias parameter, a model have the following objective function and parameter gradient:

$$\tilde{J}(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y}) = \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w} + J(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y})$$
$$\nabla_{\boldsymbol{w}}\tilde{J}(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y}) = \alpha\boldsymbol{w} + \nabla_{\boldsymbol{w}}J(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y})$$

On a single gradient step, the update is as follows:

$$\boldsymbol{w} \leftarrow (1 - \epsilon\alpha)\boldsymbol{w} - \epsilon\nabla_{\boldsymbol{w}}J(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y})$$

The addition of weight decay has modified learning rule to multiplicatively shrink weight vector by constant factor on each step before gradient update.
Using quadratic approximation to objective function at minimal unregularised training cost, approximation is

$$\hat{J}(\bar{\theta}) = J(\boldsymbol{w}^*) + \frac{1}{2}(\boldsymbol{w} - \boldsymbol{w}^*)\boldsymbol{H}(\boldsymbol{w} - \boldsymbol{w}^*), \qquad \boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} J(\boldsymbol{w})$$

where $\boldsymbol{H}$ is Hessian matrix of J with respect to $\boldsymbol{w}$ evaluated at $\boldsymbol{w}^*$. Minimum of $\hat{J}$ occurs where gradient is

$$\nabla_{\boldsymbol{w}}\hat{J}(\boldsymbol{w}) = \boldsymbol{H}(\boldsymbol{w} - \boldsymbol{w}^*) = \boldsymbol{0}$$

Adding weight decay gradient, solve for minimum of regularised version of $\hat{J}$. Let $\tilde{\boldsymbol{w}}$ be minimum, then

$$\alpha\tilde{\boldsymbol{w}} + \boldsymbol{H}(\tilde{\boldsymbol{w}} - \boldsymbol{w}^*) = 0$$
$$(\boldsymbol{H} + \alpha\boldsymbol{I})\tilde{\boldsymbol{w}} = \boldsymbol{H}\boldsymbol{w}^*$$
$$\tilde{\boldsymbol{w}} = (\boldsymbol{H} + \alpha\boldsymbol{I})^{-1}\boldsymbol{H}\boldsymbol{w}^*$$

As $\alpha \to 0$, regularised solution $\tilde{\boldsymbol{w}} \to \boldsymbol{w}^*$. Note $\boldsymbol{H}$ is real and symmetric, hence decompose into diagonal matrix $\boldsymbol{A}$ and orthonormal basis of eigenvectors $\boldsymbol{Q}$ such that $\boldsymbol{H} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T$, to get

$$\tilde{\boldsymbol{w}} = (\boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T + \alpha\boldsymbol{I})^{-1}\boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T\boldsymbol{w}^* = \boldsymbol{Q}(\boldsymbol{\Lambda} + \alpha\boldsymbol{I})^{-1}\boldsymbol{\Lambda}\boldsymbol{Q}^T\boldsymbol{w}^*$$

The effect of weight decay rescale $\boldsymbol{w}^*$ along axes defined by eigenvectors of $\boldsymbol{H}$. The component of $\boldsymbol{w}^*$ aligned with i-th eigenvector of $\boldsymbol{H}$ is rescaled by factor $\frac{\lambda_i}{\lambda_i + \alpha}$. For components where $\lambda_i \gg \alpha$, the effects of regularisation is relatively small. For components where $\lambda_i \ll \alpha$, components will be shrunk to nearly zero magnitude.

Definition 4.3.4. *L^1 Regularisation*
The L^1 regularisation is the sum of absolute values of individual parameters.

$$\Omega(\boldsymbol{\theta}) = \|\boldsymbol{w}\|_1 = \sum_i |w_i|$$

Definition 4.3.5. *Behaviour of L^1 Regularisation*
Assuming no bias parameter, a model has the following objective function and parameter gradient:

$$\tilde{J}(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y}) = \alpha\|\boldsymbol{w}\|_1 + J(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y})$$
$$\nabla_{\boldsymbol{w}}\tilde{J}(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y}) = \alpha\ \text{sign}(\boldsymbol{w}) + \nabla_{\boldsymbol{w}}J(\boldsymbol{w}; \boldsymbol{X}, \boldsymbol{y})$$

Note that the regularisation contribution to gradient is a constant factor with sign equal to $\text{sign}(w_i)$. Minimum of $\tilde{J}$ occurs at where

$$\nabla_{\boldsymbol{w}}\tilde{J}(\boldsymbol{w}) = \boldsymbol{H}(\boldsymbol{w} - \boldsymbol{w}^*)$$

Assuming the Hessian is diagonal, $\boldsymbol{H} = \text{diag}([H_{1,1}, \ldots, H_{n,n}])$, where each $H_{i,i} > 0$. The quadratic approxima-

tion of regularised objective function is then

$$\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y}) = J(\boldsymbol{w}^*;\boldsymbol{X},\boldsymbol{y}) + \sum_i \left[\frac{1}{2}H_{i,i}(\boldsymbol{w}_i - \boldsymbol{w}_i^*)^2 + \alpha\,|w_i|\right]$$

$$w_i = \text{sign}(w_i^*)\max\left\{|w_i^*| - \frac{\alpha}{H_{i,i}}, 0\right\}$$

For the case where $w_i^* > 0$ for all i, then

i. if $w_i^* \leq \frac{\alpha}{H_{i,i}}$, the optimal value is $w_i = 0$ as contribution of $J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$ to regularised objective $\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$ is overwhelmed in direction i by L^1 regularisation which pushes w_i to zero.
ii. if $w_i^* > \frac{\alpha}{H_{i,i}}$, regularisation shifts optimal value of w_i in the direction by $\frac{\alpha}{H_{i,i}}$.

For the case where $w_i^* < 0$, this happens similarly, but with L^1 penalty decreasing w_i by $\frac{\alpha}{H_{i,i}}$, with min value 0. Note that L^1 produces a more sparse solution, which is a feature selection mechanism.

Remark 4.3.6. *Norm Penalties as Constrained Optimisation*
Let cost function regularised by parameter norm penalty be

$$\tilde{J}(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) = J(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) + \alpha\Omega(\boldsymbol{\theta})$$

Construct a generalised generalised Lagrange function,

$$\mathcal{L}(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) = J(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) + \alpha(\Omega(\boldsymbol{\theta}) - k)$$

The solution to the constrained problem is then

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \max_{\alpha,\alpha\geq 0} \mathcal{L}(\boldsymbol{\theta},\alpha)$$

Note that optimal value α^* will shrink $\Omega(\boldsymbol{\theta})$, but not such that it is less than k.
Fixing α^*, the problem is then a function of $\boldsymbol{\theta}$,

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta},\alpha^*) = \arg\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) + \alpha^*\Omega(\boldsymbol{\theta})$$

This is the regularised training problem of minimising $\tilde{J}$. The parameter norm penalty is imposing a constraint on the weights. If explicit constraints are to be used rather than penalties, use stochastic gradient descent on $J(\boldsymbol{\theta})$, then take the projected $\boldsymbol{\theta}$ to the nearest point that satisfies $\Omega(\boldsymbol{\theta}) < k$. This method is used if k is predefined by user, and time is not to be spent on searching for α value that corresponds to this k.
Explicit constraints and re-projection work better in circumstances where non-convex optimisation is involved, and this may avoid getting stuck in local minima. Explicit constraints also impose stability, and together with high learning rate allows rapid exploration of parameter space.

Remark 4.3.7. *Dataset Augmentation*
Effective technique for object classification. Operations such as translation of images by few pixels in each direction can greatly improved generalisation, even if the model has been designed to be partially translation invariant using convolution and pooling techniques. Rotation and scaling of images are also quite effective.
For speech recognition tasks, noice injection will improve performance. To improve robustness, train models with random noise applied to inputs. Noise injection also works when applied to the hidden units (data augmentation at multiple levels of abstraction). Enabling dropout constructs new inputs by multiplying by noise.

Remark 4.3.8. *Noise Robustness*
For some models, addition of noise with infinitesimal variance at input of model is equivalent to imposing penalty on norm of the weights. Generally, noise injection can be much more powerful than simply shrinking the parameters, especially when noise is added to hidden units.
Noise may also be added to weights, which is primarily used in recurrent neural networks. This is a stochastic implementation of Bayesian inference over the weights (model weights are uncertain and representable via a probability distribution that reflects this uncertainty).
Adding noise to weights may also result in more stability in function to be learned. Let the cost function be

$$J = \mathbb{E}_{p(x,y)}[(\hat{y}(\boldsymbol{x}) - y)^2]$$

where $\hat{y}(\boldsymbol{x})$ is function to be trained that maps set of features $\boldsymbol{x}$ to a scalar using least-squares cost function J. Assume each input presentation include a random perturbation $\epsilon_{\boldsymbol{W}} \sim \mathcal{N}(\boldsymbol{\epsilon};\boldsymbol{0},\eta\boldsymbol{I})$ in the network weight of a standard l-layer MLP. Let the perturbed model be $\hat{y}_{\boldsymbol{\epsilon_W}}(\boldsymbol{x})$. With injection o noise, objective function becomes

$$\tilde{J}_{\boldsymbol{W}} = \mathbb{E}_{p(\boldsymbol{x},y,\boldsymbol{\epsilon_W})}\left[(\hat{y}_{\boldsymbol{\epsilon_W}}(\boldsymbol{x}) - y)^2\right] = \mathbb{E}_{p(\boldsymbol{x},y,\boldsymbol{\epsilon_W})}\left[\hat{y}^2_{\boldsymbol{\epsilon_W}}(\boldsymbol{x}) - 2y\hat{y}_{\boldsymbol{\epsilon_W}} + y^2\right]$$

For small η, the minimisation of J with added noise (with covariance $\eta\boldsymbol{I}$) is equivalent to minimisation of J with additional regularisation term $\eta\mathbb{E}_{p(\boldsymbol{x}),y}[\|\nabla_{\boldsymbol{W}}\hat{y}(\boldsymbol{x})\|^2]$. This encourages parameters to go to regions of parameter space where small perturbations of weights have relatively small influence on output.

Remark 4.3.9. *Label Smoothing*
Datasets may have some mistake in y labels. Regularise a model based on softmax with k output values by replacing hard 0 and 1 classification targets with targets of $\frac{\epsilon}{k-1}$ and $1-\epsilon$ respectively, assuming that the training set label y is correct with probability $1-\epsilon$. Cross-entropy can then be used on these soft targets.

Remark 4.3.10. *Semi-Supervised Learning*
Unlabelled examples $P(\boldsymbol{x})$ and $P(\boldsymbol{x}, \boldsymbol{y})$ are used to estimate $P(\boldsymbol{y} \mid \boldsymbol{x})$ or predict $\boldsymbol{y}$ from $\boldsymbol{x}$.
Construct a generative model of either $P(\boldsymbol{x})$ or $P(\boldsymbol{x}, \boldsymbol{y})$ which shares parameters with a discriminative model of $P(\boldsymbol{y} \mid \boldsymbol{x})$. The supervised criterion $\log P(\boldsymbol{y} \mid \boldsymbol{x})$ may then be traded with the unsupervised or generative one ($\log P(\boldsymbol{x})$ or $\log P(\boldsymbol{x}, \boldsymbol{y})$). The generative criterion then expresses a particular form of prior belief about the solution to the supervised learning problem (structure of $P(\boldsymbol{x})$ is connected to structure of $P(\boldsymbol{y} \mid \boldsymbol{x})$ such that is is captured by the shared parametrisation).

Remark 4.3.11. *Multi-Task Learning*
Improve generalisations by pooling examples (soft constraints imposed on parameters) from several tasks.
Different supervised tasks (predicting $\boldsymbol{y}^{(i)}$ given $\boldsymbol{x}$) share the same input $\boldsymbol{x}$, as well as some intermediate-level representation $\boldsymbol{h}^{(\text{shared})}$ capturing a common pool of factors. Model can be divided in two parts:

i. Task-specific parameters
ii. Generic parameters, shared across all the tasks

Improved generalisation and associated error bounds can be achieved due to shared parameters, but this only happens if some assumptions about statistical relationship between different tasks are valid.

Remark 4.3.12. *Early Stopping*
To obtain a model with better validation set error by returning to the parameter setting at the point in time with lowest validation set error. Model validation is done in parallel to model training.
Early stopping has the effect of restricting the optimisation procedure to a relatively small volume of parameter space in neighbourhood of initial parameter value $\boldsymbol{\theta}_o$. Assume τ optimisation steps are taken with learning rate ϵ. The product $\epsilon\tau$ is a measure of effective capacity. Assuming gradient is bounded, restricting $\epsilon\tau$ limits the volume of parameter space reachable from $\boldsymbol{\theta}_o$, hence $\epsilon\tau$ behaves as the reciprocal of weight decay coefficients.

Algorithm 2: Early Stopping Algorithm

Require:
Number of steps n between evaluations
Patience p, the number of times to observe worsening validation error before stopping
Initial parameters $\boldsymbol{\theta}_o$

$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_o, \quad i \leftarrow 0, \quad j \leftarrow 0, \quad v \leftarrow \infty, \quad \boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}, \quad i^* \leftarrow i$
while $j < p$ **do**
  Update $\boldsymbol{\theta}$ by running training algorithm for n steps
  $i \leftarrow i + n, \quad v' \leftarrow \text{ValidationSetError}(\boldsymbol{\theta})$
  if v' < v **then**
    $j \leftarrow 0, \quad \boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}, \quad i^* \leftarrow i, \quad v \leftarrow v'$
  end if
  $j \leftarrow j + 1$
end while
Best parameters are $\boldsymbol{\theta}^*$, best number of training steps is i^*

Remark 4.3.13. *Parameter Tying and Parameter Sharing*
Used to express prior knowledge on suitable values of model parameters (precise values are not known, but there should be some dependencies between parameters). To regularise parameters to be close, may use:

i. Parameter norm penalty: assume model A with parameters $\boldsymbol{w}^{(A)}$ and model B with parameters $\boldsymbol{w}^{(B)}$. Two models map input to two different but related outputs $\hat{y}^{(A)} = f(\boldsymbol{w}^{(A)}, \boldsymbol{x})$ and $\hat{y}^{(B)} = g(\boldsymbol{w}^{(B)}, \boldsymbol{x})$. The model parameters $w_i^{(A)}$ should be close to $w_i^{(B)}$ $\forall i$. Using regularisation, a parameter norm penalty may be imposed, such as using L^2 penalty to get $\Omega(\boldsymbol{w}^{(A)}, \boldsymbol{w}^{(B)}) = \left\| \boldsymbol{w}^{(A)} - \boldsymbol{w}^{(B)} \right\|_2^2$.
ii. Parameter sharing: using constraints. Only a subset of the parameters (input set) need to be stored in memory. May lead to significant reduction in memory footprint of the model (i.e., in CCNs)

Remark 4.3.14. *Sparse Representations*
Representation sparsity is where elements of the representation are zero, accomplished by same sort of mechanisms used in parameter regularisation.

Given a linear regression $\boldsymbol{y} = \boldsymbol{B}\boldsymbol{h}$, where $\boldsymbol{y} \in \mathbb{R}^m, \boldsymbol{B} \in \mathbb{R}^{m\times n}, \boldsymbol{h} \in \mathbb{R}^n$, and $\boldsymbol{h}$ is the representation of data $\boldsymbol{x}$, add to loss function J a norm penalty on the representation ($\Omega(\boldsymbol{h})$)):

$$\tilde{J}(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) = J(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) + \alpha\Omega(\boldsymbol{h})$$

where $\alpha \in [0, \infty)$ weights relative contribution of norm penalty, with larger values of α for more regularisation. L^1 penalty on elements of representation induces sparsity, $\Omega(\boldsymbol{h}) = \|\boldsymbol{h}\|_1 = \sum_i |h_i|$. For representations with elements constrained to lie on unit interval, may use Student-t prior or KL divergence penalties.
Other approaches such as hard constraint on activation values may also work. *Orthogonal matching pursuit* encodes an input $\boldsymbol{x}$ with representation $\boldsymbol{h}$ that solves the constrained optimisation problem

$$\arg \min_{\boldsymbol{h}, \|\boldsymbol{h}\|_0 < k} \|\boldsymbol{x} - \boldsymbol{W}\boldsymbol{h}\|^2$$

where $\|\boldsymbol{h}\|_0$ is the number of non-zero entries of $\boldsymbol{h}$. If $\boldsymbol{W}$ is constrained to be orthogonal, this can be solved efficiently with OMP-k method, where k is number of non-zero features allowed.

Remark 4.3.15. *Bootstrap Aggregating (Bagging)*
Construct k different datasets, each with same number of examples as original dataset, but constructed by sampling with replacement from original dataset. Each dataset may miss some examples but also contain duplicate examples. Each model i is then trained on the dataset i. Differences in random initialisation, random selection of mini-batches, differences in hyperparameters, or different outcomes of non-deterministic implementation of neural networks are enough to allow each model to make partially independent errors.

Remark 4.3.16. *Boosting*
Construct k different datasets from original dataset. Using ith dataset, train ith weak learner. Test the ith weak learner with validation dataset, and each datapoint with wrong prediction is sent to $(i+1)$th dataset. The $(i+1)$th dataset is then used to train the $(i+1)$th weak learner. Continue until reaching the total number of subsets for the total prediction.

Remark 4.3.17. *Dropout*
Dropout trains the ensemble consisting of all sub-networks that can be formed by removing non-output units from the base network by multiplying its output value by zero. To train a neural network with dropout, use minibatch-based learning algorithms (i.e., Stochastic Gradient Descent). Each time an example is loaded into minibatch, randomly (and independently) sample a different binary mask $\boldsymbol{\mu}$ to apply to all of input and hidden units in the network, where the probability of sampling a mask value of one is the dropout rate.
Let $J(\boldsymbol{\theta}, \boldsymbol{\mu})$ be cost of model defined by parameters $\boldsymbol{\theta}$ and mask $\boldsymbol{\mu}$. Dropout training is minimising $\mathbb{E}_{\boldsymbol{\mu}} J(\boldsymbol{\theta}, \boldsymbol{\mu})$. Each model inherits different subset of parameters from parent neural network, and a tiny fraction of possible sub-networks are trained for a single step. Parameter sharing allows sub-networks to arrive at good parameter settings. Each sub-network given by $\boldsymbol{\mu}$ defines a probability distribution $p(y \mid \boldsymbol{x}, \boldsymbol{\mu})$. The arithmetic mean over all masks is then given by

$$\sum_{\boldsymbol{\mu}} p(\boldsymbol{\mu}) p(y \mid \boldsymbol{x}, \boldsymbol{\mu})$$

where $p(\boldsymbol{\mu})$ is the distribution used to sample $\boldsymbol{\mu}$ during training.
Using dropout as a regularisation technique reduces effective capacity of a model. To offset this, increase the size of the model. Typically optimal validation error is much lower when using dropout, but this comes at cost of larger model and more iterations of training.
For very large datasets, costs of using dropout and larger models may outweigh benefits of regularisation. For sparse datasets (extremely few labelled training examples available), Bayesian neural networks outperform dropout. When additional unlabelled data is available, unsupervised learning may gain advantage over dropout.

Remark 4.3.18. *Adversarial Training*
Adversarial example may be constructed by optimisation to search for an input $\boldsymbol{x}'$ near $\boldsymbol{x}$ such that the model output is very different at $\boldsymbol{x}'$.
The primary cause is excessive linearity in some neural networks. Changing input by ϵ causes linear function with weights $\boldsymbol{w}$ to change by as much as $\epsilon \|\boldsymbol{w}\|_1$, which can be very large if $\boldsymbol{w}$ is high-dimensional.
Adversarial examples assists in semi-supervised learning. At point $\boldsymbol{x}$ that is not associated with a label, model assigns label $\hat{y}$ (may not be true label). Seek adversarial example $\boldsymbol{x}'$ that causes classifier to output label $y' \neq \hat{y}$. Classifier may then be trained to assign same label to $\boldsymbol{x}$ and $\boldsymbol{x}'$, learning function robust to small changes.

Remark 4.3.19. *Tangent Distance Algorithm*
Non-parametric nearest-neighbour algorithm, where examples on the same manifold share the same category. The nearest-neighbour distance between points $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ is the distance between manifolds M_1 and M_2. Computationally cheap alternative is to approximate M_i by its tangent plane at $\boldsymbol{x}_i$ and measure the distance between two tangents, or between a tangent plane and a point.

Remark 4.3.20. *Tangent Propagation Algorithm*
Trains neural net classifier with extra penalty to make each output $f(\boldsymbol{x})$ of neural net locally invariant to factors of variation, which correspond to movement along manifold which examples from same class concentrate. Local invariance is achieved by requiring $\nabla_{\boldsymbol{x}} f(\boldsymbol{x})$ to be orthogonal to known manifold tangent vectors $\boldsymbol{v}^{(i)}$ at $\boldsymbol{x}$, or equivalently the directional derivative of f at $\boldsymbol{x}$ in directions $\boldsymbol{v}^{(i)}$ be small by adding regularisation penalty:

$$\Omega(f) = \sum_i \left(\left(\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) \right)^T \boldsymbol{v}^{(i)} \right)^2$$

Tangent vectors are derived a priori from effect of transformations (i.e., translation, rotation, scaling).
Note this regularises model to resist infinitesimal perturbation, and the infinitesimal approach poses difficulties for models based on rectified linear units.
Data augmentation is the non-infinitesimal version of Tangent Propagation Algorithm, where network is explicitly trained to correctly classify distinct inputs created from more than infinitesimal transformation.

Remark 4.3.21. *Double Back-propagation Algorithm*
Algorithm regularises Jacobian to be small. Require model to be invariant to all directions of change in input so long as the change is small. Adversarial training is the non-infinitesimal version of Double Back-propagation Algorithm, where inputs were found near original inputs and model is trained to produce same input.

Remark 4.3.22. *Manifold Tangent Classifier*
Eliminates need to know tangent vectors a priori. Auto-encoders may estimate manifold tangent vectors, and the classifier makes use of this technique to avoid requiring user-specified tangent vectors. The steps are:

i. Use auto-encoder to learn manifold structure by unsupervised learning
ii. Use these tangents to regularise a neural net classifier as in Tangent Propagation Algorithm.

4.3.1 Optimisation for Deep Learning

Remark 4.3.23. *Optimisation*
A cost function $J(\boldsymbol{\theta})$ is reduced to improve some performance measure P.
Optimisation algorithms include some specialisation on specific structure of machine learning objective functions.
Cost function is average over training set,

$$J(\boldsymbol{\theta}) = \mathbb{E}_{(\boldsymbol{x},y)\sim\hat{p}_{\text{data}}} L(f(\boldsymbol{x};\boldsymbol{\theta}), y)$$

where L is per-example loss function, $f(\boldsymbol{x};\boldsymbol{\theta})$ is predicted output given input $\boldsymbol{x}$, $\hat{p}_{\text{data}}$ is empirical distribution.
For supervised learning, y is target output.
The goal is to usually to minimise objective function where expectation is taken on data generating distribution p_{data} rather than over the finite training set:

$$J^*(\boldsymbol{\theta}) = \mathbb{E}_{(\boldsymbol{x},y)\sim p_{\text{data}}} L(f(\boldsymbol{x};\boldsymbol{\theta}), y)$$

Note that machine learning algorithm usually minimises a surrogate loss function, and halts when convergence criterion based on early stopping is satisfied.

Definition 4.3.24. *Empirical Risk Minimisation*
To minimise expected loss on training set, where $\hat{p}(\boldsymbol{x}, y)$ is the empirical distribution. The empirical risk is

$$\mathbb{E}_{\boldsymbol{x},y\sim\hat{p}_{\text{data}}(\boldsymbol{x},y)}[L(f(\boldsymbol{x},\boldsymbol{\theta}), y)] = \frac{1}{m}\sum_{i=1}^{m} L(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)})$$

where m is number of training samples. Note that empirical risk minimisation is prone to overfitting, as models with high capacity may memorise the training set.

Remark 4.3.25. *Batch and Minibatch Algorithms*
Objective algorithms for machine learning computes each update to the parameters based on expected value of cost function using only a subset of terms of full cost function.
For maximum likelihood estimation problems, note that

$$\boldsymbol{\theta}_{\text{ML}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{m} \log p_{\text{model}}(\boldsymbol{x}^{(i)}, y^{(i)}; \boldsymbol{\theta})$$
$$J(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y\sim\hat{p}_{\text{data}}} \log p_{\text{model}}(\boldsymbol{x}, y; \boldsymbol{\theta})$$

The gradient is then

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y\sim\hat{p}_{\text{data}}} \nabla_{\boldsymbol{\theta}} \log\log p_{\text{model}}(\boldsymbol{x}, y; \boldsymbol{\theta})$$

If the optimisation algorithm uses the entire training set, then it is a batch gradient method.
If a single sample is used at a time, then this is a stochastic/online gradient method.
For algorithms that uses more than one but less than all samples, then this is a minibatch gradient method.

Remark 4.3.26. *Challenges: Ill-Conditioning of Hessian Matrix H*
May cause stochastic gradient descent to be stuck where very small step increases the cost function.
Note that a gradient descent step of $-\boldsymbol{\epsilon g}$ will add

$$\frac{1}{2}\epsilon^2 \boldsymbol{g}^T \boldsymbol{H}\boldsymbol{g} - \epsilon \boldsymbol{g}^T \boldsymbol{g}$$

to the cost. This will be a problem when $\frac{1}{2}\epsilon^2 \boldsymbol{g}^T \boldsymbol{H}\boldsymbol{g} > \epsilon \boldsymbol{g}^T \boldsymbol{g}$.
Monitor squared gradient norm $\boldsymbol{g}^T\boldsymbol{g}$ and $\boldsymbol{g}^T\boldsymbol{H}\boldsymbol{g}$ term. When ill-conditioning happens, gradient norm does not shrink significantly but $\boldsymbol{g}^T\boldsymbol{H}\boldsymbol{g}$ term grows more than order of magnitude, and learning rate shrinks.

Remark 4.3.27. *Challenges: Local Minimums*
Neural networks and models with multiple parametrised latent variables have multiple local minima due to *model identifiability* problem. Model is identifiable if a sufficiently large training set can rule out all but one setting of model parameters. Models with latent variables are not identifiable as equivalent models are obtained by exchanging latent variables with each other (*weight space symmetry*). If there are m layers with n units each, then there are $n!^m$ ways to arrange the hidden units, making the model unidentifiable.
There may be an extremely large or uncountably infinite amount of local minima in a cost function. If the local minima have a high cost compared to global minimum, then this is problematic.

Remark 4.3.28. *Challenges: Plateaus, Saddle Points, Flat Regions*
For a function $f : \mathbb{R}^n \to \mathbb{R}$, expected ratio of saddle points to local minima grows exponentially with n. Hessian matrix at a saddle point has a mixture of positive and negative eigenvalues.
Degenerate regions with flat gradient may correspond to high value of objective function.

Remark 4.3.29. *Challenges: Cliffs and Exploding Gradients*
On extremely steep cliff structure, gradient update step may move parameters extremely far, losing most of

the optimisation work done previously. The consequences may be mitigated using gradient clipping heuristics, which reduces the step size to be small enough.

Remark 4.3.30. *Challenges: Long-Term Dependencies*
Occurs when computational graph becomes very deep. Let a computational graph contain a path that repeatedly multiply a matrix $\boldsymbol{W}$. After t steps, this is $\boldsymbol{W}^t$. Let the eigen-decomposition of $\boldsymbol{W}$ be $\boldsymbol{W} = \boldsymbol{V}\text{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}$. Then we can see that $\boldsymbol{W}^t = (\boldsymbol{V}\text{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1})^t = \boldsymbol{V}\text{diag}(\lambda)^t\boldsymbol{V}^{-1}$.
There will be a vanishing and exploding gradient problem if gradients on the graph are scaled according to $\text{diag}(\boldsymbol{\lambda})^t$. Note that recurrent networks uses same $\boldsymbol{W}$ at each time step, hence may face the problem.

Remark 4.3.31. *Challenges: Inexact Gradients*
Objective function for minimisation is intractable due to noisy/biased estimate of these quantities; in this case its gradient is intractable as well. Choose surrogate loss function that is easier to approximate than true loss.

Remark 4.3.32. *Challenges: Poor Correspondence between Local and Global Structure*
As gradient descent algorithms make small local moves, optimisation on local downhill moves may fail if local surface does not point toward the global solution. In higher dimensional space, learning algorithms may incur high costs or long training time to circumnavigate the manifold. Hence to find good initial points for problems.

Remark 4.3.33. *Stochastic Gradient Descent (SGD)*
Crucial parameter is the learning rate, which may be changed over time; at iteration k this is ϵ_k.
The SGD gradient estimation introduces a source of noise that does not vanish even at minimum. A sufficient condition to guarantee converge of SGD is

$$\sum_{k=1}^{\infty} \epsilon_k = \infty, \quad \sum_{k=1}^{\infty} \epsilon_k^2 < \infty$$

In practice, it is common to decay learning rate linearly until iteration τ:

$$\epsilon_k = (1-\alpha)\epsilon_0 + \alpha\epsilon_\tau, \quad \alpha = \frac{k}{\tau}$$

After iteration τ, leave ϵ constant.
Compute time per update does not growth with number of training examples.

Algorithm 3: Stochastic Gradient Descent (SGD) Update at Training Iteration k

Require:
Learning rate ϵ_k, Initial parameter $\boldsymbol{\theta}$

while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient estimate $\hat{\boldsymbol{g}} \leftarrow +\frac{1}{m}\nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \epsilon\hat{\boldsymbol{g}}$
end while

Remark 4.3.34. *Momentum Algorithm*
Algorithm introduces variable $\boldsymbol{v}$ that is set to an exponentially decaying average of negative gradient.
Hyperparameter $\alpha \in [0, 1)$ determines how quickly the contributions of previous gradients exponentially decay.
The update rule is given by

$$\boldsymbol{v} \leftarrow \alpha\boldsymbol{v} - \epsilon\nabla_{\boldsymbol{\theta}} \left(\frac{1}{m} \sum_{i=1}^{m} L(\boldsymbol{f}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)}) \right)$$
$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$$

The larger α is relative to ϵ, the more previous gradients affect current direction.
If gradient is $\boldsymbol{g}$, it will accelerate in direction of $-\boldsymbol{g}$ until reaching terminal velocity where size of each step is

$$\frac{\epsilon \|\boldsymbol{g}\|}{1-\alpha}$$

Algorithm 4: Stochastic Gradient Descent (SGD) with Momentum

Require:
Learning rate ϵ_k, Momentum parameter α

while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$

Compute gradient estimate $\boldsymbol{g} \leftarrow +\frac{1}{m}\nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute velocity update: $\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \boldsymbol{g}$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$
end while

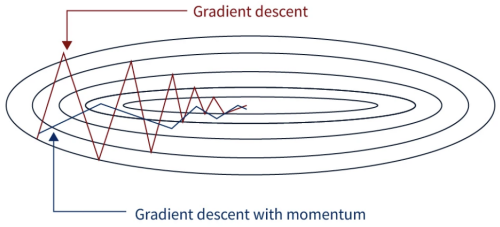


Figure 54: Stochastic Gradient Descent (SGD) Algorithm with and without Momentum

Remark 4.3.35. *Nesterov Momentum*
Inspired by Nesterov's accelerated gradient method, the update rules are:

$$\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \nabla_{\boldsymbol{\theta}} \left(\frac{1}{m} \sum_{i=1}^{m} L(\boldsymbol{f}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta} + \alpha \boldsymbol{v}), \boldsymbol{y}^{(i)}) \right)$$
$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$$

Note, gradient is evaluated after current velocity is applied.
Hence, Nesterov momentum is adding a correction factor to the standard method of momentum.

Algorithm 5: Stochastic Gradient Descent (SGD) with Nesterov Momentum

Require:
Learning rate ϵ_k, Momentum parameter α
Initial parameter $\boldsymbol{\theta}$, Initial velocity $\boldsymbol{v}$

while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Apply interim update: $\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta} + \alpha \boldsymbol{v}$
Compute gradient (at interim point): $\boldsymbol{g} \leftarrow \frac{1}{m}\nabla_{\tilde{\boldsymbol{\theta}}} \sum_i L(f(\boldsymbol{x}^{(i)}; \tilde{\boldsymbol{\theta}}), \boldsymbol{y}^{(i)})$
Compute velocity update: $\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \boldsymbol{g}$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$
end while

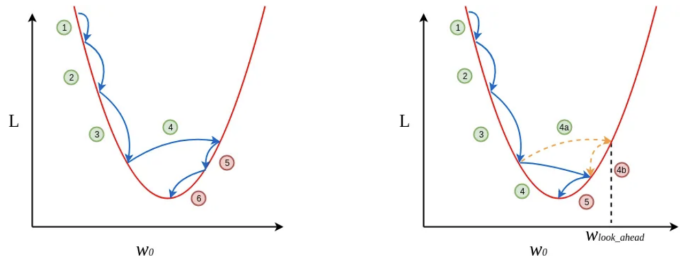


Figure 55: Momentum vs Nesterov Gradient Descent

Remark 4.3.36. *Parameter Initialisation Challenges*
Training deep neural networks is difficult as most algorithms are strongly affected by choice of initialisation.
Designing initialisation strategies is difficult as neural network optimisation is not yet well understood, as the nice properties from initialisation may not be preserved as training proceeds.
Some initial points may be beneficial for optimisation, but detrimental for generalisation.

Remark 4.3.37. *Parameter Initialisation Strategies*
Initial parameters need to 'break symmetry' between different units. May explicitly search for large set of basis functions mutually different from each other, but this may incur noticeable computational cost.

Random initialisation from high-entropy distribution over high-dimensional space is computationally cheaper. Larger initial weights will yield stronger symmetry breaking effect and avoid redundant units. However, if weights are set too large, may result in chaos (extreme sensitivity to small perturbations of input). Exploding gradient may be mitigated by gradient clipping. The bias (i.e., parameters encoding conditional variance of a prediction) for each unit is set to heuristically chosen constraints.

Remark 4.3.38. *Weight Initialisation Heuristics*
Initialising parameters of neural network $\boldsymbol{\theta}$ to $\boldsymbol{\theta}_0$ is similar to imposing a Gaussian prior $p(\boldsymbol{\theta})$ with mean $\boldsymbol{\theta}_0$, hence it make sense to choose $\boldsymbol{\theta}_0$ near 0. Initialisation of $\boldsymbol{\theta}_0$ to large values means prior specifies which units should interact with each other, and how they should interact.
For a fully connected neural network with m inputs and n outputs, weights may be initialised by:

i. Sampling from $U\left(-\frac{1}{\sqrt{m}}, \frac{1}{\sqrt{m}}\right)$
ii. Sampling from normalised initialisation $W_{i,j} \sim U\left(-\sqrt{\frac{6}{m+n}}, \sqrt{\frac{6}{m+n}}\right)$ (Glorot and Bengio). Compromise between goal of initialising all layers to have same activation variance and that of same gradient variance. Derived from assumption that network is chain of matrix multiplications with no nonlinearities.
iii. Sampling from random orthogonal matrices, with scaling or gain factor g that accounts for nonlinearity at each layer, with specific values for different types of nonlinear activation functions. Assumes neural network is sequence of matrix multiplications without nonlinearities. Guarantees the total number of training iterations required to reach convergence is independent of depth.
Increasing scaling factor g pushes network toward regime where activations increase in norm in forward propagation, and gradients increase in norm in backward propagation.
For scaling rules that set all of initial weights to have same standard deviation, individual weight becomes extremely small when layers become large.
iv. Sparse initialisation (Martens), where each unit is initialised to have exactly k non-zero weight. Keep total number of input to unit independent from number of inputs m without making magnitude of individual weight elements shrink with m.
Imposes very strong prior on weights that are chosen to have large Gaussian values.

If computational resources allow, treat initial scale of weights for each layer as a hyper-parameter, and use hyper-parameter search algorithm. Choice of dense or sparse initialisation may also be a hyper-parameter.

Remark 4.3.39. *Bias Initialisation Heuristics*
Setting bias to zero is compatible with most weight initialisation schemes. The few other situations are:

i. If bias is for output unit, initialise bias to obtain right marginal statistics of the output. Assume initial weights are small enough that output of unit is determined only by bias. The bias will be inverse of activation function applied to marginal statistics of the output in training set.
If output is a highly skewed distribution over classes with marginal probability of class i given by element c_i of some vector $\boldsymbol{c}$, then bias vector $\boldsymbol{b}$ is such that $\text{softmax}(\boldsymbol{b}) = \boldsymbol{c}$. The models have output layers that should resemble input data $\boldsymbol{x}$, hence bias should match marginal distribution over $\boldsymbol{x}$.
ii. If bias is to be set to avoid causing too much saturation at initialisation (i.e., with ReLU). Not compatible with weight initialisation schemes that do not expect strong input form bias (i.e., with random walk init).
iii. If a unit controls whether other units are able to participate in a function. There is unit with output u and another unit $h \in [0, 1]$, multiplied together to produce output uh. Then h is a gate that determines if $uh \approx u$ or $uh \approx 0$. Hence, to set bias for h such that $h \approx 1$ at initialisation, else u does not learn.

Remark 4.3.40. *Variance or Precision Parameter Initialisation Heuristics*
For linear regression with conditional variance estimate using model

$$p(y \mid \boldsymbol{x}) = \mathcal{N}(y \mid \boldsymbol{w}^T\boldsymbol{x} + b, 1/\beta)$$

where β is a precision parameter. The variance of precision parameters may be initialised to 1.
Another approach is to assume initial weights are close enough to zero, and bias may be set while ignoring effect of the weights, then set bias to produce the correct marginal mean of the output, and set variance parameters to the marginal variance of the output in the training set.

Remark 4.3.41. *Model Parameter Initialisation with Machine Learning*
Initialise supervised model with parameters learned by unsupervised model trained on same inputs.
Alternatively, perform supervised training on a related task, which may yield initialisation that offers faster convergence than random initialisation. Information on distribution is encoded into initial parameters of model.

Remark 4.3.42. *Adaptive Gradient (AdaGrad) Algorithm*
Algorithm individually adapts learning rates of all model parameters by scaling them inversely proportional to square root of sum of all of their historical squared values. Has greater progress in more gently sloped directions

of parameter space. On deep neural networks, accumulation of squared gradients from beginning of training can result in premature and excessive decrease in effective learning rate.

Algorithm 6: Adaptive Gradient (AdaGrad) Algorithm

Require:
Global Learning rate ϵ
Initial parameter $\boldsymbol{\theta}$
Small constant δ, perhaps 10^{-7} for numerical stability

Initialise gradient accumulation variable $\boldsymbol{r} = \mathbf{0}$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Accumulate squared gradient: $\boldsymbol{r} \leftarrow \boldsymbol{r} + \boldsymbol{g} \odot \boldsymbol{g}$
Compute update: $\Delta\boldsymbol{\theta} \leftarrow -\frac{\epsilon}{\delta + \sqrt{\boldsymbol{r}}} \odot \boldsymbol{g}$ (Division and square root applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Remark 4.3.43. *Root Mean Squared Propagation (RMSProp) Algorithm*
Modifies AdaGrad to perform better in non-convex setting by changing gradient accumulation into exponentially weighted moving average, which discards history from extreme past so it can converge rapidly after finding a convex bowl. Use of moving average introduces a new hyper-parameter ρ that controls length scale of the moving average. RMSProp is effective and practical for deep neural networks.

Algorithm 7: Root Mean Squared Propagation (RMSProp) Algorithm

Require:
Global Learning rate ϵ
Decay rate ρ
Initial parameter $\boldsymbol{\theta}$
Small constant δ, perhaps 10^{-6} to stabilise division by small numbers

Initialise accumulation variable $\boldsymbol{r} = \mathbf{0}$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Accumulate squared gradient: $\boldsymbol{r} \leftarrow \rho\boldsymbol{r} + (1-\rho)\boldsymbol{g} \odot \boldsymbol{g}$
Compute update: $\Delta\boldsymbol{\theta} \leftarrow -\frac{\epsilon}{\sqrt{\delta + \boldsymbol{r}}} \odot \boldsymbol{g}$ ($\frac{\epsilon}{\sqrt{\delta + \boldsymbol{r}}}$ applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Algorithm 8: Root Mean Squared Propagation (RMSProp) Algorithm

Require:
Global Learning rate ϵ
Decay rate ρ
Momentum coefficient α
Initial parameter $\boldsymbol{\theta}$
Initial velocity $\boldsymbol{v}$

Initialise accumulation variable $\boldsymbol{r} = \mathbf{0}$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute interim update $\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta} + \alpha\boldsymbol{v}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\tilde{\boldsymbol{\theta}}} \sum_i L(f(\boldsymbol{x}^{(i)}; \tilde{\boldsymbol{\theta}}), \boldsymbol{y}^{(i)})$
Accumulate gradient: $\boldsymbol{r} \leftarrow \rho\boldsymbol{r} + (1-\rho)\boldsymbol{g} \odot \boldsymbol{g}$
Compute velocity update: $\boldsymbol{v} \leftarrow \alpha\boldsymbol{v} - \frac{\epsilon}{\sqrt{\boldsymbol{r}}} \odot \boldsymbol{g}$ ($\frac{1}{\sqrt{\boldsymbol{r}}}$ applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$
end while

Remark 4.3.44. *Adaptive Moment Estimation (Adam) Algorithm*
Variant on combination of RMSProp and momentum. Note, momentum is incorporated directly as an estimate

of first order moment of gradient; bias corrections are made to estimates of both first-order moments and second-order moments to account for initialisation at origin.
Algorithm is fairly robust to choice of hyper-parameters.

Algorithm 9: Adaptive Moment Estimation (Adam) Algorithm

Require:
Step size ϵ (Suggested default: 0.001)
Exponential decay rates for moment estimates ρ_1, ρ_2 in $[0, 1)$ (Suggested defaults: 0.9 and 0.999)
Small constant δ used for numerical stabilisation (Suggested default: 10^{-8})
Initial parameters $\boldsymbol{\theta}$

Initialise 1st and 2nd momentum variables $\boldsymbol{s} = \mathbf{0}$, $\boldsymbol{r} = \mathbf{0}$
Initialise time step $t = 0$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Update time step $t \leftarrow t + 1$
Update biased first and second moment estimate: $\boldsymbol{s} \leftarrow \rho_1 \boldsymbol{s} + (1 - \rho_1)\boldsymbol{g}$ and $\boldsymbol{r} \leftarrow \rho_2 \boldsymbol{r} + (1 - \rho_2)\boldsymbol{g} \odot \boldsymbol{g}$
Correct bias in first and second moment: $\hat{\boldsymbol{s}} \leftarrow \frac{\boldsymbol{s}}{1-\rho_1^t}$ and $\hat{\boldsymbol{r}} \leftarrow \frac{\boldsymbol{r}}{1-\rho_2^t}$
Compute update: $\Delta\boldsymbol{\theta} = -\epsilon \frac{\hat{\boldsymbol{s}}}{\sqrt{\hat{\boldsymbol{r}}} + \delta}$ (Operations applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Remark 4.3.45. *Newton's Method*
Based on second-order Taylor's series approximating $J(\boldsymbol{\theta})$ near the point $\boldsymbol{\theta}_0$, ignoring higher derivatives

$$J(\boldsymbol{\theta}) \approx J(\boldsymbol{\theta}_0) + (\boldsymbol{\theta} - \boldsymbol{\theta}_0)^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_0) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)^T \boldsymbol{H} (\boldsymbol{\theta} - \boldsymbol{\theta}_0)$$

where $\boldsymbol{H}$ is the Hessian of J with respect to $\boldsymbol{\theta}$ evaluated at $\boldsymbol{\theta}_0$.
Solving for critical point will get the Newton parameter update rule

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 - \boldsymbol{H}^{-1} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_0)$$

Note that for deep learning, the surface of objective function is non-convex. Regularising the Hessian will solve many of the issues such as saddle points etc.

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 - [H(f(\boldsymbol{\theta}_0)) + \alpha \boldsymbol{I}]^{-1} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_0)$$

Due to requirement for inversion of the $k \times k$ matrix with time complexity $O(k^3)$ at every training iteration, only networks with very small number of parameters may be trained with this method.

Algorithm 10: Newton's Method with Objective $J(\boldsymbol{\theta}) = \frac{1}{m} \sum_{i=1}^{m} L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), y^{(i)})$

Require:
Initial parameter $\boldsymbol{\theta}_0$
Training set of m examples

while stopping criterion not met **do**
Compute gradient $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute Hessian: $\boldsymbol{H} \leftarrow \frac{1}{m} \nabla^2_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute Hessian Inverse: $\boldsymbol{H}^{-1}$
Compute update: $\Delta\boldsymbol{\theta} = -\boldsymbol{H}^{-1}\boldsymbol{g}$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Remark 4.3.46. *Conjugate Gradient Method*
Method effectively avoids computation of inversion Hessian by descending conjugate directions, which is a search direction conjugate to the previous line search direction (will not undo progress made in that direction).
Let current and previous search directions be $\boldsymbol{d}_t, \boldsymbol{d}_{t-1}$. At training iteration t, the next search direction $\boldsymbol{d}_t$ is

$$\boldsymbol{d}_t = \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) + \beta_t \boldsymbol{d}_{t-1}$$

note that β_t may be computed via Fletcher Reeves:

$$\beta_t = \frac{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t)^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t)}{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})}$$

or may be computed with Polak-Ribiere:

$$\beta_t = \frac{(\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1}))^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t)}{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})}$$

The non-linear conjugate gradient method is a variation which uses occasional resets where the method is restarted with line search along with the unaltered gradient.

Algorithm 11: Conjugate Gradient Method

Require:
Initial parameter $\boldsymbol{\theta}_0$
Training set of m examples
Initialise $\boldsymbol{\rho}_0 = 0$
Initialise $g_0 = 0$
Initialise $t = 1$

while stopping criterion not met **do**
Initialise the gradient $\boldsymbol{g}_t = \mathbf{0}$
Compute gradient $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute Polak-Ribiere metric: $\beta_t = \frac{(\boldsymbol{g}_t - \boldsymbol{g}_{t-1})^T \boldsymbol{g}_t}{\boldsymbol{g}_{t-1}^T \boldsymbol{g}_{t-1}}$
(Nonlinear conjugate gradient: optionally reset $\beta_t = 0$, i.e., if t is a multiple of some constant k)
Compute search direction: $\boldsymbol{\rho}_t = -\boldsymbol{g}_t + \beta_t \boldsymbol{\rho}_{t-1}$
Perform line search to find: $\epsilon^* = \arg\min_\epsilon \frac{1}{m} \sum_{i=1}^m L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}_t + \epsilon \boldsymbol{\rho}_t), \boldsymbol{y}^{(i)})$
(On truly quadratic cost function, analytically solve for ϵ^*)
Apply update: $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \epsilon^{\theta} \boldsymbol{\rho}_t$
Apply time step $t \leftarrow t + 1$
end while

Remark 4.3.47. *Nonlinear Conjugate Gradients*
Method of conjugate gradients is still applicable for non-linear objective function.
Conjugate directions are no longer assumed to remain at minimum of the objective for previous directions. Has occasional resets where method of conjugate gradients is restarted with line search along unaltered gradient.

Remark 4.3.48. *Broyden-Fletcher-Goldfarb-Shanno (BGFS) Algorithm*
Similar to conjugate gradient, but uses more direct approach to approximation of newton's update.
Method approximate the inverse of hessian $\boldsymbol{H}$ with matrix $\boldsymbol{M}_t$, iteratively refined by low rank updates to better approximate $\boldsymbol{H}^{-1}$. Direction of gradient descent is then determined by $\boldsymbol{\rho}_t = \boldsymbol{M}_t \boldsymbol{g}_t$. Line search is performed in this direction to determine step size ϵ^*. Final update of parameter is then $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \epsilon^* \boldsymbol{\rho}_t$.
Note, BGFS store matrix $\boldsymbol{M}$ which requires $O(n^2)$ memory, making it impractical for most modern models.

Remark 4.3.49. *Limited Memory BFGS (L-BFGS) Algorithm*
Memory costs of BFGS may be decreased by not storing complete inverse Hessian approximation $\boldsymbol{M}$. L-BFGS computes approximation $\boldsymbol{M}$ with same method as BFGS, but assumes at beginning that $\boldsymbol{M}^{(t-1)}$ is the identity matrix, rather than storing the approximation between steps.
If used with exact line searches, directions defined by L-BFGS are mutually conjugate, and this procedure remains well-behaved when minimum of line search is reached only approximately.
L-BFGS may be generalised to store some vectors used to update $\boldsymbol{M}$ at each time step, costing $O(n)$ per step.

Remark 4.3.50. *Meta-Algorithm: Batch Normalisation*
An update from optimisation algorithm may result in multiple-order effects, and in very deep networks, even higher-order interactions can be significant.
Batch normalisation provides an elegant way of re-parametrising any deep network, which reduces the problem of coordinating updates across multiple layers. Batch normalisation may be applied to any input/hidden layer.
Let $\boldsymbol{H}$ be minibatch of activations of layer to normalise, with activations for each example appearing in a row. To normalise $\boldsymbol{H}$, replace it with $\boldsymbol{H}' = \frac{\boldsymbol{H} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}$ where $\boldsymbol{\mu}$ is a vector of mean of each unit and σ is a vector of standard deviation of each unit. Then H_{ij} is normalised by subtracting μ_j and dividing σ_j. Rest of network then operates on $\boldsymbol{H}'$ just as in original network on $\boldsymbol{H}$.
At training, the mean and standard deviations are computed as

$$\boldsymbol{\mu} = \frac{1}{m} \sum_i \boldsymbol{H}_{i,:}, \quad \boldsymbol{\sigma} = \sqrt{\delta + \frac{1}{m} \sum_i (\boldsymbol{H} - \boldsymbol{\mu})_i^2}$$

where δ is a small positive value (i.e., 10^{-8}) to avoid undefined gradient of $\sqrt{z}$ at $z = 0$.
These are then back-propagated to normalise $\boldsymbol{H}$, hence gradient will never increase the standard deviation or

mean of h_i. Batch normalisation re-parametrises the model to make some units always standardised.
At test time, μ and σ may be replaced by running averages from training time. This allows model to be evaluated on a single example, without needing to use definitions of μ and $\boldsymbol{\sigma}$ that depends on entire minibatch.

Remark 4.3.51. *Batch Normalisation but Maintain Network Express Power*
Replace batch of hidden activations $\boldsymbol{H}$ with $\boldsymbol{\gamma H'} + \boldsymbol{\beta}$ rather than $\boldsymbol{H'}$, where $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are learned parameters to allow new variable to have any mean and standard deviation. With original $\boldsymbol{H'}$, mean of $\boldsymbol{H}$ was determined by complicated interaction between parameters in layers below $\boldsymbol{H}$. With $\boldsymbol{\gamma H'} + \boldsymbol{\beta}$, the mean is determined solely by $\boldsymbol{\beta}$, which is much easier to learn with gradient descent.
Most neural network layers are of form $\phi(\boldsymbol{XW} + \boldsymbol{b})$, where ϕ is some fixed nonlinear activation function. Then $\boldsymbol{XW} + \boldsymbol{b}$) should be replaced by normalised version of $\boldsymbol{XW}$, and bias term omitted as it becomes redundant with β parameter applied from batch normalisation re-parametrisation. Statistics of input are non-Gaussian and less amenable to standardisation by linear operations.

Remark 4.3.52. *Meta-Algorithm: Coordinate Descent*
Solving optimisation problems by minimising the objective function $f(\boldsymbol{x})$ with respect to a single variable x_i, and rapidly cycling through all variables. *Block coordinate descent* refers to minimising with respect to a subset of variables simultaneously.
Coordinate descent works when different variables can be clearly separated into groups that play relatively isolated roles, or when optimisation with respect to one group of variables is significantly more efficient than optimisation with respect to all of the variables.

Example 4.3.53. *Sparse Encoding with Coordinate Descent*
Consider the cost function

$$J(\boldsymbol{H}, \boldsymbol{W}) = \sum_{i,j} |H_{ij}| + \sum_{i,j} (\boldsymbol{X} - \boldsymbol{W}^T \boldsymbol{H})^2_{i,j}$$

To find weight matrix $\boldsymbol{W}$ that can linearly decode a matrix of activation values $\boldsymbol{H}$ to reconstruct training set $\boldsymbol{X}$. Most applications also include weight decay or constraint on norms of columns of $\boldsymbol{W}$ to prevent the pathological solution with extremely small $\boldsymbol{H}$ and large $\boldsymbol{W}$.
Note J is not convex, but inputs may be divided into the dictionary parameters $\boldsymbol{W}$ and the code representations $\boldsymbol{H}$. Minimising object function with respect to either set is a convex problem.

Remark 4.3.54. *Polyak Averaging*
Average several points in trajectory through parameter space visited by optimisation algorithm.
If t iterations of gradient descent visit points $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(t)}$, then the output is $\hat{\boldsymbol{\theta}}^{(t)} = \frac{1}{t} \sum_i \boldsymbol{\theta}^{(i)}$.
The optimisation algorithm may leap back and forth across a valley several times without visiting the minimum point, and the average of all points on either side of the valley should be close to minimum value.
This is developed solely for stochastic gradient descent (SGD) to stabilise the estimates (Granziol et al. (2020)).
For non-convex problems, an exponentially decaying running average is used:

$$\hat{\boldsymbol{\theta}}^{(t)} = \alpha \hat{\boldsymbol{\theta}}^{(t-1)} + (1 - \alpha) \boldsymbol{\theta}^{(t)}$$

Remark 4.3.55. *Supervised Pre-training*
Pre-training refers to training simple models on simple tasks before training the desired model on desired task. Greedy algorithms break problem into many components, then solve for optimal version of each component in isolation, which is must cheaper than algorithms that solve for best joint solution. Fine-tuning on joint optimisation algorithm searches for optimal solution to the full problem.
Greedy supervised training breaks supervised learning problems into other simpler supervised learning problems.
Transfer learning refers to pre-training a deep network on a set of tasks, then initialise a same-size network with first k layers of first net. The layers of second networks (upper layers initialised randomly) are then jointly trained to perform a different set of tasks with fewer training examples than for first set of tasks.
FitNets approach begins by training a network with low depth and high width, which becomes a teacher for a second network (the student), which is much deeper and thinner. Student network is trained to not only predict the output on original task, but also on value of middle layer of teacher network.

Remark 4.3.56. *Neural Network Design Choice: Linear Transformation*
Model units have generally moved towards using more linear functions, which allow model to have nice properties that make optimisation easier. The model's local gradient information corresponds reasonably well to moving towards a distant solution.

Remark 4.3.57. *Neural Network Design Choice: Linear Paths and Skip Connections*
Linear paths and skip connections between layers reduce length of shortest path from lower layer's parameters to the output, mitigating vanishing gradient problem.
Adding extra copies of output that are attached to intermediate hidden layers is an alternative to pre-training,

where these 'auxiliary heads' are trained to perform the same task as primary output at the top of the network to ensure that the lower layers receive a large gradient. After training, these may be discarded.

Remark 4.3.58. *Continuation Methods*
Family of strategies that can make optimisation easier by choosing initial points to ensure that local optimisation spends most of its time in well-behaved regions. Method constructs series of objective functions $\{J^{(0)}, \ldots, J^{(n)}\}$ which are increasingly difficult to minimise (well-behaved over less regions of $\boldsymbol{\theta}$ space) for the true cost function $J(\boldsymbol{\theta})$. The cost functions are designed such that a solution to one is a good initial point for the next.
Continuation methods aim to reach a global minimum despite many local minima. Easier cost functions are constructed through the blurring operation, which approximates

$$J^{(i)}(\boldsymbol{\theta}) = \mathbb{E}_{\theta' \sim \mathcal{N}(\boldsymbol{\theta}'; \boldsymbol{\theta}, \sigma^{(i)2})} J(\boldsymbol{\theta})'$$

via sampling. Some non-convex functions become approximately convex when blurred, but preserved enough information about local global minimum. However, approach might not work when

i. so many incremental cost functions are required that cost of entire procedure remains high;
ii. NP-hard optimisation problems remain NP-hard, even when continuation methods are applicable;
iii. function might not become convex, no matter how much it is blurred;
iv. function may become convex due to blurring, but minimum of this blurred function may track a local rather than global minimum of original cost function.

Remark 4.3.59. *Curriculum Learning*
Planning a learning process to begin by learning simple concepts, and progress to learning more complex topics that depend on these simpler concepts. Earlier objective functions $J^{(i)}$ are made easier by increasing the influence of simpler examples.
Stochastic curriculum is a variation of the approach where a random mix of easy and difficult examples is always presented to the learner, but average proportion of more difficult examples is gradually increased.

4.3.2 Convolutional Neural Networks

Definition 4.3.60. *Convolutional Neural Networks (CNNs)*
Neural network for processing data with a known, grid-like topology. Uses convolution instead of general matrix multiplication in at least one of their layers.

Definition 4.3.61. *Convolution*
The convolution of two functions f and g, denoted by $*$, is the integral of product of two functions after one is reflected about the y-axis and shifted.

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

The first argument $f(\tau)$ is the input, and second argument $g(t-\tau)$ is the kernel. The output is feature map.

Definition 4.3.62. *Discrete Convolution*
For complex-valued functions f and g defined on set of $\mathbb{Z}$ integers, the discrete convolution of f and g is

$$(f * g)[n] = \sum_{m=-\infty}^{\infty} f[m]g[n-m]$$

If function g_N is periodic with period N, then the convolution is also periodic and identical to

$$(f * g_N)[n] = \sum_{m=0}^{N-1} \left(\sum_{k=-\infty}^{\infty} f[m+kN] \right) g_N[n-m]$$

Note that convolution is commutative, as the kernel is flipped relative to the input (as m increases, the index into input increases, but index into kernel decreases), but this is not an important property of neural network implementation. Neural network libraries implement cross-correlation instead.

Definition 4.3.63. *Cross-Correlation*
A measure of similarity of two series as a function of the displacement of one relative to the other. Also known as *sliding inner or dot product*. This is defined as

$$(f \star g)(\tau) = \int_{-\infty}^{\infty} \overline{f(\tau)}g(t+\tau)dt$$

This is related to convolution by

$$[f(t) \star gt](t) = [\overline{f(-t)} * g(t)](t)$$

Remark 4.3.64. *Motivation: Sparse Interactions*
The kernel is smaller than input, but is able to detect meaningful features. Hence fewer parameters are to be stored, reducing memory requirements of the model. Units in deeper layers may indirectly interact with a larger portion of the input, hence efficiently describe complicated interactions between many variables.

Remark 4.3.65. *Motivation: Parameter Sharing*
Each member of kernel is used at every position of the input except at boundary, which depends on design decisions regarding the boundary. Only one set of parameter is required to be learned. The runtime of forward propagation is at $O(k \times n)$ for $k \lll m$ connections for each input, given m input and n outputs.

Remark 4.3.66. *Motivation: Equivariant Representations*
An *equivariant* function is such that if the input changes, the output changes in the same way.
Applying a transformation, then convolution, is equivalent to applying convolution, then the transformation.

Remark 4.3.67. *Typical CNN Layers*
The layers of a typical CNN are:

i. Convolution: layer performs several convolutions in parallel to produce set of linear activations
ii. Detector: each linear activation is run through a non-linear activation function
iii. Pooling: modify the output of the layer further with summary statistics of nearby outputs

Example 4.3.68. *Example Types of Pooling*

i. Max Pooling: reports maximum output for each region of the feature map
ii. Average Pooling: reports average for each region of the feature map
iii. L^2 Pooling: calculates Euclidean norm (L^2 norm) for each region of the feature map

Remark 4.3.69. *Pooling on Invariance*
Pooling makes the representation approximately invariant to small translations of input.

Use of pooling is analogous to adding an infinitely strong prior that the function the layer learns must be invariant to small translations. As pooling summarises responses over whole neighbourhood, fewer pooling units than detector units may be used. To handle inputs of varying size, the size of offset between pooling regions may be changed so that the classification layer always receives the same number of summary statistics.

Remark 4.3.70. *Multi-Channel Operations*
Not commutative unless each operation has the same number of output and input channels.
Assume there exists 4-D kernel tensor $\mathcal{K}$ with element $K_{i,j,k,l}$ giving connection between strength between a unit in channel i of the output and a unit in channel j of the input, with offset of k rows and l columns between the output unit and input unit. Assume input consists of observed data $\mathcal{V}$ with element $V_{i,j,k}$ giving the value of input unit within channel i at row j and column k. Assume output consists of $\mathcal{Z}$ in same format as $\mathcal{V}$. If $\mathbb{Z}$ is produced by convolving $\mathcal{K}$ across $\mathcal{V}$ without flipping $\mathcal{K}$, then

$$Z_{i,j,k} = \sum_{l,m,n} V_{l,j+m-1,k+n-1} K_{i,l,m,n}$$

where summation over l, m, n is over all values for which tensor indexing operations inside summation is valid.

Remark 4.3.71. *Downsampled Convolution*
To reduce computational cost by sampling only every s pixels (*stride*) in each direction of output, then

$$Z_{i,j,k} = c(\mathcal{K}, \mathcal{V}, s)_{i,j,k} = \sum_{l,m,n} [\mathcal{V}_{l,(j-1)\times s+m,(k-1)\times s+n} \mathcal{K}_{i,l,m,n}]$$

Remark 4.3.72. *Zero Padding*
Zero-padding is adding zeroes to end of input sequence $\mathcal{V}$ so that the total number of samples is equal to the next higher multiple of two.

i. Valid Convolution: no zero-padding used. If input image has width m and kernel has width k, the output will be width $m - k + 1$. Rate of shrinkage can be dramatic if kernel used are large.
ii. Same Convolution: just enough zero-padding is added to keep size of output equal to size of input. Border pixels may be underrepresented in the model.
iii. Full Convolution: zeroes are added for every pixel to be visited k times in each direction, resulting in an input image of width $m + k - 1$.

Remark 4.3.73. *Locally Connected Layers (Unshared Convolution)*
Similar operation to discrete convolution with small kernel, but without parameter sharing. Adjacency matrix of MLP is the same, but every connection has its own weight as specified by 6-D tensor $\mathcal{W}$, where the indices of $\mathcal{W}$ are the output channel i, the output row j, the output column k, the input channel l, the row offset within the the input m, and the column offset within the input n. The liner part of the layer is then

$$Z_{i,j,k} = \sum_{l,m,n} [\mathcal{V}_{l,j+m-1,k+n-1} w_{i,j,k,l,m,n}]$$

Locally connected layers are useful when it is known that each feature should be function of a small part of space but would not occur across all of space.

Remark 4.3.74. *Tiled Convolution*
Compromise between convolutional layer and locally connected layer. A set of kernels will be rotated through space. Immediate neighbouring locations will have different filters, but memory requirements for storing the parameters will increase only by a factor of the size of this set of kernels.
Let k be a 6-D tensor, where two of the dimensions correspond to different locations in output matrix. Output locations cycle through a set of t different choices of kernel stack in each direction.

$$Z_{i,j,k} = \sum_{l,m,n} \mathcal{V}_{l,j+m-1,k+n-1} \mathcal{K}_{i,l,m,n,j \bmod t+1,k \bmod t+1}$$

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